T2K 6.57 $\times 10^{20}$-POT Joint 3-Flavour Oscillation Analysis

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Abstract

The results of the *VALOR* joint 3-flavour oscillation analysis on the combined Run 1+2+3+4 dataset are reported in this technical note. This dataset corresponds to an integrated JPARC neutrino beam exposure of 6.57 $\times 10^{20}$ POT.

This analysis is an update of the work presented in Ref. [1] based on the *VALOR* ν_μ-disappearance analysis presented in Refs. [2], [3], [4] and [5], with improved methods and an enlarged dataset. Furthermore, this updated joint analysis also contains the correlations between the SK efficiencies for the single μ-like ring and the single e-like ring event samples that were not available in the previous analysis [1].

In this analysis, the atmospheric squared-mass splitting $|\Delta m^2|$ (the Fogli-Lisi convention is used, see main text for details), the mixing angles $sin^2\theta_{23}$, $sin^2\theta_{13}$ and the CP-invariance violating phase δ_{CP}, are determined jointly through a simultaneous oscillation fit to the reconstructed energy spectrum of the SuperK single μ-like ring and single e-like ring event samples. The oscillation fit was performed in a 3-flavour framework including matter effects in constant density matter. Official T2K 2013 analysis inputs were used. All 64 systematics considered in this analysis were allowed to float in the fit.

Assuming normal hierarchy (NH), the joint 3-flavour oscillation fit to the observed reconstructed energy spectrum of single μ-like ring and single e-like ring events of the combined Run 1+2+4 dataset, with all mixing angles constrained to the physical region, yields $|\Delta m^2_{32}|$ = 2.512 $\times 10^{-3}$ eV^2/c^4, $\sin^2\theta_{23}$ = 0.524, $\sin^2\theta_{13}$ = 0.0422 and δ_{CP} = 1.909 as the best-fit values (p-value = 0.921). The corresponding results for inverted hierarchy (IH) are $|\Delta m^2_{13}|$ = 2.488 $\times 10^{-3}$ eV^2/c^4, $\sin^2\theta_{23}$ = 0.523, $\sin^2\theta_{13}$ = 0.0491 and δ_{CP} = 1.005 (p-value = 0.952). Both fits used 2012 PDG values for $\sin^2 2\theta_{12}$ and the solar squared-mass splitting δm^2 [30].

Confidence regions are constructed using only the constant-$\Delta\chi^2$ method in the present technical note, alternative methods, such as Feldman-Cousins [56], will be evaluated in following versions. For NH, the 90% (68%) CL allowed region for the four oscillation parameters can be summarized as 2.320 $\times 10^{-3}$ (2.394$\times 10^{-3}$) $eV^2/c^4 < |\Delta m^2_{32}| <$ 2.697$\times 10^{-3}$ (2.623$\times 10^{-3}$) eV^2/c^4, 0.435 (0.465) $< sin^2\theta_{23} <$ 0.608 (0.581), 0.018 (0.021) $< sin^2\theta_{13} <$ 0.063 (0.055) and -π (-π) $< \delta_{CP} < \pi$ (π) For IH, the corresponding 90% (68%) CL allowed regions are 2.292 $\times 10^{-3}$ (2.370$\times 10^{-3}$) $eV^2/c^4 < |\Delta m^2_{13}| <$ 2.672 $\times 10^{-3}$ (2.605$\times 10^{-3}$) eV^2/c^4, 0.434 (0.458) $< sin^2\theta_{23} <$ 0.578 (0.596), 0.022 (0.028) $< sin^2\theta_{13} <$ 0.076 (0.064) and -π (-π) $< \delta_{CP} < \pi$ (π)

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1. Introduction

In this technical note, the results of a joint 3-flavour oscillation analysis performed by the *VALOR* group on the combined Run 1+2+3+4 (6.57×10^{20} POT) dataset are reported. This dataset was collected in the following run periods: in January - June 2010 (Run1, 0.323×10^{20} POT), in November 2010 - March 2011 (Run2, 1.108×10^{20} POT) and, following the recovery work from the damage of the 2011 Tohoku earthquake [23], in March - June 2012 (Run3, 1.579×10^{20} POT) and in October 2012 - 8th May 2013 (Run 4, 3.560×10^{20} POT).

This analysis is an update of the work presented in Ref. [1], based on the *VALOR* ν_μ-disappearance analysis presented in Refs. [2], [3], [4] and [5], with improved methods and an enlarged dataset. The baseline of this analysis has been explained in detail in the Run 1+2+3 technical note (Ref. [1]); a summary of the updates in the analysis inputs and structural changes concerning the fitting method, such as adding together SK efficiencies and FSI+SI systematics, are summarized in Sec.2. Among these changes, the most remarkable one is that this updated joint analysis contains the correlations between the SK efficiencies for the single μ-like ring and the single e-like ring event samples that were not available in the previous analysis for Run 1+2+3 dataset.

In the joint oscillation analysis described in this technical note, the atmospheric squared-mass splitting $|\Delta m^2|$, the mixing angles $\sin^2\theta_{23}$, $\sin^2\theta_{13}$ and the CP-violating phase δ_{CP}, are determined together through a simultaneous fit of the reconstructed energy spectra of both the SuperK single μ-like ring and single e-like ring event samples. In contrast to the stand-alone ν_e-appearance and ν_μ-disappearance fits, the joint 3-flavour oscillation fit presented in this work does not have to assume prior values for any of these four oscillation parameters and therefore the dependency between all of them is taken into account. The 12-sector (solar) oscillation parameters, $\sin^2\theta_{12}$ and $|\delta m^2|$ are kept fixed since their effect was demonstrated to be negligible in B.

This analysis is a continuation of the work of the *VALOR* ν_μ-disappearance analysis presented in Refs. [2], [3], [4] and [5]. The *VALOR* 3-flavour stand-alone ν_μ-disappearance analysis framework was extended to include a new 3-flavour ν_e-appearance analysis. Thus, the updated framework is able to perform both stand-alone ν_μ-disappearance and ν_e-appearance analyses and a complete 3-flavour joint oscillation analysis. A validation of the new ν_e-appearance analysis implemented, compared with the official results, is presented in Appendix A.

Neutrino oscillation probabilities are computed for this analysis in a 3-flavour framework including constant-density matter effects developed by the *VALOR* group. Details of the oscillation probability calculations, estimates of the numerical accuracy of such calculations and comparisons with other oscillation probability codes are given in Appendices A and B of Ref. [5]. The Fogli-Lisi convention is used for the atmospheric mass-squared splitting, as explained in Sec.3.

A total of 64 efficiency, energy scale, neutrino flux, neutrino cross-section, initial-state nuclear environment and final-state re-scattering systematics are considered for this analysis. These systematic parameters and their effects on the extrapolated single μ-like ring event and single e-like ring event reconstructed energy spectrum are described in Sec. 5.

As in previous analyses performed by the *VALOR* group, the oscillation fit was made using the likelihood-ratio method [58]. All 64 systematics considered in this analysis were allowed to float in the fit. The oscillation fitting method is presented in detail in Sec. 6, including the description of the chosen goodness-of-fit test, suitable for the Run 1+2+4 statistics and already tested for the Run 1+2+3 dataset in Ref. [1].

The fitter performance has been tested in detail using toy MC experiments and the results of these tests are presented Sec. C. Joint 3-flavour oscillation fits of fake datasets have been studied in detail and the results are presented in Sec. D. These fits were performed using the complete *VALOR* analysis with an integrated exposure of 6.57×10^{20} POT.

The results of the joint measurement of $|\Delta m^2|$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$ and δ_{CP}, and the systematic parameter pulls, using the Run 1+2+3+4 dataset are given in Sec. 7.1.1, and the goodness-of-fit tests are presented in Sec. 7.1.3. The best-fit single μ-like ring and single e-like ring event reconstructed energy spectra are shown in Sec. 7.1.1. In Sec. 7.1.2, profile $\Delta\chi^2$ distributions are presented for each of $|\Delta m^2|$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$, δ_{CP} and 68% and 90% confidence regions are presented in following four 2-dimensional oscillation parameter spaces: ($\sin^2\theta_{23}$, $|\Delta m^2|$), ($\sin^2\theta_{13}$, $|\Delta m^2|$), ($\sin^2\theta_{13}$, δ_{CP}) and ($\sin^2\theta_{23}$, $\sin^2\theta_{13}$). Comparisons of the Run 1+2+3+4 joint 3-flavour oscillation fit results in both normal and inverted hierarchies were made. In Sec. 7.2 results are presented for the Run1+2+3+4 dataset fit including the reactor constraint: profile $\Delta\chi^2$ distributions are presented for each of $|\Delta m^2|$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$, δ_{CP} and 68% and 90% confidence regions are presented in following six 2-dimensional oscillation parameter spaces: ($\sin^2\theta_{23}$, $|\Delta m^2|$), ($\sin^2\theta_{13}$, $|\Delta m^2|$), ($\sin^2\theta_{13}$, δ_{CP}), ($\sin^2\theta_{23}$, $\sin^2\theta_{13}$) ($\sin^2\theta_{23}$, δ_{CP}) and ($|\Delta m^2|$, δ_{CP}), comparison also results in both normal and inverted hierarchies.

Finally, a summary is given in Sec. 8.

2. Summary of updates with respect to Run1+2+3 joint 3-flavour oscillation analysis in TN154

The following changes constitute the updates that have been implemented in the present analysis compared with the Run1+2+3 joint 3-flavour oscillation analysis described in Ref. [1].

2.1. Changes in the analysis inputs

- The dataset has increased from an exposure of 3.01×10^{20} POT corresponding to Run 1+2+3 to an exposure of 6.57×10^{20} POT corresponding to Run 1+2+3+4, when adding Run4 with 3.56×10^{20} POT. This addition means a change in the observed number of single μ-like ring events from 58 for the Run 1+2+3 dataset to 120 for the Run 1+2+3+4 dataset, and the observed number of single e-like ring events from 11 for the Run 1+2+3 dataset to 28 for the Run 1+2+3+4 dataset.

- An updated SuperK MC based on NEUT has been used in this analysis, being the current SuperK software version 13a (previous version used in Ref. [1] was 11c). The JNUBEAM flux tuning version used for the creation of this new MC was *11btunedv3.2*. Consequently, MC templates, systematic response functions (splines) and systematic covariance matrices were redone with the updated SuperK MC.

- The new fiTQun (v3r1) cut for rejection of NC π^0 events has been used in the SuperK MC and data for single ring e-like sample [29].

- A new BANFF version is used in this analysis [7], [8]. The actual file used is *postfit_banff_2013_joint_v5.root*. The list of parameters included in the BANFF post-fit files varies depending on the different versions produced for each oscillation analysis. Table 8 in Sec. 5 presents a summary of the different lists of parameters and binning used in each oscillation analysis in the latest version. In Sec. 5 the best-fit values used for the BANFF parameters will be shown.

- As a consequence of the updated SuperK MC and the new BANFF version, the response functions (splines) necessary to evaluate some of the cross-section systematics had to be regenerated. The true energy binning used to create these splines is the BANFF flux binning for both samples. However, for the oscillated ν_e modes the true energy binning used is the same one used for ν_μ (11 bins) for the single μ-like ring event sample and the one used for ν_e (7 bins) for the single e-like ring event sample in order to match both stand-alone disappearance and appearance analyses respectively. The reconstructed energy binning used in the splines is different for the single ring μ-like and the single ring e-like samples. For the single ring μ-like sample the following reconstructed energy binning is used: an initial bin from 0 to 0.3 GeV followed by 50 MeV bins up to 1 GeV (14 bins), 100 MeV bins up to 3 GeV (20 bins), 3.0-3.5 GeV, 3.5-4.0 GeV, 4.0-5.0 GeV, 5.0-7.0 GeV, 7.0-10.0 GeV and 10.0-30.0 GeV (41 bins in total). For the single ring e-like sample a uniform reconstructed binning was used with 50 MeV bins from 0 to 1.25 GeV (25 bins).

- The covariance matrix used to evaluate the FSI+SI systematic errors was redone for both ν_μ-disappearance and ν_e-appearance analyses [6], [20]. Built on those covariance matrices, a new FSI+SI covariance matrix has been produced for the joint analysis including correlations between the single ring μ-like and the single ring e-like samples [24].

- Systematic errors due to Photo-Nuclear (PN) effects have been also considered in the ν_e-appearance analyses. Those errors were added to the FSI+SI ones in the corresponding covariance matrix for the single ring e-like sample.

- The SuperK detector error matrix was recreated for both ν_μ-disappearance and ν_e-appearance analyses [9],[10]. In addition, a new SuperK detector error matrix has been implemented including correlations between the single ring μ-like and the single ring e-like samples [25]. The analysis presented in this technical note was performed using the latest SuperK detector error matrix including correlations between the two samples. More details are given in Sec. 5.

- After discussion with the T2K-SK group for the ν_μ disappearance analysis, now SuperK events with reconstructed energy larger than 30 GeV are included in the 10-30 GeV bin.

- Following the recommendations agreed after discussion within the oscillation analysis and NIWG groups for the ν_μ disappearance analysis, the spectral function systematic is tweaked using a Gaussian distribution centered at 0 and bounded in the range [0,1] when generating toy MC experiments, and it is restricted in the fit to the range [0,1] with a Gaussian constraint (penalty term) that favours 0 (relativistic Fermi gas) with respect to 1 (spectral function) in the present analysis. The difference with respect to the previous analysis in [1] is that this parameter was previously thrown uniformly within the allowed region when generating toy MC experiments.

- The value of the oscillation parameter $sin^2\theta_{12}$ has been updated to be consistent with the official 2013 ν_e-appearance analyses. The value used in the present analysis is shown in Tab. 1.

- The value and error for $sin^2 2\theta_{13}$ from the reactor experiments to be used as a constraint has been updated. The new value used, coming from PDG2013, is $sin^2 2\theta_{13} = 0.095 \pm 0.010$ and it is a weighted average of the results of the three reactor experiments Daya Bay, RENO and Double Chooz [28].

- The SuperK reconstructed energy scale error has changed from 2.3% to 2.4%[12].

- The value of the CC $\nu/\bar{\nu}$ cross section normalisation systematic has changed from 40% to 20%

Parameter	Value
$\sin^2 2\theta_{12}$	0.8495
$\sin^2 \theta_{12}$	0.306
Δm^2_{21}	7.5×10^{-5} eV^2/c^4

Table 1: Values used for $\sin^2 2\theta_{12}$ and Δm^2_{21} in the joint oscillation analysis

2.2. Changes in the methodology

- Since the missing correlations between the single ring μ-like and the single ring e-like samples concerning the FSI+SI and SuperK detector systematic errors were calculated, it was possible for this analysis to combine the SuperK detector and FSI+SI errors like the rest of oscillation analyses do. Therefore the present analysis followed this prescription and used SK+FSI combined errors, reducing in this way the number of total systematic errors by 18.

- The oscillated $\nu_e \to \nu_\mu$, $\bar{\nu_e} \to \bar{\nu_\mu}$ and $\bar{\nu_\mu} \to \bar{\nu_e}$ templates were introduced. The way they were constructed, re-weighting the ν_μ SuperK MC with the ν_e flux for $\nu_e \to \nu_\mu$ for instance, will be explained in Sec. 4. The $\bar{\nu_e} \to \bar{\nu_\mu}$ templates were also considered, but finally not included since their contribution is negligible in both samples as it can be clearly seen in Tab. 3.

- After some discussion within the oscillation analysis group about using $\sin^2 2\theta_{13}$ or $\sin^2 \theta_{13}$ for the final results, the framework has been updated to fit $\sin^2 \theta_{23}$ but $\sin^2 2\theta_{13}$ as well. However, tests made using toy datasets or the final result for Run 1+2+3+4 dataset fit show no difference at all between results obtained fitting $\sin^2 2\theta_{13}$ or $\sin^2 \theta_{13}$ compared after the proper re-scale.

3. Choice of oscillation parameters

This analysis departs from choices which have been followed historically in the T2K stand-alone ν_e-appearance and ν_μ-disappearance analyses by changing the oscillation mixing parameters and the mass-squared splitting parameters. There is a strong motivation for this change in the context of a joint 3-flavour analysis, and it is outlined below.

Neutrino mixing parameters

In the stand-alone ν_μ-disappearance analysis, $\sin^2 2\theta_{23}$ was the variable typically measured since it appears in the first-order approximate formula for the ν_μ survival probability:

$$P(\nu_\mu \to \nu_\mu) \approx 1 - sin^2 2\theta_{23} sin^2 \left(\frac{\Delta m^2_{32} L}{4E_\nu} \right) \quad (1)$$

In the latest stand-alone ν_μ-disappearance analysis [6], however, it was found that the choice of octant (first octant with $\theta_{23} < \pi/4$ or second octant with $\theta_{23} > \pi/4$) had a significant effect in the confidence regions. Therefore, the latest T2K stand-alone ν_μ-disappearance analysis already switched to fit $\sin^2 \theta_{23}$ instead of $\sin^2 2\theta_{23}$.

Looking at the leading-order approximation of $P(\nu_\mu \to \nu_e)$, the 23-sector mixing parameter appears as $\sin^2 \theta_{23}$:

$$P(\nu_\mu \to \nu_e) \approx sin^2 2\theta_{13} sin^2 \theta_{23} sin^2 \left(\frac{\Delta m^2_{31} L}{4E_\nu} \right) \quad (2)$$

So in the context of the joint fit it makes even more sense to use $\sin^2 \theta_{23}$ since it is possible to lift the octant degeneracy by combining the ν_μ-disappearance and ν_e-appearance analyses, as shown graphically in Fig. 1.

Thus, the natural choice for the joint 3-flavour oscillation analysis to fit for $\sin^2 \theta_{ij}$ rather than $\sin^2 2\theta_{ij}$, and this is the choice that has been used in the VALOR joint analysis since the previous analysis described in [1].

Recently the question of fitting $\sin^2 2\theta_{13}$ instead of $\sin^2 \theta_{13}$ was pointed out within the oscillation analysis group. On the one hand, $\sin^2 2\theta_{13}$ appears in the leading term or the $P(\nu_\mu \to \nu_e)$ oscillation probability formula 2. But there $\cos^2 \theta_{13}$ appears in the approximate formula for $P(\nu_\mu \to \nu_\mu)$ oscillation probability:

$$
\begin{aligned}
P(\nu_\mu \to \nu_\mu) &\approx 1 - 4\sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right) c^2_{13} s^2_{23} \left[s^2_{12} c^2_{23} + c^2_{12} s^2_{13} s^2_{23} + c^2_{12} c^2_{23} + s^2_{12} s^2_{13} s^2_{23}\right] \\
&= 1 - 4\sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right) c^2_{13} s^2_{23} \left(s^2_{12} + c^2_{12}\right) \left[c^2_{23} + s^2_{13} s^2_{23}\right] \\
&= 1 - 4\sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right) c^2_{13} s^2_{23} \left[c^2_{23} + s^2_{13} s^2_{23}\right] \\
&= 1 - \sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right) \left[\cos^2\theta_{13} \sin^2 2\theta_{23} + \sin^4\theta_{23} \sin^2 2\theta_{13}\right]
\end{aligned}
\tag{3}
$$

Therefore, although $P(\nu_\mu \to \nu_e)$ is approximately linear with $\sin^2 2\theta_{13}$, $P(\nu_\mu \to \nu_\mu)$ is not, so it is not clear that switching to $\sin^2 2\theta_{13}$ would be the natural choice for the joint fit. Furthermore, after several checks, it was found that there was no visible difference in the results obtained fitting $\sin^2 2\theta_{13}$ directly or fitting $\sin^2\theta_{13}$ and making the appropriate re-scaling that could indicate an easier fit when using $\sin^2 2\theta_{13}$. Examples are given in Sec. D and 7.

On the other hand, in the context of global fits like the ones in Ref. [31] or [33], the variable used is $\sin^2\theta_{13}$, as it was in this variable that MINOS reported its joint fit's results in NNN 2013 [34].

For these reasons, results of the VALOR joint oscillation analysis will be given mainly presented in $\sin^2\theta_{13}$. In any case, the VALOR framework has been updated to fit the three options:

- ($\sin^2\theta_{23}$,$\sin^2\theta_{13}$)
- ($\sin^2 2\theta_{23}$,$\sin^2 2\theta_{13}$)
- ($\sin^2\theta_{23}$,$\sin^2 2\theta_{13}$)

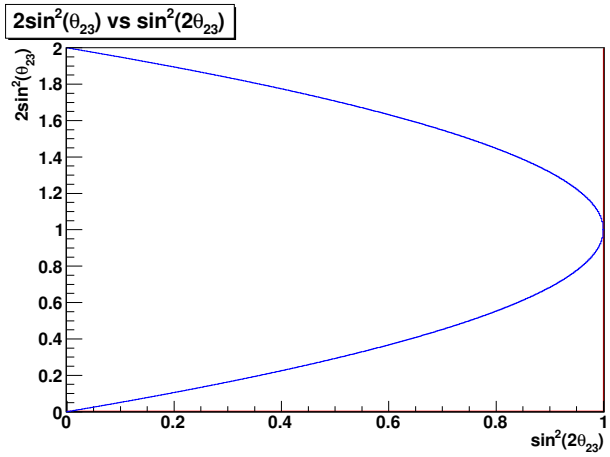


Figure 1: $\sin^2\theta_{23}$ (appears in leading-order term of P($\nu_\mu \to \nu_e$) vs $\sin^2 2\theta_{23}$(appears in leading-order term of P($\nu_\mu \to \nu_\mu$).

Mass-squared splitting parameters

Previous T2K analyses have used the following mass-squared splitting parameters:

$$\left(|\Delta m^2_{32}|, \Delta m^2_{21}\right)$$

which are defined as $\Delta m^2_{ij} = m^2_i - m^2_j$ according to the PDG notation [30]. The mass hierarchy is set through the sign of Δm^2_{32} i.e. $\Delta m^2_{32} = |\Delta m^2_{32}|$ in the normal hierarchy and $\Delta m^2_{32} = -|\Delta m^2_{32}|$ in the inverted hierarchy. The remaining mass-squared splitting is defined as $\Delta m^2_{31} = \Delta m^2_{32} + \Delta m^2_{21}$ for both hierarchies.

In the normal hierarchy (NH) $m^2_1 < m^2_2 << m^2_3$, while in the inverted hierarchy (IH) $m^2_3 << m^2_1 < m^2_2$, i.e. $|\Delta m^2_{31}|$ is the largest mass-squared splitting in the NH, whereas $|\Delta m^2_{32}|$ is the largest in the IH. This means that it is not possible to compare the two hierarchies if the atmospheric mass-squared difference is input as $|\Delta m^2_{32}|$. This has been pointed out by several authors and different schemes are typically employed in global oscillation fits to circumvent this issue. In the latest

stand-alone ν_μ-disappearance analysis [6], for instance, results comparing both mass hierarchies were presented in such a way that the Y axis showing $|\Delta m^2|$ had a different meaning for normal hierarchy, Δm^2_{32}, and inverted hierarchy, Δm^2_{13}.

The VALOR joint oscillation analysis uses the scheme defined by Fogli and Lisi in [31] [32] in which $\left(|\Delta m^2_{32}|, \Delta m^2_{21}\right)$ are replaced by:

$$\left(|\Delta m^2|, \delta m^2\right)$$

where $\Delta m^2 = \left(m^2_3 - \frac{m^2_1+m^2_2}{2}\right) = \left(\frac{\Delta m^2_{31}+\Delta m^2_{32}}{2}\right)$ and $\delta m^2 = \Delta m^2_{21}$. $|\Delta m^2|$ is the "atmospheric" squared-mass splitting, which has the same absolute value for both mass hierarchies. δm^2 is the "solar" squared-mass splitting which is known to be positive. When the mass-squared splittings are input as $\left(\Delta m^2, \delta m^2\right)$, the mass hierarchy is set through the sign of $|\Delta m^2|$, and Δm^2_{32} and Δm^2_{31} are calculated as

$$\Delta m^2_{32} = \Delta m^2 - \frac{\delta m^2}{2}$$
$$\Delta m^2_{31} = \Delta m^2 + \frac{\delta m^2}{2}$$

for both mass hierarchies. The ability to input the mass-squared differences as $\left(|\Delta m^2_{32}|, \Delta m^2_{21}\right)$ has also been retained for backward compatibility and to allow various validation checks.

The effects of changing the convention for the inputs of the mass-squared splitting parameters on P($\nu_\mu \rightarrow \nu_\mu$) and P($\nu_\mu \rightarrow \nu_e$) are shown in Fig. 2.

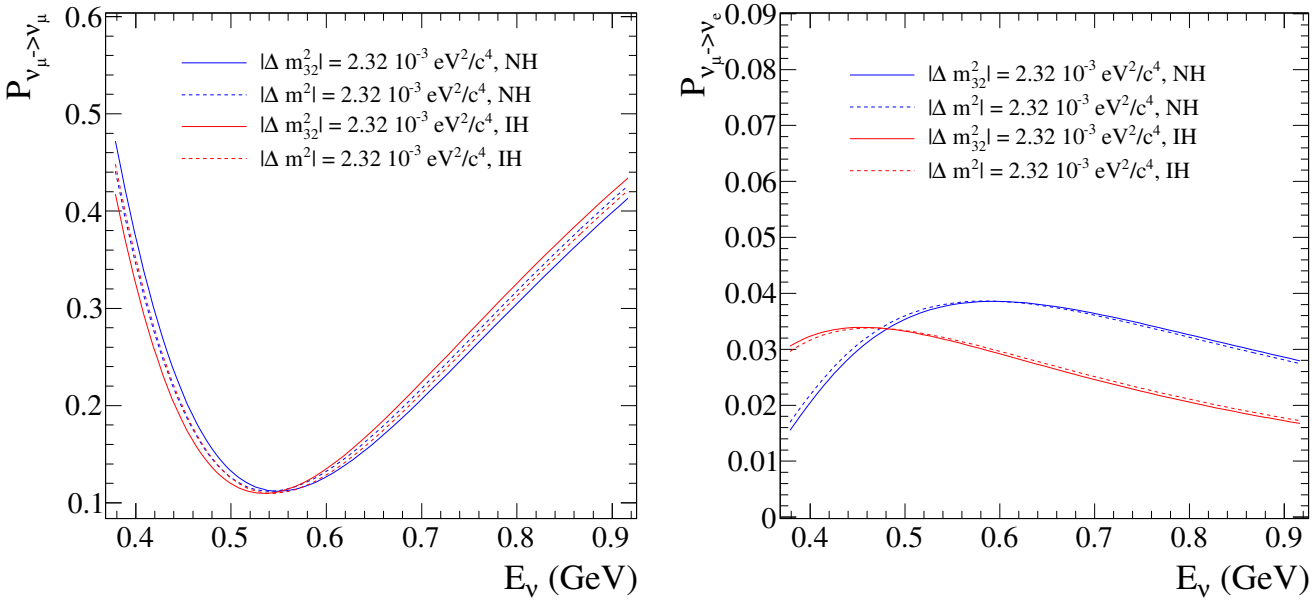


Figure 2: Effects of changing the convention for the inputs of the mass-squared splitting parameters on P($\nu_\mu \rightarrow \nu_\mu$) (left) and P($\nu_\mu \rightarrow \nu_e$) (right): $|\Delta m^2_{32}|$ = 2.32×10^{-3} eV^2/c^4 (solid lines) and $|\Delta m^2|$ = 2.32×10^{-3} eV^2/c^4 (dashed lines) for the normal hierarchy (blue) and inverted hierarchy (red). The other oscillation parameters are $\sin^2 2\theta_{12}$ = 0.857, δm^2 = 7.5×10^{-5} eV^2/c^4, $\sin^2 2\theta_{13}$ = 0.098 (2012 PDG values), δ_{CP} = 0 and $\sin^2 2\theta_{23}$ = 0.9.

Section H of the previous VALOR joint oscillation analysis technical note [1] illustrates the difference in the comparison between results in the two mass hierarchies when using the Fogli and Lisi convention $\left(|\Delta m^2|, \delta m^2\right)$ and the previous convention $\left(|\Delta m^2_{32}|, \Delta m^2_{21}\right)$.

Despite of the new convention for mass-squared splitting parameters discussed here, that will be use for the main part of this technical note, for the final results in Sec. 7 it was requested by the oscillation analysis conveners to produce the plots using $|\Delta m^2| = |\Delta m^2_{32}|(NH), |\Delta m^2_{13}|(IH)$ when comparing results in the two hierarchies and Δm^2_{32} for both mass hierarchies (with opposite signs) for comparisons with MINOS results.

Treatment of solar (12-sector) oscillation parameters

A study has been made in which $\sin^2 2\theta_{12}$ and Δm^2_{21} were allowed to float in the fit of the single μ-like ring Super K sample, and the results are shown in Appendix B and Appendix F of [1]. It was found that when these parameters were allowed to float

freely, the fitter returned extreme values of these parameter because there is no sensitivity in the single μ-like ring sample nor the single e-like ring sample to constrain them, being those extreme values far away from the values established by different experiments. When they were allowed to float including the constraint known from well established measurements instead, there was no effect in the results obtained (details of these tests are given in Appendix B). For this reason, these parameters were fixed throughout this analysis to the values in table 1.

4. Predictions of single μ-like ring and single e-like ring event reconstructed energy spectra in SuperK

4.1. Methodology

In this analysis, the oscillation parameters are determined by fitting the reconstructed energy (E_r) spectra of the single μ-like ring and single e-like ring events observed at SuperK with the predicted spectra of each sample. The predicted number $N_{SK;s,r}$ of events in the rth reconstructed energy bin in the SuperK sample s (single μ-like ring or single e-like ring) is computed as follows:

$$N^{exp}_{SK;s,r} = \sum_{m}\sum_{t}\sum_{r\prime} P_{m,t} \cdot T_{s,r,r\prime,f_{\Delta E},s} \cdot S_{s,m,t,r\prime,\mathbf{f}} \cdot N^{MC}_{SK;s,m,r\prime,t} \quad (4)$$

In Eq. 4, $N^{MC}_{SK;s,m,r\prime,t}$ is the input SuperK Monte Carlo (MC) template containing the number of events in the sample s (single μ-like ring or single e-like ring), with true reaction mode m, and found in the true energy bin t and in the reconstructed energy bin $r\prime$. Notice that $r\prime$ index indicates a SuperK reconstructed energy bin in absence of a shift in reconstructed energy scale and the index r indicates a SuperK reconstructed energy bin after a shift in the reconstructed energy scale is applied. The construction of the nominal MC templates is presented in detail in Sec. 4.2. The nominal MC templates are reweighted using both flux-tuning weights based on NA61 hadron-production data (described in Sec.4.3) and flux and cross-section tuning weights based on ND280 data (described in Sec.4.4).

The term $S_{s,m,t,r\prime,\vec{f}}$ in Eq. 4 is an overall, multiplicative, systematic error factor depending on the sample s, the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$ and a vector of nuisance (systematic) parameters $\vec{f}$. A complete description of the nuisance parameters considered in this analysis is given in Sec. 5 of this technical note. $T_{r;r\prime;f_{\Delta E}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to the uncertainty in the SuperK reconstructed energy scale, expressed here in terms of the nuisance parameter $f_{\Delta E}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied in the true energy bin t of the SuperK MC template which corresponds to mode m; application of the term $P_{m;t}$ is discussed in Sec. 4.5.

4.2. Construction of nominal SuperK Monte Carlo templates

The nominal SuperK single μ-like ring (single e-like ring) MC templates $N^{MC}_{SK;s,m,r\prime,t}$, indicating s if the sample is single μ-like ring or single e-like ring, are constructed by applying the corresponding selection cuts for the single μ-like ring (single e-like ring) events to the official SuperK MC files. Details of the input official MC files are given in Sec. 4.2.1. The calculation of the normalization of each MC file is described in Sec. 4.2.2. The official single μ-like ring (single e-like ring) selection cuts are listed in Sec. 4.2.3. In Sec. 4.2.4, the list of MC templates used in this analysis is presented and the true and reconstructed energy binning of the templates is discussed in Sec. 4.2.5.

4.2.1. Input Monte Carlo files

This analysis uses the official 2013a SuperK MC files, based on NEUT and generated using the 11btunedv3.2 JNUBEAM flux. There are five official MC files: ν_μ,$\bar{\nu_\mu}$, ν_e and $\bar{\nu_e}$ selections generated using the nominal JNUBEAM flux without oscillations, and a fifth file of ν_e interactions generated with the ν_μ flux ($\nu_\mu \rightarrow \nu_e$). The latest selection will be denoted the "oscillated ν_e" sample (it corresponds to the sample called "signal-ν_e" in the ν_e-appearance analyses). The addition of the other interaction channels ($\bar{\nu_\mu} \rightarrow \bar{\nu_e}$, $\nu_e \rightarrow \nu_\mu$ and $\bar{\nu_e} \rightarrow \bar{\nu_\mu}$) has been studied for this analysis. In the Run1+2+3+4 stand-alone disappearance analysis [6], the templates for the channel $\nu_e \rightarrow \nu_\mu$ were already included. In this joint analysis, the other two, $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$ and $\bar{\nu_e} \rightarrow \bar{\nu_\mu}$ were also considered. Since separate official SuperK files are not available for these extra channels, they were emulated by reweighting the SuperK MC sample of the final flavour with the flux corresponding to the initial flavour: for instance, templates for the $\nu_e \rightarrow \nu_\mu$ channel were obtained reweighting the ν_μ SuperK MC selection with the ν_e flux. The contribution of the extra templates at typical oscillation parameters can be found in Tab. 3. From this table it is clear that the contribution from the channel $\bar{\nu_e} \rightarrow \bar{\nu_\mu}$ is negligible in both the single μ-like ring and the single e-like ring samples, and therefore it was decided to keep only the new templates for $\nu_e \rightarrow \nu_\mu$, whose contribution is not negligible in the single μ-like ring sample, and for $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$, whose contribution is not negligible in the single e-like ring sample.

All samples were generated including the neutrino flux estimates up to 30 GeV.

Sample	POT normalization
ν_μ	4.47598 $\times 10^{23}$
$\bar{\nu}_\mu$	6.57643 $\times 10^{24}$
ν_e	2.23833 $\times 10^{25}$
$\bar{\nu}_e$	1.06961 $\times 10^{26}$
Oscillated ν_e	2.14913 $\times 10^{23}$

Table 2: Normalization of input SuperK MC neutrino species.

4.2.2. *Normalization of Monte Carlo samples*

The normalization (integrated exposure in terms of POT) of each event selection (ν_μ,$\bar{\nu_\mu}$, ν_e, $\bar{\nu_e}$ and oscillated ν_e) is calculated from the number of events with a MC truth interaction vertex within the 22.5 kt fiducial volume, defined by:

$$N = \int dSdIdE \cdot \frac{d^3\Phi_{SK}}{dSdIdE_\nu} \cdot \sigma_{H_2O}(E_\nu) \cdot \frac{N_A}{A} \cdot \rho \cdot L \tag{5}$$

where $d^3\Phi_{SK}/dSdIdE$ is the number of flux particles for the given neutrino species per neutrino energy bin dE_ν, per unit area dS and per POT, $\sigma_{H_2O}(E_\nu)$ is the total interaction cross section in water for the given neutrino species, I is the beam intensity in terms of POT, N_A is Avogadro's number, A is the mass number for water, ρ is the water density and L is the neutrino path length in the water volume. The derived POT normalization of each MC file is shown in Tab. 2. This normalization depends only on the neutrino specie, being the same for each neutrino specie (ν_μ,$\bar{\nu_\mu}$, ν_e, $\bar{\nu_e}$ or oscillated ν_e) for the two event samples (single μ-like ring or single e-like ring). After being generated from a MC selection corresponding to a calculated integrated beam exposure I, each MC template is normalized to the integrated beam exposure of the Run 1+2+3+4 dataset (6.57×10^{20} POT) by scaling the bin contents of the template with 6.57×10^{20}/I.

4.2.3. *single μ-like ring and single e-like ring event selection cuts*

The SuperK MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e, oscillated ν_μ and oscillated $\bar{\nu}_e$ neutrino species passing the standard SuperK selection cuts for single μ-like ring events or single e-like ring events. The following cuts are the ones for the single μ-like ring events:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts (`evclass == 1`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`wall > 200`),
- the visible energy in the event is above 30 MeV (`evis > 30`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is μ-like (`ip[0] == 3`),
- the reconstructed muon momentum is above 200 MeV (`amomm[0] > 200`), and
- there is at most 1 decay e^- in the event (`nmue <= 1`).

For the single e-like ring events, the cuts are the following ones:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts (`evclass == 1`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`wall > 200`),
- the visible energy in the event is above 100 MeV (`evis > 100`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is e-like (`ip[0] == 2`),
- number of detected decay e^- in the event is zero (`nmue == 0`),
- reconstructed neutrino energy is less than 1250 MeV ($E_\nu^{rec} < 1250$) and

- fiTQun π^0 rejection cut (`ln` $(L_{\pi^0}/L_e < 175 - 0.875 \times m_{\gamma\gamma})$.

Notice that the first four cuts in the single μ-like ring selection are common for both single μ-like ring and single e-like ring events, being the first three ones grouped under the name FCFV (full contained in fiducial volume) cut. A study was made to remove the cut in the reconstructed neutrino energy in the single e-like ring event selection in [20], but it was finally decided not to include the high energy tail events ($E_\nu^{rec} > 1250$ MeV) for this Run1+2+3+4 analysis. More details about the new fiTQun algorithm to reject π^0 are given in [29].

4.2.4. List of Monte Carlo templates

For each SuperK MC sample, a number of different MC templates is constructed corresponding to different true reaction modes.

The template granularity depends on the type of oscillation analysis and the specific systematic parameters considered in the analysis. For the *VALOR* stand-alone disappearance analyses, different number of templates were used depending on the analysis: 12 MC templates were used for the 2010 Run 1+2 2-flavour disappearance analysis [3], 15 MC templates were used for the 2010 Run 1+2 3-flavour disappearance analysis [4], 28 MC templates were used for the 2012 Run 1+2+3 3-flavour disappearance analysis [5] and 32 MC templates were used for the 2013 Run 1+2+3+4 3-flavour disappearance analysis [6].

This joint oscillation analysis uses a different number of MC templates depending on the sample, single μ-like ring or single e-like ring, due to the different separation of NC modes.

For the single μ-like ring sample, the following 36 MC templates are used:

- ν_μ CCQE
- ν_μ CC1π,
- ν_μ CC coherent,
- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e, \nu_\tau$) NC1$\pi^\pm$,
- ν_μ (and $\rightarrow \nu_e, \nu_\tau$) NC other,
- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e, \bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e, \bar{\nu}_\tau$) NC other,
- ν_e CCQE,
- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu, \nu_\tau$) NC1$\pi^\pm$,
- ν_e (and $\rightarrow \nu_\mu, \nu_\tau$) NC other,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu, \bar{\nu}_\tau$) NC1$\pi^\pm$,

- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu, \bar{\nu}_\tau$) NC other,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1π,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC coherent,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC other,
- oscillated $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$ CCQE,
- oscillated $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$ CC1π,
- oscillated $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$ CC coherent,
- oscillated $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$ CC other,
- oscillated $\nu_e \rightarrow \nu_\mu$ CCQE,
- oscillated $\nu_e \rightarrow \nu_\mu$ CC1π,
- oscillated $\nu_e \rightarrow \nu_\mu$ CC coherent, and
- oscillated $\nu_e \rightarrow \nu_\mu$ CC other.

For the single e-like ring sample, the following 40 MC templates are used:

- ν_μ CCQE
- ν_μ CC1π,
- ν_μ CC coherent,
- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e, \nu_\tau$) NC1$\pi^0$,
- ν_μ (and $\rightarrow \nu_e, \nu_\tau$) NC coherent,
- ν_μ (and $\rightarrow \nu_e, \nu_\tau$) NC other,
- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e, \bar{\nu}_\tau$) NC1$\pi^0$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e, \bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e, \bar{\nu}_\tau$) NC other,
- ν_e CCQE,
- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu, \nu_\tau$) NC1$\pi^0$,
- ν_e (and $\rightarrow \nu_\mu, \nu_\tau$) NC coherent,

- ν_e (and $\rightarrow \nu_\mu, \nu_\tau$) NC other,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu, \bar{\nu}_\tau$) NC1$\pi^0$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu, \bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu, \bar{\nu}_\tau$) NC other,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1π,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC coherent,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC other,
- oscillated $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$ CCQE,
- oscillated $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$ CC1π,
- oscillated $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$ CC coherent,
- oscillated $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$ CC other,
- oscillated $\nu_e \rightarrow \nu_\mu$ CCQE,
- oscillated $\nu_e \rightarrow \nu_\mu$ CC1π,
- oscillated $\nu_e \rightarrow \nu_\mu$ CC coherent, and
- oscillated $\nu_e \rightarrow \nu_\mu$ CC other.

4.2.5. Binning of Monte Carlo templates

Each of the MC templates previously described in Sec. 4.2.4 for the single μ-like ring sample has 6132 2-dimensional bins (= 84 true energy bins $\times$ 73 reconstructed energy bins), and each of the MC templates for the single e-like ring sample has 2100 2-dimensional bins (= 84 true energy bins $\times$ 25 reconstructed energy bins). The 84 true energy bins, common for both the single μ-like ring and the single e-like ring samples, are arranged as follows:

- 6 50-MeV bins from 0-0.3 GeV,
- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV.

For the single μ-like ring sample the 73 reconstructed energy bins are the following:

- 60 50-MeV bins from 0-3 GeV,

- 4 250-MeV bins from 3-4 GeV,
- 4 500-MeV bins from 4-6 GeV,
- 4 1000-MeV bins from 6-10 GeV and
- 1 bin from 10-30 GeV.

while for the single e-like ring sample the 25 reconstructed energy bins used are uniform 50-MeV bins from 0 to 1.25 GeV.

The same reconstructed neutrino energy binning is used for the fit of the single μ-like ring reconstructed energy spectrum (73 reconstructed energy bins) and for the single e-like ring reconstructed energy spectrum (25 reconstructed energy bins).

A study was performed to decide the optimum binning scheme balancing the needs for accuracy and speed. Details are given in Appendix I of Ref. [5].

The binning described here is unchanged with respect to the Run 1+2+3 analysis, but, after discussion with T2K-SK, events with reconstructed energy $>$30 GeV are now included in the 10-30 GeV bin.

4.3. Applying flux tuning based on NA61 data

In this analysis, as in the *VALOR* stand-alone disappearance analysis in [6], the nominal MC templates, which were constructed using a MC event sample generated with the 11a flux, are reweighted to the 11bv3.2 flux which was tuned on NA61 hadron production measurements [13][14]. The reweighted MC templates are denoted the *NA61-tuned* MC templates in this document. The applied weights were obtained from [15] and are shown as a function of the true neutrino energy in Figs. 1-4 in Ref. [5]. The effect of this flux tune on the predicted reconstructed energy spectrum of single μ-like ring events and single e-like ring events is discussed in Sec. 4.6

4.4. Applying flux and cross-section (BANFF) tuning based on ND280 data

In this analysis, the BANFF flux and cross-section parameter tuning is used to reweight the NA61-tuned MC templates. The MC templates obtained after this second reweighting will be denoted the *BANFF-tuned* MC templates in this document. Updated BANFF weights were used in this analysis with respect to the Run 1+2+3 one reported in [1]- For the 2013 analyses, the division made to the ND280 CC events was updated, such that now those events are divided into the following samples: CC0π-like, CC1π-like and CC other. For the 2012 analyses (Run 1+2+3 analyses), the division of ND280 CC events was into the two samples: CCQE-like and CCnonQE. In the tracker, the ionisation energy loss in the TPC and/or FGD can be used to identify a track as a pion. Also the presence of a Michel electron in the FGD1 serves to identify a pion track. Using this information, CC0π-like and CC1π-like samples are selected such as they have exactly 0 and 1 track consistent with a π^+ respectively. If more than 1 track is consistent with a π^+, the event is included in the CC other sample. Events with at least one negative track or with a track consistent with an electron are also included in the CC other sample. More details about the selection are given in [7].

The fundamental update on the BANFF fit with respect to the previous 2012 analysis is the use of these three ND280 samples (CC0π-like, CC1π-like and CC other) to find the best-fit values of the flux normalization parameters and the cross-section parameters and their correlations [8]. The postfit values of these parameters (taken from version 5 of the 2013 Run 1+2+3+4 BANFF fit in `postfit_banff_2013_disappearance_v5.root`) are used in turn to calculate the nominal predicted single μ-like ring and single e-like ring reconstructed energy spectra in SuperK. A list of BANFF parameters and their best-fit values is shown in Tab. 9.

The effect of the BANFF tune on the predicted reconstructed energy spectra of single μ-like ring and single e-like ring events is discussed in Sec. 4.6

4.5. Applying neutrino oscillations

Oscillation probabilities are computed in a 3-flavour framework including matter effects in constant-density matter (assuming an Earth crust density of 2.6 g/cm^3). Custom oscillation probability calculation code was developed within the *VALOR* analysis framework. Details and estimates of the numerical accuracy of this code were first presented in [4], and is also shown in Appendix A of [5]. Comparisons between the *VALOR* oscillation probability calculation code and Prob3++ [62], which is typically used by the alternative T2K oscillation analyses, are shown in Appendix B of [5].

Oscillations are applied as a function of true energy to the ν_μ CCQE, ν_μ CC1π, ν_μ CC coherent, ν_μ CC other, $\bar{\nu}_\mu$ CCQE, $\bar{\nu}_\mu$ CC1π, $\bar{\nu}_\mu$ CC coherent, $\bar{\nu}_\mu$ CC other, ν_e CCQE, ν_e CC1π, ν_e CC coherent, ν_e CC other, $\bar{\nu}_e$ CCQE, $\bar{\nu}_e$ CC1π, $\bar{\nu}_e$ CC coherent, $\bar{\nu}_e$ CC other, oscillated $\nu_\mu \to \nu_e$ CCQE, oscillated $\nu_\mu \to \nu_e$ CC1π, oscillated $\nu_\mu \to \nu_e$ CC coherent, oscillated $\nu_\mu \to \nu_e$ CC other, oscillated $\bar{\nu_\mu} \to \bar{\nu_e}$ CCQE, oscillated $\bar{\nu_\mu} \to \bar{\nu_e}$ CC1π, oscillated $\bar{\nu_\mu} \to \bar{\nu_e}$ CC coherent, oscillated $\bar{\nu_\mu} \to \bar{\nu_e}$ CC other, oscillated $\nu_e \to \nu_\mu$ CCQE, oscillated $\nu_e \to \nu_\mu$ CC1π, oscillated $\nu_e \to \nu_\mu$ CC coherent and oscillated $\nu_e \to \nu_\mu$ CC other MC templates.

The MC templates constructed from the unoscillated MC samples are weighted with the corresponding survival probability: The ν_μ templates are weighted with P($\nu_\mu \to \nu_\mu$), the $\bar{\nu}_\mu$ templates with P($\bar{\nu}_\mu \to \bar{\nu}_\mu$), the ν_e templates with P($\nu_e \to \nu_e$) and the $\bar{\nu}_e$ templates with P($\bar{\nu}_e \to \bar{\nu}_e$). The MC templates made from the oscillated $\nu_\mu \to \nu_e$ MC sample are weighted with P($\nu_\mu \to \nu_e$) (the sample was generated assuming 100% of ν_μ transform to ν_e), the MC templates made from the oscillated $\bar{\nu_\mu} \to \bar{\nu_e}$ MC sample are weighted with P($\bar{\nu_\mu} \to \bar{\nu_e}$) and the MC templates made from the oscillated $\nu_e \to \nu_\mu$ MC sample are weighted with P($\nu_e \to \nu_\mu$).

In the standard 3-flavour oscillation framework, oscillations of ν_e and ν_μ can yield ν_τ, while oscillations of $\bar{\nu}_e$ and $\bar{\nu}_\mu$ can yield $\bar{\nu}_\tau$. In this analysis, contributions from ν_τ-CC and $\bar{\nu}_\tau$-CC were neglected as their energy threshold is around 3.5 GeV and their effect is negligible[1]. Accordingly, this analysis, similarly to the stand-alone disappearance analysis [6], uses no ν_τ-CC and $\bar{\nu}_\tau$-CC MC templates (and, in fact, no ν_τ and $\bar{\nu}_\tau$ SuperK MC samples are available).

As it has been emphasized in [6], the ν_μ NC MC templates for a mode m are proxies for the NC MC templates for the mixture of ν_e+ν_μ+ν_τ resulting from 3-flavour ν_μ oscillations for that mode m. The same applies to the $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$ NC MC templates. These NC MC templates are unchanged under standard 3-flavour oscillations.

Also it should be noted that there are no explicit NC MC templates made from the oscillated ν_e sample. If they were used, the oscillated ν_e (i.e. ν_e coming from ν_μ oscillations) would be double counted since they are already included in the ν_μ NC MC templates.

The same oscillation parameters are used for neutrinos and antineutrinos.

4.6. Predictions of nominal and tuned-MC single μ-like ring and single e-like ring spectra

In this section, the expected numbers of events and single μ-like ring and single e-like ring reconstructed energy spectra in SuperK are presented for various oscillation scenarios.

Estimates of the effect of the NA61 and BANFF tunings and of assumptions made in the 3-flavour neutrino oscillation framework used in this analysis are also presented.

Unless otherwise stated, the normal hierarchy is assumed. All plots were generated for an integrated exposure of 6.57×10^{20} POT.

The expected number of single μ-like ring events and single e-like ring events is shown in Fig. 3 as a function of the oscillation parameters $\sin^2\theta_{23}$ and Δm^2 with $\sin^2\theta_{13}$ = 0.0251 and δ_{CP} = 0 and in Fig. 4 as a function of the oscillation parameters $\sin^2\theta_{13}$ and δ_{CP} with $\sin^2\theta_{23}$ = 0.513 and $\Delta m^2 = 2.4375 \times 10^{-3} eV^2/c^4$. In both figures the solar parameters were fixed to the values in Tab. 1 and the normal hierarchy was assumed.

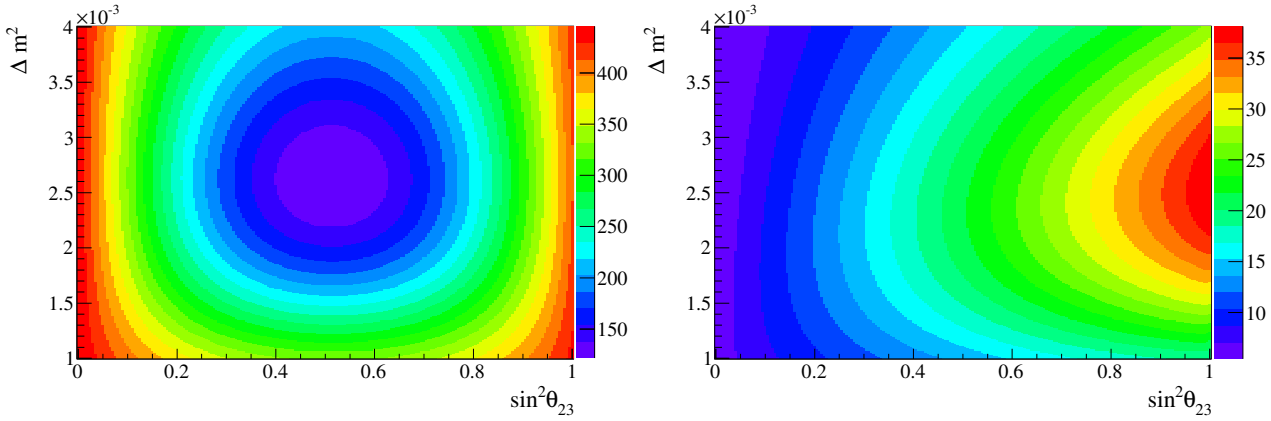


Figure 3: Predicted number of single μ-like ring events (left) and single e-like ring events (right) as a function of the oscillation parameters $\sin^2\theta_{23}$ and Δm^2 with $\sin^2\theta_{13}$ = 0.0251 and δ_{CP} = 0. The solar parameters were fixed to the values in Tab. 1 and the normal hierarchy was assumed.

Predicted single μ-like ring and single e-like ring SuperK reconstructed energy spectra are shown in Fig. 5 for an exposure of 6.57×10^{20} POT after BANFF tuning applied to the MC templates, for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2|$=2.4375×10^{-3} eV^2/c^4, $\sin^2\theta_{13}$=0.0251, δ_{CP}=0 and the values for the solar parameters in Tab. 1, and normal mass hierarchy assumed, and in Fig. 6 for no oscillations.

Table 3 shows the predicted number of single μ-like ring and single e-like ring events for an exposure of 6.57×10^{20} POT after BANFF tuning applied to the MC templates, for oscillations with $sin^2(\theta_{12}) = 0.306$, $\Delta m^2_{21} = 7.5 \times 10^{-5}$,

[1] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [5] that gives $\ll 1\%$ ν_τ contamination in the 1-ring μ-like SuperK sample. A SuperK MC study is required to determine whether the SuperK efficiency assumptions made there are valid.

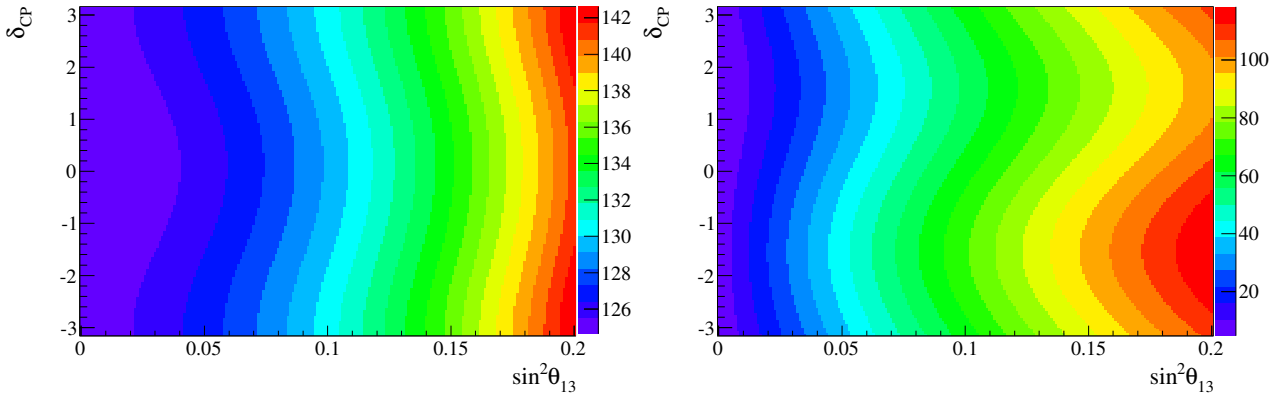


Figure 4: Predicted number of single μ-like ring events (left) and single e-like ring events (right) as a function of the oscillation parameters $\sin^2\theta_{13}$ and δ_{CP} with $\sin^2\theta_{23}$ = 0.513 and $\Delta m^2 = 2.4375 \times 10^{-3} eV^2/c^4$. The solar parameters were fixed to the values in Tab. 1 and the normal hierarchy was assumed.

$sin^2(\theta_{23}) = 0.5$, $|\Delta m^2| = 2.4375 \times 10^{-3}$, $sin^2(\theta_{13}) = 0.0251$,$\delta_{CP} = 0$ and normal hierarchy assumed. All the extra templates considered, as explained in Sec. 2 are included in this table, although for the analysis the templates corresponding to $\bar{\nu_e} \rightarrow \bar{\nu_\mu}$ were finally not used due to its negligible contribution to both samples as can be observed in Tab. 3.

A MC tune derived using NA61 data is applied to the nominal MC templates, and another tune derived using the ND280 data is applied to the NA61-tuned templates as described in Secs. 4.3 and 4.4. The effects of the NA61 and BANFF tunes on the total expected number of single μ-like ring events and single e-like ring events, and in the expected number for each interaction mode of each sample are shown in Tab. 4 and Tab. 5 respectively for single μ-like ring events and single e-like ring events with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2|$=2.4375$\times 10^{-3}$ eV^2/c^4, $\sin^2\theta_{13}$=0.0251, δ_{CP}=0 and the values for the solar parameters in Tab. 1, and normal mass hierarchy assumed. Tables 6 and 7 show the same effect for no oscillations.

Finally, the effects of the NA61 and BANFF tunes on the reconstructed energy spectrum of single μ-like ring events and single e-like ring events are shown in Figs. 7 and 8 respectively for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2|$=2.4375$\times 10^{-3}$ eV^2/c^4, $\sin^2\theta_{13}$=0.0251, δ_{CP}=0 and the values for the solar parameters in Tab. 1, and normal mass hierarchy assumed. For the case of no oscillations the effects are shown in Figs. 9 and 10. Figures 7 and 9 show the ratio of flux-tuned spectra to nominal spectra, and Figs. 8 and 10 show the ratio of BANFF-tuned spectra to flux-tuned spectra.

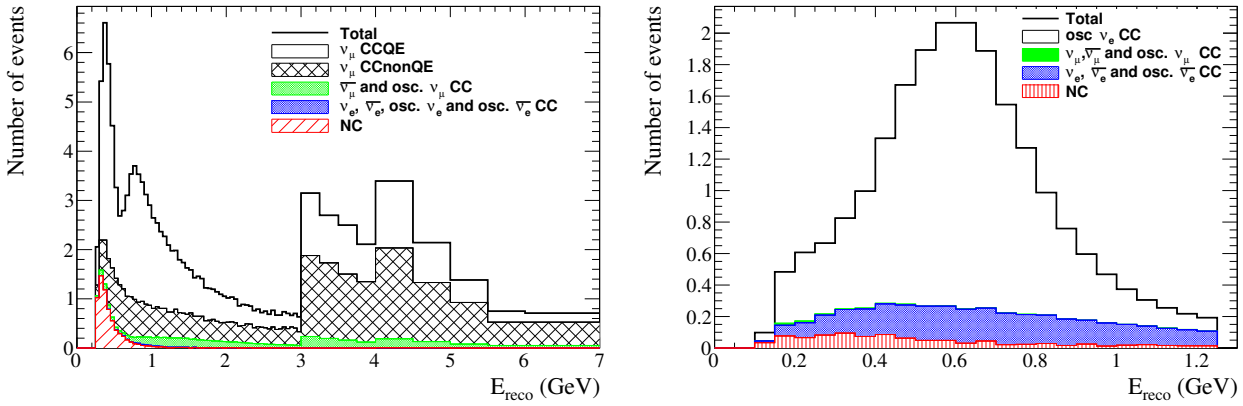


Figure 5: Predicted single μ-like ring (left) and single e-like ring (right) SuperK reconstructed energy spectra showing contributions from various true neutrino interaction modes, for an exposure of 6.57 $\times 10^{20}$ POT after BANFF tuning applied to the MC templates, for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2|$=2.4375$\times 10^{-3}$ eV^2/c^4, $\sin^2\theta_{13}$=0.0251, δ_{CP}=0 and the values for the solar parameters in Tab. 1, and normal mass hierarchy assumed. Notice that they are stacked plots.

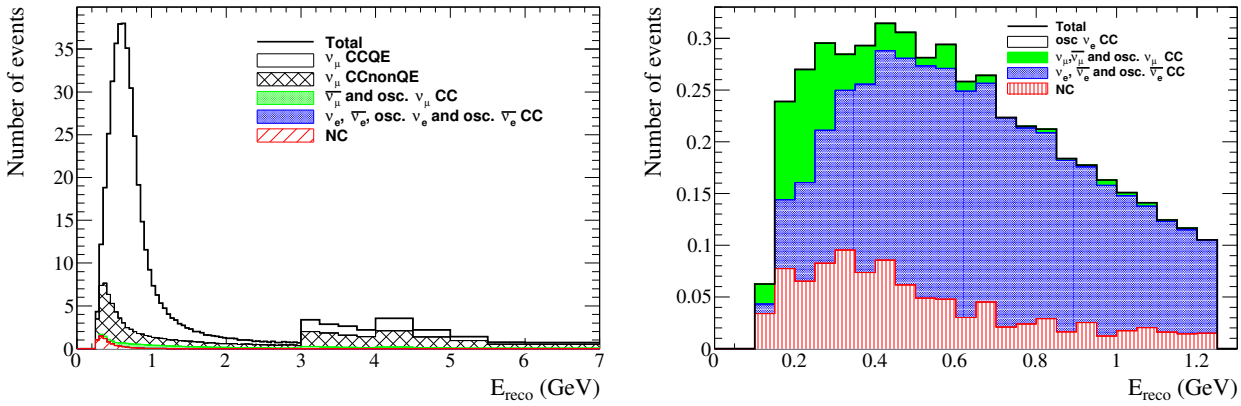


Figure 6: Predicted single μ-like ring (left) and single e-like ring (right) SuperK reconstructed energy spectra showing contributions from various true neutrino interaction modes, for an exposure of 6.57 $\times 10^{20}$ POT after BANFF tuning applied to the MC templates, for no oscillations. Notice that they are stacked plots.

Events	$1R_\mu$	$1R_e$
Total	**125.009**	**21.575**
ν_μ	116.473	0.944
$\nu_e \rightarrow \nu_\mu$	0.162	0.0002
$\bar{\nu_\mu}$	7.808	0.055
$\bar{\nu_e} \rightarrow \bar{\nu_\mu}$	0.009	2.8×10^{-6}
ν_e	0.261	3.127
$\nu_\mu \rightarrow \nu_e$	0.268	17.069
$\bar{\nu_e}$	0.024	0.152
$\bar{\nu_\mu} \rightarrow \bar{\nu_e}$	0.003	0.229

Table 3: Predicted number of single μ-like ring and single e-like ring events for an exposure of 6.57 $\times 10^{20}$ POT after BANFF tuning applied to the MC templates with oscillations using $sin^2(\theta_{12}) = 0.306$, $\Delta m^2_{21} = 7.5 \times 10^{-5}$, $sin^2(\theta_{23}) = 0.5$, $|\Delta m^2| = 2.4375 \times 10^{-3}$, $sin^2(\theta_{13}) = 0.0251$,$\delta_{CP} = 0$ and normal hierarchy assumed. All the extra templates considered are included in this table, although for the analysis the templates corresponding to $\bar{\nu_e} \rightarrow \bar{\nu_\mu}$ were finally not used due to its negligible contribution to both samples.

	N_{SK} $(1R_\mu)$ (nominal MC)	$N_{SK}(1R_\mu)$ (NA61-tuned MC)	$N_{SK}(1R_\mu)$ (BANFF-tuned MC)
Total	116.3010	131.068468	124.999646
ν_μ CCQE	66.1414	73.4564	71.001
ν_μ CC 1π	27.7397	31.9485	29.8022
ν_μ CC coherent	0.777604	0.896377	0.908341
ν_μ CC other	7.07256	8.58636	8.59818
ν_μ/ν_τ NC $1\pi^{+/-}$	3.40391	3.88813	2.93482
ν_μ/ν_τ NC other (ν_μ disap)	2.91219	3.42837	3.22896
Osc. ν_μ CCQE	0.11739	0.129868	0.127455
Osc. ν_μ CC 1π	0.0273768	0.0304843	0.0308567
Osc. ν_μ CC coherent	0.000820945	0.00091295	0.000912363
Osc. ν_μ CC other	0.0023429	0.00264651	0.00262636
$\bar{\nu_\mu}$ CCQE	4.4296	4.74618	4.62207
$\bar{\nu_\mu}$ CC 1π	2.08038	2.22023	2.13314
$\bar{\nu_\mu}$ CC coherent	0.228793	0.244564	0.24827
$\bar{\nu_\mu}$ CC other	0.460575	0.490922	0.485532
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.188316	0.203257	0.143433
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other (ν_μ disap)	0.176427	0.188866	0.17509
ν_e CCQE	0.032855	0.0354703	0.0350708
ν_e CC 1π	0.0204747	0.0226596	0.0227531
ν_e CC coherent	0.000987329	0.00108351	0.0010907
ν_e CC other	0.00571593	0.00680668	0.00662014
ν_e NC $1\pi^{+/-}$	0.0914324	0.105093	0.0751162
ν_e NC other (ν_μ disap)	0.112184	0.13168	0.121272
$\bar{\nu_e}$ CCQE	0.00208505	0.00209649	0.00202569
$\bar{\nu_e}$ CC 1π	0.00133653	0.00134519	0.00129645
$\bar{\nu_e}$ CC coherent	0.000130205	0.000132438	0.000131732
$\bar{\nu_e}$ CC other	0.000438235	0.000442349	0.000414588
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.0122542	0.012354	0.00823658
$\bar{\nu_e}$ NC other (ν_μ disap)	0.0135318	0.0135915	0.0120993
Osc. $\bar{\nu_e}$ CCQE	0.00208423	0.00211068	0.00212723
Osc. $\bar{\nu_e}$ CC 1π	0.000736528	0.000743867	0.00083112
Osc. $\bar{\nu_e}$ CC coherent	0.000105411	0.0001078	0.000111042
Osc. $\bar{\nu_e}$ CC other	$4.14101e-05$	$4.03833e-05$	$3.99067e-05$
Osc. ν_e CCQE	0.179706	0.198075	0.189984
Osc. ν_e CC 1π	0.0593724	0.0657117	0.0709648
Osc. ν_e CC coherent	0.00474813	0.00523373	0.00509285
Osc. ν_e CC other	0.00140532	0.00157693	0.00155758

Table 4: Calculated numbers of 1 μ-like ring events using the nominal MC templates, the NA61-tuned MC templates and the BANFF-tuned MC templates. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 6.5700×10^{20} POT and oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2|$=2.4375×10^{-3} eV^2/c^4, $\sin^2\theta_{13}$=0.0251, δ_{CP}=0 and the values for the solar parameters in Tab. 1. The normal mass hierarchy is assumed.

	N_{SK} $(1R_e)$ (nominal MC)	$N_{SK}(1R_e)$ (NA61-tuned MC)	$N_{SK}(1R_e)$ (BANFF-tuned MC)
Total	20.127070	22.09764937	21.57522709
ν_μ CCQE	0.0453741	0.0496952	0.0480461
ν_μ CC 1π	0.0156793	0.0173959	0.017129
ν_μ CC coherent	$6.10445e-06$	$6.76407e-06$	$6.72862e-06$
ν_μ CC other	0.000911687	0.00100829	0.000955159
ν_μ/ν_τ NC $1\pi^0$	0.388976	0.439891	0.392396
ν_μ/ν_τ NC coherent	0.137977	0.155442	0.153895
ν_μ/ν_τ NC other	0.362555	0.41513	0.331377
Osc. ν_μ CCQE	0.00012324	0.000135933	0.000134193
Osc. ν_μ CC 1π	$3.11875e-05$	$3.46067e-05$	$3.57263e-05$
Osc. ν_μ CC coherent	$7.09391e-07$	$7.86044e-07$	$7.96507e-07$
Osc. ν_μ CC other	$1.72422e-06$	$1.90727e-06$	$1.80658e-06$
$\bar{\nu_\mu}$ CCQE	0.000814438	0.000877578	0.000876255
$\bar{\nu_\mu}$ CC 1π	0.000216456	0.000234442	0.000248752
$\bar{\nu_\mu}$ CC coherent	$3.90421e-06$	$4.19644e-06$	$4.32443e-06$
$\bar{\nu_\mu}$ CC other	0.000116914	0.000121656	0.000124034
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$	0.0230774	0.0247496	0.0199007
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent	0.0145857	0.015653	0.0158927
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.0189814	0.0203247	0.0175157
ν_e CCQE	2.14935	2.28619	2.26432
ν_e CC 1π	0.686132	0.740416	0.770455
ν_e CC coherent	0.0086466	0.00922557	0.00923746
ν_e CC other	0.0523994	0.0581934	0.0580978
ν_e NC $1\pi^0$	0.0114767	0.0130331	0.010705
ν_e NC coherent	0.00361033	0.00412855	0.0041174
ν_e NC other	0.0106842	0.0123403	0.010105
$\bar{\nu_e}$ CCQE	0.0966976	0.0977422	0.0972983
$\bar{\nu_e}$ CC 1π	0.0399394	0.0399544	0.0417812
$\bar{\nu_e}$ CC coherent	0.00649861	0.00656767	0.00666227
$\bar{\nu_e}$ CC other	0.00369702	0.00364819	0.00364705
$\bar{\nu_e}$ NC $1\pi^0$	0.00154176	0.00155016	0.0011913
$\bar{\nu_e}$ NC coherent	0.000755522	0.000754551	0.000745645
$\bar{\nu_e}$ NC other	0.00118549	0.00119532	0.000967519
Osc. $\bar{\nu_e}$ CCQE	0.172551	0.174851	0.176732
Osc. $\bar{\nu_e}$ CC 1π	0.0364425	0.0367514	0.0397291
Osc. $\bar{\nu_e}$ CC coherent	0.010185	0.0103078	0.0106222
Osc. $\bar{\nu_e}$ CC other	0.00148974	0.00148497	0.00151454
Osc. ν_e CCQE	13.586	14.9802	14.4317
Osc. ν_e CC 1π	2.16905	2.40173	2.56318
Osc. ν_e CC coherent	0.0398694	0.0440768	0.0427846
Osc. ν_e CC other	0.029388	0.0326214	0.0310565

Table 5: Calculated numbers of 1 e-like ring events using the nominal MC templates, the NA61-tuned MC templates and the BANFF-tuned MC templates. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 6.5700×10^{20} POT and oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2|$=2.4375×10^{-3} eV^2/c^4, $\sin^2\theta_{13}$=0.0251, δ_{CP}=0 and the values for the solar parameters in Tab. 1. The normal mass hierarchy is assumed.

	N_{SK} $(1R_\mu)$ (nominal MC)	$N_{SK}(1R_\mu)$ (NA61-tuned MC)	$N_{SK}(1R_\mu)$ (BANFF-tuned MC)
Total	418.1312196	464.2725328	445.97965006
ν_μ CCQE	332.239	367.167	351.657
ν_μ CC 1π	56.1902	63.533	62.2279
ν_μ CC coherent	1.89644	2.13608	2.10816
ν_μ CC other	7.96301	9.60025	9.61597
ν_μ/ν_τ NC $1\pi^{+/-}$	3.40391	3.88813	2.93482
ν_μ/ν_τ NC other (ν_μ disap)	2.91219	3.42837	3.22896
Osc. ν_μ CCQE	0	0	0
Osc. ν_μ CC 1π	0	0	0
Osc. ν_μ CC coherent	0	0	0
Osc. ν_μ CC other	0	0	0
$\bar{\nu_\mu}$ CCQE	9.00809	9.66424	9.52196
$\bar{\nu_\mu}$ CC 1π	2.90935	3.11417	3.06154
$\bar{\nu_\mu}$ CC coherent	0.428081	0.458785	0.466122
$\bar{\nu_\mu}$ CC other	0.519092	0.553722	0.54821
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.188316	0.203257	0.143433
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other (ν_μ disap)	0.176427	0.188866	0.17509
ν_e CCQE	0.0353695	0.0381661	0.0377479
ν_e CC 1π	0.0215739	0.0238398	0.0240314
ν_e CC coherent	0.00105668	0.00115808	0.00116602
ν_e CC other	0.00581175	0.00691363	0.00672625
ν_e NC $1\pi^{+/-}$	0.0914324	0.105093	0.0751162
ν_e NC other (ν_μ disap)	0.112184	0.13168	0.121272
$\bar{\nu_e}$ CCQE	0.00217443	0.00218673	0.00211503
$\bar{\nu_e}$ CC 1π	0.00137591	0.00138477	0.00133868
$\bar{\nu_e}$ CC coherent	0.000135134	0.000137458	0.000136806
$\bar{\nu_e}$ CC other	0.000442257	0.000446286	0.000418341
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.0122542	0.012354	0.00823658
$\bar{\nu_e}$ NC other (ν_μ disap)	0.0135318	0.0135915	0.0120993
Osc. $\bar{\nu_e}$ CCQE	0	0	0
Osc. $\bar{\nu_e}$ CC 1π	0	0	0
Osc. $\bar{\nu_e}$ CC coherent	0	0	0
Osc. $\bar{\nu_e}$ CC other	0	0	0
Osc. ν_e CCQE	0	0	0
Osc. ν_e CC 1π	0	0	0
Osc. ν_e CC coherent	0	0	0
Osc. ν_e CC other	0	0	0

Table 6: Calculated numbers of 1 μ-like ring events using the nominal MC templates, the NA61-tuned MC templates and the BANFF-tuned MC templates. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 6.5700×10^{20} POT and no oscillations (therefore setting $\sin^2\theta_{23}$=0, $\sin^2\theta_{13}$=0.0 and $\sin^2\theta_{12}$ = 0, rest of parametes as in Tab. 4).

	N_{SK} $(1R_e)$ (nominal MC)	$N_{SK}(1R_e)$ (NA61-tuned MC)	$N_{SK}(1R_e)$ (BANFF-tuned MC)
Total	4.731432425	5.12253318	4.972703646
ν_μ CCQE	0.397783	0.438134	0.421466
ν_μ CC 1π	0.0616491	0.068337	0.0701801
ν_μ CC coherent	0.00293567	0.00325288	0.00323584
ν_μ CC other	0.00293567	0.00325059	0.00281718
ν_μ/ν_τ NC $1\pi^0$	0.388976	0.439891	0.392396
ν_μ/ν_τ NC coherent	0.137977	0.155442	0.153895
ν_μ/ν_τ NC other	0.362555	0.41513	0.331377
Osc. ν_μ CCQE	0	0	0
Osc. ν_μ CC 1π	0	0	0
Osc. ν_μ CC coherent	0	0	0
Osc. ν_μ CC other	0	0	0
$\bar{\nu_\mu}$ CCQE	0.00249756	0.00268942	0.00268219
$\bar{\nu_\mu}$ CC 1π	0.000499511	0.000537927	0.000576688
$\bar{\nu_\mu}$ CC coherent	$9.99023e-05$	0.00010738	0.000110655
$\bar{\nu_\mu}$ CC other	0.000199805	0.000208933	0.000212396
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$	0.0230774	0.0247496	0.0199007
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent	0.0145857	0.015653	0.0158927
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.0189814	0.0203247	0.0175157
ν_e CCQE	2.33832	2.48769	2.46376
ν_e CC 1π	0.729551	0.786826	0.82008
ν_e CC coherent	0.00930467	0.00992705	0.00994043
ν_e CC other	0.0541256	0.0600755	0.059969
ν_e NC $1\pi^0$	0.0114767	0.0130331	0.010705
ν_e NC coherent	0.00361033	0.00412855	0.0041174
ν_e NC other	0.0106842	0.0123403	0.010105
$\bar{\nu_e}$ CCQE	0.103599	0.104734	0.104258
$\bar{\nu_e}$ CC 1π	0.0417871	0.0418131	0.0437492
$\bar{\nu_e}$ CC coherent	0.00693483	0.0070092	0.00711015
$\bar{\nu_e}$ CC other	0.00380218	0.00375253	0.00375165
$\bar{\nu_e}$ NC $1\pi^0$	0.00154176	0.00155016	0.0011913
$\bar{\nu_e}$ NC coherent	0.000755522	0.000754551	0.000745645
$\bar{\nu_e}$ NC other	0.00118549	0.00119532	0.000967519
Osc. $\bar{\nu_e}$ CCQE	0	0	0
Osc. $\bar{\nu_e}$ CC 1π	0	0	0
Osc. $\bar{\nu_e}$ CC coherent	0	0	0
Osc. $\bar{\nu_e}$ CC other	0	0	0
Osc. ν_e CCQE	0	0	0
Osc. ν_e CC 1π	0	0	0
Osc. ν_e CC coherent	0	0	0
Osc. ν_e CC other	0	0	0

Table 7: Calculated numbers of 1 e-like ring events using the nominal MC templates, the NA61-tuned MC templates and the BANFF-tuned MC templates. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 6.5700×10^{20} POT and no oscillations (therefore setting $\sin^2\theta_{23}$=0, $\sin^2\theta_{13}$=0.0 and $\sin^2\theta_{12}$ = 0, rest of parametes as in Tab. 4).

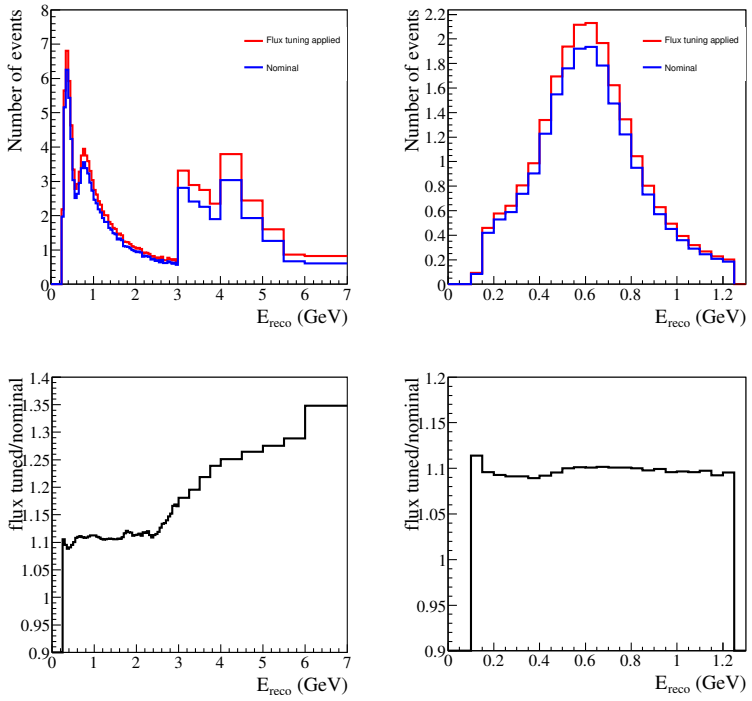


Figure 7: Top: Reconstructed-energy spectrum of single μ-like ring events (left) and single e-like ring events (right), for an exposure of 6.57×10^{20} POT, both with and without the effect of the NA61 flux tuning of the nominal MC templates. The spectra are shown for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2|$=2.4375$\times10^{-3}$ eV^2/c^4, $\sin^2\theta_{13}$=0.0251, δ_{CP}=0 and the values for the solar parameters in Tab. 1, and normal mass hierarchy assumed. Bottom: Ratio of flux-tuned spectrum to the nominal MC spectrum.

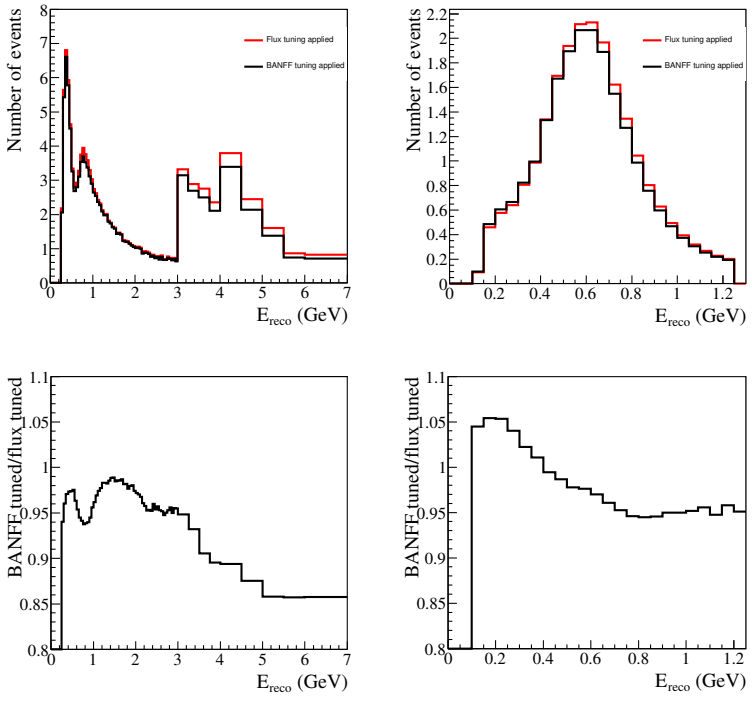


Figure 8: Top: Reconstructed-energy spectrum of single μ-like ring events (left) and single e-like ring events (right), for an exposure of 6.57×10^{20} POT, both with and without the effect of BANFF flux and cross-section tuning on the NA61-tuned flux MC templates. The spectra are shown for oscillations with $\sin^2\theta_{23}$=0.5 and $|\Delta m^2|$=2.4375$\times10^{-3}$ eV^2/c^4, $\sin^2\theta_{13}$=0.0251, δ_{CP}=0 and the values for the solar parameters in Tab. 1, and normal mass hierarchy assumed. Bottom: Ratio of flux-tuned spectrum to the nominal MC spectrum.

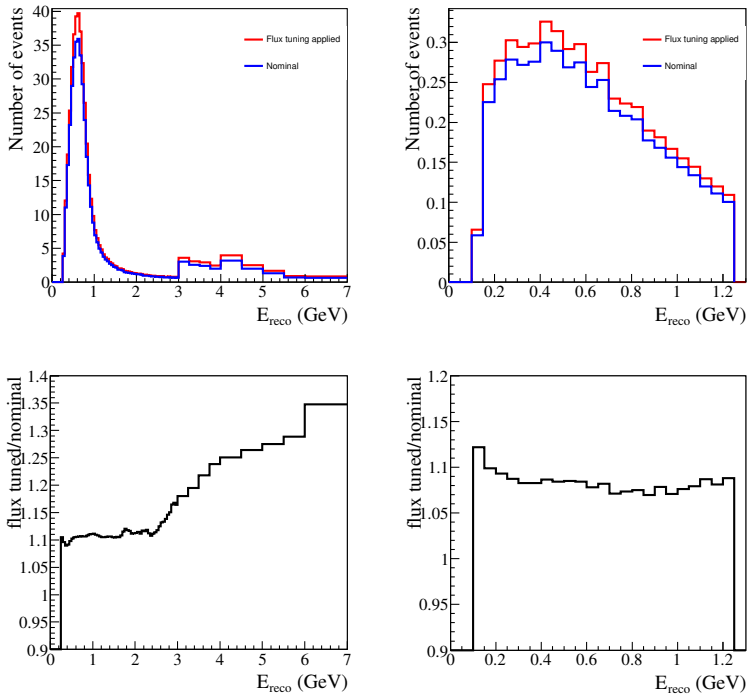


Figure 9: Top: Reconstructed-energy spectrum of single μ-like ring events (left) and single e-like ring events (right), for an exposure of 6.57×10^{20} POT, both with and without the effect of the NA61 flux tuning of the nominal MC templates. The spectra are shown for no oscillations (therefore setting $\sin^2\theta_{23}$=0, $\sin^2\theta_{13}$=0.0 and $\sin^2\theta_{12}$ = 0, rest of parametes as in Fig. 7). Bottom: Ratio of flux-tuned spectrum to the nominal MC spectrum.

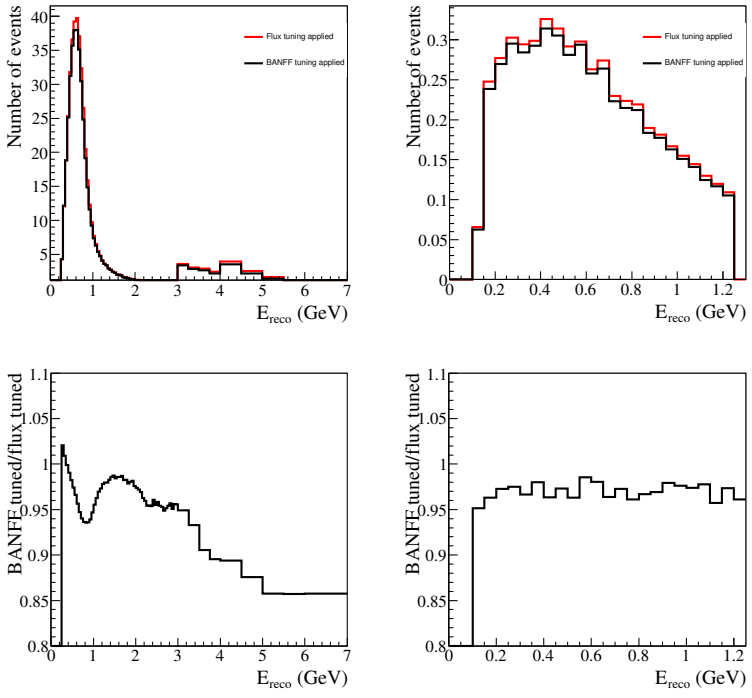


Figure 10: Top: Reconstructed-energy spectrum of single μ-like ring events (left) and single e-like ring events (right), for an exposure of 6.57×10^{20} POT, both with and without the effect of BANFF flux and cross-section tuning on the NA61-tuned flux MC templates. The spectra are shown for no oscillations (therefore setting $\sin^2\theta_{23}$=0, $\sin^2\theta_{13}$=0.0 and $\sin^2\theta_{12}$ = 0, rest of parametes as in Fig. 7). Bottom: Ratio of flux-tuned spectrum to the nominal MC spectrum.

5. Systematic effects in the prediction of the single μ-like ring and single e-like ring event reconstructed energy spectra in SuperK

There are 64 systematic parameters considered in this analysis. They will be described now, grouped into 4 categories:

1. **SuperK energy scale** (1 parameter).
   The systematic parameter $f^{SK}_{E;r}$ is included to account for the energy scale uncertainty, which is estimated to be 2.4% [12]. This systematic parameter is applied differently to the other systematics, since it is not applied as a weight in the number of events in each bin. The effect of the uncertainty in the reconstructed energy scale is computed by scaling the bin edges of the MC templates and, assuming uniform distribution of events within the bins, calculating the number of events gained from / lost to neighbouring bins. The effect of the energy scale uncertainty is propagated to all MC templates as shown in Tab. 11 and 12.

2. **SuperK efficiencies & effects of intranuclear and secondary re-interactions & Photo nuclear effects** (6+12 parameters).

   There are uncertainties in the efficiencies of the cuts used to select single ring events at SuperK: the fiducial volume and reduction chain, and the OD, ring-counting, PID, momentum and decay-electron cuts. These uncertainties are evaluated comparing SK-IV atmospheric data with atmospheric neutrino Monte Carlo. For the 2013 joint oscillation analysis, these uncertainties were evaluated taking into account correlations between the two single μ-like ring and single e-like ring event samples [25]. Very pure samples of atmospheric neutrinos with either single μ-like ring or single e-like ring events at the final state were selected. Those samples are fitted to the data including constraints from control samples (stopping cosmic muons and hybrid π^0 samples), using a Markov-Chain Monte Carlo method with the efficiencies as fit parameters (see details in [26]). Finally, the covariance matrix is calculated using toy MC experiments as explained in [27].
   Intranuclear final-state interactions (FSI) have significant effects in the T2K energy range and result in event topologies, for scattering from nuclear targets, which are drastically different from those created in neutrino scattering from free nucleons. Uncertainties due to FSI were estimated by simultaneously varying the NEUT parameters that scale the interaction probabilities for quasi-elastic scattering, absorption, charge exchange and pion production using the method described in Sec. 2 in [16]. Uncertainties due to secondary pion interactions (SI) were also evaluated. Since the same model is used for FSI and SI, it is possible to evaluate the uncertainties in both FSI and SI simultaneously. For each stand-alone analysis, ν_μ disappearance or ν_e appearance analysis, the same set of 8 FS+SI parameters is used. The effect of FSI+SI uncertainties is estimated by reweighting at the same time single μ-like ring and single e-like selected SK MC events with the same 8 parameters set for both samples using T2KReweight, and finding the number of events in each one of the 18 event categories (6 categories in ν_μ disappearance and 12 categories ν_e appearance, the same as the final parametrization of SK+FSI+SI parameters).
   The reweighting method for FSI is described in Sec. 8 in [43]. Variations in the SI interaction probabilities were included for each FSI parameter set. Subsequently a covariance matrix was calculated using equation 1 in [16]; the quantity p in that equation is the average event weight for a FSI parameter set, and the sum is over the parameter sets. This covariance matrix contains correlations between the two single μ-like ring and single e-like ring events samples [24].
   For this analysis, systematic errors related to Photo Nuclear effects (interactions between photons and nuclei) were taken into account following the work done for the stand-alone appearance analyses. The identification of a π^0 relies on the detection of the two photons produced by this particle, but a photon can be absorbed before emitting Cerenkov light, being therefore undetected, and this would create an additional uncertainty for the interaction modes where a π^0 is produced. A covariance matrix has been calculated for the PN uncertainty using the fiTQun π^0 cut, in the same binning used for the stand-alone appearance analysis SK and FSI systematic errors [19], [20].
   In this analysis,the SuperK detector efficiency and FSI+SI systematics are considered together, adding their covariance matrices linearly, what it is possible since the same binning is used for both covariance matrices. Then, the PN covariance matrix is added linearly to the SK+FSI submatrix (12 x 12) related to the single e-like ring sample. Details of these uncertainties are given in Tab. 9, and the final SK+FSI+SI+PN covariance matrix and fractional errors are presented in 5 and 12 respectively. These efficiency/FSI/SI systematics are applied to the MC templates as shown in Tab. 11 and 12.

3. **Correlated flux and cross-section (BANFF) parameters** (33 parameters).

   The postfit uncertainties of all BANFF flux and cross-section parameters and their correlations are taken into account. The version of the BANFF fit is slightly different for the stand-alone disappearance, stand-alone appearance and joint oscillation analyses. Concerning the flux parameters, for the stand-alone disappearance analysis the version of the BANFF

fit considers only the ν_μ and $\bar{\nu}_\mu$ flux parameters, whereas for the stand-alone appearance flux parameters are considered for ν_μ, $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$, with a smaller number of $\bar{\nu}_\mu$ flux bins than the stand-alone disappearance analysis. The BANFF version for the joint analysis includes flux bins for ν_μ, $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$, with the number of flux bins for $\bar{\nu}_\mu$ from the stand-alone disappearance analysis. There are also differences in the number of correlated cross-section parameters included in the different BANFF versions for the stand-alone analyses and the joint analysis. For the stand-alone disappearance analysis, there are three CCQE normalization parameters (0-1.5 GeV, 1.5-3.5 GeV and 3.5-30 GeV) and two for the CC1π normalization (1.5-3.5 GeV and 3.5-30 GeV), whereas for the stand-alone appearance analysis only one parameter is considered for each normalization, the one in the lower energy range. There is also one extra parameter in the stand-alone appearance analysis with respect to the stand-alone disappearance analysis: this is the NC1π^0 normalization parameter. The BANFF version produced for the joint analysis considers all the cross-section parameters included in the stand-alone disappearance and stand-alone appearance versions (8 in total), however, the NC1π^0 normalization parameter is only applied to the single e-like ring sample templates, because there are no single μ-like ring NC1π^0 templates.
A summary of the list of parameters included in the BANFF postfit files for each oscillation analysis is given in Tab. 8

Parameter	ν_μ-disappearance analysis	ν_e-appearance analysis	joint analysis
ν_μ flux	0-0.4, 0.4-0.5, 0.5-0.6, 0.6-0.7, 0.7-1.0, 1.0-1.5, 1.5-2.5, 2.5-3.5, 3.5-5.0, 5.0-7.0, 7.0-30.0 **(11 bins)**	0-0.4, 0.4-0.5, 0.5-0.6, 0.6-0.7, 0.7-1.0, 1.0-1.5, 1.5-2.5, 2.5-3.5, 3.5-5.0, 5.0-7.0, 7.0-30.0 **(11 bins)**	0-0.4, 0.4-0.5, 0.5-0.6, 0.6-0.7, 0.7-1.0, 1.0-1.5, 1.5-2.5, 2.5-3.5, 3.5-5.0, 5.0-7.0, 7.0-30.0 **(11 bins)**
$\bar{\nu_\mu}$ flux	0-0.7, 0.7-1.0, 1.0-1.5, 1.5-2.5, 2.5-30.0 **(5 bins)**	0-1.5, 1.5-30.0 **(2 bins)**	0-0.7, 0.7-1.0, 1.0-1.5, 1.5-2.5, 2.5-30.0 **(5 bins)**
ν_e flux	–	0-0.5, 0.5-0.7, 0.7-0.8, 0.8-1.5, 1.5-2.5, 2.5-4.0, 4.0-30.0 **(7 bins)**	0-0.5, 0.5-0.7, 0.7-0.8, 0.8-1.5, 1.5-2.5, 2.5-4.0, 4.0-30.0 **(7 bins)**
$\bar{\nu_e}$ flux	–	0-2.5, 2.5-30.0 **(2 bins)**	0-2.5, 2.5-30.0 **(2 bins)**
M_A^{QE}	✓	✓	✓
M_A^{RES}	✓	✓	✓
CCQE normalization (0-1.5 GeV)	✓	✓	✓
CCQE normalization (1.5-3.5 GeV)	✓	–	✓
CCQE normalization (3.5-30. GeV)	✓	–	✓
CC1π normalization (0-2.5 GeV)	✓	✓	✓
CC1π normalization (2.5-30 GeV)	✓	–	✓
NC1π^0 normalization	–	✓	✓

Table 8: List of parameters included in the BANFF postfit files, varying depending on the different versions produced for each oscillation analysis.

A list of the BANFF parameters, their best-fit values and their prefit and postfit errors is given in Tab. 9 and Fig. 13.
The BANFF parameters are applied to the MC templates as specified in Tab. 11 and 12. Among the BANFF cross-section parameters, all but two are simple normalization factors. However, the parameters $f^{banff}_{25;t,r}$ and $f^{banff}_{26;t,r}$, the ones corresponding to the axial masses (CC quasi-elastic, M_A^{QE}, and resonance-production, M_A^{RES} respectively), have a more complicated, often non-linear, dependence on true and reconstructed neutrino energy. For this reason, event reweighting methods were used to obtain the fractional changes of the number of events in bins of true and reconstructed neutrino energy, for a range of values (-3.0σ, -2.5σ, -2.0σ, ..., +3.0σ) for the $f^{banff}_{25;t,r}$ and $f^{banff}_{26;t,r}$ parameters. In our p.d.f. calculation, cubic splines
[2] are used to interpolate between these event weights to obtain the fractional change for any value of these systematic

[2] In this analysis, as in the *VALOR* stand-alone disappearance analysis [6], the cubic splines are computed for each non-linear systematic and for each MC template, in bins of reconstructed and true energy. To produce these splines, the SK13a SuperK MC, the 11bv3.2 flux weights are used and the following oscillation parameters: $\sin^2(2\theta_{23})$ = 1.0, $\sin^2(2\theta_{13})$ = 0.098, $\sin^2(2\theta_{12})$ = 0.8495, $|\Delta m^2_{32}|$ = 2.44×10^{-3} eV^2/c^4, Δm^2_{21} = 7.5×10^{-5} eV^2/c^4, δ_{CP} = 0 and the normal mass hierarchy. The splines for the single μ-like ring sample were made using 41 bins in reconstructed energy: 1 bin from 0-0.3 GeV, 14 50-MeV bins from 0.3-1.0 GeV, 20 100-MeV bins from 1.0-3.0 GeV, 2 500-MeV bins from 3.0-4.0 GeV, 1 bin from 4.0-5.0 GeV, 1 bin from 5.0-7.0 GeV, 1 bin from 7.0-10.0 GeV and 1 bin from 10.0-30.0 GeV. The ν_μ, $\bar{\nu}_\mu$ and oscillated ν_e selections for the single μ-like ring sample have 11 bins in true energy: 1 bin from 0-0.4 GeV, 3 100-MeV bins from 0.4-0.7 GeV, 1 bin from 0.7-1.0 GeV, 1 bin from 1.0-1.5 GeV, 2 1-GeV bins from 1.5-3.5 GeV, 1 bin from 3.5-5.0 GeV, 1 bin from 5.0-7.0 GeV and 1 bin from 7.0-30.0 GeV, and the ν_e and oscillated ν_μ samples have 7 bins in true energy: 1 bin from 0-0.05 GeV, 1 bin from 0.5-0.7 GeV, 1 bin from 0.7-0.8 GeV, 1 bin from 0.8-1.5 GeV, 1 bin from 1.5-2.5 GeV, 1 bin from 2.5-4.0 GeV and 1 bin from 4.0-30.0 GeV, and finally, the $\bar{\nu}_e$ sample have 2 bins in true energy: 1 bin from 0-2.5 GeV and 1 bin from 2.5-30.0 GeV. The binning of the cubic splines for the single μ-like ring events was studied, and the results were presented in Appendix I of [5].

parameters in the (-3.0σ, +3.0σ) range. An example of one cubic spline is presented in Fig. 11.

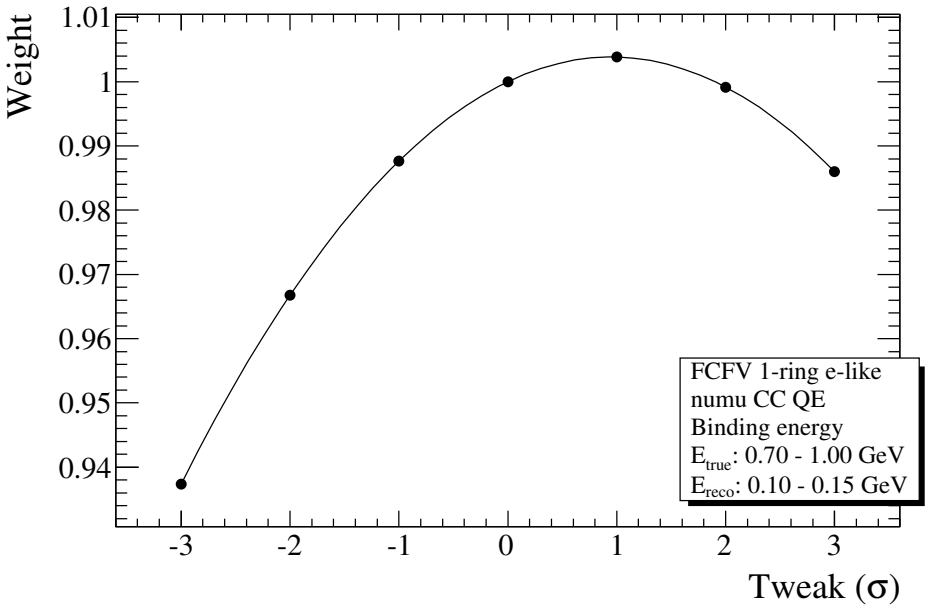


Figure 11: Example of response function (cubic spline) showing the weight vs the tweak for the Binding Energy systematic parameter for the 1R$_e$ sample, ν_μ CCQE interaction mode, E_{true} between 0.7-1.0 GeV and E_{reco} between 0.10-0.15 GeV.

4. **Independent cross-section parameters**(12 parameters).

In addition to the BANFF cross-section systematics, another 12 cross-section systematics are taken into account; these additional systematics are considered to be independent between themselves and to have no cancellation between the near and far detectors [16] [46] [18].
There are some differences in the list of independent cross-section parameters between the different oscillation analyses. There is one systematic parameter which is only considered in the stand-alone appearance analysis, but not in the stand-alone disappearance analysis: this is the NC coherent cross-section parameter. On the other hand, there are three cross-section systematics that are used in the stand-alone disappearance analysis but not in the stand-alone appearance analysis: the systematic for the Binding Energy, the NC1$\pi^\pm$ cross-section and the $\sigma_{\bar{\nu}}/\sigma_\nu$ ratio. The joint oscillation analysis considers them all (12 parameters) although the NC coherent cross-section parameter is only applied to the single e-like ring NC coherent templates (there are no single μ-like ring NC coherent templates) and the NC1$\pi^\pm$ cross-section parameter is only applied to the single μ-like ring NC1$\pi^\pm$ templates (there are no single e-like ring NC1$\pi^\pm$ templates).
Three of these additional systematics relate to the initial-state nuclear environment modeling for CCQE. Neutrino Monte Carlo generators typically use Relativistic Fermi Gas (RFG) models, with or without extensions to take nucleon-nucleon correlation into account, and with some, often simplistic, prescription for the treatment of off-shell kinematics. The RFG model implementation depends on two parameters, the Fermi momentum level (225 MeV for ^{16}O) and the nucleon binding energy (27 MeV for ^{16}O), which are extracted from quasi-elastic electron scattering data. The systematic parameter $f_{pF;t,r}$ takes into account the effect of a 30 MeV 1σ uncertainty in the Fermi momentum level for ^{16}O, while the parameter $f_{bindE;t,r}$ takes into account the effect of a 9 MeV 1σ uncertainty in the nucleon binding energy for for ^{16}O. The implementation details of the nuclear model can have a significant effect and, therefore, the effect of switching between two such models was also included as a systematic uncertainty. This analysis considers the effect of switching between the RFG model and a better-motivated Spectral Function (SF) model. The latter does not assume a fixed binding energy

For the single e-like ring sample splines the reconstructed and true energy bins used were the same ones used for the stand-alone appearance analyses. Therefore, reconstructed energy bins of 50 MeV were used from 0-1.25 GeV for ν_μ, $\bar{\nu}_\mu$,ν_e, $\bar{\nu}_e$ and oscillated ν_e. The binning for true energy for the single e-like ring events coincide with the one used for the single μ-like ring events, with the exception of the oscillated ν_e sample, which uses in the single e-like ring sample the following 7 bins in true energy: 1 bin from 0-0.05 GeV, 1 bin from 0.5-0.7 GeV, 1 bin from 0.7-0.8 GeV, 1 bin from 0.8-1.5 GeV, 1 bin from 1.5-2.5 GeV, 1 bin from 2.5-4.0 GeV and 1 bin from 4.0-30.0 GeV. The spline spectra produced using the cubic splines are very similar to those produced using the CPU-intensive event-by-event reweighting, especially for the dominant components of the spectra.

and includes the correlations between nucleon momentum and binding energy. The systematic parameter $f_{SF;t,r}$ includes the effect of switching between the two models. Thus, the systematic parameter $f_{SF;t,r}$ needs an special treatment. After discussions with the NIWG group, the latest recommendation is followed in this analysis (see below), and the effect of the change in its treatment means a big difference as it is shown in Sec. A.

Simple, energy-independent, generator-level uncertainties are considered for the CC coherent, NC, $\sigma_{\nu_e}/\sigma_{\nu_\mu}$ ratio and $\sigma_{\bar{\nu}}/\sigma_{\nu}$ ratio cross sections. This analysis uses a 100% uncertainty for the CC coherent cross-section, a 30% uncertainty for both the NC1$\pi^{\pm}$, NC coherent and NC "other" cross section, a 3% uncertainty on the ν_e / ν_μ CC cross-section ratio and a 20% uncertainty on the $\bar{\nu}$ / ν CC cross-section ratio.

The effect of the variation of the above cross sections is included in the analysis through the systematic parameters $f_{CCcoh;t}$, $f_{NC1\pi^{\pm};t}$, $f_{NCcoh;t}$, $f_{NCoth;t}$, $f_{CC\nu_e;t}$ and $f_{CC\bar{\nu};t}$ respectively.

The treatment of Δ resonances in the nuclear environment has been controversial in the neutrino Monte Carlo generator community. Certain models include a 20%-level of π-less Δ decays in order to include the effect of $\Delta + N \rightarrow N + N$ scattering, in which the Δ interacts before it can decay. In contrast, other models include no π-less Δ decays on the basis that this would amount to double counting since, whatever the size of the effect, it is already included in hadron-nucleus scattering data to which the intranuclear cascade Monte Carlos are tuned. This analysis uses NEUT, in which 20% of Δs have a π-less decay by default, and takes the uncertainty in this branching fraction to be 100%. The systematic parameter $f_{\pi-less\Delta;t,r}$ includes the effect of varying the fraction of π-less Δ decays to the corresponding MC templates.

The $f_{CCothShape;t,r}$ systematic parameterizes the uncertainty in CC multi-pion, CC deep inelastic scattering and CC η, K and photon production. From MINOS data, this is known to be $\approx$10% at 4 GeV. Using this as a reference point, the uncertainty is estimated as 0.4/E_ν, where E_ν is in GeV. It might appear that this error could diverge at low energy, but this does not happen since the threshold is $\approx$0.6 GeV and there are very few interactions below 1 GeV.

The $f_{Wshape;t,r}$ parameter modifies the shape of the initial pion momentum distribution (before final-state interactions) by changing the distribution of the pion/nucleon invariant mass while keeping the total cross section constant. As the W shape parameter decreases, the pion momentum shifts to lower values, and this increases the probability of pion absorption in a final-state interaction. If such an absorption occurs, the event is migrated from CC1π (its true initial state) and appears to be CCQE.

The effect of the $f_{CCcoh;t}$, $f_{NC1\pi^{\pm};t}$, $f_{NCcoh;t}$, $f_{NCoth;t}$, $f_{CC\nu_e;t}$ and $f_{CC\bar{\nu};t}$ systematic parameters is a simple normalisation change for the corresponding MC templates and thus easily implemented. Systematic parameters $f_{pF;t,r}$, $f_{SF;t,r}$, $f_{bindE;t,r}$, $f_{\pi-less\Delta;t,r}$, $f_{CCothShape;t,r}$ and $f_{Wshape;t,r}$ have a complicated, often non-linear, dependence on true and reconstructed energy. These are treated in a similar way to the BANFF parameters $f^{banff}_{25;t,r}$ and $f^{banff}_{26;t,r}$. Event reweighting methods were used to obtain the fractional changes of the number of events, in bins of true and reconstructed neutrino energy, for a range of values (-3.0σ, -2.5σ, -2.0σ, ..., +3.0σ) of the systematic parameters. Again cross-section splines with polynomials of order 3 were used to interpolate between these event weights to obtain the fractional change for any value of these systematic parameters in the (-3.0σ, +3.0σ) range.

Some of these systematics are special cases:

- $f_{SF;t,r}$ is discrete; it can be either spectral function off (RFG) or spectral function on (SF). Splines are generated with SF off as nominal (0), and SF on as +1σ. This parameter is restricted to the range between 0 and +1σ, and linear interpolation is made between the weights for nominal and +1σ.
- For $f_{Wshape;t,r}$ only $\pm$1σ errors are evaluated using event reweighting. Weights for positive tweaks are interpolated or extrapolated from the nominal and +1σ weights, and weights for negative tweaks are interpolated or extrapolated from the nominal and -1σ weights.
- For $f_{\pi-less\Delta;t,r}$ only $\pm$1.0 sigma errors are evaluated using event reweighting. Weights for positive tweaks are interpolated or extrapolated from the nominal and +1σ weights, and weights for tweaks between nominal and -1σ are interpolated from the nominal and -1σ weights. Weights for tweaks more negative than -1σ are taken to be the same as the -1σ weight; this is done since 20% of Δs have a π-less decay by default in NEUT, and the error is 20%. For this reason, it is unphysical to tweak $f_{\pi-less\Delta;t,r}$ to be more negative than -1σ.

The 12 independent cross-section systematics considered in this analysis are summarised in Tab. 9. These cross-section systematics are applied to the MC templates as shown in Tab. 11 and 12.

Table 10 presents a summary of which systematic parameter is applied to each sample (single μ-like ring or single e-like ring) for the stand-alone and the joint analyses.

SuperK energy scale systematics (1)

$f^{SK}_{E;r}$	Reconstructed energy scale systematic. The uncertainty is estimated to be 2.4% [12].

SuperK detector efficiency & intranuclear and secondary interaction and phononuclear systematics (18)

SuperK detector efficiency (ring counting, fiducial volume, reduction chain, OD cut, PID cut, momentum cut and decay electron cut efficiency uncertainties), FSI (final state interaction / intranuclear hadron transport) and SI (secondary interaction / detector hadron transport) and PN (photo nuclear/ interactions between photons and nuclei) systematic parameters. Eighteen correlated parameters are included (i=0,...,17) corresponding to different modes and reconstructed energy bins.

$f^{SK+FSI}_{i;r}$

Parameter	Description	1σ fractional error
0	1R$_\mu$, ν_μ & $\bar{\nu}_\mu$ CCQE, $E_{reco} <$ 0.4 GeV	0.0199
1	1R$_\mu$, ν_μ & $\bar{\nu}_\mu$ CCQE, 0.4 GeV $\leq E_{reco} \leq$ 1.1 GeV	0.0193
2	1R$_\mu$, ν_μ & $\bar{\nu}_\mu$ CCQE, $E_{reco} >$ 1.1 GeV	0.0199
3	1R$_\mu$, ν_μ & $\bar{\nu}_\mu$ CCnonQE	0.0933
4	1R$_\mu$, ν_e & $\bar{\nu}_e$ CC	1.005
5	1R$_\mu$, All NC	0.599
6	1R$_e$, $osc.\nu_e$ CC$E_{reco} <$ 0.35 GeV	0.110
7	1R$_e$, $osc.\nu_e$ CC0.35 GeV $\leq E_{reco} \leq$ 0.8 GeV	0.0309
8	1R$_e$, $osc.\nu_e$ CC$E_{reco} >$ 0.8 GeV	0.0347
9	1R$_e$, ν_μ & $\bar{\nu}_\mu$ CC$E_{reco} <$ 0.35 GeV	1.244
10	1R$_e$, ν_μ & $\bar{\nu}_\mu$ CC0.35 GeV $\leq E_{reco} \leq$ 0.8 GeV	1.216
11	1R$_e$, ν_μ & $\bar{\nu}_\mu$ CC$E_{reco} >$ 0.8 GeV	1.229
12	1R$_e$, ν_e & $\bar{\nu}_e$ CC$E_{reco} <$ 0.35 GeV	0.0683
13	1R$_e$, ν_e & $\bar{\nu}_e$ CC0.35 GeV $\leq E_{reco} \leq$ 0.8 GeV	0.0421
14	1R$_e$, ν_e & $\bar{\nu}_e$ CC$E_{reco} >$ 0.8 GeV	0.0628
15	1R$_e$, All NC$E_{reco} <$ 0.35 GeV	0.524
16	1R$_e$, All NC0.35 GeV $\leq E_{reco} \leq$ 0.8 GeV	0.247
17	1R$_e$, All NC$E_{reco} >$ 0.8 GeV	0.604

The covariance matrix is listed in Eq. 5 and Fig. 12 shows the values of fractional errors for each SK+FSI+PN systematic parameter.

Independent cross-section systematics (12)

$f_{pF;t,r}$	Uncertainty in Fermi momentum for ^{16}O was taken to be 225$\pm$30 MeV [46].
$f_{bindE;t,r}$	Uncertainty in binding energy for ^{16}O was taken to be 27$\pm$9 MeV [46].
$f_{SF;t,r}$	Nuclear environment modeling systematic: Switches from Fermi Gas (RFG) to Spectral Function (SF) model (0:RFG - 1:SF) [46].
$f_{\pi-less\Delta;t,r}$	Systematic due to the uncertainty (20% [46]) in the fraction of π-less Δ decays in resonance-production events.
$f_{CCothShape;t,r}$	Systematic due to the uncertainty (40% [46]) in CC multipion, CC DIS and CC resonant η, K and photon production cross section.
$f_{Wshape;t,r}$	Systematic due to the uncertainty (52% [46]) in the shape of the initial pion momentum distribution (before final-state interactions) in resonance interactions.
$f_{CCcoh;t}$	Systematic due to the uncertainty (100% [46]) on the CC coherent cross section.
$f_{NC1\pi^\pm;t}$	Systematic due to the uncertainty (30% [46]) on the NC1$\pi^\pm$ cross section.
$f_{NCcoh;t}$	Systematic due to the uncertainty (30% [46]) on the NC coherent cross section.
$f_{NCoth;t}$	Systematic due to the uncertainty (30% [46]) on the NC other (all but single $\pi^\pm$ for 1R$_\mu$ sample or all but single π^0 and coherent or 1R$_e$) cross section.

$f_{CC\nu_e;t}$	Systematic due to the uncertainty (3% [16]) on the ν_e / ν_μ CC cross-section ratio.	
$f_{CC\bar{\nu};t}$	Systematic due to the uncertainty (20% [18]) on the $\bar{\nu}$ / ν CC cross-section ratio.	
Correlated flux and cross-section (BANFF) systematics (33)		
$f^{banff}_{i;t,r}$	Flux and cross-section parameters tuned by BANFF group on ND280 data [8]. Twenty-three correlated parameters are included (i=0,...,22).	

Parameter	Description	Best-fit	1σ pre/postfit fractional error
0	ν_μ flux normalisation, E = 0.0 - 0.4 GeV	1.029	0.121/0.085
1	ν_μ flux normalisation, E = 0.4 - 0.5 GeV	1.022	0.130/0.088
2	ν_μ flux normalisation, E = 0.5 - 0.6 GeV	0.995	0.122/0.080
3	ν_μ flux normalisation, E = 0.6 - 0.7 GeV	0.966	0.115/0.076
4	ν_μ flux normalisation, E = 0.7 - 1.0 GeV	0.934	0.129/0.085
5	ν_μ flux normalisation, E = 1.0 - 1.5 GeV	0.992	0.116/0.077
6	ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.037	0.100/0.068
7	ν_μ flux normalisation, E = 2.5 - 3.5 GeV	1.054	0.095/0.065
8	ν_μ flux normalisation, E = 3.5 - 5.0 GeV	1.035	0.112/0.072
9	ν_μ flux normalisation, E = 5.0 - 7.0 GeV	0.975	0.152/0.073
10	ν_μ flux normalisation, E = 7.0 - 30.0 GeV	0.943	0.187/0.082
11	$\bar{\nu}_\mu$ flux normalisation, E = 0.0 - 0.7 GeV	1.030	0.133/0.102
12	$\bar{\nu}_\mu$ flux normalisation, E = 0.7 - 1.0 GeV	1.011	0.117/0.090
13	$\bar{\nu}_\mu$ flux normalisation, E = 1.0 - 1.5 GeV	1.007	0.119/0.094
14	$\bar{\nu}_\mu$ flux normalisation, E = 1.5 - 2.5 GeV	1.026	0.123/0.104
15	$\bar{\nu}_\mu$ flux normalisation, E = 2.5 -30.0 GeV	1.008	0.122/0.107
16	ν_e flux normalisation, E = 0.0 - 0.5 GeV	1.033	0.131/0.098
17	ν_e flux normalisation, E = 0.5 - 0.7 GeV	1.013	0.125/0.093
18	ν_e flux normalisation, E = 0.7 - 0.8 GeV	0.976	0.141/0.110
19	ν_e flux normalisation, E = 0.8 - 1.5 GeV	0.9978	0.105/0.073
20	ν_e flux normalisation, E = 1.5 - 2.5 GeV	1.021	0.105/0.072
21	ν_e flux normalisation, E = 2.5 - 4.0 GeV	1.003	0.123/0.067
22	ν_e flux normalisation, E = 4.0 - 30.0 GeV	0.947	0.168/0.077
23	$\bar{\nu}_e$ flux normalisation, E = 0.0 - 2.5 GeV	1.014	0.192/0.180
24	$\bar{\nu}_e$ flux normalisation, E = 2.5 - 30.0 GeV	0.9552	0.142/0.080
25	CCQE axial-mass scaling factor	1.025	0.372/0.059
26	Resonance-production axial-mass scaling factor	0.797	0.183/0.056
27	CCQE normalisation, E = 0.0 - 1.5 GeV	0.966	0.110/0.076
28	CCQE normalisation, E = 1.5 - 3.5 GeV	0.931	0.300/0.103
29	CCQE normalisation, E = 3.5 - 30.0 GeV	0.852	0.300/0.114
30	$CC1\pi$ normalisation, E = 0.0 - 2.5 GeV	1.265	0.317/0.163
31	$CC1\pi$ normalisation, E = 2.5 - 30.0 GeV	1.122	0.400/0.172
32	$NC1\pi^0$ normalisation	1.135	0.328/0.248

The covariance matrix is listed in Eq. 5 and Fig. 13 shows the post-fit values and post-fit fractional errors for each BANFF systematic parameter.

Table 9: Summary of systematics included the *VALOR* Run 1+2+3+4 joint oscillation analysis.

$$C^{SK+FSI} = \tag{6}$$

	$f_{0;r}^{SK+FSI}$	$f_{1;r}^{SK+FSI}$	$f_{2;r}^{SK+FSI}$	$f_{3;r}^{SK+FSI}$	$f_{4;r}^{SK+FSI}$	$f_{5;r}^{SK+FSI}$	$f_{6;r}^{SK+FSI}$	$f_{7;r}^{SK+FSI}$	$f_{8;r}^{SK+FSI}$	$f_{9;r}^{SK+FSI}$	$f_{10;r}^{SK+FSI}$	$f_{11;r}^{SK+FSI}$	$f_{12;r}^{SK+FSI}$	$f_{13;r}^{SK+FSI}$	$f_{14;r}^{SK+FSI}$	$f_{15;r}^{SK+FSI}$	$f_{16;r}^{SK+FSI}$	$f_{17;r}^{SK+FSI}$
$f_{0;r}^{SK+FSI}$	0.00040	0.00037	0.00031	0.00028	0.00039	0.00022	0.00021	0.00032	0.00006	0.00020	0.00019	0.00020	0.00020	0.00027	-0.00017	0.00019	0.00018	0.00018
$f_{1;r}^{SK+FSI}$	0.00037	0.00037	0.00033	0.00037	0.00383	0.00026	0.00029	0.00030	0.00010	0.00015	0.00015	0.00014	0.00026	0.00028	-0.00007	0.00020	0.00019	0.00024
$f_{2;r}^{SK+FSI}$	0.00031	0.00033	0.00040	0.00079	0.00408	0.00030	0.00055	0.00027	0.00022	0.00027	0.00022	0.00017	0.00041	0.00033	0.00026	0.00019	0.00020	0.00032
$f_{3;r}^{SK+FSI}$	0.00028	0.00037	0.00079	0.00871	0.00345	0.00226	0.00927	0.00124	0.00075	0.00299	0.00122	0.00036	0.00558	0.00308	0.00296	-0.00113	-0.00123	0.00219
$f_{4;r}^{SK+FSI}$	0.00039	0.00383	0.00408	0.00345	1.00961	0.00724	0.00156	0.00035	-0.00001	-0.00593	-0.00552	-0.00594	0.00099	0.00060	0.00011	-0.00069	0.00128	0.00669
$f_{5;r}^{SK+FSI}$	0.00022	0.00026	0.00030	0.00226	0.00724	0.35896	0.00036	0.00012	0.00021	0.00125	0.00511	0.00271	0.00012	-0.00004	0.00018	0.01622	0.00577	0.00064
$f_{6;r}^{SK+FSI}$	0.00021	0.00029	0.00055	0.00927	0.00156	0.00036	0.01218	0.00173	0.00106	0.00360	0.00196	0.00033	0.00744	0.00407	0.00388	-0.00050	-0.00050	0.00323
$f_{7;r}^{SK+FSI}$	0.00032	0.00030	0.00027	0.00124	0.00035	0.00012	0.00173	0.00095	0.00057	0.00054	0.00038	0.00019	0.00117	0.00110	0.00050	0.00015	0.00009	0.00051
$f_{8;r}^{SK+FSI}$	0.00006	0.00010	0.00022	0.00075	-0.00001	0.00021	0.00106	0.00057	0.00120	0.00028	0.00020	0.00011	0.00079	0.00072	0.00182	0.00026	0.00030	0.00047
$f_{9;r}^{SK+FSI}$	0.00020	0.00015	0.00027	0.00299	-0.00593	0.00125	0.00360	0.00054	0.00028	1.54760	1.51082	1.52756	0.00218	0.00119	0.00097	0.00056	-0.00024	-0.00273
$f_{10;r}^{SK+FSI}$	0.00019	0.00015	0.00022	0.00122	-0.00552	0.00511	0.00196	0.00038	0.00020	1.51082	1.47783	1.49290	0.00119	0.00061	0.00044	0.00161	0.00252	0.00080
$f_{11;r}^{SK+FSI}$	0.00020	0.00014	0.00017	0.00036	-0.00594	0.00271	0.00033	0.00019	0.00011	1.52756	1.49290	1.50974	0.00028	0.00016	0.00003	-0.00060	0.00056	-0.00240
$f_{12;r}^{SK+FSI}$	0.00020	0.00026	0.00041	0.00558	0.00099	0.00012	0.00744	0.00117	0.00079	0.00218	0.00119	0.00028	0.00466	0.00259	0.00251	-0.00035	-0.00028	0.00188
$f_{13;r}^{SK+FSI}$	0.00027	0.00028	0.00033	0.00308	0.00060	-0.00004	0.00407	0.00110	0.00072	0.00119	0.00061	0.00016	0.00259	0.00178	0.00139	-0.00014	-0.00017	0.00110
$f_{14;r}^{SK+FSI}$	-0.00017	-0.00007	0.00026	0.00296	0.00011	0.00018	0.00388	0.00050	0.00182	0.00097	0.00044	0.00003	0.00251	0.00139	0.00395	-0.00005	0.00015	0.00124
$f_{15;r}^{SK+FSI}$	0.00019	0.00020	0.00019	-0.00113	-0.00069	0.01622	-0.00050	0.00015	0.00026	0.00056	0.00161	-0.00060	-0.00035	-0.00014	-0.00005	0.27476	0.06449	0.05220
$f_{16;r}^{SK+FSI}$	0.00018	0.00019	0.00020	-0.00123	0.00128	0.00577	-0.00050	0.00009	0.00030	-0.00024	0.00252	0.00056	-0.00028	-0.00017	0.00015	0.06449	0.06079	0.09676
$f_{17;r}^{SK+FSI}$	0.00018	0.00024	0.00032	0.00219	0.00669	0.00064	0.00323	0.00051	0.00047	-0.00273	0.00080	-0.00240	0.00188	0.00110	0.00124	0.05220	0.09676	0.36527

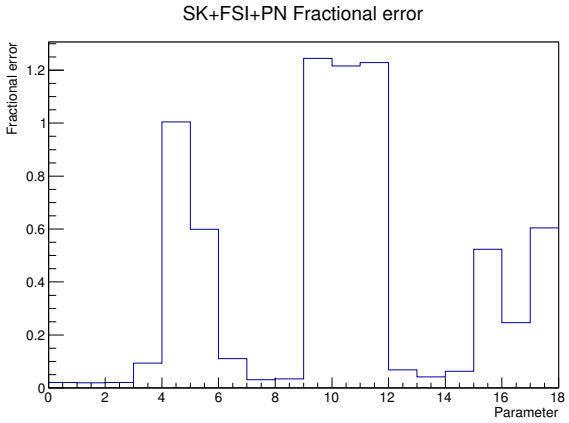


Figure 12: Values of the fractional errors for each SK+FSI+PN systematic parameter.

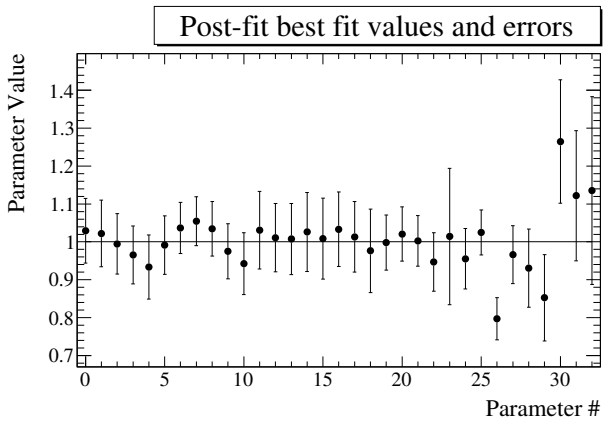


Figure 13: Post-fit values and post-fit fractional errors for each BANFF systematic parameter.

$$C^{banff} = \tag{7}$$

$f^{banff}_{0;t,r}$	$f^{banff}_{0;t,r}$	$f^{banff}_{1;t,r}$	$f^{banff}_{2;t,r}$	$f^{banff}_{3;t,r}$	$f^{banff}_{4;t,r}$	$f^{banff}_{5;t,r}$	$f^{banff}_{6;t,r}$	$f^{banff}_{7;t,r}$	$f^{banff}_{8;t,r}$	$f^{banff}_{9;t,r}$	$f^{banff}_{10;t,r}$	$f^{banff}_{11;t,r}$	$f^{banff}_{12;t,r}$	$f^{banff}_{13;t,r}$	$f^{banff}_{14;t,r}$	$f^{banff}_{15;t,r}$	$f^{banff}_{16;t,r}$	$f^{banff}_{17;t,r}$	$f^{banff}_{18;t,r}$	$f^{banff}_{19;t,r}$	$f^{banff}_{20;t,r}$	$f^{banff}_{21;t,r}$	$f^{banff}_{22;t,r}$	$f^{banff}_{23;t,r}$	$f^{banff}_{24;t,r}$	$f^{banff}_{25;t,r}$	$f^{banff}_{26;t,r}$	$f^{banff}_{27;t,r}$	$f^{banff}_{28;t,r}$	$f^{banff}_{29;t,r}$	$f^{banff}_{30;t,r}$	$f^{banff}_{31;t,r}$	$f^{banff}_{32;t,r}$
$f^{bonff}_{0;t,r}$	0.007	0.007	0.006	0.005	0.004	0.004	0.004	0.003	0.002	0.002	0.001	0.008	0.006	0.005	0.005	0.003	0.007	0.006	0.006	0.005	0.005	0.003	0.002	0.001	0.001	0.006	0.002	0.001	0.000	0.004	0.002	0.001	0.005	0.002	0.004
$f^{bonff}_{1;t,r}$	0.008	0.007	0.006	0.005	0.005	0.004	0.004	0.003	0.002	0.002	0.001	0.006	0.005	0.004	0.004	0.003	0.007	0.006	0.006	0.005	0.005	0.003	0.002	0.001	0.001	0.005	0.002	0.001	0.000	0.005	0.002	0.001	0.006	0.001	0.004
$f^{bonff}_{2;t,r}$	0.007	0.006	0.006	0.006	0.005	0.004	0.004	0.003	0.002	0.002	0.001	0.005	0.005	0.004	0.004	0.003	0.006	0.006	0.006	0.004	0.004	0.003	0.002	0.001	0.001	0.005	0.002	0.001	0.000	0.005	0.003	0.001	0.005	0.001	0.004
$f^{bonff}_{3;t,r}$	0.006	0.005	0.006	0.006	0.005	0.005	0.004	0.003	0.002	0.002	0.001	0.005	0.005	0.005	0.005	0.003	0.005	0.005	0.005	0.004	0.004	0.003	0.002	0.001	0.001	0.004	0.002	0.001	0.000	0.005	0.003	0.001	0.005	0.001	0.004
$f^{bonff}_{4;t,r}$	0.005	0.005	0.005	0.005	0.007	0.006	0.004	0.003	0.002	0.002	0.001	0.004	0.004	0.004	0.003	0.002	0.004	0.004	0.004	0.003	0.003	0.002	0.001	0.001	0.001	0.004	0.002	0.001	0.000	0.004	0.002	0.001	0.004	0.001	0.004
$f^{bonff}_{5;t,r}$	0.004	0.004	0.004	0.004	0.006	0.006	0.005	0.003	0.002	0.002	0.001	0.005	0.005	0.005	0.004	0.003	0.005	0.005	0.005	0.004	0.004	0.003	0.002	0.001	0.001	0.004	0.002	0.001	0.000	0.004	0.002	0.001	0.004	0.001	0.004
$f^{bonff}_{6;t,r}$	0.004	0.004	0.004	0.004	0.004	0.005	0.005	0.003	0.002	0.002	0.002	0.005	0.005	0.004	0.004	0.003	0.004	0.004	0.004	0.003	0.003	0.002	0.002	0.001	0.001	0.004	0.002	0.001	0.000	0.004	0.002	0.001	0.004	0.001	0.004
$f^{bonff}_{7;t,r}$	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.004	0.003	0.003	0.002	0.004	0.003	0.003	0.003	0.002	0.003	0.003	0.003	0.003	0.003	0.002	0.001	0.001	0.001	0.003	0.002	0.001	0.000	0.003	0.002	0.001	0.003	0.001	0.003
$f^{bonff}_{8;t,r}$	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.003	0.005	0.004	0.003	0.002	0.002	0.002	0.002	0.001	0.002	0.002	0.002	0.001	0.002	0.001	0.001	0.001	0.001	0.002	0.001	0.000	0.000	0.002	0.001	0.000	0.002	0.000	0.002
$f^{bonff}_{9;t,r}$	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.004	0.005	0.005	0.002	0.002	0.002	0.002	0.000	0.004	0.003	0.003	0.003	0.003	0.003	0.002	0.002	0.002	0.003	0.002	0.000	0.000	0.003	0.002	0.000	0.003	0.000	0.003
$f^{bonff}_{10;t,r}$	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.003	0.005	0.007	0.001	0.002	0.001	0.001	0.002	0.001	0.001	0.002	0.002	0.002	0.002	0.001	0.001	0.001	0.002	0.001	0.000	0.000	0.002	0.001	0.000	0.002	0.000	0.002
$f^{bonff}_{11;t,r}$	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.003	0.003	0.001	0.010	0.008	0.008	0.008	0.003	0.007	0.007	0.007	0.004	0.004	0.004	0.003	0.003	0.003	0.005	0.003	0.002	0.000	0.005	0.003	0.001	0.005	0.001	0.005
$f^{bonff}_{12;t,r}$	0.006	0.005	0.005	0.005	0.004	0.005	0.005	0.003	0.002	0.002	0.002	0.008	0.008	0.008	0.008	0.004	0.008	0.007	0.007	0.006	0.005	0.004	0.003	0.002	0.002	0.004	0.002	0.001	0.000	0.004	0.002	0.001	0.004	0.001	0.004
$f^{bonff}_{13;t,r}$	0.005	0.004	0.004	0.004	0.004	0.004	0.004	0.003	0.002	0.002	0.001	0.008	0.008	0.009	0.007	0.004	0.007	0.006	0.006	0.005	0.005	0.003	0.002	0.002	0.002	0.003	0.002	0.001	0.000	0.003	0.002	0.001	0.003	0.001	0.003
$f^{bonff}_{14;t,r}$	0.005	0.004	0.004	0.004	0.003	0.004	0.005	0.003	0.002	0.003	0.003	0.007	0.007	0.011	0.009	0.004	0.005	0.005	0.006	0.004	0.004	0.003	0.003	0.003	0.003	0.004	0.003	0.002	0.000	0.004	0.002	0.001	0.004	0.001	0.004
$f^{bonff}_{15;t,r}$	0.003	0.002	0.002	0.002	0.001	0.002	0.003	0.001	0.002	0.004	0.005	0.004	0.004	0.009	0.011	0.002	0.002	0.002	0.004	0.004	0.002	0.003	0.003	0.002	0.002	0.003	0.003	0.002	0.000	0.003	0.002	0.001	0.003	0.001	0.003
$f^{bonff}_{16;t,r}$	0.007	0.006	0.005	0.005	0.005	0.005	0.004	0.003	0.003	0.004	0.003	0.008	0.007	0.007	0.004	0.002	0.010	0.006	0.006	0.005	0.006	0.003	0.002	0.001	0.001	0.002	0.001	0.000	0.000	0.002	0.001	0.000	0.002	0.000	0.002
$f^{bonff}_{17;t,r}$	0.006	0.007	0.006	0.005	0.004	0.005	0.004	0.003	0.003	0.001	0.001	0.007	0.006	0.006	0.007	0.005	0.006	0.009	0.006	0.005	0.005	0.004	0.002	0.001	0.001	0.002	0.001	0.000	0.000	0.002	0.001	0.000	0.002	0.000	0.002
$f^{bonff}_{18;t,r}$	0.006	0.006	0.006	0.005	0.005	0.004	0.004	0.003	0.003	0.001	0.002	0.006	0.005	0.005	0.005	0.004	0.005	0.005	0.006	0.012	0.006	0.005	0.003	0.002	0.002	0.003	0.002	0.001	0.000	0.003	0.002	0.001	0.003	0.001	0.003
$f^{bonff}_{19;t,r}$	0.005	0.005	0.004	0.004	0.004	0.004	0.003	0.003	0.003	0.002	0.002	0.005	0.005	0.005	0.005	0.004	0.002	0.006	0.005	0.006	0.005	0.004	0.003	0.002	0.002	0.003	0.002	0.001	0.000	0.003	0.002	0.001	0.003	0.001	0.003
$f^{bonff}_{20;t,r}$	0.003	0.003	0.003	0.003	0.002	0.003	0.003	0.003	0.004	0.003	0.003	0.004	0.003	0.003	0.003	0.003	0.003	0.004	0.003	0.003	0.003	0.003	0.004	0.003	0.003	0.004	0.003	0.002	0.000	0.004	0.002	0.001	0.004	0.002	0.004
$f^{bonff}_{21;t,r}$	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.003	0.004	0.004	0.004	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.002	0.003	0.003	0.003	0.004	0.004	0.004	0.004	0.003	0.002	0.000	0.004	0.002	0.001	0.004	0.002	0.004
$f^{bonff}_{22;t,r}$	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.008	0.008	0.009	0.007	0.003	0.007	0.006	0.006	0.002	0.002	0.003	0.004	0.004	0.004	0.005	0.004	0.002	0.000	0.005	0.003	0.001	0.005	0.002	0.005
$f^{bonff}_{23;t,r}$	0.006	0.005	0.004	0.004	0.004	0.004	0.004	0.003	0.003	0.002	0.002	0.007	0.007	0.008	0.007	0.004	0.009	0.009	0.009	0.008	0.009	0.007	0.003	0.002	0.002	0.032	0.006	0.001	0.000	0.003	0.002	0.000	0.005	0.001	0.004
$f^{bonff}_{24;t,r}$	0.002	0.002	0.002	0.002	0.001	0.001	0.002	0.002	0.003	0.004	0.005	0.003	0.003	0.003	0.003	0.003	0.004	0.003	0.002	0.002	0.003	0.004	0.004	0.004	0.005	0.006	0.006	0.000	0.000	0.001	0.000	0.000	0.001	0.000	0.000
$f^{bonff}_{25;t,r}$	-	-	-	-	-	-	-	-	-	-	0.000	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.004	0.001	0.000	0.001	0.001	0.002	0.001	0.002	0.001
$f^{bonff}_{26;t,r}$	0.001	0.001	0.001	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.003	0.000	0.000	0.000	0.000	0.000	0.000	0.000
$f^{bonff}_{27;t,r}$	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
$f^{bonff}_{28;t,r}$	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
$f^{bonff}_{29;t,r}$	0.004	0.005	0.005	0.005	0.004	0.003	0.003	0.002	0.002	0.002	0.002	0.003	0.003	0.003	0.003	0.002	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003	0.003
$f^{bonff}_{30;t,r}$	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
$f^{bonff}_{31;t,r}$	0.005	0.006	0.005	0.006	0.007	0.007	0.006	0.003	0.002	0.002	0.001	0.006	0.006	0.006	0.006	0.002	0.002	0.006	0.006	0.006	0.006	0.005	0.004	0.004	0.004	0.005	0.004	0.004	0.004	0.005	0.004	0.004	0.005	0.004	0.005
$f^{bonff}_{32;t,r}$	0.004	0.004	0.004	0.004	0.005	0.005	0.004	0.002	0.001	0.001	0.001	0.004	0.004	0.004	0.004	0.002	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.003	0.003	0.004	0.003	0.002	0.004	0.003	0.004

Type	Systematics	Comment	Stand-alone $1R_\mu$	Stand-alone $1R_e$	Joint OA $1R_\mu$	Joint OA $1R_e$	N_{syst}
BANFF	$f^{banff}_{0;t,r}$ - $f^{banff}_{24;t,r}$	ν_μ flux	11	11	11	11	25
		$\bar{\nu_\mu}$ flux	5	2	5	5	
		ν_e flux	–	7	7	7	
		$\bar{\nu_e}$ flux	–	2	2	2	
	$f^{banff}_{25;t,r}$	CCQE axial mass	✓	✓	✓	✓	8
	$f^{banff}_{26;t,r}$	resonance-production axial mass	✓	✓	✓	✓	
	$f^{banff}_{27;t,r}$ - $f^{banff}_{29;t,r}$	CCQE normalization	3	1	3	3	
	$f^{banff}_{30;t,r}$ - $f^{banff}_{31;t,r}$	CC1π normalization	2	1	2	2	
	$f^{banff}_{32;t,r}$	NC1π^0	–	✓	–	✓	
Independent cross-sections	$f_{pF;t,r}$	Fermi momentum	✓	✓	✓	✓	12
	$f_{bindE;t,r}$	Binding energy	✓	–	✓	✓	
	$f_{SF;t,r}$	Nuclear modeling	✓	✓	✓	✓	
	$f_{Wshape;t,r}$	Shape of the π momentum distribution	✓	✓	✓	✓	
	$f_{\pi-less\Delta;t,r}$	π-less Δ decay	✓	✓	✓	✓	
	$f_{CCothShape;t,r}$	CC other shape	✓	✓	✓	✓	
	$f_{CCcoh;t}$	CC coherent cross-section	✓	✓	✓	✓	
	$f_{NC1\pi^\pm;t}$	NC1$\pi^\pm$ cross-section	✓	–	✓	–	
	$f_{NCcoh;t}$	NC coherent cross-section	–	✓	–	✓	
	$f_{NCoth;t}$	NC other cross-section	✓	✓	✓	✓	
	$f_{CC\nu_e;t}$	σ_{ν_e}/ σ_{ν_μ} ratio	✓	✓	✓	✓	
	$f_{CC\bar{\nu};t}$	$\sigma_{\bar{\nu}}$/ σ_{ν} ratio	✓	–	✓	✓	
SK + FSI	$f^{SK}_{E;r}$	Reconstructed energy scale	✓	✓	✓	✓	1
	$f^{SK+FSI}_{0;r}$ - $f^{SK+FSI}_{5;r}$	$1R_\mu$ efficiencies	✓	–	✓	–	6
	$f^{SK+FSI}_{6;r}$ - $f^{SK+FSI}_{17;r}$	$1R_e$ efficiencies	–	✓	–	✓	12
Total			**41**	**49**	**64**		

Table 10: List of all systematic parameters included in the joint oscillation analysis. A ✓symbol indicates whether the systematic parameter is applied on the single μ-like ring and single e-like ring MC templates.

	ν_μ CCQE	ν_μ CC1π	ν_μ CC coherent	ν_μ CC other	ν_μ NC1$\pi^\pm$	ν_μ NC other	$\bar{\nu_\mu}$ CCQE	$\bar{\nu_\mu}$ CC1π	$\bar{\nu_\mu}$ CC coherent	$\bar{\nu_\mu}$ CC other	$\bar{\nu_\mu}$ NC1$\pi^\pm$	$\bar{\nu_\mu}$ NC other	ν_e CCQE	ν_e CC1π	ν_e CC coherent	ν_e CC other	ν_e NC1$\pi^\pm$	ν_e NC other	$\bar{\nu_e}$ CCQE	$\bar{\nu_e}$ CC1π	$\bar{\nu_e}$ CC coherent	$\bar{\nu_e}$ CC other	$\bar{\nu_e}$ NC1$\pi^\pm$	$\bar{\nu_e}$ NC other	Osc-ν_e CCQE	Osc-ν_e CC1π	Osc-ν_e CC coherent	Osc-ν_e CC other	Osc-ν_μ CCQE	Osc-ν_μ CC1π	Osc-ν_μ CC coherent	Osc-ν_μ CC other	Osc-$\bar{\nu_e}$ CCQE	Osc-$\bar{\nu_e}$ CC1π	Osc-$\bar{\nu_e}$ CC coherent	Osc-$\bar{\nu_e}$ CC other
$f^{SK}_{E;r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{0;t,r}$ - $f^{banff}_{10;t,r}$	✓	✓	✓	✓	✓	✓																			✓	✓	✓	✓								
$f^{banff}_{11;t,r}$ - $f^{banff}_{15;t,r}$							✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓
$f^{banff}_{16;t,r}$ - $f^{banff}_{22;t,r}$													✓	✓	✓	✓	✓	✓											✓	✓	✓	✓				
$f^{banff}_{23;t,r}$ - $f^{banff}_{24;t,r}$																			✓	✓	✓	✓	✓	✓												
$f^{banff}_{25;t,r}$	✓						✓						✓						✓						✓				✓				✓			
$f^{banff}_{26;t,r}$		✓		✓	✓	✓		✓		✓	✓	✓		✓		✓	✓	✓		✓		✓	✓	✓		✓		✓		✓		✓		✓		✓
$f^{banff}_{27;t,r}$	✓						✓						✓						✓						✓				✓				✓			
$f^{banff}_{28;t,r}$	✓						✓						✓						✓						✓				✓				✓			
$f^{banff}_{29;t,r}$	✓						✓						✓						✓						✓				✓				✓			
$f^{banff}_{30;t,r}$		✓						✓						✓						✓						✓				✓				✓		
$f^{banff}_{31;t,r}$		✓						✓						✓						✓						✓				✓				✓		
$f^{banff}_{32;t,r}$																																				
$f_{pF;t,r}$	✓						✓						✓						✓						✓				✓				✓			
$f_{SF;t,r}$	✓						✓						✓						✓						✓				✓				✓			
$f_{bindE;t,r}$	✓						✓						✓						✓						✓				✓				✓			
$f_{\pi-less\Delta;t,r}$		✓			✓	✓		✓			✓	✓		✓			✓	✓		✓			✓	✓		✓			✓				✓			
$f_{CCothShape;t,r}$				✓						✓						✓						✓						✓				✓				✓
$f_{Wshape;t,r}$					✓	✓					✓	✓					✓	✓					✓	✓												
$f_{CCcoh;t}$			✓						✓						✓						✓						✓				✓			✓	
$f_{NC1\pi^\pm;t}$					✓						✓						✓						✓												
$f_{NCcoh;t}$																																				
$f_{NCoth;t}$						✓						✓						✓						✓												
$f_{CC\nu_e;t}$													✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓					✓	✓	✓	✓
$f_{CC\bar{\nu};t}$							✓	✓	✓	✓									✓	✓	✓	✓											✓	✓	✓	✓
$f^{SK+FSI}_{0;r}$	✓						✓																						✓							
$f^{SK+FSI}_{1;r}$	✓						✓																						✓							
$f^{SK+FSI}_{2;r}$	✓						✓																						✓							
$f^{SK+FSI}_{3;r}$		✓	✓	✓				✓	✓	✓																				✓	✓	✓				
$f^{SK+FSI}_{4;r}$													✓	✓	✓	✓			✓	✓	✓	✓			✓	✓	✓	✓					✓	✓	✓	✓
$f^{SK+FSI}_{5;r}$					✓	✓					✓	✓					✓	✓					✓	✓												
$f^{SK+FSI}_{6;r}$ - $f^{SK+FSI}_{17;r}$																																				

Table 11: A ✓symbol denotes that the given single μ-like ring MC template is modified when the given systematic parameter is tweaked. The table lists all systematic parameters considered in this analysis (first column), and all 36 MC templates used for constructing the single μ-like ring event reconstructed energy spectrum p.d.f (first row).

	ν_μ CCQE	ν_μ CC1π	ν_μ CC coherent	ν_μ CC other	ν_μ NC1π^0	ν_μ NC coherent	ν_μ NC other	$\bar{\nu}_\mu$ CCQE	$\bar{\nu}_\mu$ CC1π	$\bar{\nu}_\mu$ CC coherent	$\bar{\nu}_\mu$ CC other	$\bar{\nu}_\mu$ NC1π^0	$\bar{\nu}_\mu$ NC coherent	$\bar{\nu}_\mu$ NC other	ν_e CCQE	ν_e CC1π	ν_e CC coherent	ν_e CC other	ν_e NC1π^0	ν_e NC coherent	ν_e NC other	$\bar{\nu}_e$ CCQE	$\bar{\nu}_e$ CC1π	$\bar{\nu}_e$ CC coherent	$\bar{\nu}_e$ CC other	$\bar{\nu}_e$ NC1π^0	$\bar{\nu}_e$ NC coherent	$\bar{\nu}_e$ NC other	Osc-ν_e CCQE	Osc-ν_e CC1π	Osc-ν_e CC coherent	Osc-ν_e CC other	Osc-ν_μ CCQE	Osc-ν_μ CC1π	Osc-ν_μ CC coherent	Osc-ν_μ CC other	Osc-$\bar{\nu}_e$ CCQE	Osc-$\bar{\nu}_e$ CC1π	Osc-$\bar{\nu}_e$ CC coherent	Osc-$\bar{\nu}_e$ CC other
$f^{SK}_{E;r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{banff}_{0;t,r}$ - $f^{banff}_{10;t,r}$	✓	✓	✓	✓	✓	✓	✓																							✓	✓	✓	✓								
$f^{banff}_{11;t,r}$ - $f^{banff}_{15;t,r}$								✓	✓	✓	✓	✓	✓	✓																						✓	✓	✓	✓
$f^{banff}_{16;t,r}$ - $f^{banff}_{22;t,r}$															✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓				
$f^{banff}_{23;t,r}$ - $f^{banff}_{24;t,r}$																						✓	✓	✓	✓	✓	✓	✓												
$f^{banff}_{25;t,r}$	✓							✓							✓							✓							✓				✓				✓			
$f^{banff}_{26;t,r}$		✓		✓	✓		✓		✓		✓	✓		✓		✓		✓	✓		✓		✓		✓	✓		✓		✓		✓		✓		✓		✓		✓
$f^{banff}_{27;t,r}$	✓							✓							✓							✓							✓				✓				✓			
$f^{banff}_{28;t,r}$	✓							✓							✓							✓							✓				✓				✓			
$f^{banff}_{29;t,r}$	✓							✓							✓							✓							✓				✓				✓			
$f^{banff}_{30;t,r}$		✓							✓							✓							✓							✓				✓				✓		
$f^{banff}_{31;t,r}$		✓							✓							✓							✓							✓				✓				✓		
$f^{banff}_{32;t,r}$					✓							✓							✓						✓														
$f_{pF;t,r}$	✓							✓							✓							✓							✓				✓				✓			
$f_{SF;t,r}$	✓							✓							✓							✓							✓				✓				✓			
$f_{bindE;t,r}$	✓							✓							✓							✓							✓				✓				✓			
$f_{\pi-less\Delta;t,r}$		✓			✓		✓		✓			✓		✓		✓			✓		✓		✓			✓		✓			✓				✓				✓		
$f_{CCothShape;t,r}$				✓							✓							✓							✓							✓							✓
$f_{Wshape;t,r}$					✓		✓					✓		✓					✓		✓					✓		✓												
$f_{CCcoh;t}$			✓							✓							✓							✓							✓							✓		
$f_{NC1\pi^\pm;t}$																																								
$f_{NCcoh;t}$						✓							✓							✓						✓														
$f_{NCoth;t}$							✓							✓							✓						✓													
$f_{CC\nu_e;t}$															✓	✓	✓	✓				✓	✓	✓	✓				✓	✓	✓	✓					✓	✓	✓	✓
$f_{CC\bar{\nu};t}$								✓	✓	✓	✓											✓	✓	✓	✓												✓	✓	✓	✓
$f^{SK+FSI}_{0;r}$ - $f^{SK+FSI}_{5;r}$																																								
$f^{SK+FSI}_{6;r}$ - $f^{SK+FSI}_{8;r}$																													✓	✓	✓	✓								
$f^{SK+FSI}_{9;r}$ - $f^{SK+FSI}_{11;r}$	✓	✓	✓	✓				✓	✓	✓	✓																						✓	✓	✓	✓				
$f^{SK+FSI}_{12;r}$ - $f^{SK+FSI}_{14;r}$															✓	✓	✓	✓				✓	✓	✓	✓												✓	✓	✓	✓
$f^{SK+FSI}_{15;r}$ - $f^{SK+FSI}_{17;r}$					✓	✓	✓					✓	✓	✓					✓	✓	✓					✓	✓	✓												

Table 12: A ✓symbol denotes that the given single e-like ring MC template is modified when the given systematic parameter is tweaked. The table lists all systematic parameters considered in this analysis (first column) and all 40 MC templates used for constructing the single e-like ring event reconstructed energy spectrum p.d.f (first row).

Source of uncertainty	1Rμ $\delta N_{SK}/N_{SK}$	1Re $\delta N_{SK}/N_{SK}$
SK+FSI	4.997%	3.654%
SK+FSI wo correlations	5.610%	3.472%
BANFF (prefit)	21.748%	26.044%
BANFF (postfit)	2.743%	3.147%
Independent XSec	4.999%	4.686%
Total (BANFF postfit)	7.648%	6.752%
Total (BANFF postfit) wo SK+FSI correlations	8.086%	6.653%
Total (BANFF prefit)	23.448%	26.804%

Table 13: Effect of 1σ systematic parameter variation on the number of single μ-like ring and single e-like ring events, calculated throwing 10^6 toy experiments, varying only the systematics in the selected category (fixing the rest to their nominal value) and finding the RMS/mean of the distribution of number of events of the 10^6 toy experiments. The numbers shown were computed for oscillations with $\sin^2\theta_{12}$=0.306, $\sin^2\theta_{13}$=0.0251, $\sin^2\theta_{23}$=0.5, δm^2=7.5$\times10^{-5}$ eV^2/c^4, $|\Delta m^2|$=2.4375$\times10^{-3}$ eV^2/c^4, δ_{CP}=0 and with the normal mass hierarchy assumed, for an exposure of 6.57$\times10^{20}$ POT and BANFF-tuned MC templates were used.

	68% CL	90% CL
$sin^2\theta_{23}$ (NH)	[0.458,0.568]	[0.428,0.598]
$sin^2\theta_{23}$ (IH)	[0.456,0.566]	[0.427,0.596]
θ_{23} (NH)	[42.6,48.9]	[40.9,50.7]
θ_{23} (IH)	[42.5,48.8]	[40.8,50.5]
$\Delta m^2_{32}(10^3 eV^2/c^4)$ (NH)	[2.41,2.61]	[2.34,2.68]
$\Delta m^2_{13}(10^3 eV^2/c^4)$ (IH)	[2.38,2.58]	[2.31,2.64]

5.1. Evaluation of effects of systematic parameters on the predicted single μ-like ring and single e-like ring reconstructed energy spectra and number of events

In this section, the effects of the systematics described in Sec. 5 on the predicted number of single μ-like ring events and single e-like ring events, and the predicted reconstructed energy spectra in SuperK will be evaluated.

The effects of systematic uncertainties on the predicted number of events are summarised in Tab. 13, for oscillations with $\sin^2\theta_{12}$=0.306, $\sin^2\theta_{13}$=0.0251, $\sin^2\theta_{23}$=0.5, δm^2=7.5$\times10^{-5}$ eV^2/c^4, $|\Delta m^2|$=2.4375$\times10^{-3}$ eV^2/c^4, δ_{CP}=0 and with the normal mass hierarchy assumed. In this table, systematic parameters are grouped into four categories (BANFF, independent cross sections, SK+FSI errors and total) and all parameter correlations are taken into account. Percentage uncertainties are given, computed throwing 10^6 toy experiments, varying only the systematics in the selected category (fixing the rest to their nominal value) and finding the RMS/mean of the distribution of number of events of the 10^6 toy experiments. Results were generated for an integrated exposure of 6.57$\times10^{20}$ POT and the BANFF-tuned MC templates were used.

The effects of the systematic uncertainties on the reconstructed energy spectrum of single μ-like ring events and single e-like ring events are illustrated in Figs. 14-17, for the typical oscillation scenario used. Figure 14 shows the effect of the BANFF systematics, Fig. 15 shows the effect of the independent cross-section systematics, Fig. 16 shows the effect of the SK+FSI systematics and Fig. 17 shows the total error envelope combining all systematic uncertainties. These plots showing the error envelope were calculated as the $\pm1\sigma$ spread of bin contents from 20k toy MC experiments generated with randomized systematic parameters (the ones in each category for 14-16 and all of them for 17), taking into account all correlations between them.

The error envelope in Figs. 16 and 17 spikes in the last bin of the single e-like ring spectra (notice that the last reconstructed energy bin in the single μ-like ring spectra is not included in the plots, whose range is limited between 0 and 7 GeV) due to the effect of tweaking the reconstructed energy scale systematic parameter: a positive (negative) tweak of the reconstructed energy scale systematic produces that events are gained (lost) in the last bin from (to) the previous bin, but no events are lost (gained) to (from) the next bin since it does not exist, producing an increment of the error envelope in that last bin. This can be easily tested if repeating the study in plot 16 without tweaking the reconstructed energy scale systematic: this is done in Fig. 18 and it shows that this effect disappears when not tweaking this systematic.

Figure 19 shows the ratio of the error envelopes in Figs. 14-16 to the total error envelope in Fig. 17 for the maximum (top plots) and minimum (bottom plots) spread.

Detailed studies showing the effects of individual systematics on the expected number of single μ-like ring and single e-like ring events and spectra are presented in [36].

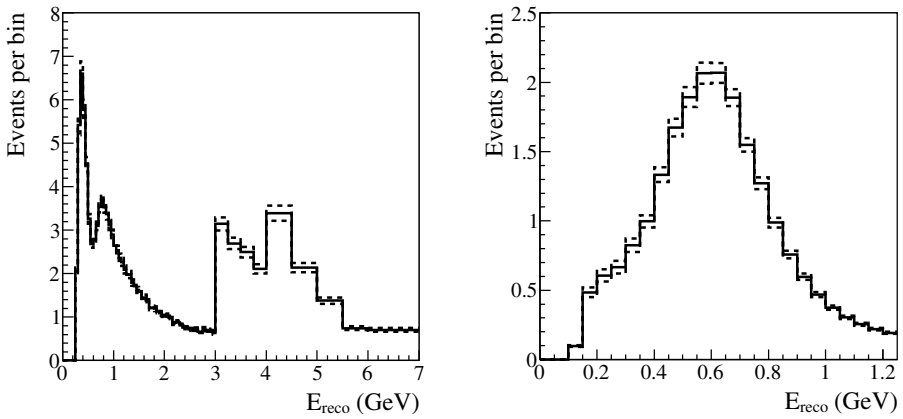


Figure 14: Error envelope of the single μ-like ring and single e-like event reconstructed energy spectra, for oscillations with typical values and the normal mass hierarchy assumed, for the BANFF systematic parameters. The numbers shown were calculated for an exposure of 6.57×10^{20} POT. BANFF-tuned MC templates were used. The error envelope was calculated as the $\pm1\sigma$ spread of bin contents using an ensemble of 20k toy MC experiments generated with randomized BANFF systematic parameters, taking into account their correlations.

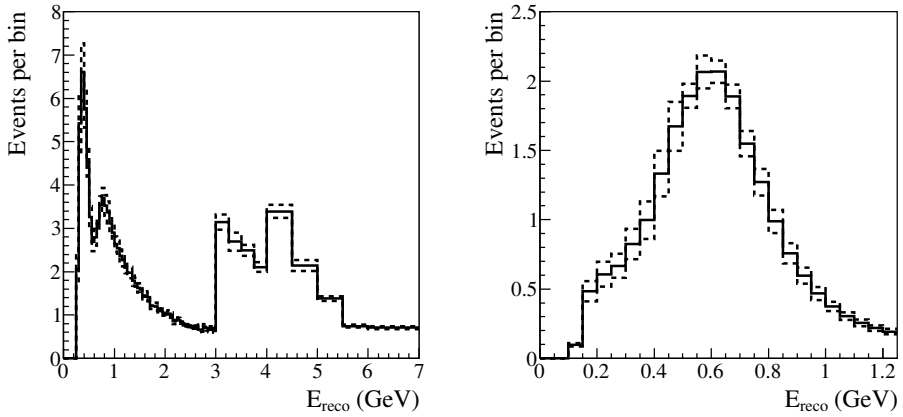


Figure 15: Error envelope of the single μ-like ring and single e-like event reconstructed energy spectra, for oscillations with typical values and the normal mass hierarchy assumed, for the independent cross-section systematic parameters. The numbers shown were calculated for an exposure of 6.57×10^{20} POT. BANFF-tuned MC templates were used. The error envelope was calculated as the $\pm1\sigma$ spread of bin contents using an ensemble of 20k toy MC experiments generated with randomized independent cross-section systematic parameters.

6. Neutrino oscillation fitting method

6.1. Fit method

Several choices exist for the oscillation parameters that could be fitted. As it has been already introduced in Sec. 3, this analysis is fitting for $\sin^2(2\theta_{23})$ in order to take into account the effect of the octant of this parameter (as it was already discussed, during the review process of the T2K Run 1+2+3 ν_μ-disappearance PRL [22] it was found that the choice of the octant had a large effect on the confidence contour in $\sin^2(2\theta_{23})$).

There are also different choices for the mass-squared splitting parameter: typically $|\Delta m^2_{32}|$ is used in ν_μ-disappearance analyses, but, as explained in Sec.3, the value of this parameter is different depending on the mass hierarchy, since it is the largest mass-squared splitting for the inverted mass hierarchy and the second largest for the normal mass hierarchy. Thus, it can be confusing to compare results obtained under the different mass hierarchy hypotheses. For this reason, this analysis is fitting for the mass-squared difference defined as $\Delta m^2 = \left(m^2_3 - \frac{m^2_1+m^2_2}{2}\right)$, whose absolute value is the same for both mass hierarchies [32]. Another option, used in the *VALOR* Run 1+2+3+4 ν_μ-disappearance analysis [6], is to fit $|\Delta m^2_{32}|$ for the normal hierarchy and $|\Delta m^2_{31}|$ for the inverted hierarchy.

The *VALOR* joint oscillation analysis, like the *VALOR* disappearance analysis, uses a binned likelihood-ratio method. Mea-

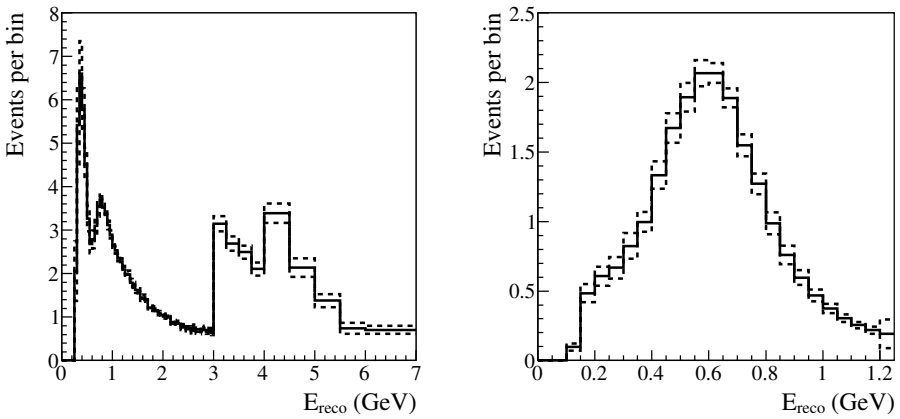


Figure 16: Error envelope of the single μ-like ring and single e-like event reconstructed energy spectra, for oscillations with typical values and the normal mass hierarchy assumed, for the SK+FSI systematic parameters. The numbers shown were calculated for an exposure of 6.57×10^{20} POT. BANFF-tuned MC templates were used. The error envelope was calculated as the $\pm1\sigma$ spread of bin contents using an ensemble of 20k toy MC experiments generated with randomized SK+FSI systematic parameters, taking into account their correlations.

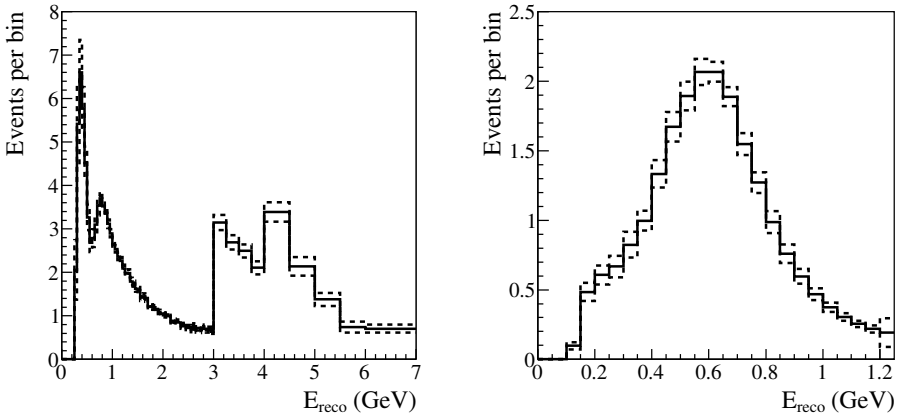


Figure 17: Total error envelope of the single μ-like ring and single e-like event reconstructed energy spectra, for oscillations with typical values and the normal mass hierarchy assumed. The numbers shown were calculated for an exposure of 6.57×10^{20} POT. BANFF-tuned MC templates were used. The error envelope was calculated as the $\pm1\sigma$ spread of bin contents using an ensemble of 20k toy MC experiments generated with all systematic parameters randomized, taking into account their correlations.

surements of the oscillation parameters $\sin^2\theta_{23}$, $|\Delta m^2|$, $\sin^2\theta_{13}$ and δ_{CP}[3] are obtained by comparing the observed and predicted SuperK reconstructed energy spectra for single μ-like ring events and single e-like ring events.

Let N_μ and N_e be the number of reconstructed energy bins considered for the single μ-like ring and single e-like ring samples respectively, and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the SuperK reconstructed energy spectrum prediction of either sample (e.g. parameters that control the beam, neutrino interaction simulations and the detector response). Best-fit values are obtained by minimizing:

[3] As already mentioned, a study has been made in which $\sin^2\theta_{12}$ and δm^2 were allowed to float in the fit and the results are shown in Appendix B. The effect was negligible and it was decided to keep them fixed to the values in Tab.1.

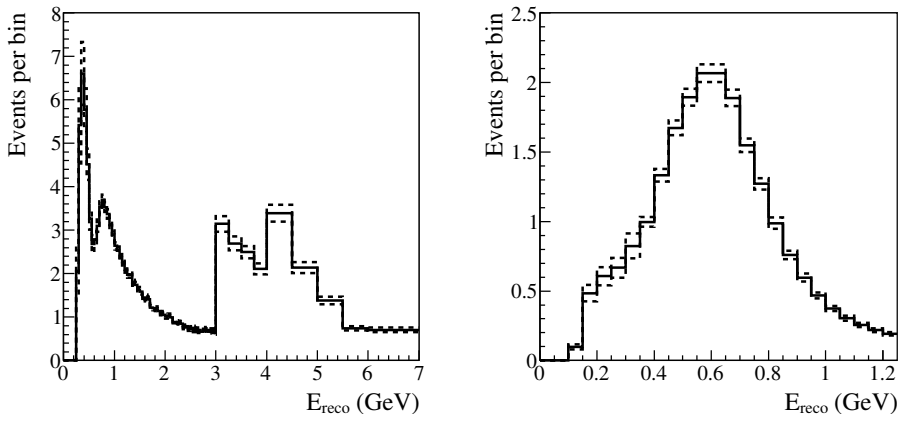


Figure 18: As in Fig. 16, but not tweaking the reconstructed energy scale systematic.

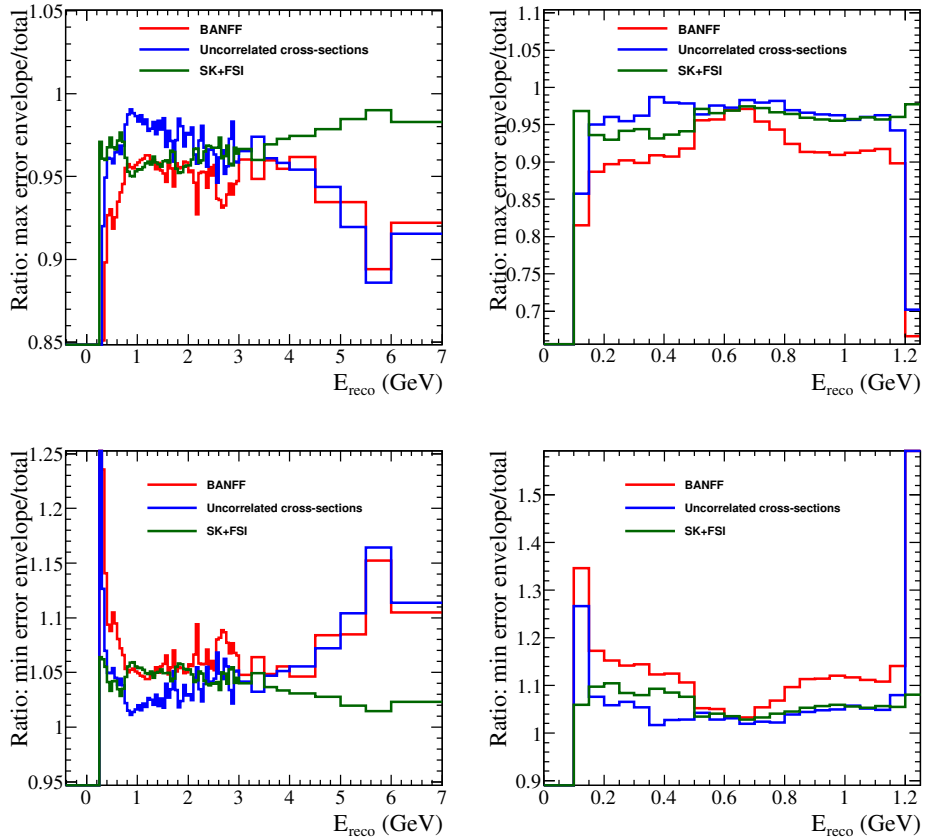


Figure 19: Ratios of the error envelopes in Figs. 14-16 to the total error envelope in Fig. 17 for the maximum (top plots) and minimum (bottom plots) spread.

$$
\begin{aligned}
-2\ln\lambda(|\Delta m^2|, sin^2\theta_{23}, \sin^2\theta_{13}, \delta_{CP}; \mathbf{a}) = 2 \cdot \sum_{r=0}^{N_\mu - 1} \left[(N^{exp}_{SK;\mu,r} - N^{obs}_{SK;\mu,r}) + N^{obs}_{SK;\mu,r} \cdot ln\left(\frac{N^{obs}_{SK;\mu,r}}{N^{exp}_{SK;\mu,r}}\right)\right] \\
+ 2 \cdot \sum_{r=0}^{N_e - 1} \left[(N^{exp}_{SK;e,r} - N^{obs}_{SK;e,r}) + N^{obs}_{SK;e,r} \cdot ln\left(\frac{N^{obs}_{SK;e,r}}{N^{exp}_{SK;e,r}}\right)\right] \\
+ (\mathbf{a} - \mathbf{a_0})^T \cdot \mathbf{C}^{-1} \cdot (\mathbf{a} - \mathbf{a_0})
\end{aligned}
\tag{8}
$$

where $N^{obs}_{SK;\mu,r}$ ($N^{obs}_{SK;e,r}$) is the observed number of single μ-like ring (single e-like ring) events in the rth bin, and $N^{exp}_{SK;\mu,r}$ ($N^{exp}_{SK;e,r}$) is the corresponding expected number of events; this expected number is a function of $|\Delta m^2|$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$ and δ_{CP} and the nuisance parameter vector $\mathbf{a}$. Also $\mathbf{a_0}$ is a $1 \times N_s$ - dimensional array containing the nominal values of the systematic parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, and $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$. For the stand-alone ν_μ-disappearance (ν_e-appearance) fit, only the single μ-like ring (single e-like ring) sample is used and the corresponding term in Eq. 8 is dropped.

The likelihood-ratio method used in this analysis is equivalent to the extended maximum likelihood method used in the alternative T2K oscillation analysis (see [2, 58] for more details). The advantage of the likelihood ratio method, compared with the extended maximum-likelihood method, is that in the large-sample limit, the quantity $-2\ln\lambda(\sin^2\theta_{23}, |\Delta m^2_{32}|$ or $|\Delta m^2_{31}|; \mathbf{a})$ in Eq.8 has a χ^2 distribution and it can therefore be used as a goodness-of-fit test. In this note, $-2\ln\lambda(\sin^2\theta_{23}, |\Delta m^2_{32}|$ or $|\Delta m^2_{31}|; \mathbf{a})$ and $\chi^2(\sin^2\theta_{23}, |\Delta m^2_{32}|$ or $|\Delta m^2_{31}|; \mathbf{a})$ are used interchangeably. The minimization is performed with Minuit [59], using the Migrad method. If Migrad does not converge, the starting values of the oscillation parameters are re-initialised: the new initial values are randomly drawn from uniform distributions in the ranges 2×10^{-3} eV^2/c^4 $\leq |\Delta m^2| \leq 3 \times 10^{-3}$ eV^2/c^4, $0.015 \leq \sin^2\theta_{13} \leq 0.035$ (or $0.05 \leq \sin^2 2\theta_{13} \leq 0.15$), $-\pi \leq \delta_{CP} \leq \pi$ and either $0.35 \leq \sin^2\theta_{23} \leq 0.50$ or $0.53 \leq \sin^2\theta_{23} \leq 0.68$ (the new initial value of $\sin^2\theta_{23}$ is in the same octant as the previous initial value). The fit is retried with the new initial values, and this is repeated up to 25 times. If the fit still fails to converge after 25 attempts, the Simplex algorithm is used.

The search for the minimum value of $-2\ln\lambda(|\Delta m^2|, \sin^2\theta_{23}, \sin^2\theta_{13}, \delta_{CP}; \mathbf{a})$ is performed in the range 1.0×10^{-3} eV^2/c^4 $\leq |\Delta m^2| \leq 6.0 \times 10^{-3}$ eV^2/c^4, $0.0 \leq \sin^2\theta_{23} \leq 1.0$, $0.0 \leq \sin^2\theta_{13} \leq 1.0$ and $-\pi \leq \delta_{CP} \leq \pi$. After a successful fit, The Hesse algorithm is called to improve Minuit's estimates of the errors on the fitted parameters.

In order to find a solution in each octant for $\sin^2\theta_{23}$, and since it is unlikely that Minuit jumps to the minimum in the octant mirror to the one it was initilized in, each fit is repeated twice: in the first fit, the initial value is $\sin^2\theta_{23} = 0.6$, while the second fit uses as initial value the mirror point in the other octant of the best fit of $\sin^2\theta_{23}$ from the first fit (taking into account the best-fit value of $\sin^2\theta_{13}$ from the first fit to calculate the correct mirror point: $(\sin^2\theta_{23})_{mirror}$ = center $\pm|$ center - $(\sin^2\theta_{23})_{bf}|$ where the center of symmetry (value of maximal disappearance) is calulated as center = $(2(1 - (\sin^2\theta_{13})_{bf}))^{-1}$ as explained in Appendix E of [1]). An simpler method, setting initial values $\sin^2\theta_{23} = 0.6$ and $\sin^2\theta_{23} = 0.4$ for the two fits, was also tested, resulting as well in a solution in each octant as expected.

All the 64 systematic parameters are allowed to float in the fit, and are restricted to the range [-5σ_s, +5σ_s] where σ_s is the one standard deviation error assigned to each systematic parameter[4]. Values of systematic parameters that give a negative predicted number of events in any reconstructed energy bin are not allowed. If this scenario arises, the number of events is changed to +1E-8 in that bin. This is done for the sum of all the interaction modes rather than for each mode separately.

6.2. *Adding reactor constraint*

In this technical note, the results of the Run 1+2+3+4 dataset will be presented in two ways: one was done fitting only T2K dataset and one will be including the reactor constraint in the fit. This constraint works adding a penalty term to the χ^2 function, proportional to the difference between the best-fit value of $\sin^2\theta_{13}$ and the one used as constraint. The $\chi^2_{reactor}$ is calculated following Eq. 9, where the values of $(\sin^2\theta_{13})_{reactor}$ and $\sigma_{reactor}$ used were the latest 2013 PDG values, coming from a weighted average of the results from the three reactor experiments Daya Bay, RENO and Double Chooz [28]: $(\sin^2\theta_{13})_{reactor}$ = 0.095 and $\sigma_{reactor}$ = 0.01.

$$\chi^2_{reactor} = \left(\frac{\sin^2\theta_{13} - (\sin^2\theta_{13})_{reactor}}{\sigma_{reactor}}\right)^2 \quad (9)$$

6.3. *Goodness-of-fit tests*

In this analysis, the procedure to be used for computing the goodness-of-fit will be the same one used in the *VALOR* Run 1+2+3 joint oscillation analysis [1].

The problem of defining a general goodness-of-fit for sparsely populated distributions was discussed in detail during the development of the Run 1+2+3 stand-alone ν_μ-disappearance and ν_e-appearance analyses. See for example [60]. It was pointed out that, for the Run 1+2+3 statistics, the shape information is not as helpful in forming p-values as the rate information. It was decided that p-values will be calculated using the rate and the reconstructed energy distribution, and that the distribution will use a very coarse binning scheme to ensure that there are sufficient numbers of events in each bin.

[4] Such an error is defined for all but one of the systematic parameters considered in this analysis. The exception is the systematic parameters f_{SF} which parameterizes the uncertainty on nuclear modelling, switching between the Relativistic Fermi Gas (RFG) and Spectral Function (SF) models. The allowed range of the parameter f_{SF} in the oscillation fit is [0,1].

A similar approach is the one used for the joint 3-flavour analysis presented in [1]. For the single μ-like ring sample, 5 reconstructed energy bins were used: 0-0.4, 0.4-0.7, 0.7-1.0, 1.0-2.0 and 2.0-30.0 GeV. For the single e-like ring sample, 3 reconstructed energy bins were used: 0-0.35, 0.35-0.80, 0.80-1.25 GeV. In this note, p-values for the joint 3-flavour fit are computed following the 5+3 binning scheme described above. When p-values were quoted for the stand-alone ν_μ-disappearance (ν_e-appearance) analysis, the corresponding 5 (3) bin scheme of the single μ-like ring (single e-like ring) event sample was used.

The goodness-of-fit tests that will be shown in this technical note are built using 1000 toy MC experiments created at the best-fit point to be tested, with statistical and systematic fluctuations. Then, each toy MC experiment is fitted in the coarse binning described above, including the four oscillation parameters and all systematics in the joint 3-flavour fit.

6.4. Construction of confidence regions

In this technical note, confidence regions are constructed using the constant $\Delta\chi^2$ method [63] (pp. 390-401). 68% and 90% CL limits are reported for each of $|\Delta m^2|$, $sin^2\theta_{23}$, $sin^2\theta_{13}$ and δ_{CP} oscillation parameters. In addition, 2-dimensional confidence regions for 68% and 90% CL will be shown for the following combinations of parameters: ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2\theta_{13}$). When introducing the reactor constraint, the following additional 2-dimensional confidence regions are also presented: ($|\Delta m^2|$,δ_{CP}) and ($sin^2\theta_{23}$,δ_{CP}).

The first step to construct confidence contours in an oscillation parameter space is to build the $\Delta\chi^2$ surface in that space. The oscillation parameter space is divided into a regular grid. The grid limits and spacing was used in all confidence regions is presented in Tab. 14.

Parameter	Min	Max	Step	Bins
$sin^2\theta_{23}$	0.3	0.7	0.01	41
$\|\Delta m^2\|$ ($10^{-3}eV^2/c^4$)	2.0	3.0	0.05	21
$sin^2\theta_{13}$	0.0	0.1	0.005	21
$sin^2 2\theta_{13}$	0.0	0.4	0.02	21
$sin^2\theta_{13}$ (reactor)	0.015	0.035	0.001	21
$sin^2 2\theta_{13}$ (reactor)	0.06	0.14	0.004	21
δ_{CP}	$-\pi$	π	$2\pi/50$	51

Table 14: Grid limits and spacing used in all confidence regions.

For the joint 3-flavour analysis confidence regions, the χ^2 is minimized at each grid point with respect to all 64 nuisance parameters considered in this analysis and all oscillation parameters not shown on the plot (and not considered to be fixed).

The confidence regions for the joint analysis are constructed by finding the global best-fit point (fitting all oscillation parameters and all systematic parameters at the same time) and calculating the lines of *constant* $\Delta\chi^2$ from this point[5]. A special procedure is followed to construct the confidence regions in ($sin^2\theta_{13}$, δ_{CP}) space, called *raster scan*, where a best-fit value of $sin^2\theta_{13}$ is found for each fixed value of δ_{CP}, resulting in a line of best-fit points instead of a single best-fit point. Then, confidence intervals in $sin^2\theta_{13}$ at a fixed δ_{CP} are found with respect to each best-fit $sin^2\theta_{13}$ using the 1D $\Delta\chi^2$ critical values [56].

The method of Feldman and Cousins could be used to find the critical values of $\Delta\chi^2$ for the confidence regions of the joint oscillation analysis. In that case, the prescription to find the Feldman-Cousins confidence regions would be imitating the one used for constructing 1D confidence regions in the *VALOR* ν_μ disappearance analysis [6]. Independent Feldman-Cousins confidence regions would be computed for each one of the 2-dimensional parameter spaces separately, using the results in this analysis to select the true input values of the other two oscillation parameters which are fitted for each 2-dimensional parameter space, following for them the $\Delta\chi^2$ surface obtained from the Run 1+2+3+4 dataset fit obtained in the parameter space defined by those other two oscillation parameters. Using these $\Delta\chi^2$ surfaces, the other two oscillation parameters would be varied following the preference of T2K Run 1+2+3+4 data.

For the inclusion of systematics in the Feldman and Cousins method, the same procedure as used for the *VALOR* ν_μ disappearance analysis would be followed. To construct a 2-dimensional confidence region, for instance ($\sin^2\theta_{23}$, $|\Delta m^2|$), where N systematic parameters are included in the p.d.f. and all of these N systematics are allowed to float in the fit, the value of $\Delta\chi^2_{critical}$ at each true ($\sin^2\theta_{23}$, $|\Delta m^2|$) grid point would be calculated from an ensemble of toy MC experiments, each one of which is generated with the true oscillation parameters of that point and with all N systematics randomised. Then, for each toy MC experiment, both $\sin^2\theta_{23}$, $|\Delta m^2|$ and all N nuisance parameters along with $\sin^2\theta_{13}$ and δ_{CP} are fitted simultaneously to obtain the value of χ^2(MC, best fit). A separate fit, with $\sin^2\theta_{23}$ and $|\Delta m^2|$ fixed to their true values and

[5] For 1 fit parameter, the critical values of $\Delta\chi^2$ used are 1.00 (68% CL), 2.71 (90% CL), 3.84 (95% CL), 6.63 (99% CL). For 2 parameters fitted simultaneously, the critical values of $\Delta\chi^2$ used are 2.30 (68% CL), 4.61 (90% CL), 5.99 (95% CL), 9.21 (99% CL).

with all N nuisance parameters allowed to float along with $\sin^2\theta_{13}$ and δ_{CP}, is performed to obtain the value of χ^2(MC, true) (i.e. χ^2 is *independently* minimised, for *each* toy MC experiment, with respect to the nuisance parameters).

Feldman-Cousins values of critical $\Delta\chi^2$ vs δ_{CP} have already been calculated and presented in Sec. 7.2.3. The procedure used to calculate them is explained in that section.

7. Results of the joint oscillation analysis with the Run 1+2+3+4 (6.570 $\times$ 10^{20} POT) dataset

Despite the new convention for mass-squared splitting parameters discussed in 3 and used in the rest of this technical note, for the final results it was requested by the oscillation analysis conveners to produce the plots using $|\Delta m^2| = |\Delta m^2_{32}|(NH), |\Delta m^2_{13}|(IH)$ when comparing results in the two hierarchies and Δm^2_{32} for both mass hierarchies (with opposite signs) for comparisons with MINOS results.

7.1. Results with T2K data only

In this subsection the results of the joint oscillation analysis with the Run 1+2+3+4 (6.570 $\times$ 10^{20} POT) dataset without using reactor constraint will be presented, i.e. using only T2K data. Best-fit oscillation parameters, systematic parameter pulls and best-fit single e-like ring and single μ-like ring event reconstructed energy spectra are presented in 7.1.1, confidence regions in one dimensional and two dimensional parameter spaces are presented in 7.1.2 and goodness-of-fit tests in 7.1.3.

7.1.1. Best-fit oscillation parameters, systematic parameter pulls and best-fit single e-like ring and single μ-like ring event reconstructed energy spectra

A joint 3-flavour oscillation fit, using the method described in Sec. 6 was performed simultaneously on the single e-like ring and single μ-like ring event reconstructed energy spectra of the 6.570$\times10^{20}$ POT Run 1+2+3+4 dataset. This fit allowed all 64 nuisance parameters considered in this work to float. This analysis found $|\Delta m^2_{32}|$ = 2.512 $\times10^{-3}$ eV^2/c^4, $\sin^2\theta_{23}$ = 0.524, $\sin^2\theta_{13}$ = 0.0422 and δ_{CP} = 1.909 (p-value = 0.921) to be the joint 3-flavour oscillation best-fit values for the Run 1+2+3+4 dataset for Normal Hierarchy and $|\Delta m^2_{13}|$ = 2.488 $\times10^{-3}$ eV^2/c^4, $\sin^2\theta_{23}$ = 0.523, $\sin^2\theta_{13}$ = 0.0491 and δ_{CP} = 1.005 (p-value = 0.952) for Inverted Hierarchy. Between the two best-fit points, the one with a smaller value of χ^2 is the one with Inverted Hierarchy, so the two dimensional confidence regions presented in 7.1.2 were built with a common best-fit point for the two mass hierarchies which is the one for the Inverted Hierarchy assumption. Table 15 summarizes the best-fit values for the two mass hierarchies.

Analysis	Hierarchy	$\|\Delta m^2_{32}\|(NH)$ $\|\Delta m^2_{13}\|(IH)$ $10^{-3}eV^2/c^4$	$\sin^2\theta_{23}$	$\sin^2\theta_{13}$/$\sin^2 2\theta_{13}$	δ_{CP}	$N^{1R\mu}_{obs}$	$N^{1R\mu}_{exp}$	N^{1Re}_{obs}	N^{1Re}_{exp}	χ^2/ndf
$\sin^2\theta_{13}$	NH	2.512	0.524	0.0422	1.909	120	119.915	28	27.999	84.5395/94
$\sin^2 2\theta_{13}$	NH	2.512	0.524	0.162	1.909	120	119.915	28	27.999	84.5395/94
$\sin^2\theta_{13}$	IH	2.488	0.523	0.0491	1.005	120	119.948	28	27.998	84.529/94
$\sin^2 2\theta_{13}$	IH	2.488	0.523	0.187	1.005	120	119.948	28	27.998	84.529/94

Table 15: Summary of best-fit parameters for the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset. The quoted best-fit χ^2 values were computed from the test-statistic minimized in the fit (likelihood ratio) with the reconstructed energy binning used in each fit. The table shows the best-fit values for the different mass hierarchy assumptions without applying the reactor constraint. The fit was repeated using $\sin^2 2\theta_{13}$, no difference was found with respect to the results using $\sin^2\theta_{13}$.

The pull for each systematic parameter f included in the fit was calculated as

$$\frac{f_{bf} - f_{nom}}{\sigma_{bf}}, \tag{10}$$

where f_{bf} is the best-fit value of the systematic parameter, f_{nom} is the nominal value of the parameter (corresponding to no systematic variation), and σ_{bf} is the HESSE error output from Minuit. The pulls of all systematic parameters allowed to float in the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset are shown in Figs. 20 for the fit assuming Normal Hierarchy, and in Fig. 21 for the fit assuming Inverted Hierarchy. From these figures, it is apparent that most systematic parameters barely move from their nominal values, although all relevant systematic parameters are allowed to float in these fits, few of them move by more than $\pm0.1\sigma$. When a systematic parameter that is free to move stays close to its nominal value, this means that the reduction in χ^2 from moving that parameter is less than the corresponding increase in the penalty term in equation 8. Only certain systematics are able to change the predicted reconstructed energy spectrum to a sufficient extent to offset the penalty term.

The reconstructed neutrino energy distribution of single μ-like ring and single e-like ring events in the Run 1+2+3+4 dataset is shown in Fig. 22 along with best-fit prediction of the joint 3-flavour oscillation analysis, assuming Normal Hierarchy, and

overlaid with the non-oscillation prediction. Similarly for Inverted HIerarchy in Fig. 23. The distributions are shown with the reconstructed neutrino energy binning that was used in the fitting procedure (see footnote 2 in Sec. 4). Figures 22 and 23 also present the ratio of the best-fit reconstructed neutrino energy distribution of ingle μ-like ring and single e-like ring events to the non-oscillation prediction. Fig. 24 presents the overlay of the best-fit spectra for the two assumptions of mass hierarchy.

7.1.2. Confidence regions

The marginalized $\Delta\chi^2$ distribution as function of each of $|\Delta m^2| = |\Delta m^2_{32}|(NH), |\Delta m^2_{13}|(IH)$, $sin^2\theta_{23}$, $sin^2\theta_{13}$ and δ_{CP} oscillation parameters for the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset is shown in Fig. 25 for Normal Hierarchy and in Fig. 26 for Inverted Hierarchy. In Fig. 27 the results for both mass hierarchy assumptions are compared. The vertical lines in these plots are showing the best-fit value of each parameter at the 4D best-fit point found, presented in table 15. As expected, each of the vertical lines is situated at the minimum of the marginalized $\Delta\chi^2$ distribution. From these plots, the 90% (68%) CL allowed region for the four oscillation parameters can be summarized as

- 2.320 $\times 10^{-3}$ (2.394$\times 10^{-3}$) $eV^2/c^4 < |\Delta m^2_{32}| <$ 2.697$\times 10^{-3}$ (2.623$\times 10^{-3}$) eV^2/c^4,
- 0.435 (0.465) $< sin^2\theta_{23} <$ 0.608 (0.581),
- 0.018 (0.021) $< sin^2\theta_{13} <$ 0.063 (0.055) and
- -π (-π) $< \delta_{CP} < \pi$ (π)

for Normal Hierarchy and

- 2.292 $\times 10^{-3}$ (2.370$\times 10^{-3}$) $eV^2/c^4 < |\Delta m^2_{13}| <$ 2.672 $\times 10^{-3}$ (2.605$\times 10^{-3}$) eV^2/c^4,
- 0.434 (0.458) $< sin^2\theta_{23} <$ 0.578 (0.596),
- 0.022 (0.028) $< sin^2\theta_{13} <$ 0.076 (0.064) and
- -π (-π) $< \delta_{CP} < \pi$ (π)

for Inverted Hierarchy.

The correlations between the oscillation parameters were also studied. In Figs. 28 and 29 the 68% and 90% CL regions in the following four 2-dimensional oscillation parameter spaces are presented for Normal and Inverted Hierarchy respectively: ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2\theta_{13}$). Comparisons between the confidence regions obtained from the joint 3-flavour oscillation fits under difference mass hierarchy assumptions are shown in Fig. 30. Confidence regions were constructed with respect to each independent 4D best-fit point from table 15 for each mass hierarchy in the independent Figs. 28 and 29, whereas a common best-fit point, the one from the fit with Inverted Hierarchy, as it is the one with the smallest value of χ^2, was used to build confidence regions in both mass hierarchy assumptions for Fig. 30. In the ($sin^2\theta_{13}$, δ_{CP}) parameter space the method called *raster scan* was used.

The $\Delta\chi^2$ distributions for the four oscillations parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2\theta_{13}$) is presented in Fig. 31 for the Run1+2+3+4 dataset fit with Normal Hierarchy and in Fig. 32 with Inverted Hierarchy.

A set of extra plots is included to show the best-fit values of the other oscillation parameters which are fitted in each one of the four oscillations parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2\theta_{13}$). For instance, best-fit values of $sin^2\theta_{13}$ and δ_{CP} for the Run1+2+3+4 dataset fit in the oscillation parameter space ($sin^2\theta_{23}$, $|\Delta m^2|$) with Normal Hierarchy is presented in Fig. 33. Similarly for the rest of oscillation parameter spaces in Figs. 34 - 36 and in Figs. 37 - 40 with Inverted Hierarchy.

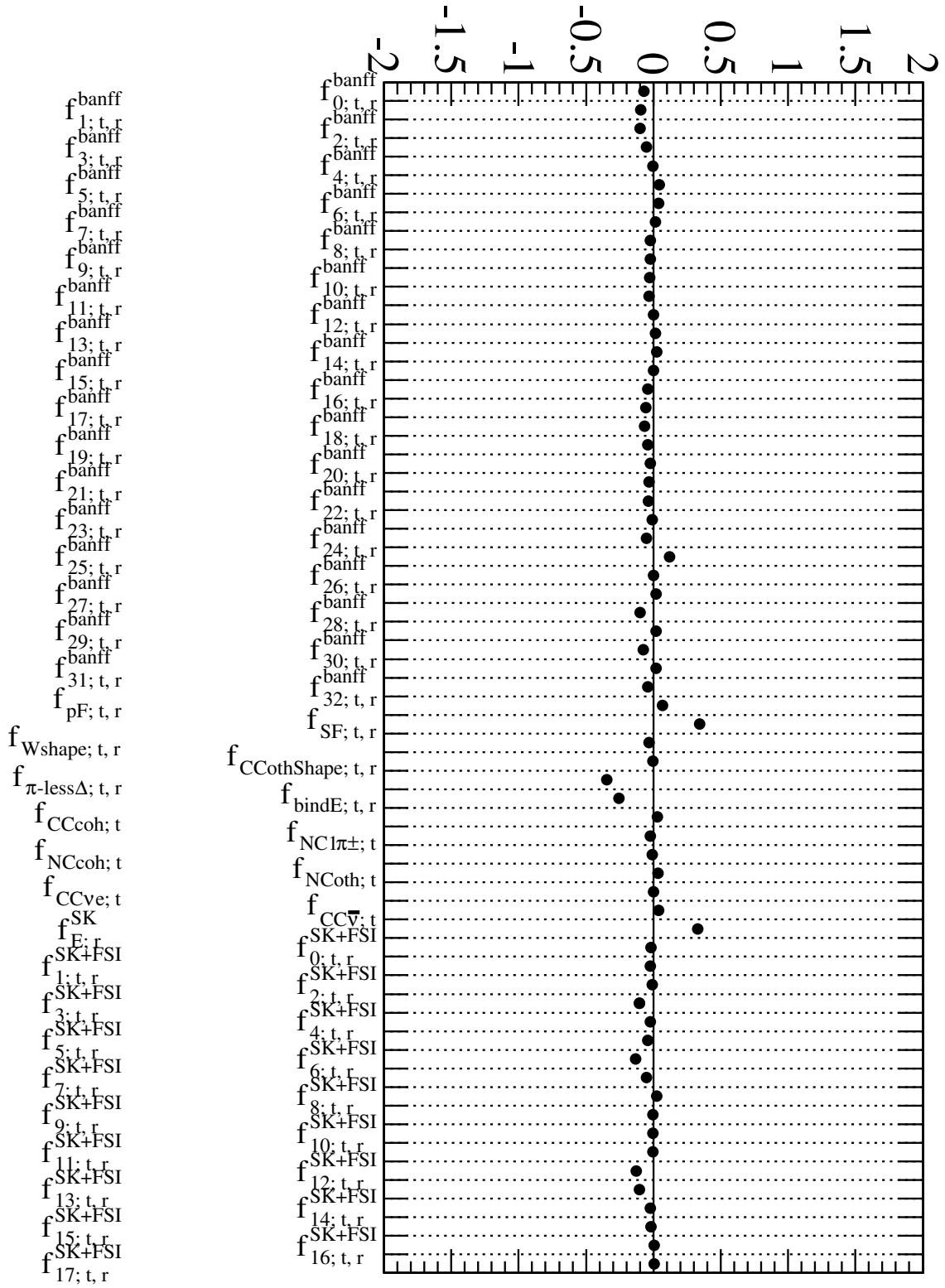


Figure 20: Systematic parameter pulls for the 64 systematics allowed to float in the Run 1+2+3+4 joint 3-flavour oscillation fit with Normal Hierarchy.

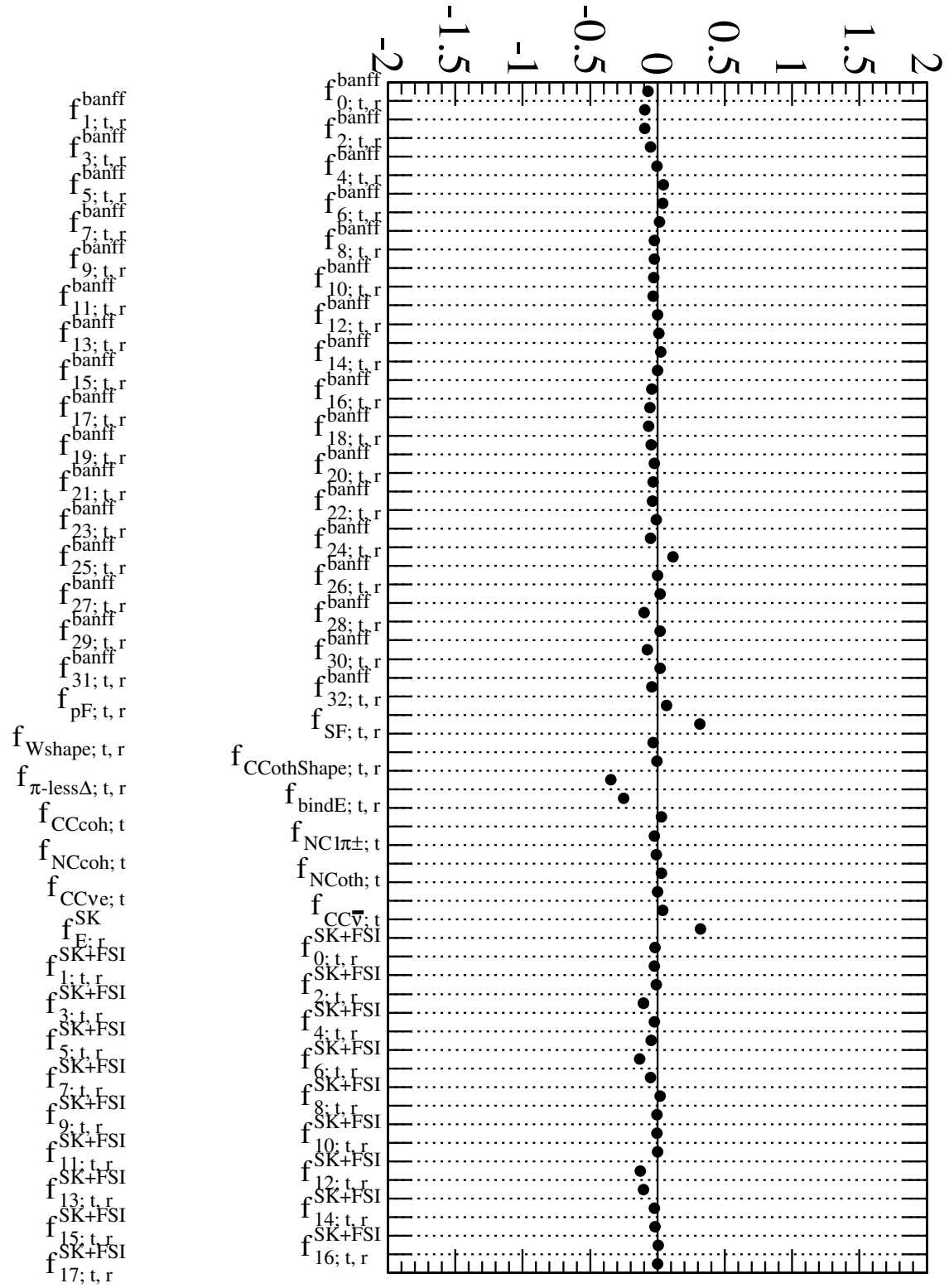


Figure 21: Systematic parameter pulls for the 64 systematics allowed to float in the Run 1+2+3+4 joint 3-flavour oscillation fit with Inverted Hierarchy.

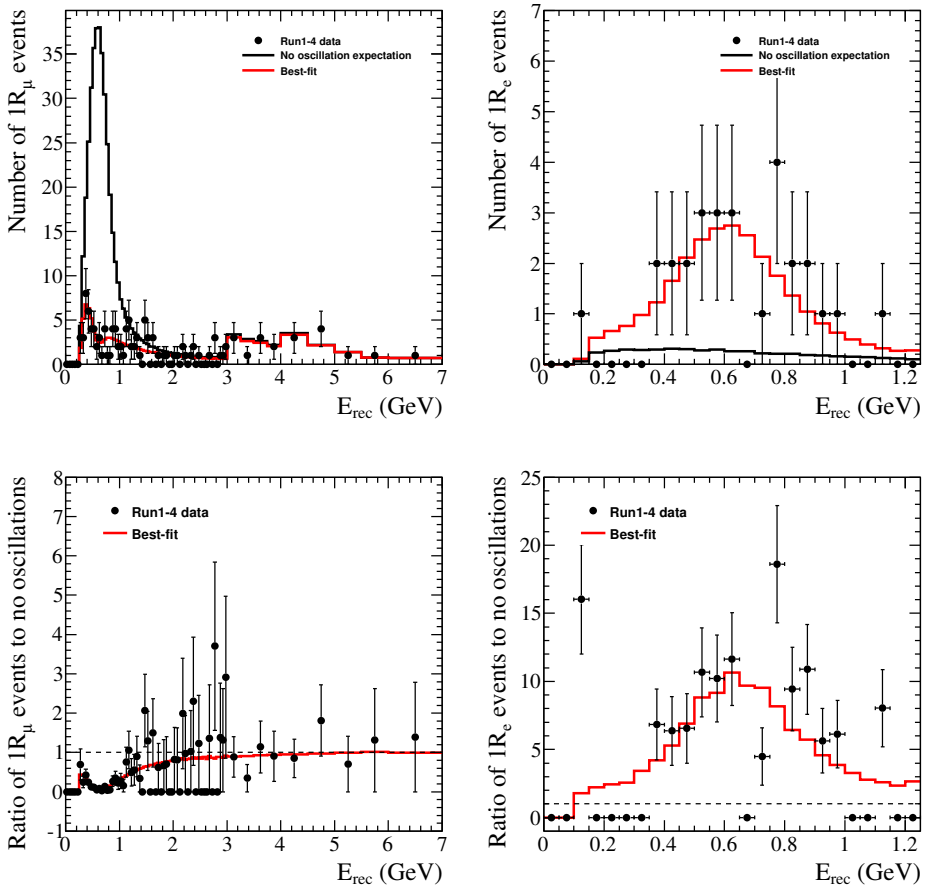


Figure 22: Reconstructed neutrino energy distribution of single μ-like ring and single e-like ring events in the Run 1+2+3+4 dataset along with best-fit prediction of the joint 3-flavour oscillation analysis, assuming Normal Hierarchy, and overlaid with the non-oscillation prediction. Bottom plots show the ratio of the best-fit reconstructed neutrino energy distribution of ingle μ-like ring and single e-like ring events to the non-oscillation prediction.

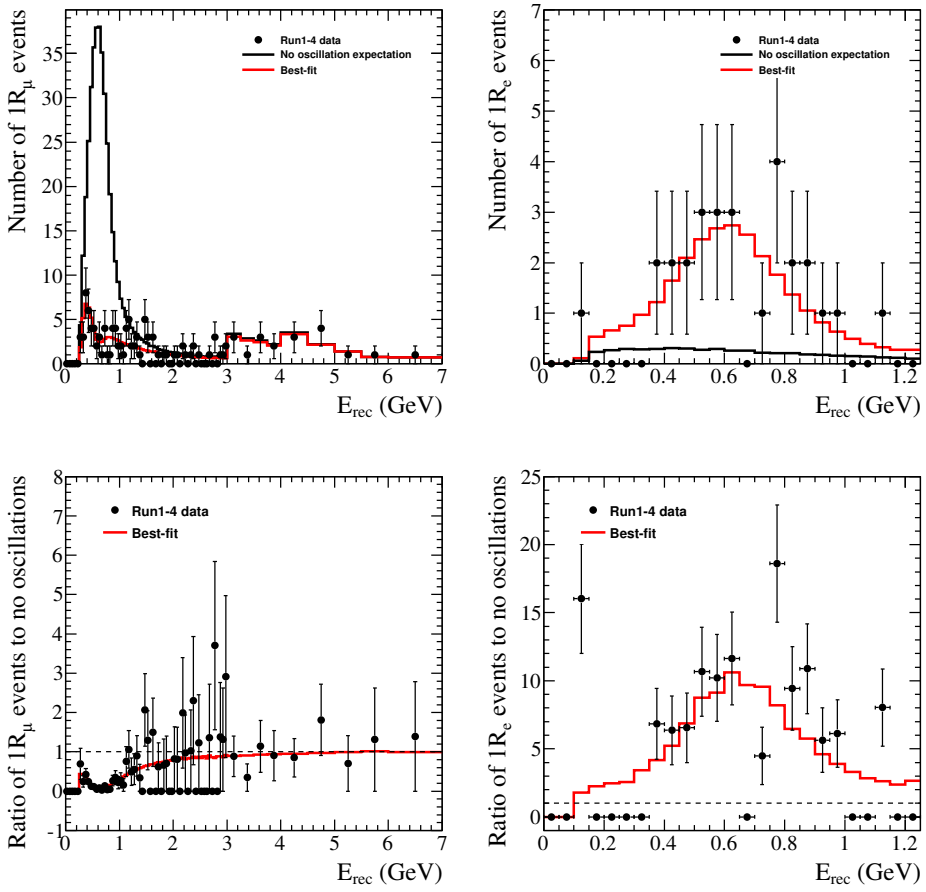


Figure 23: Reconstructed neutrino energy distribution of single μ-like ring and single e-like ring events in the Run 1+2+3+4 dataset along with best-fit prediction of the joint 3-flavour oscillation analysis, assuming Inverted Hierarchy, and overlaid with the non-oscillation prediction. Bottom plots show the ratio of the best-fit reconstructed neutrino energy distribution of ingle μ-like ring and single e-like ring events to the non-oscillation prediction.

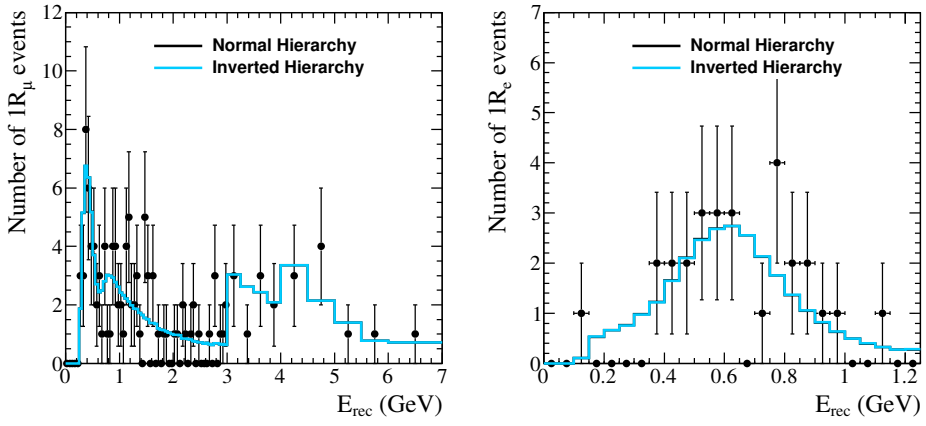


Figure 24: Comparison of the best-fit spectra of single μ-like ring and single e-like ring events for the Run 1+2+3+4 dataset for the two assumptions of mass hierarchy.

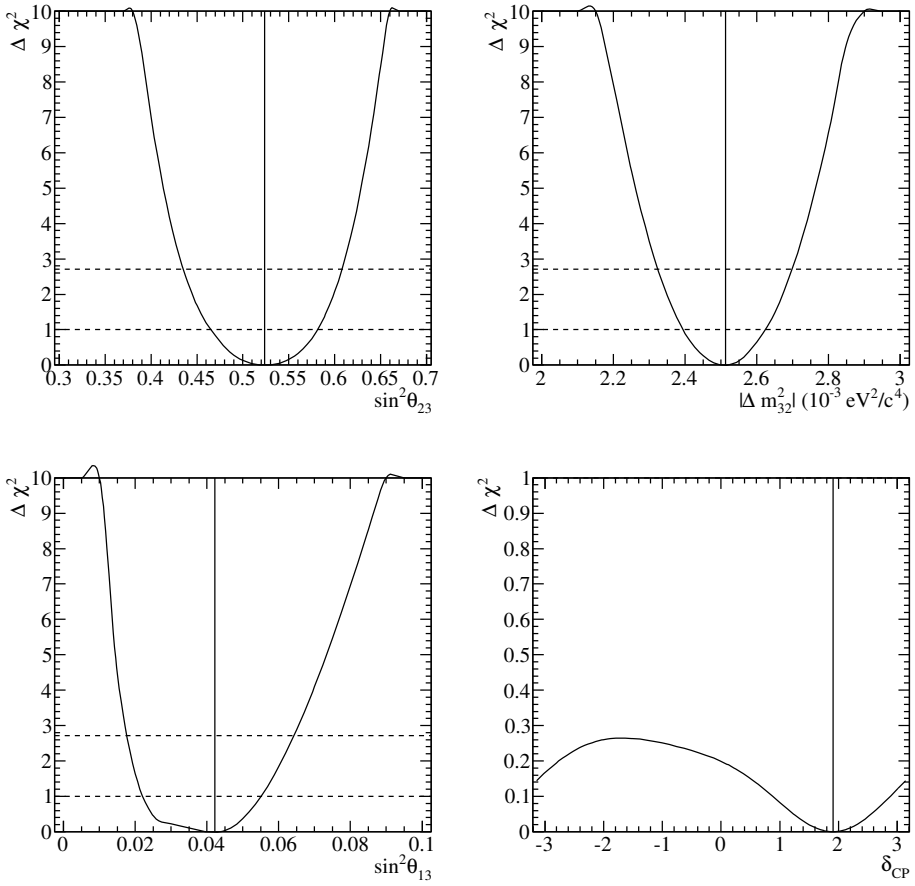


Figure 25: Marginalized $\Delta\chi^2$ distribution as function of each of $|\Delta m^2_{32}|$, $sin^2\theta_{23}$, $sin^2\theta_{13}$ and δ_{CP} oscillation parameters for the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset with Normal Hierarchy.

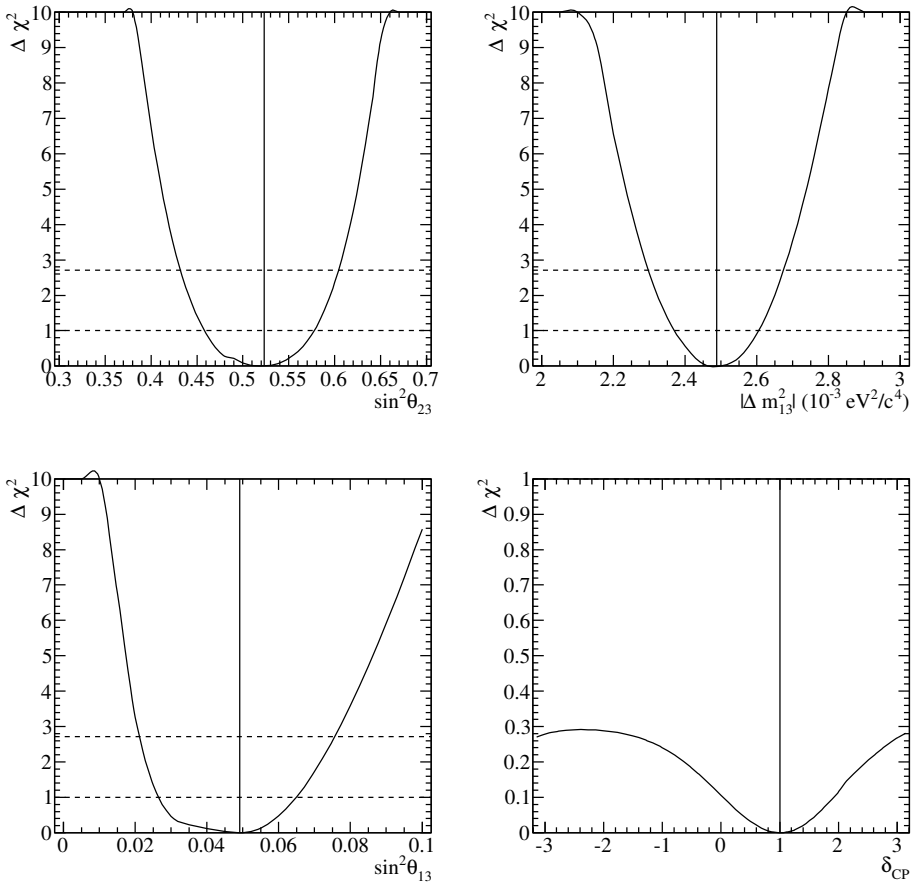


Figure 26: Marginalized $\Delta\chi^2$ distribution as function of each of $|\Delta m^2_{13}|$, $sin^2\theta_{23}$, $sin^2\theta_{13}$ and δ_{CP} oscillation parameters for the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset with Inverted Hierarchy.

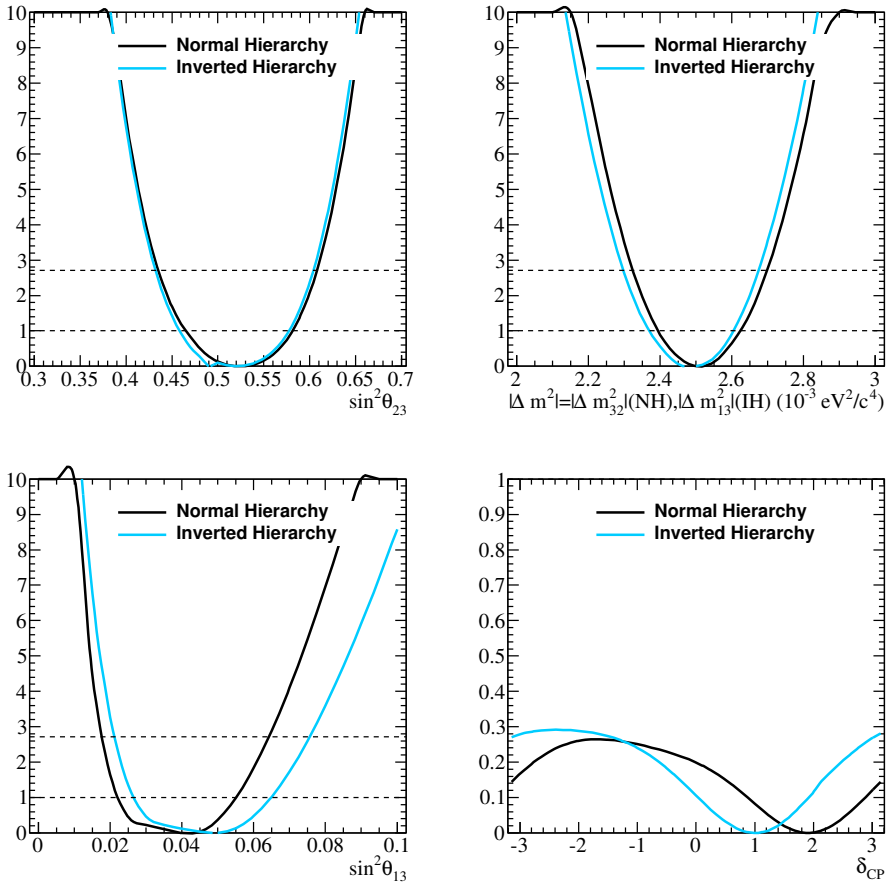


Figure 27: Comparison of marginalized $\Delta\chi^2$ distributions for the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset with Normal and Inverted Hierarchies.

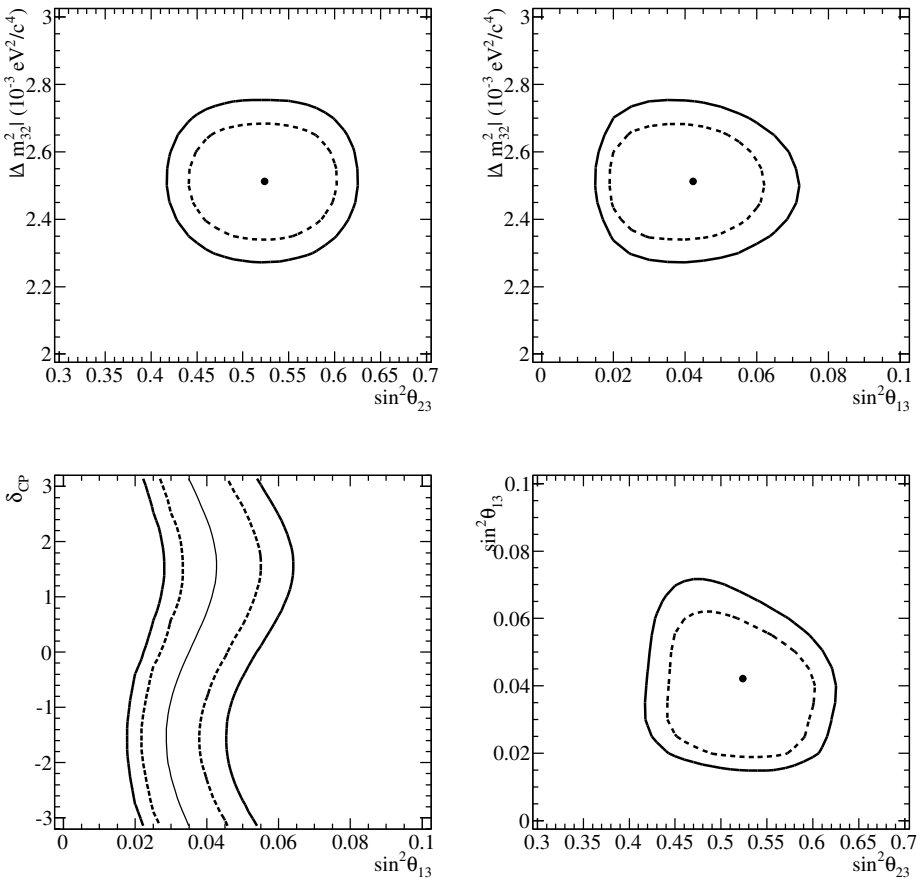


Figure 28: 68% and 90% CL regions in the following four 2-dimensional oscillation parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2\theta_{13}$) for the Run1+2+3+4 dataset fit with Normal Hierarchy. Confidence regions were constructed with respect to the 4D best-fit point with Normal Hierarchy from table 15. In the ($sin^2\theta_{13}$, δ_{CP}) parameter space the method called *raster scan* was used.

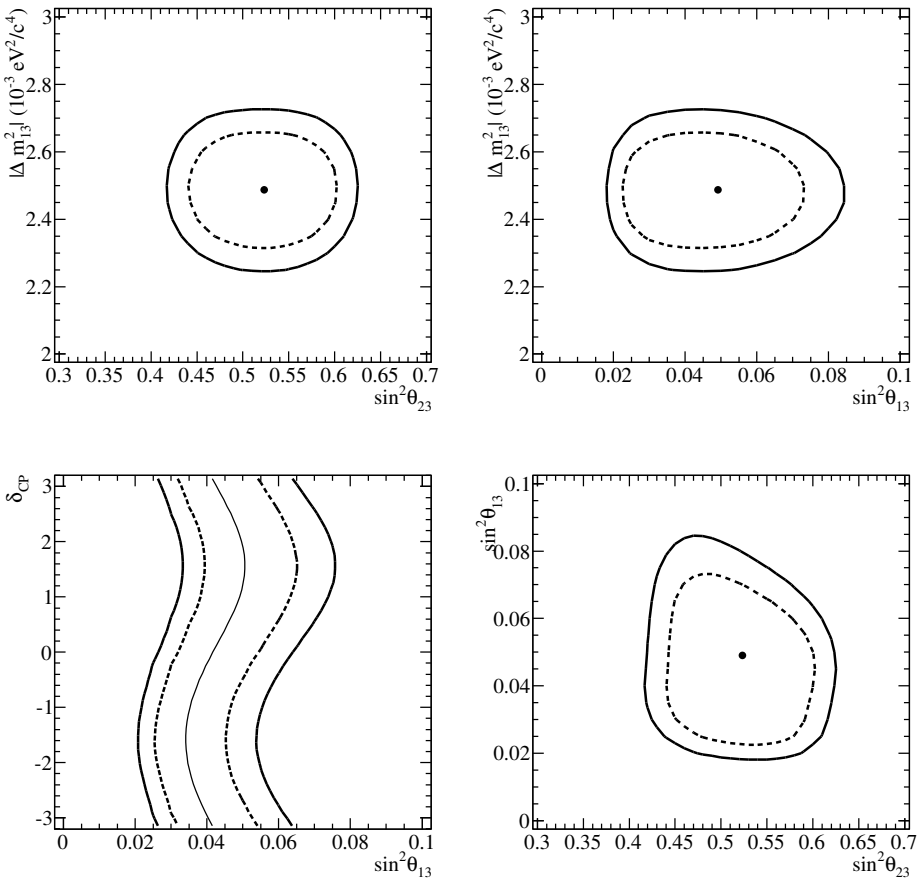


Figure 29: 68% and 90% CL regions in the following four 2-dimensional oscillation parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2\theta_{13}$) for the Run1+2+3+4 dataset fit with Inverted Hierarchy. Confidence regions were constructed with respect to the 4D best-fit point with Inverted Hierarchy from table 15. In the ($sin^2\theta_{13}$, δ_{CP}) parameter space the method called *raster scan* was used.

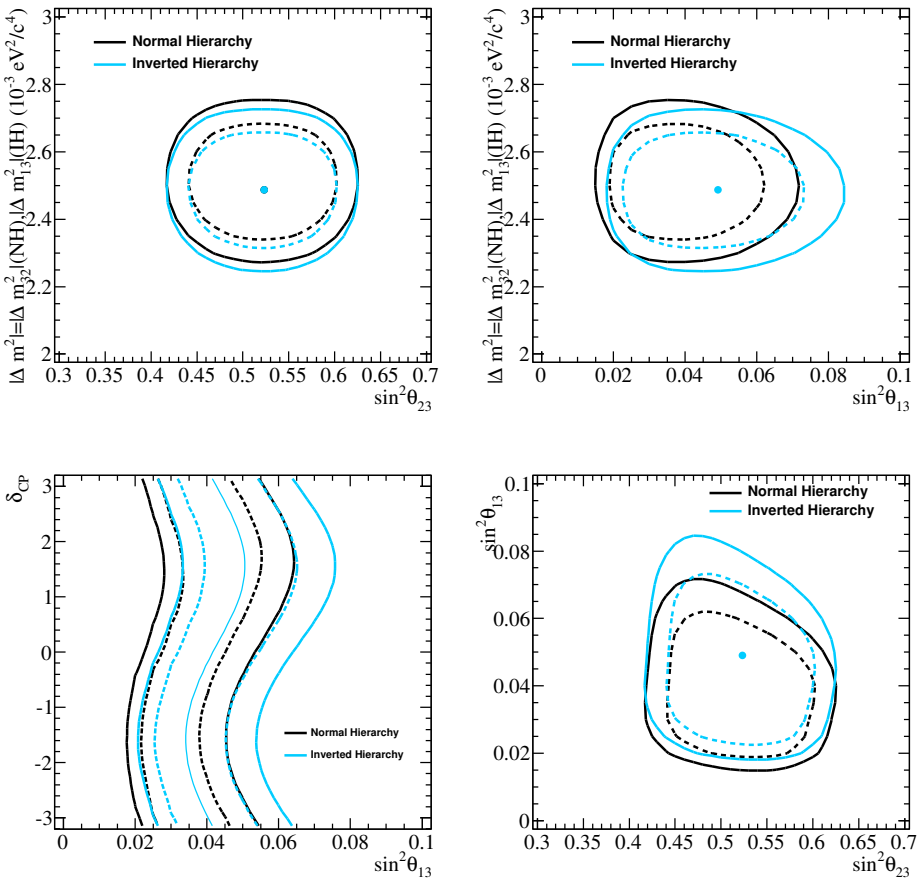


Figure 30: Comparison of 68% and 90% CL regions for the Run1+2+3+4 dataset fit with different mass hierarchy assumptions. A common best-fit point, the one from the fit with Inverted Hierarchy, as it is the one with the smallest value of χ^2, was used to build confidence regions in both mass hierarchy assumptions. n the ($sin^2\theta_{13}$, δ_{CP}) parameter space the method called *raster scan* was used.

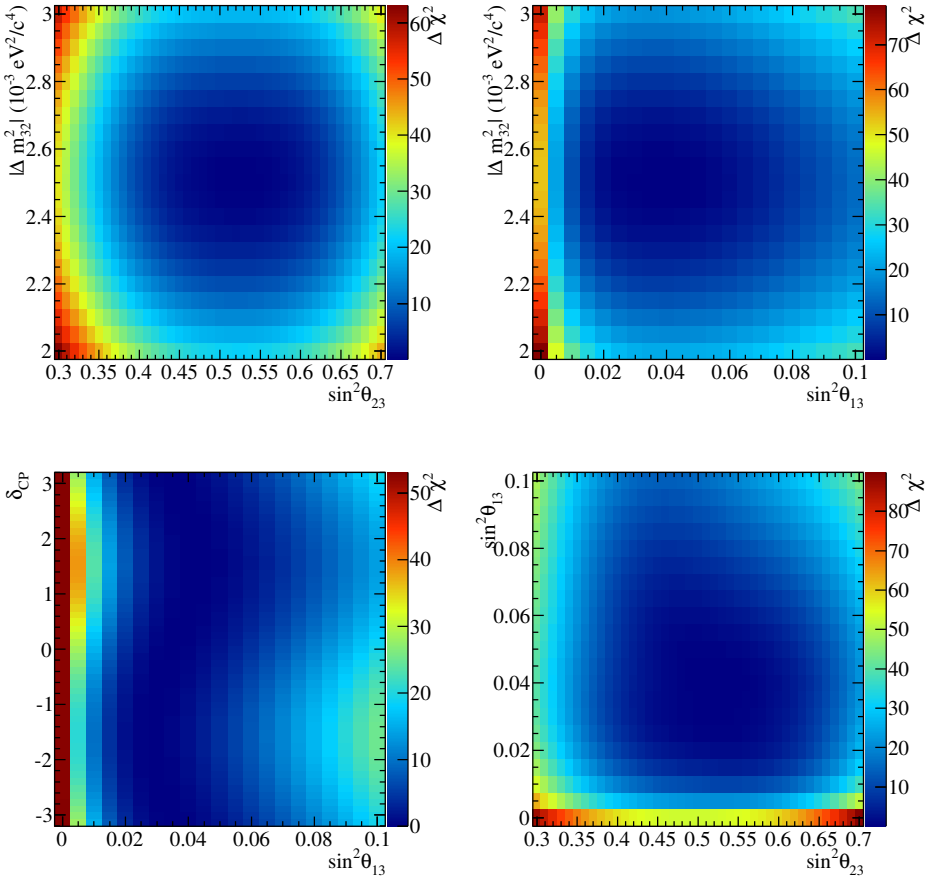


Figure 31: $\Delta\chi^2$ distributions for the four oscillations parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2\theta_{13}$) for the Run1+2+3+4 dataset fit with Normal Hierarchy.

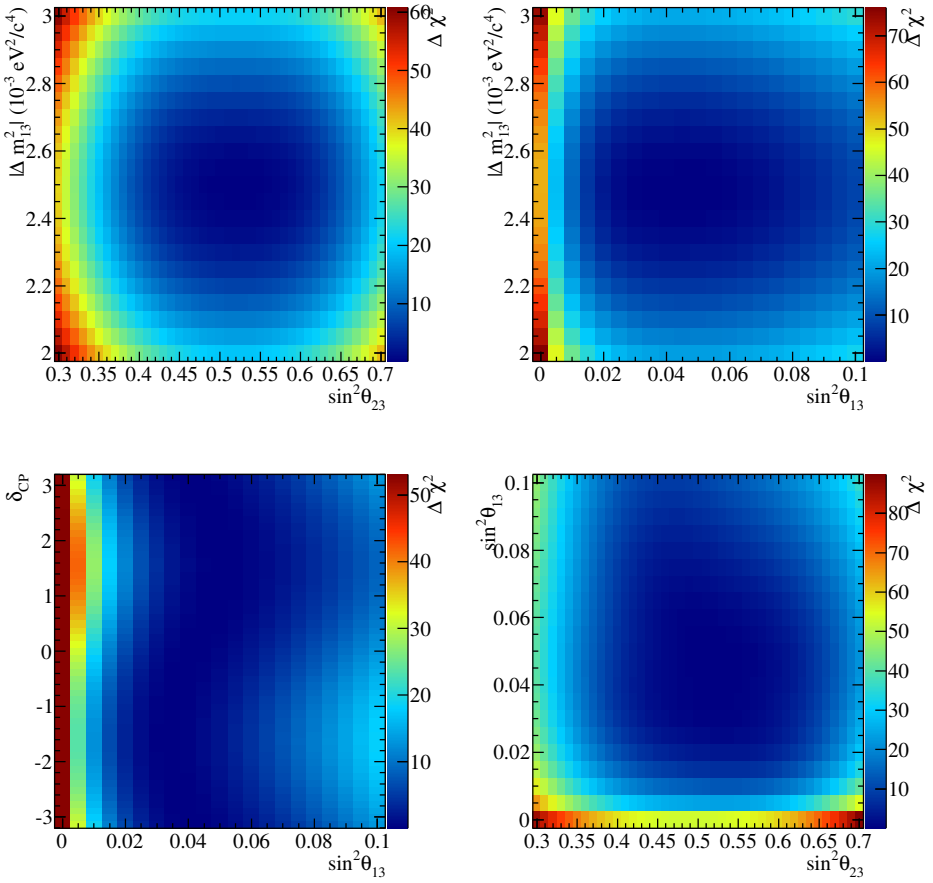


Figure 32: $\Delta\chi^2$ distributions for the four oscillations parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, $|\Delta m^2|$), ($sin^2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2\theta_{13}$) for the Run1+2+3+4 dataset fit with Inverted Hierarchy.

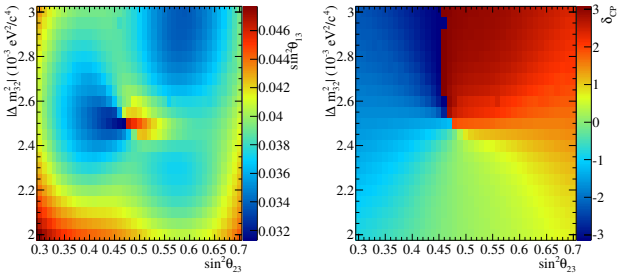


Figure 33: Best-fit values of $sin^2\theta_{13}$ and δ_{CP} for the Run1+2+3+4 dataset fit with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{23}$, $|\Delta m^2_{32}|$).

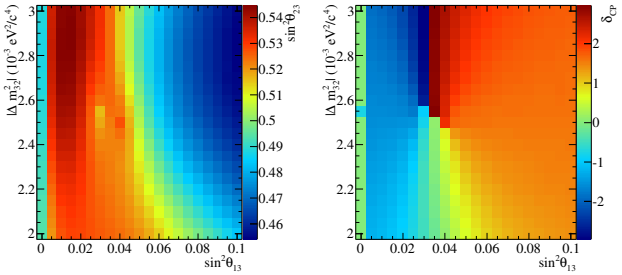


Figure 34: Best-fit values of $sin^2\theta_{23}$ and δ_{CP} for the Run1+2+3+4 dataset fit with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{13}$, $|\Delta m^2_{32}|$).

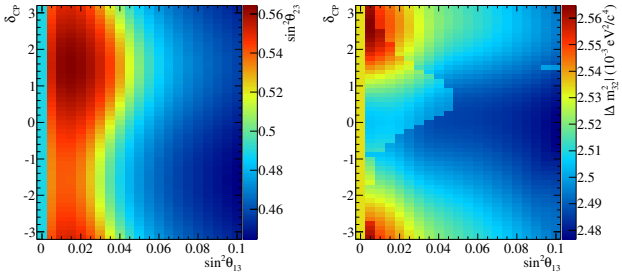


Figure 35: Best-fit values of $sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for the Run1+2+3+4 dataset fit with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{13}$,δ_{CP}).

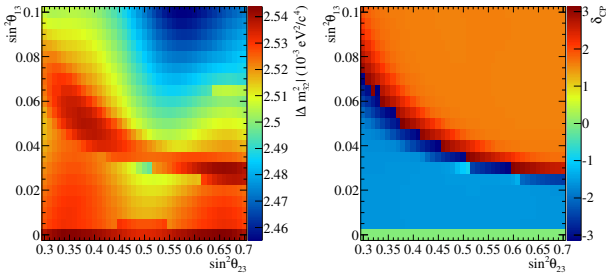


Figure 36: Best-fit values of $|\Delta m^2_{32}|$ and δ_{CP} for the Run1+2+3+4 dataset fit with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{23}$,$sin^2\theta_{13}$).

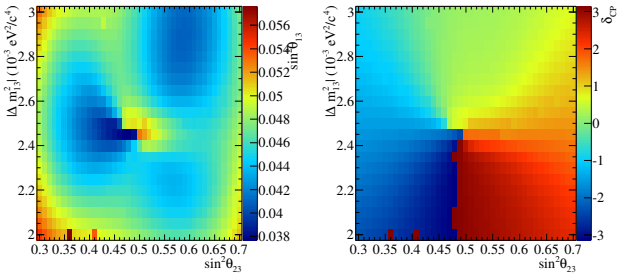


Figure 37: Best-fit values of $sin^2\theta_{13}$ and δ_{CP} for the Run1+2+3+4 dataset fit with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{23}$, $|\Delta m^2_{13}|$).

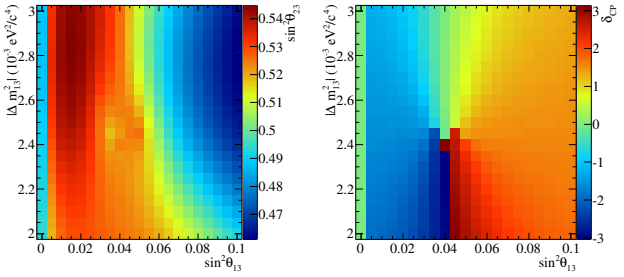


Figure 38: Best-fit values of $sin^2\theta_{23}$ and δ_{CP} for the Run1+2+3+4 dataset fit with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{13}$, $|\Delta m^2_{13}|$).

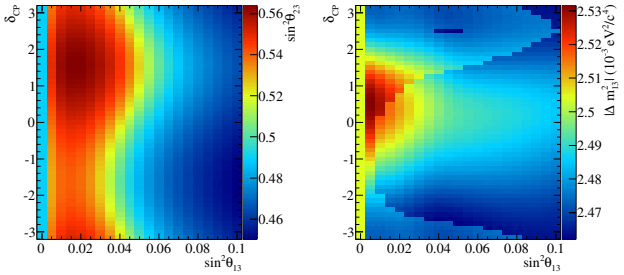


Figure 39: Best-fit values of $sin^2\theta_{23}$ and $|\Delta m^2_{13}|$ for the Run1+2+3+4 dataset fit with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{13}$,δ_{CP}).

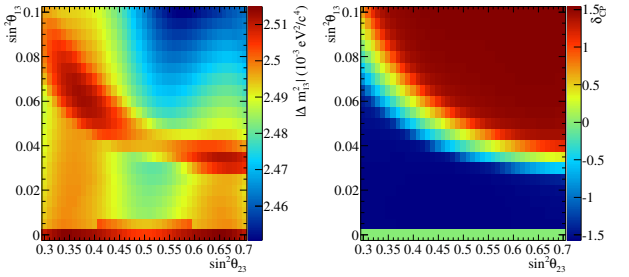


Figure 40: Best-fit values of $|\Delta m^2_{13}|$ and δ_{CP} for the Run1+2+3+4 dataset fit with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{23}$,$sin^2\theta_{13}$).

7.1.3. Goodness-of-fit tests

Goodness-of-fit tests were performed as described in Sec. 6.3, and the p-values for the joint 3-flavour oscillation fit will be presented in this section.

Fig 41 (42) shows the result of the goodness-of-fit tests with 1000 toy experiments at the best-fit point for Normal (Inverted) Hierarchy, overlaying the value of χ^2_{gof} (in the coarse binning described in Sec. 6.3) for the best-fit to the data as a red line. The p-values were calculated directly from these plots as the number of toy experiments for which $\chi^2_{gof} > (\chi^2_{gof})_{data}$ divided by the total number of toy experiments.

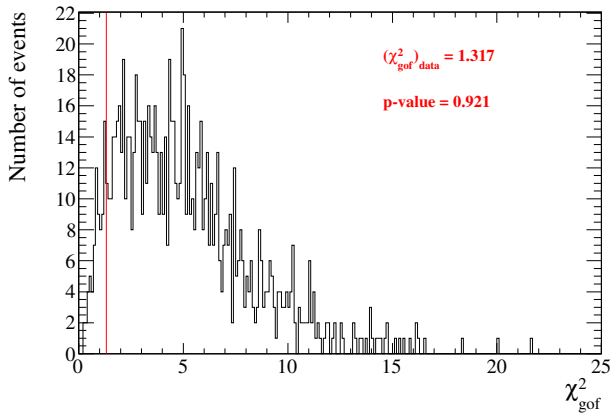


Figure 41: Result of the goodness-of-fit tests with 1000 toy experiments at the best-fit point for Normal Hierarchy, overlaying the value of χ^2_{gof} for the best-fit to the data as a red line. The p-value was calculated directly from this distribution as the number of toy experiments for which $\chi^2_{gof} > (\chi^2_{gof})_{data}$ divided by the total number of toy experiments. The procedure is described in Sec. 6.3.

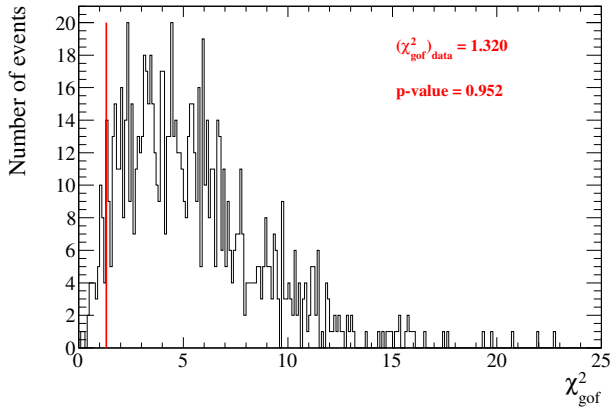


Figure 42: Result of the goodness-of-fit tests with 1000 toy experiments at the best-fit point for Inverted Hierarchy, overlaying the value of χ^2_{gof} for the best-fit to the data as a red line. The p-value was calculated directly from this distribution as the number of toy experiments for which $\chi^2_{gof} > (\chi^2_{gof})_{data}$ divided by the total number of toy experiments. The procedure is described in Sec. 6.3.

7.2. Results of the joint oscillation analysis with the Run 1+2+3+4 (6.570 $\times$ 10^{20} POT) dataset including the reactor constraint

In this subsection, results of the Run1+2+3+4 dataset joint fit will be presented including the reactor constraint as it was explained in 6.2. Since the weighted average result from reactor experiments is expressed in terms of $sin^2 2\theta_{13}$, this will be the variable used for the plots in this subsection.

7.2.1. Best-fit oscillation parameters, systematic parameter pulls and best-fit single e-like ring and single μ-like ring event reconstructed energy spectra

This analysis found $|\Delta m^2_{32}|$ = 2.509 $\times 10^{-3}$ eV^2/c^4, $\sin^2\theta_{23}$ = 0.527, $\sin^2 2\theta_{13}$ = 0.0967 and δ_{CP} = -1.554 to be the joint 3-flavour oscillation best-fit values for the Run 1+2+3+4 dataset combined with the reactor constraint for Normal Hierarchy and $|\Delta m^2_{13}|$ = 2.481 $\times 10^{-3}$ eV^2/c^4, $\sin^2\theta_{23}$ = 0.533, $\sin^2 2\theta_{13}$ = 0.0984 and δ_{CP} = -1.556 for Inverted Hierarchy.

Table 16 shows the best-fit values of the Run1+2+3+4 dataset fit with the two mass hierarchy assumptions, comparing with and without applying the reactor constraint. As expected, the value of $sin^2 2\theta_{13}$ stays close to the reactor value due to the penalty term, resulting in a variation of the values of the other oscillation parameters accordingly. The χ^2 values found at the best-fit point are larger when the reactor constraint is applied as expected, and now there is a preference for the best-fit point with Normal Hierarchy than with Inverted Hierarchy. Thus, the best-fit point with Normal Hierarchy will be the one used as common best-fit point when comparing confidence regions in the two mass hierarchies.

Analysis	Reactor	Hierarchy constraint	$\|\Delta m^2_{32}\|(NH)$ $\|\Delta m^2_{13}\|(IH)$ $10^{-3}eV^2/c^4$	$\sin^2\theta_{23}$	$\sin^2 2\theta_{13}$	δ_{CP}	$N^{1R\mu}_{obs}$	$N^{1R\mu}_{exp}$	N^{1Re}_{obs}	N^{1Re}_{exp}	χ^2/ndf
$\sin^2 2\theta_{13}$	NO	NH	2.512	0.524	0.162	1.909	120	119.915	28	27.999	84.5395/94
$\sin^2 2\theta_{13}$	YES	NH	2.509	0.527	0.0967	-1.554	120	120.383	28	25.870	85.067/94
$\sin^2 2\theta_{13}$	NO	IH	2.488	0.523	0.187	1.005	120	119.948	28	27.998	84.529/94
$\sin^2 2\theta_{13}$	YES	IH	2.481	0.533	0.0984	-1.556	120	121.204	28	23.571	85.931/94

Table 16: Best-fit parameters for the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset comparing with and without reactor constraint.

The pulls of all systematic parameters allowed to float in the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset are shown in Figs. 43 for the fit assuming Normal Hierarchy, and in Fig. 44 for the fit assuming Inverted Hierarchy.

The reconstructed neutrino energy distribution of single μ-like ring and single e-like ring events in the Run 1+2+3+4 dataset is shown in Fig. 45 along with best-fit prediction of the joint 3-flavour oscillation analysis, assuming Normal Hierarchy, and overlaid with the non-oscillation prediction. Similarly for Inverted Hierarchy in Fig. 46. These figures also present the ratio of the best-fit reconstructed neutrino energy distribution of ingle μ-like ring and single e-like ring events to the non-oscillation prediction. Fig. 47 presents the overlay of the best-fit spectra for the Normal Hierarchy with and without reactor constraint. Similarly for Inverted Hierarchy in Fig. 48.

7.2.2. Confidence regions

The marginalized $\Delta\chi^2$ distribution as function of each of $|\Delta m^2| = |\Delta m^2_{32}|(NH), |\Delta m^2_{13}|(IH)$, $sin^2\theta_{23}$, $sin^2 2\theta_{13}$ and δ_{CP} oscillation parameters for the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset with reactor constraint is shown in Fig. 49 for Normal Hierarchy and in Fig. 50 for Inverted Hierarchy. In Fig. 51 the results for both mass hierarchy assumptions are compared.

In Figs. 52 and 54 the 68% and 90% CL regions with reactor constraint in the following four 2-dimensional oscillation parameter spaces are presented for Normal and Inverted Hierarchy respectively: ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2 2\theta_{13}$). Also, since the confidence regions are interesting once the reactor constraint is added, confidence regions are also presented in the parameter spaces: ($sin^2\theta_{23}$, δ_{CP}) and ($|\Delta m^2|$, δ_{CP}) in Figs 53 and 55 for Normal and Inverted Hierarchy respectively. Comparisons between the confidence regions obtained from the joint 3-flavour oscillation fits with reactor constraint under difference mass hierarchy assumptions are shown in Figs. 56 and 57. Confidence regions were constructed with respect to each independent 4D best-fit point from table 16 for each mass hierarchy in the independent Figs. 52 and 54, whereas a common best-fit point, the one from the fit with Normal Hierarchy, as it is the one with the smallest value of χ^2, was used to build confidence regions in both mass hierarchy assumptions for Fig. 56. In the ($sin^2 2\theta_{13}$, δ_{CP}) parameter space the method *raster scan* was no longer used.

The $\Delta\chi^2$ distributions for the six oscillations parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, δ_{CP}), ($sin^2\theta_{23}$, $sin^2 2\theta_{13}$), ($sin^2\theta_{23}$, δ_{CP}) and ($|\Delta m^2|$, δ_{CP}) is presented in Fig. 58 for the Run1+2+3+4 dataset fit with reactor constraint with Normal Hierarchy and in Fig. 59 with Inverted Hierarchy.

A set of extra plots is included to show the best-fit values of the other oscillation parameters which are fitted in each one of the six oscillations parameter spaces in Figs. 60 - 65 with reactor constraint and Normal Hierarchy and in Figs. 66 - 71 with Inverted Hierarchy.

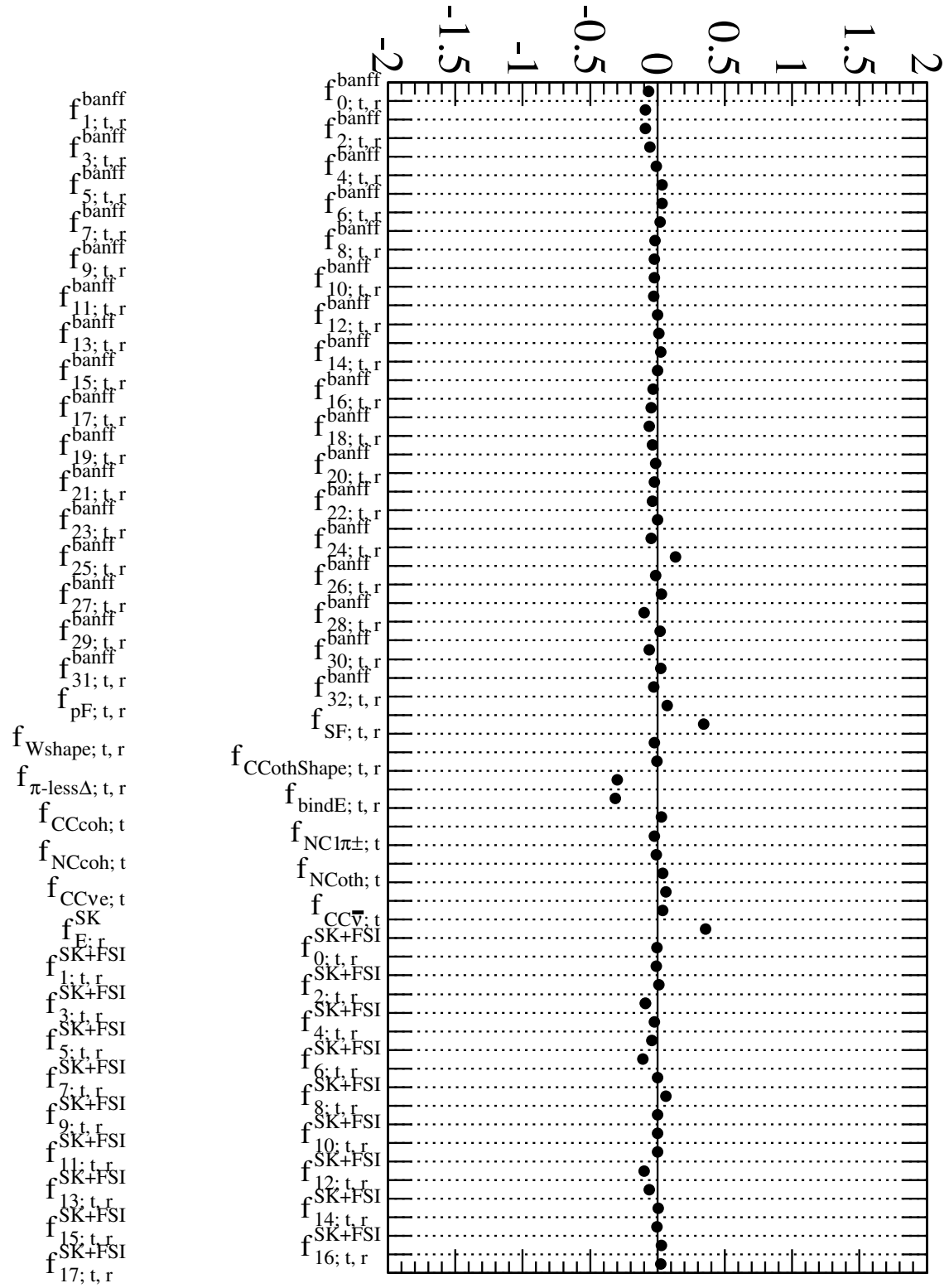


Figure 43: Systematic parameter pulls for the 64 systematics allowed to float in the Run 1+2+3+4 joint 3-flavour oscillation fit with reactor constraint with Normal Hierarchy.

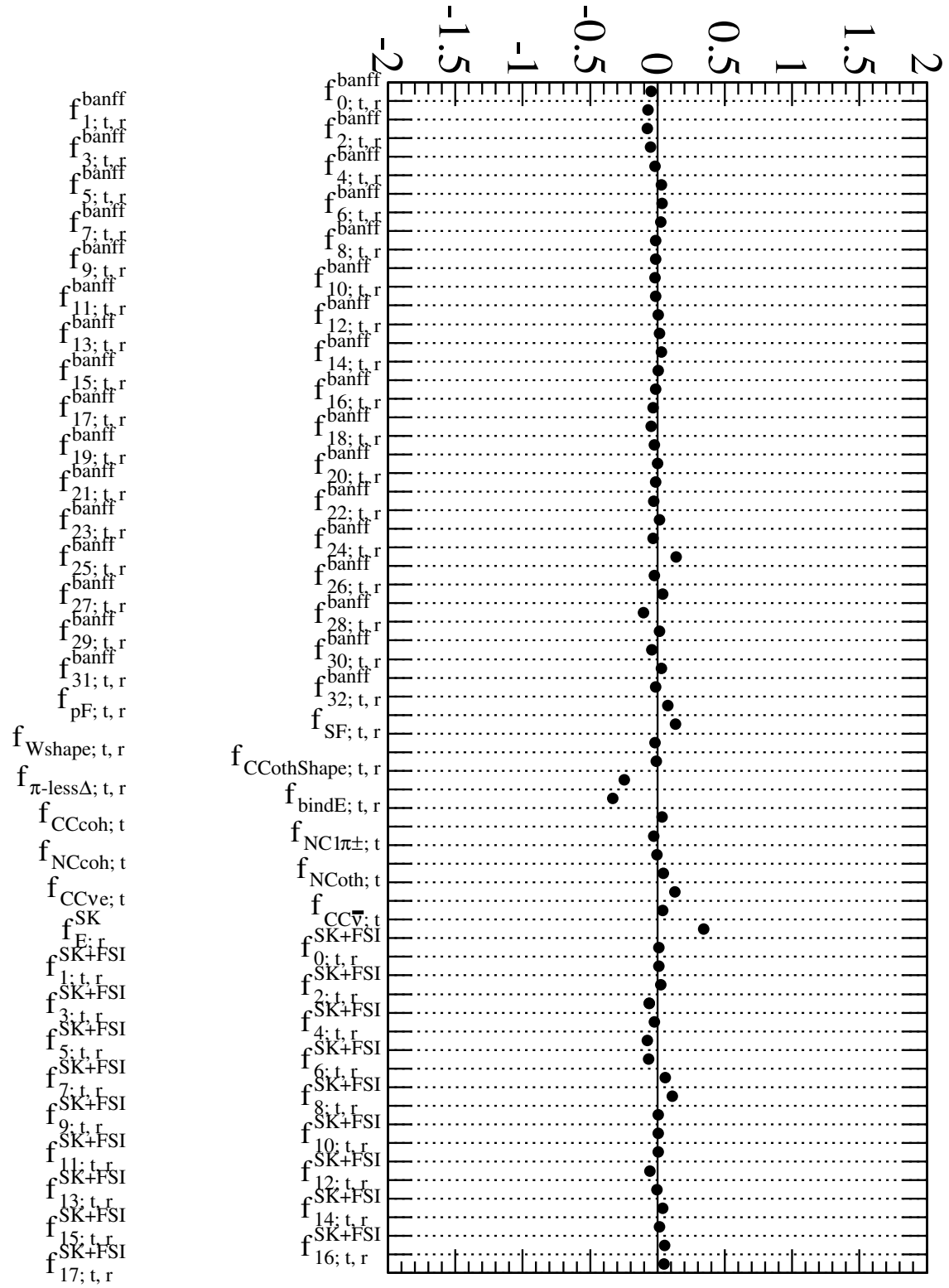


Figure 44: Systematic parameter pulls for the 64 systematics allowed to float in the Run 1+2+3+4 joint 3-flavour oscillation fit with reactor constraint with Inverted Hierarchy.

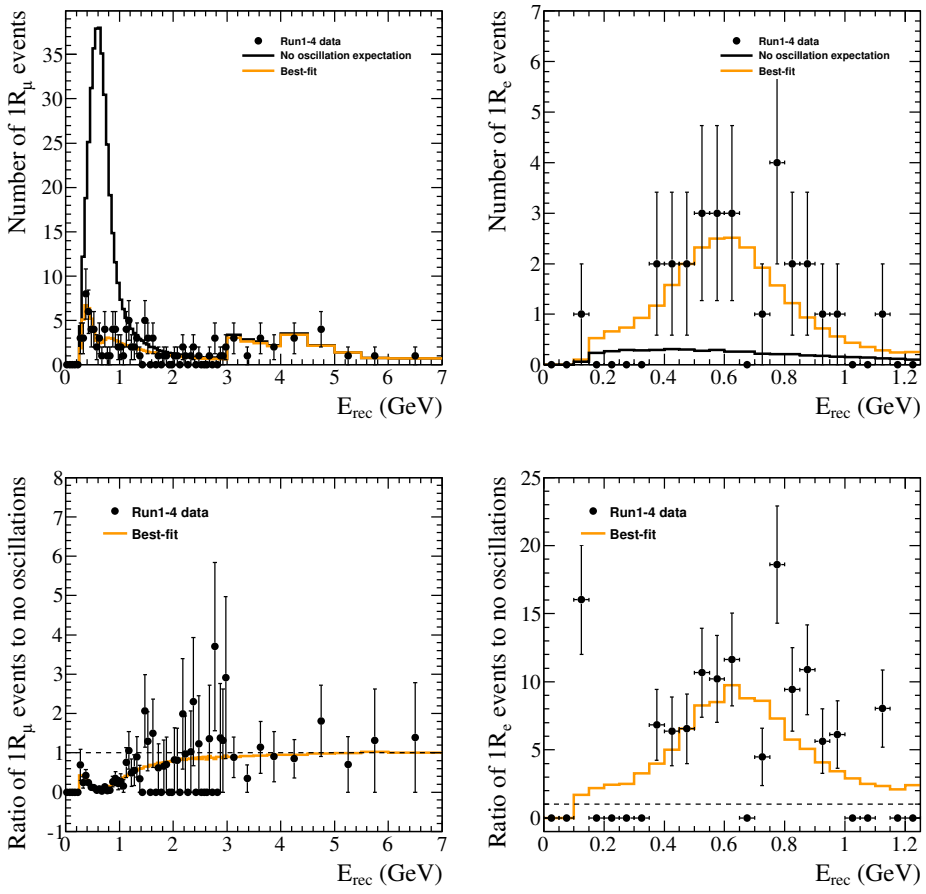


Figure 45: Reconstructed neutrino energy distribution of single μ-like ring and single e-like ring events in the Run 1+2+3+4 dataset along with best-fit prediction of the joint 3-flavour oscillation analysis with reactor constraint, assuming Normal Hierarchy, and overlaid with the non-oscillation prediction. Bottom plots show the ratio of the best-fit reconstructed neutrino energy distribution of ingle μ-like ring and single e-like ring events to the non-oscillation prediction.

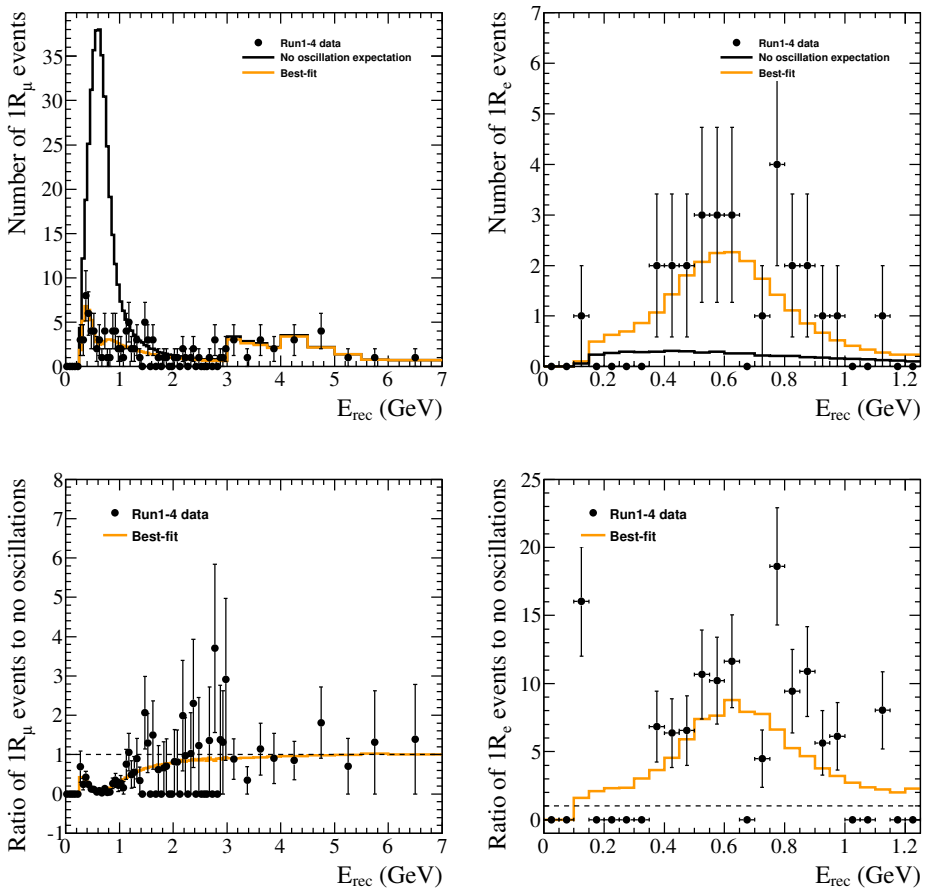


Figure 46: Reconstructed neutrino energy distribution of single μ-like ring and single e-like ring events in the Run 1+2+3+4 dataset along with best-fit prediction of the joint 3-flavour oscillation analysis with reactor constraint, assuming Inverted Hierarchy, and overlaid with the non-oscillation prediction. Bottom plots show the ratio of the best-fit reconstructed neutrino energy distribution of ingle μ-like ring and single e-like ring events to the non-oscillation prediction.

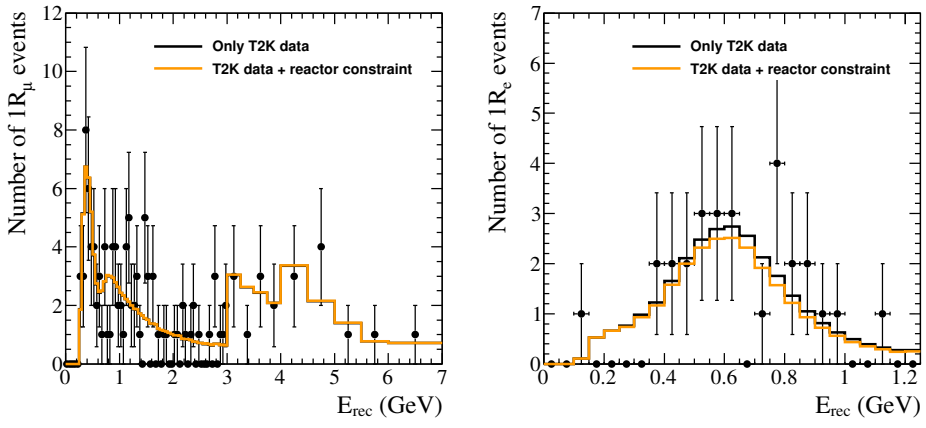


Figure 47: Comparison of the best-fit spectra of single μ-like ring and single e-like ring events for the Run 1+2+3+4 dataset for Normal Hierarchy with and without reactor constraint.

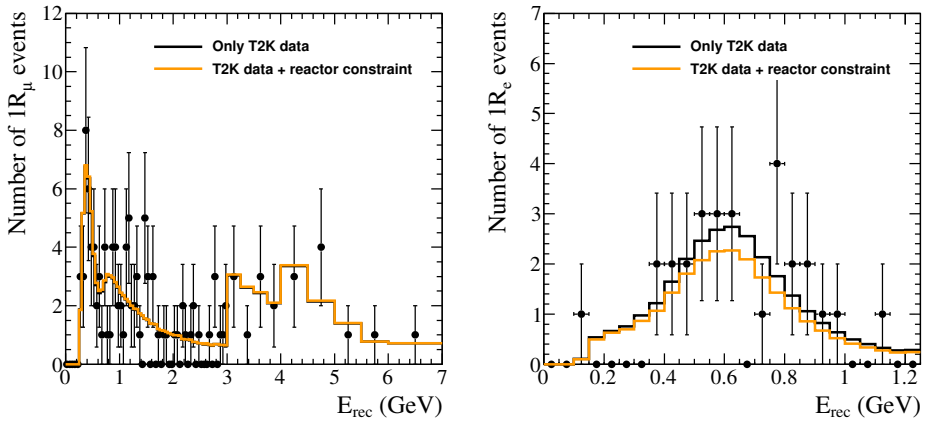


Figure 48: Comparison of the best-fit spectra of single μ-like ring and single e-like ring events for the Run 1+2+3+4 dataset for Inverted Hierarchy with and without reactor constraint.

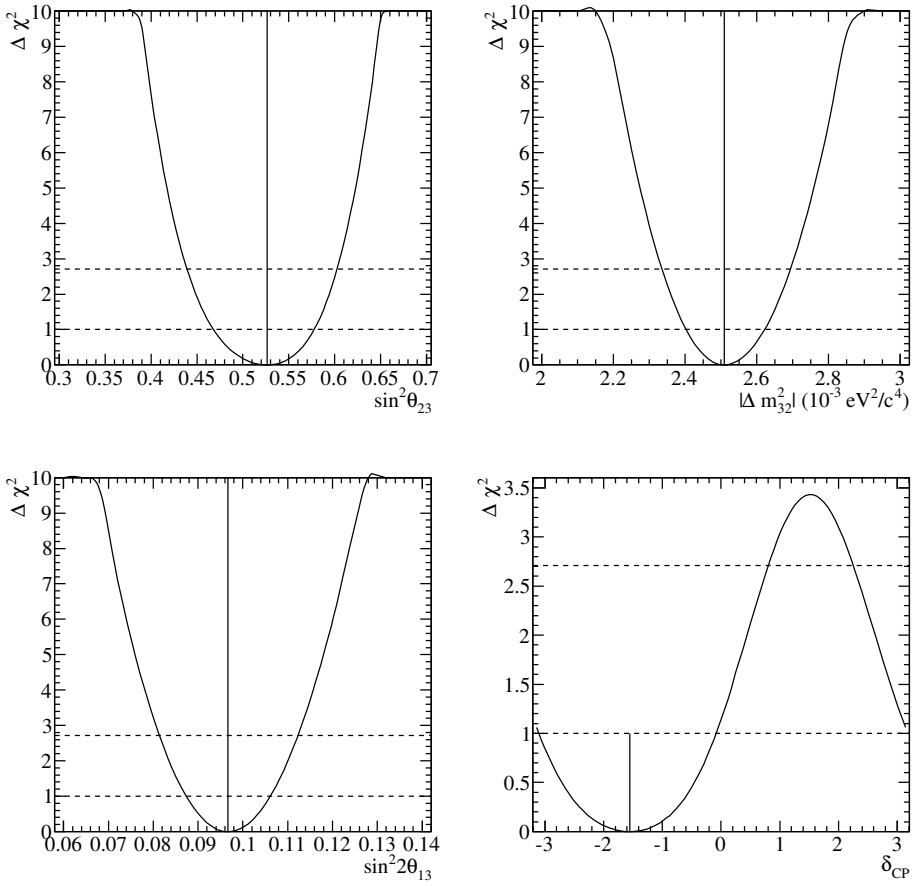


Figure 49: Marginalized $\Delta\chi^2$ distribution as function of each of $|\Delta m^2_{32}|$, $sin^2\theta_{23}$, $sin^2 2\theta_{13}$ and δ_{CP} oscillation parameters for the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset with Normal Hierarchy and reactor constraint.

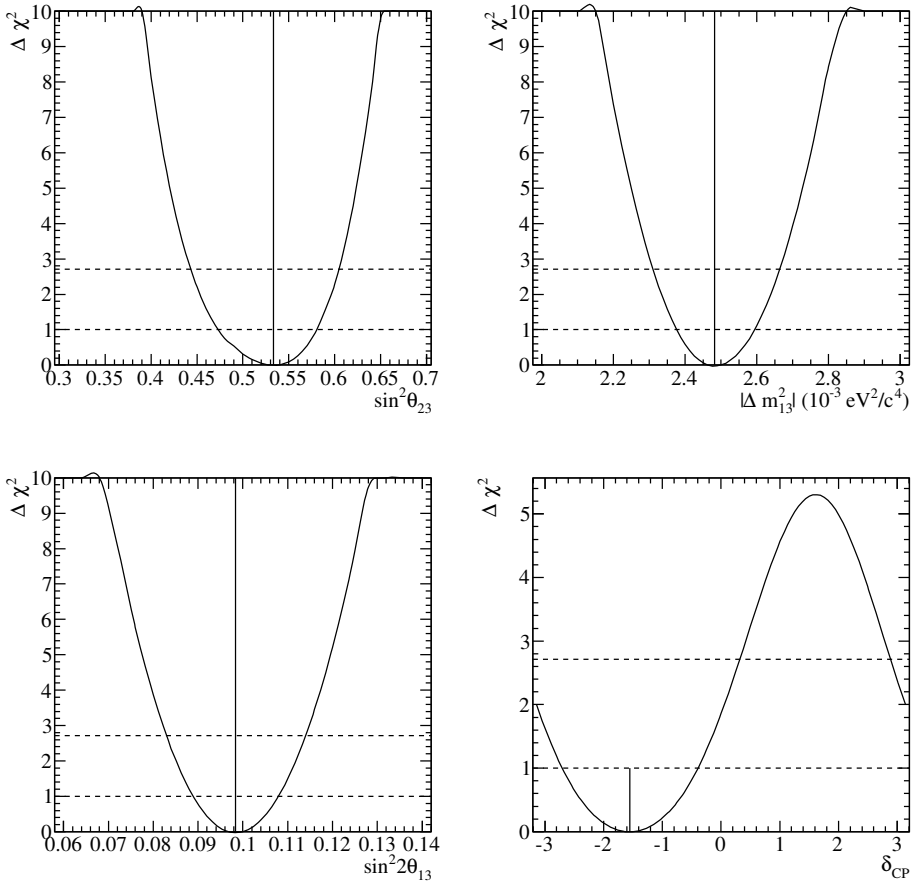


Figure 50: Marginalized $\Delta\chi^2$ distribution as function of each of $|\Delta m^2_{13}|$, $sin^2\theta_{23}$, $sin^2 2\theta_{13}$ and δ_{CP} oscillation parameters for the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset with Inverted Hierarchy and reactor constraint.

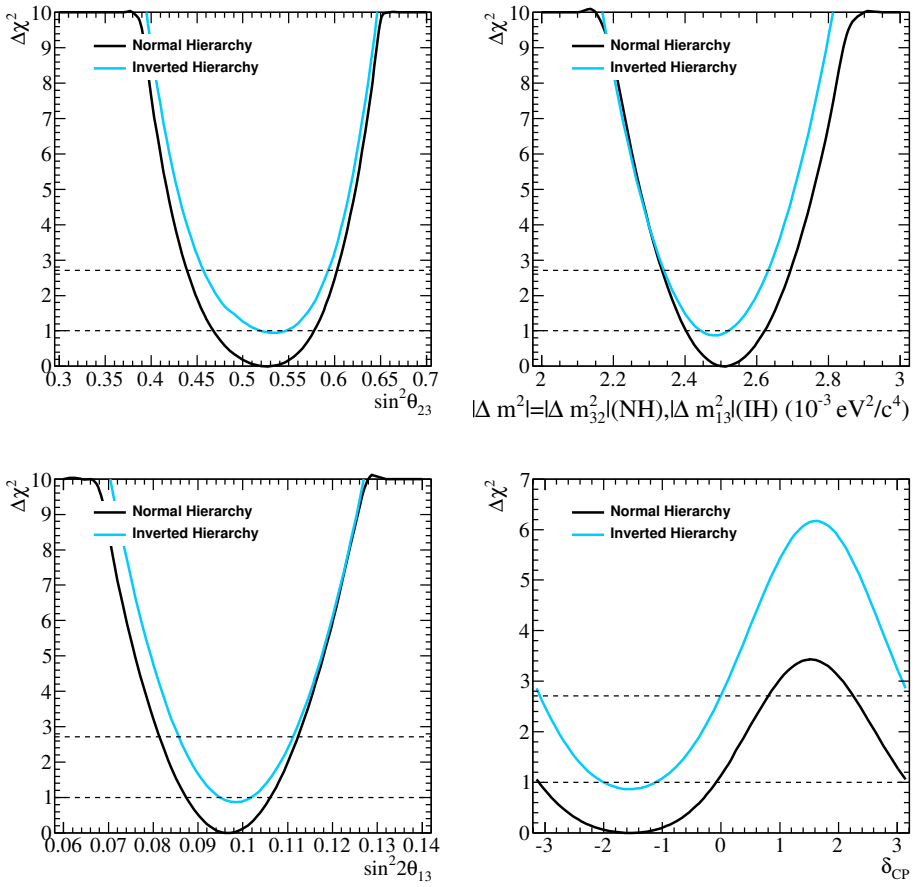


Figure 51: Comparison of marginalized $\Delta\chi^2$ distributions for the joint 3-flavour oscillation fit to the Run 1+2+3+4 dataset with Normal and Inverted Hierarchies and reactor constraint.

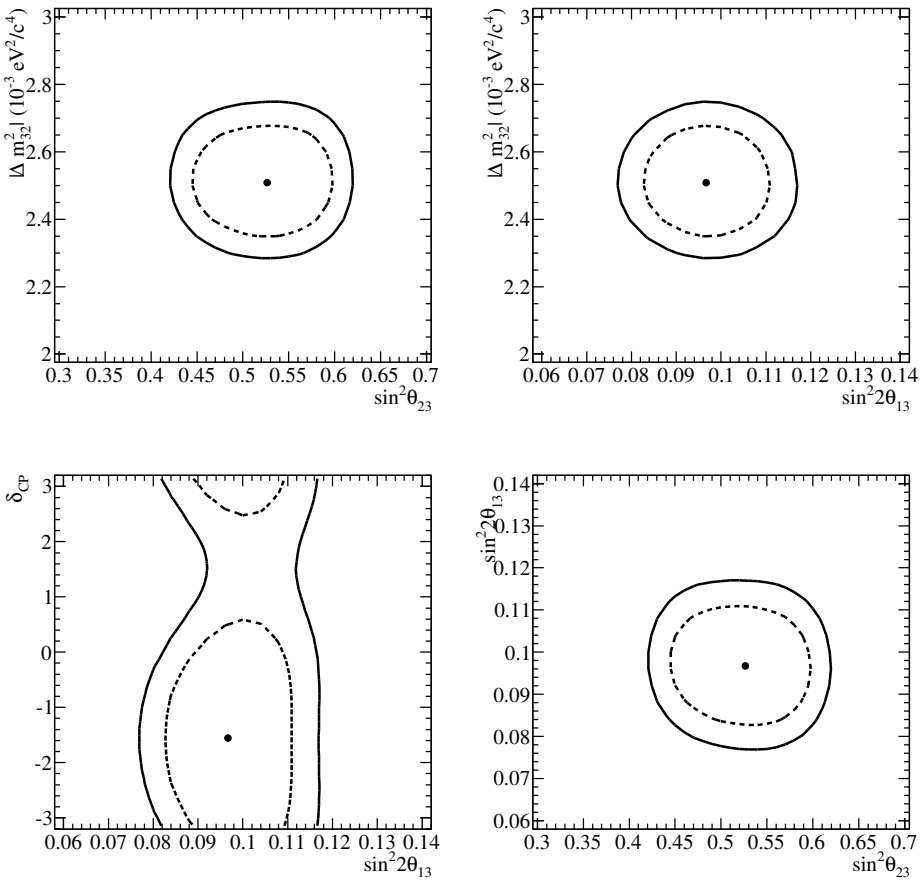


Figure 52: 68% and 90% CL regions in the following four 2-dimensional oscillation parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2 2\theta_{13}$) for the Run1+2+3+4 dataset fit with reactor constraint with Normal Hierarchy. Confidence regions were constructed with respect to the 4D best-fit point with Normal Hierarchy from table 16.

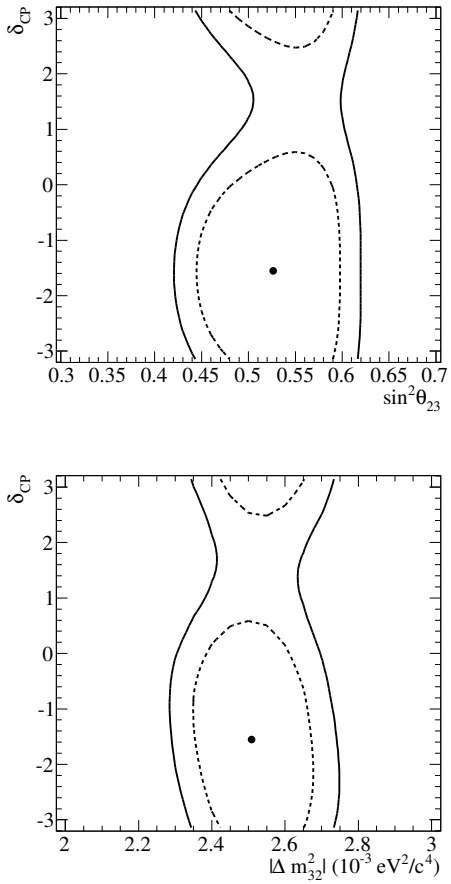


Figure 53: 68% and 90% CL regions in the following two 2-dimensional oscillation parameter spaces ($sin^2\theta_{23}$, δ_{CP}) and ($|\Delta m^2|$, δ_{CP}) for the Run1+2+3+4 dataset fit with reactor constraint with Normal Hierarchy. Confidence regions were constructed with respect to the 4D best-fit point with Normal Hierarchy from table 16.

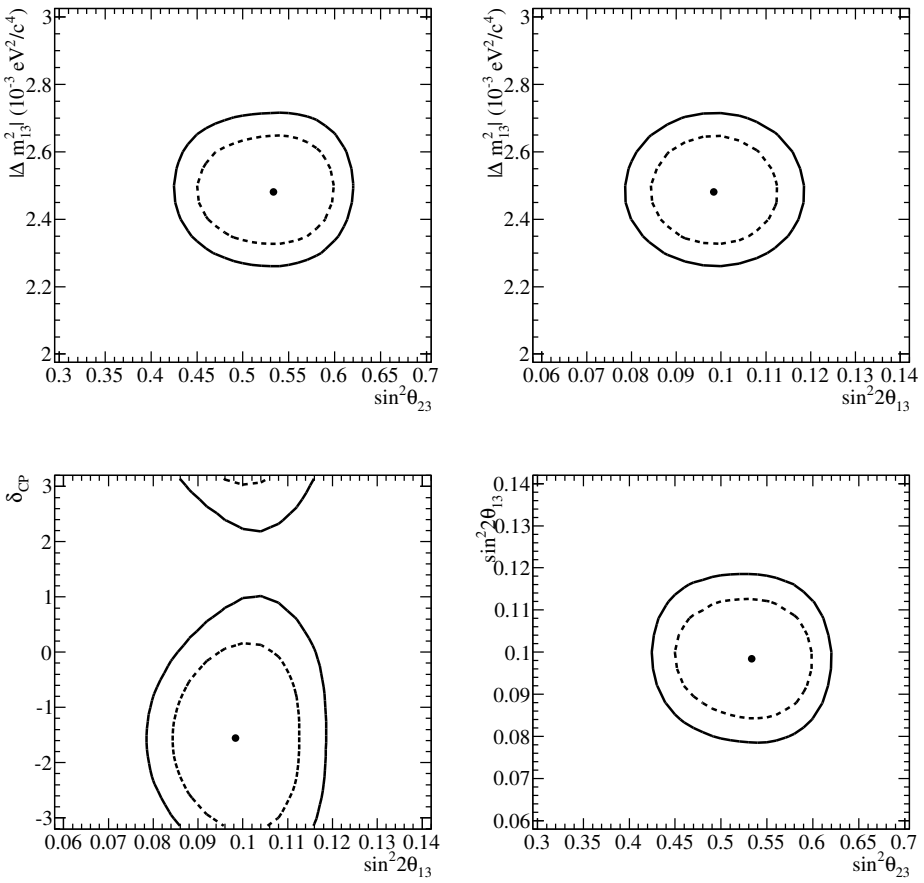


Figure 54: 68% and 90% CL regions in the following four 2-dimensional oscillation parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, δ_{CP}) and ($sin^2\theta_{23}$, $sin^2 2\theta_{13}$) for the Run1+2+3+4 dataset fit with reactor constraint with Inverted Hierarchy. Confidence regions were constructed with respect to the 4D best-fit point with Inverted Hierarchy from table 16.

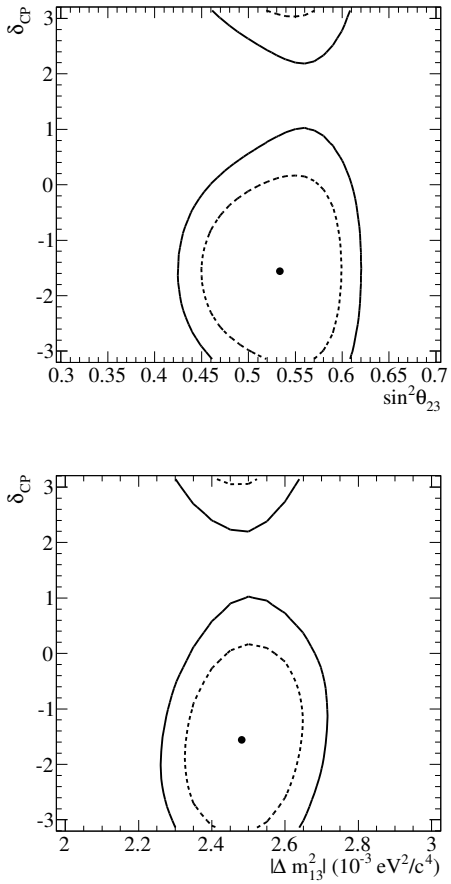


Figure 55: 68% and 90% CL regions in the following two 2-dimensional oscillation parameter spaces ($sin^2\theta_{23}$, δ_{CP}) and ($|\Delta m^2|$, δ_{CP}) for the Run1+2+3+4 dataset fit with reactor constraint with Inverted Hierarchy. Confidence regions were constructed with respect to the 4D best-fit point with Normal Hierarchy from table 16.

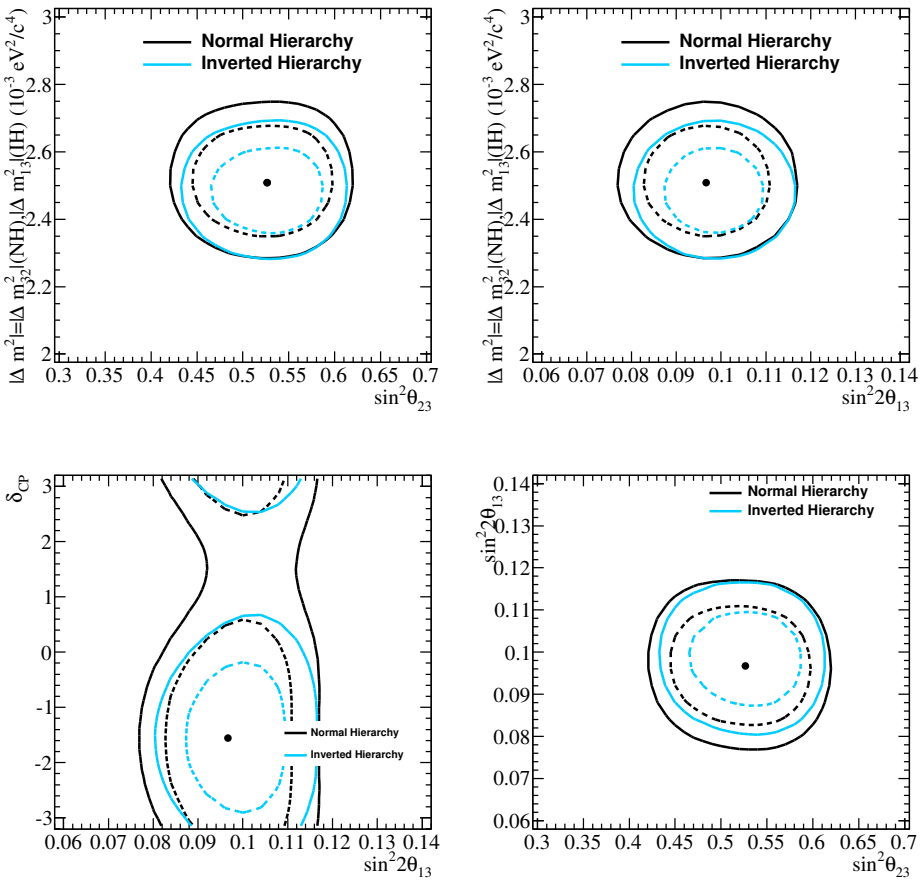


Figure 56: Comparison of 68% and 90% CL regions for the Run1+2+3+4 dataset fit with reactor constraint with different mass hierarchy assumptions. A common best-fit point, the one from the fit with Normal Hierarchy, as it is the one with the smallest value of χ^2, was used to build confidence regions in both mass hierarchy assumptions.

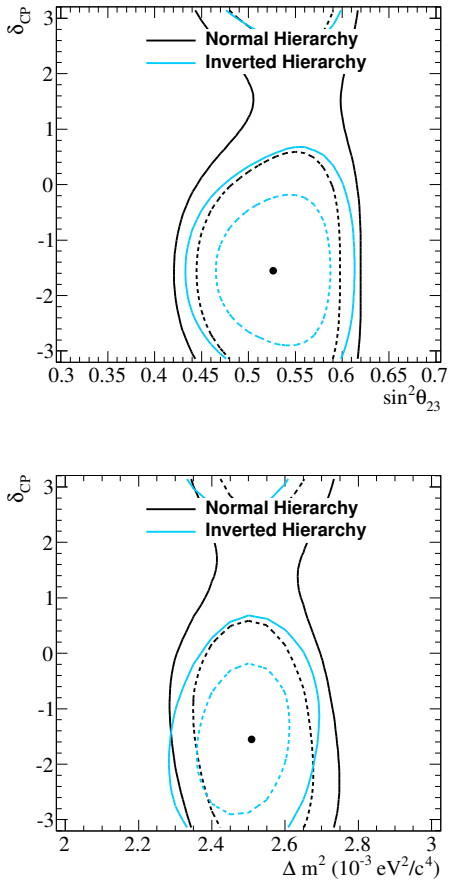


Figure 57: Comparison of 68% and 90% CL regions in the parameter spaces ($sin^2\theta_{23}$, δ_{CP}) and ($|\Delta m^2|$, δ_{CP}) for the Run1+2+3+4 dataset fit with reactor constraint with different mass hierarchy assumptions. A common best-fit point, the one from the fit with Normal Hierarchy, as it is the one with the smallest value of χ^2, was used to build confidence regions in both mass hierarchy assumptions.

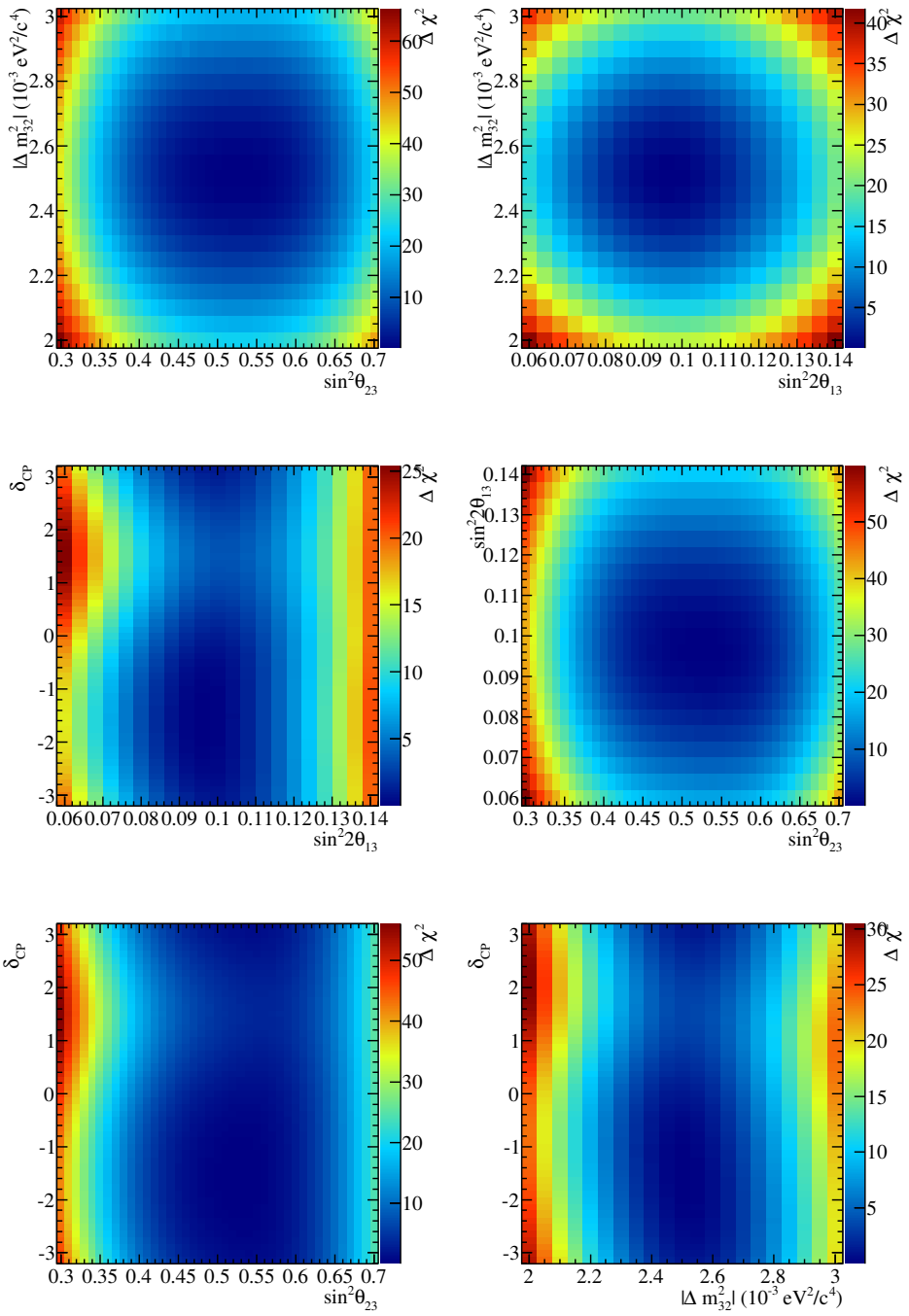


Figure 58: $\Delta\chi^2$ distributions for the six oscillations parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, δ_{CP}), ($sin^2\theta_{23}$, $sin^2 2\theta_{13}$), ($sin^2\theta_{23}$, δ_{CP}) and ($|\Delta m^2|$, δ_{CP}) for the Run1+2+3+4 dataset fit with reactor constraint with Normal Hierarchy.

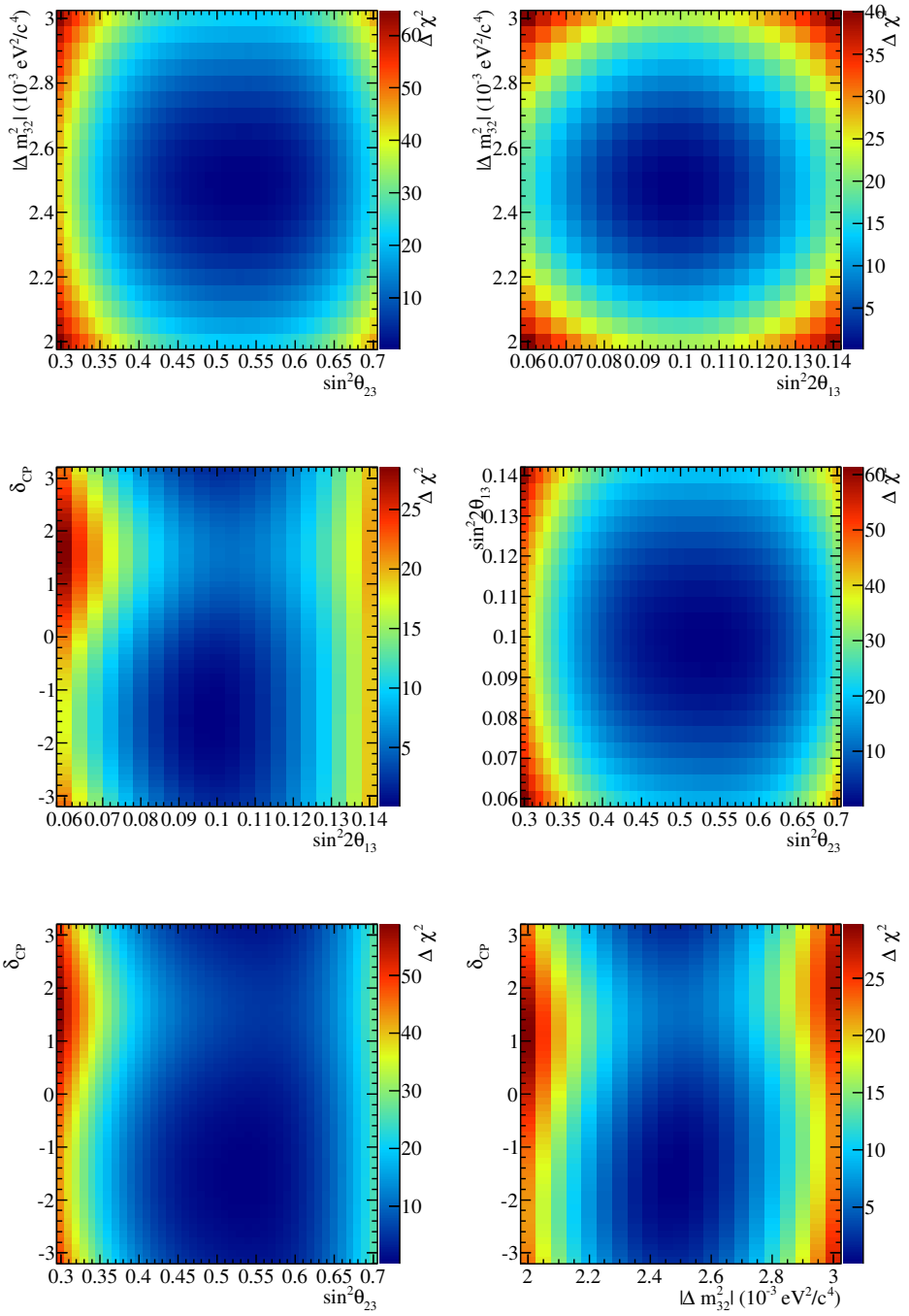


Figure 59: $\Delta\chi^2$ distributions for the six oscillations parameter spaces ($sin^2\theta_{23}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, $|\Delta m^2|$), ($sin^2 2\theta_{13}$, δ_{CP}), ($sin^2\theta_{23}$, $sin^2 2\theta_{13}$), ($sin^2\theta_{23}$, δ_{CP}) and ($|\Delta m^2|$, δ_{CP}) for the Run1+2+3+4 dataset fit with reactor constraint with Inverted Hierarchy.

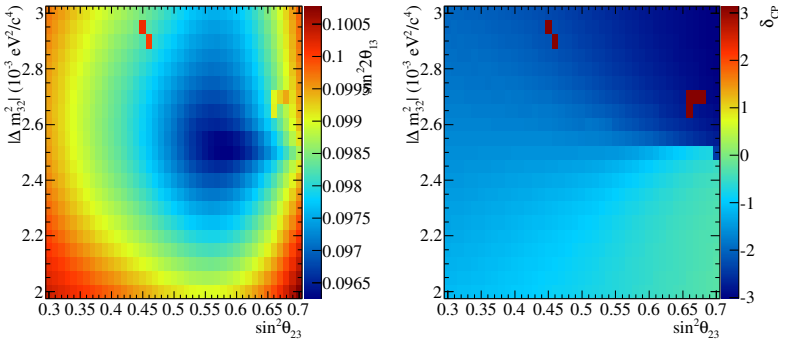


Figure 60: Best-fit values of $sin^2 2\theta_{13}$ and δ_{CP} for the Run1+2+3+4 dataset fit with reactor constraint with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{23}$, $|\Delta m^2_{32}|$).

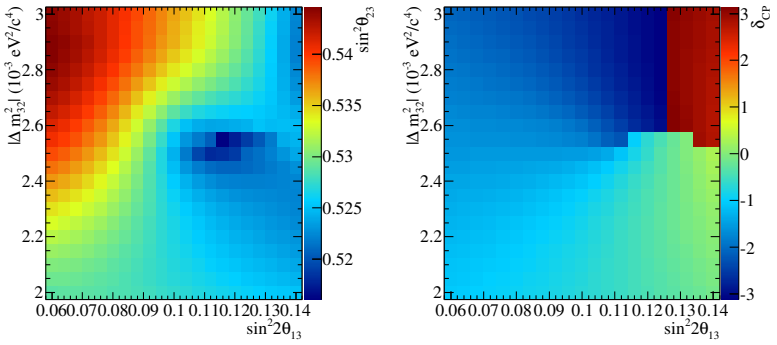


Figure 61: Best-fit values of $sin^2\theta_{23}$ and δ_{CP} for the Run1+2+3+4 dataset fit with reactor constraint with Normal Hierarchy in the oscillation parameter space ($sin^2 2\theta_{13}$, $|\Delta m^2_{32}|$).

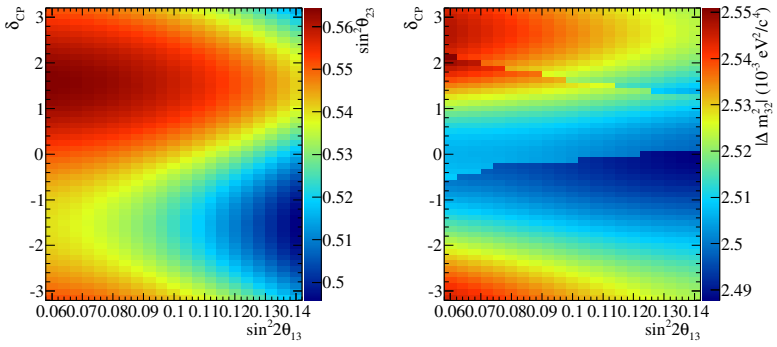


Figure 62: Best-fit values of $sin^2\theta_{23}$ and $|\Delta m^2_{32}|$ for the Run1+2+3+4 dataset fit with reactor constraint with Normal Hierarchy in the oscillation parameter space ($sin^2 2\theta_{13}$,δ_{CP}).

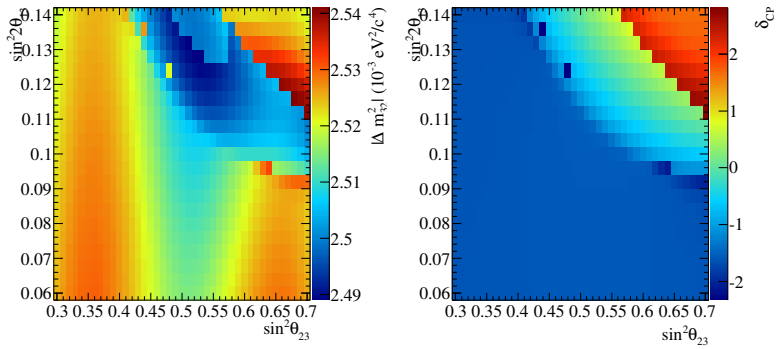


Figure 63: Best-fit values of $|\Delta m^2_{32}|$ and δ_{CP} for the Run1+2+3+4 dataset fit with reactor constraint with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{23}$,$sin^2 2\theta_{13}$).

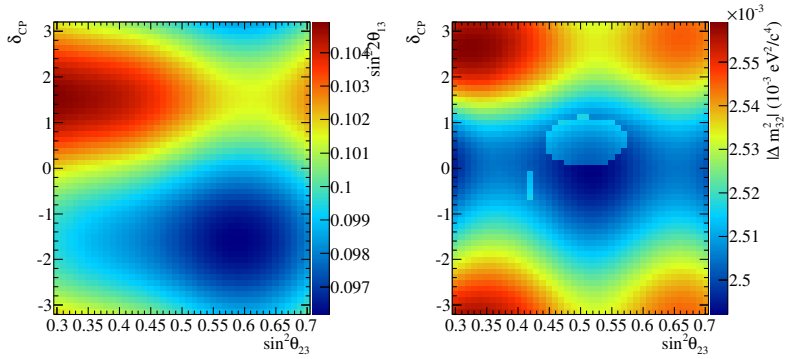


Figure 64: Best-fit values of $|\Delta m^2_{32}|$ and $sin^2 2\theta_{13}$ for the Run1+2+3+4 dataset fit with reactor constraint with Normal Hierarchy in the oscillation parameter space ($sin^2\theta_{23}$,δ_{CP}).

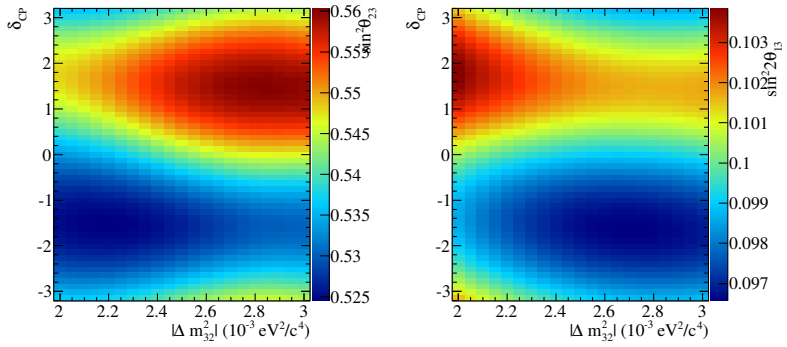


Figure 65: Best-fit values of $sin^2\theta_{23}$and $sin^2 2\theta_{13}$ for the Run1+2+3+4 dataset fit with reactor constraint with Normal Hierarchy in the oscillation parameter space ($|\Delta m^2_{32}|$,δ_{CP}).

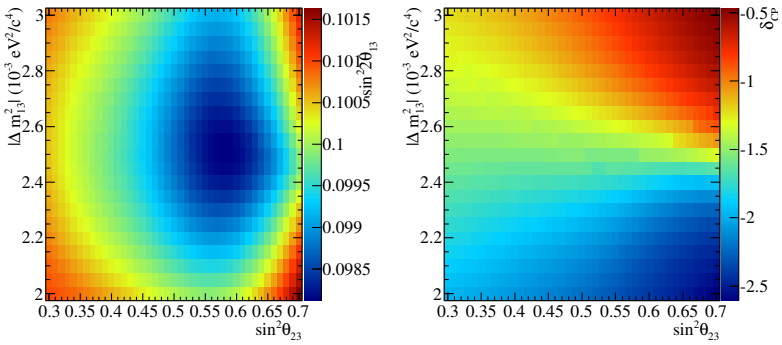


Figure 66: Best-fit values of $sin^2 2\theta_{13}$ and δ_{CP} for the Run1+2+3+4 dataset fit with reactor constraint with Inverted Hierarchy in the oscillation parameter space ($sin^2\theta_{23}$, $|\Delta m^2_{13}|$).

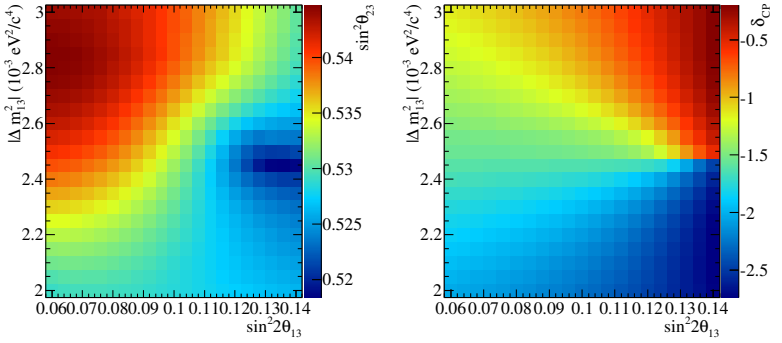


Figure 67: Best-fit values of $sin^2\theta_{23}$ and δ_{CP} for the Run1+2+3+4 dataset fit with reactor constraint with Inverted Hierarchy in the oscillation parameter space ($sin^2 2\theta_{13}$, $|\Delta m^2_{13}|$).

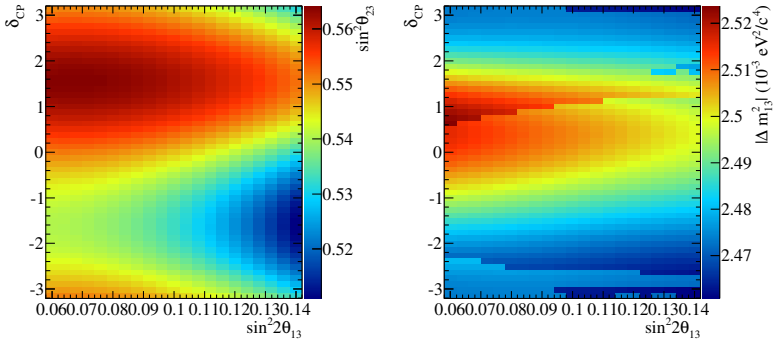


Figure 68: Best-fit values of $sin^2\theta_{23}$ and $|\Delta m^2_{13}|$ for the Run1+2+3+4 dataset fit with reactor constraint with Inverted Hierarchy in the oscillation parameter space ($sin^2 2\theta_{13}$,δ_{CP}).

Finally a comparison was made for the ($sin^2\theta_{23}$, $|\Delta m^2|$) oscillation parameter space with and without reactor constraint. Fig. 72 shows the comparison of 2D confidence regions in the ($sin^2\theta_{23}$, $|\Delta m^2|$) oscillation parameter space with and without reactor constraint for Normal Hierarchy (left) and Inverted Hierarchy (right) using independent best-fit points for each mass hierarchy and using a common best fit point (the one with Inverted Hierarchy without reactor constraint and with Normal Hierarchy with reactor constraint) in Fig. 73. From these figures, differences are small in the Normal Hierarchy assumption when adding the reactor constraint. Larger differences appear in the Inverted Hierarchy assumption, which are even larger when a common best-fit point is used for both mass hierarchies. This is to expect since the best-fit values of $sin^2\theta_{13}$ of the

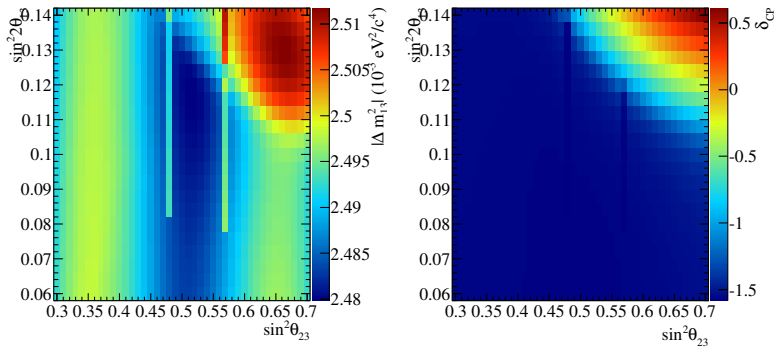


Figure 69: Best-fit values of $|\Delta m^2_{13}|$ and δ_{CP} for the Run1+2+3+4 dataset fit with reactor constraint with Inverted Hierarchy in the oscillation parameter space ($sin^2\theta_{23}$,$sin^2 2\theta_{13}$).

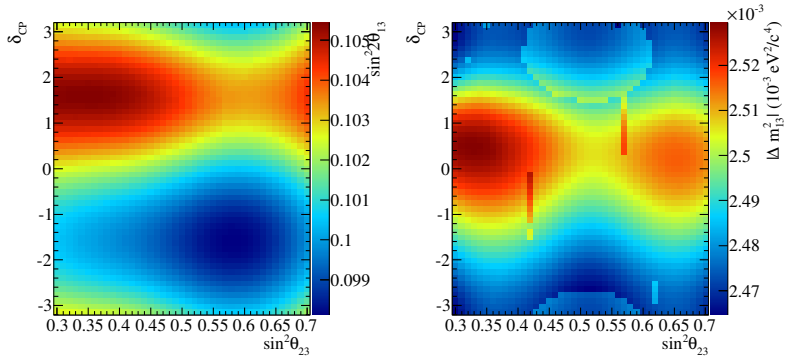


Figure 70: Best-fit values of $|\Delta m^2_{13}|$ and $sin^2 2\theta_{13}$ for the Run1+2+3+4 dataset fit with reactor constraint with Inverted Hierarchy in the oscillation parameter space ($sin^2\theta_{23}$,δ_{CP}).

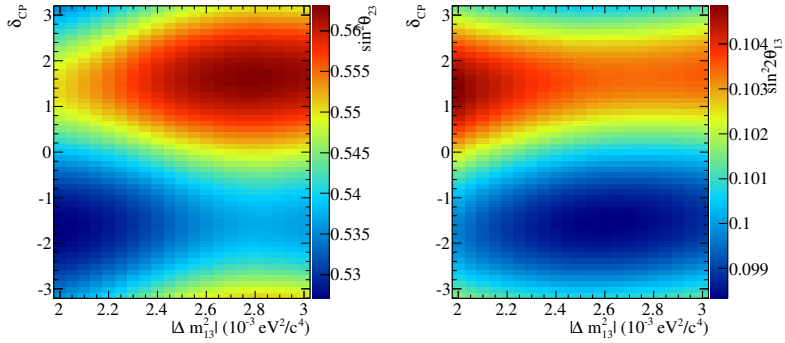


Figure 71: Best-fit values of $sin^2\theta_{23}$ and $sin^2 2\theta_{13}$ for the Run1+2+3+4 dataset fit with reactor constraint with Inverted Hierarchy in the oscillation parameter space ($|\Delta m^2_{13}|$,δ_{CP}).

Run1+2+3+4 dataset fit are larger for the Inverted Hierarchy, then when constraining this parameter larger differences appear. Furthermore, the difference in χ^2 of the best-fit values between the two mass hierarchy assumptions is very small without reactor constraint ($\approx$ 0.01), but becomes significant when adding the reactor constraint ($\approx$ 1), so larger differences are to expect between confidence regions with and without reactor constraint when using a common best-fit point.

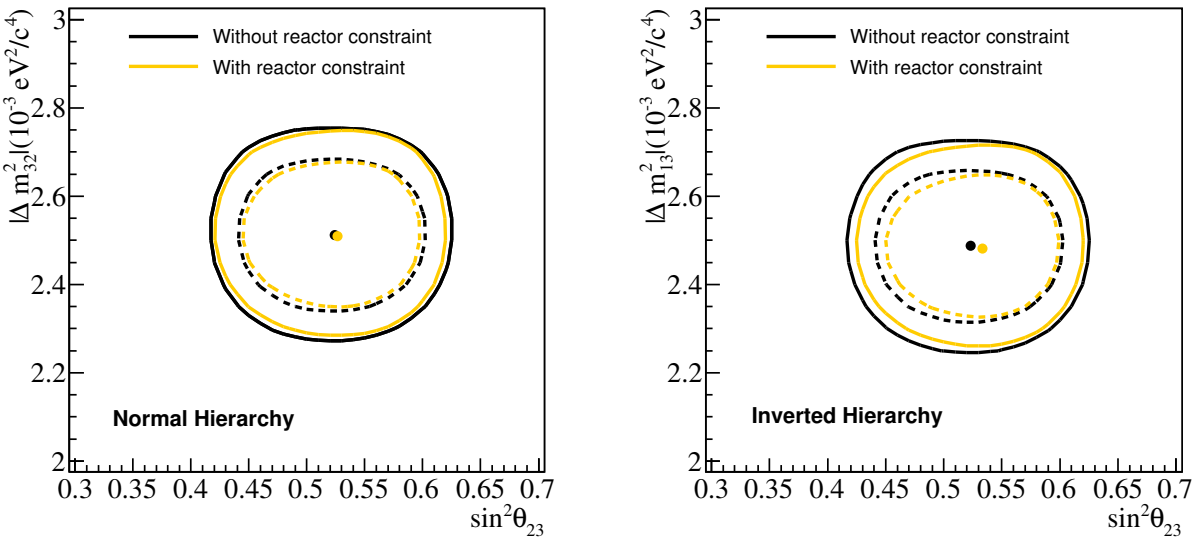


Figure 72: Confidence regions in the ($sin^2\theta_{23}$, Δm^2_{32}) parameters space for the Run1+2+3+4 dataset joint fit with and without reactor constraint for the Normal Hierarchy (left) and the Inverted Hierarchy (right) using independent best-fit points for each mass hierarchy.

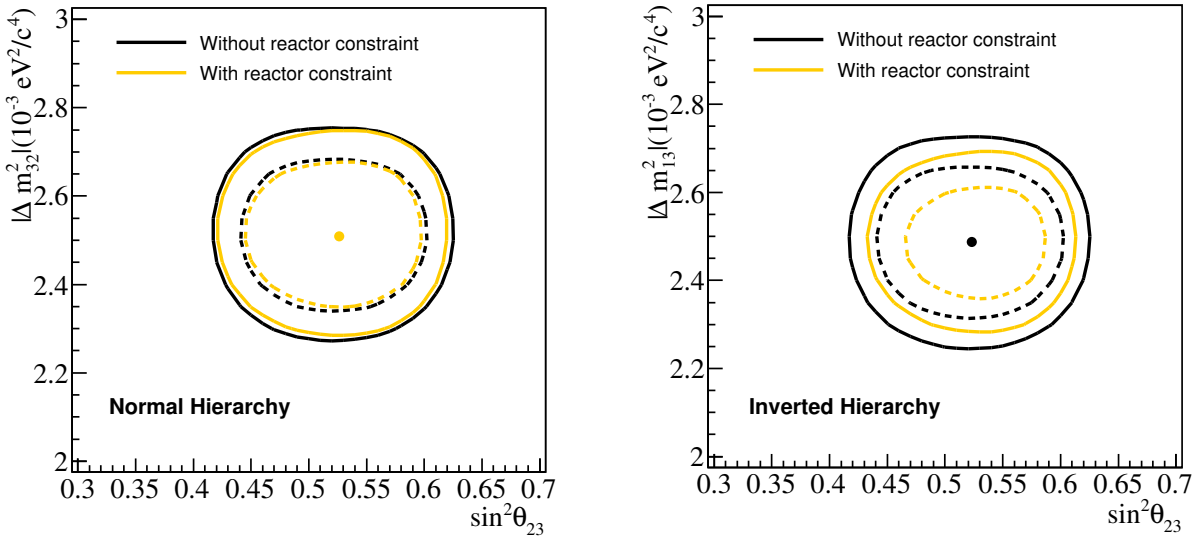


Figure 73: Confidence regions in the ($sin^2\theta_{23}$, Δm^2_{32}) parameters space for the Run1+2+3+4 dataset joint fit with and without reactor constraint for the Normal Hierarchy (left) and the Inverted Hierarchy (right) using a common best-fit point (the one with Inverted Hierarchy without reactor constraint and with Normal Hierarchy with reactor constraint).

7.2.3. $\Delta\chi^2$ vs δ_{CP} with Feldman-Cousins

In this subsection results of the Feldman-Cousins (FC) study will be presented for the plot of marginalized $\Delta\chi^2$ vs δ_{CP} with reactor constraint. This study is based in the one already done for the stand-alone ν_e appearance analysis and presented in [20] and [19], but differs in the treatment chosen to vary the other oscillation parameters. Figure 74 shows the result of the marginalized $\Delta\chi^2$ vs δ_{CP} fitting Run1+2+3+4 with reactor constraint in both mass hierarchies taking as common best-fit point the one for Normal Hierarchy. In this plot the values of the critical $\Delta\chi^2$ are overlaid. These values of critical $\Delta\chi^2$ were found using a Feldman-Cousins method as it will be explained now.

At each fixed value of δ_{CP}, 4k toy experiments were created with statistical fluctuations and systematic variations. The other oscillation parameters ($sin^2 2\theta_{13}$, $sin^2\theta_{23}$ and $|\Delta m^2|$) were also varied following the likelihood added by the 3 dimensional $\Delta\chi^2$ surface obtained for that 3D oscillation parameter space as a result of the Run1+2+3+4 dataset joint fit with reactor constraint.

Two first fits are performed for each toy experiment, assuming the true hierarchy and fitting the 4 oscillation parameters and the systematic parameters, one fit initializing in each octant for $sin^2\theta_{23}$. From these two first fits the value of χ^2_{min} is found as the minimum of the two fits. Then, another four fits are done at each toy experiment, fitting 3 oscillation parameters and systematics, but keeping δ_{CP} fixed to the true input value, repeating the fit for each mass hierarchy assumption and initializing in each octant for $sin^2\theta_{23}$ (four combinations). From these four fits, the minimum χ^2 value is selected and it will be called χ^2_{true}.

Finally for each toy experiment $\Delta\chi^2 = \chi^2_{true} - \chi^2_{min}$ is calculated. The critical $\Delta\chi^2_c$ for 68 % CL (90 % CL) is found at each value of δ_{CP} as the value for which 68 % (90 %) of the toy experiments created at that point have a $\Delta\chi^2 < \Delta\chi^2_c$.

Figure 74 shows the result of the marginalized $\Delta\chi^2$ vs δ_{CP} with the results of the critical $\Delta\chi^2_c$ values for Normal Hierarchy calculated using the method described above and presented as a set of points. In Fig. 75, results of the same study made with the stand-alone appearance analysis and using also reconstructed energy spectrum and profiling method from [20] are overlaid. The improvement of the significance for δ_{CP} with the joint fit is clear. Results presented in this Fig. 74 must be understood with the caveat that the Run1+2+3+4 $\Delta\chi^2$ surface was used for varying $sin^2\theta_{23}$ and $|\Delta m^2|$ whereas the $\Delta\chi^2$ surface used for the stand-alone apperance analysis was only including Run1+2+3 dataset. Besides, the reactor values used to constrain $sin^2 2\theta_{13}$ were updated to PDG2013 values, while PDG2012 values were used for the stand-alone appearance analysis.

As a cross-check, results of the marginalized $\Delta\chi^2$ vs δ_{CP} with the joint fit were also compared with the other stand-alone appearance analysis in [19] in Fig. 76. Results are very close, although conclusions can not be directly extracted since different marginalization methods were used in each analysis.

Finally, excluded regions can be obtained directly from Fig. 74 and they are, for Normal Hierarchy at 90 % CL: Excluded region for δ_{CP} = [0.143,0.895]π.

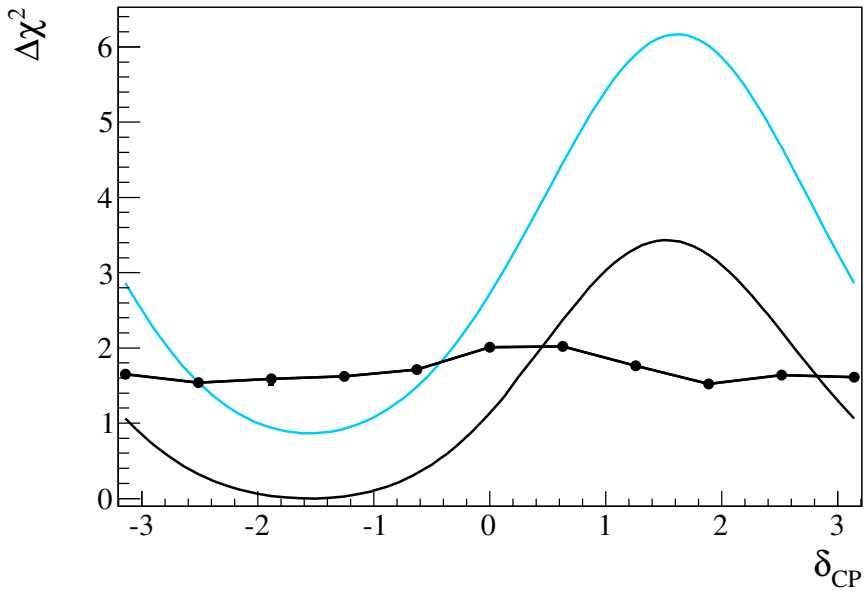


Figure 74: Marginalized $\Delta\chi^2$ vs δ_{CP} with the results of the critical $\Delta\chi^2_c$ values for Normal Hierarchy for the Run1+2+3+4 dataset joint fit with reactor constraint.

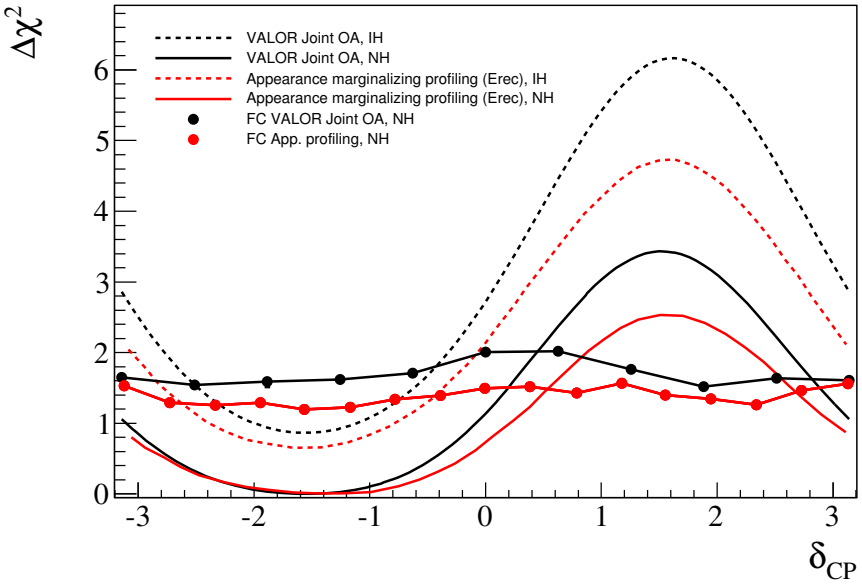


Figure 75: Marginalized $\Delta\chi^2$ vs δ_{CP} with the results of the critical $\Delta\chi^2_c$ values for Normal Hierarchy for the Run1+2+3+4 dataset joint fit with reactor constraint, compared to the same study in the stand-alone appearance analysis in [20].

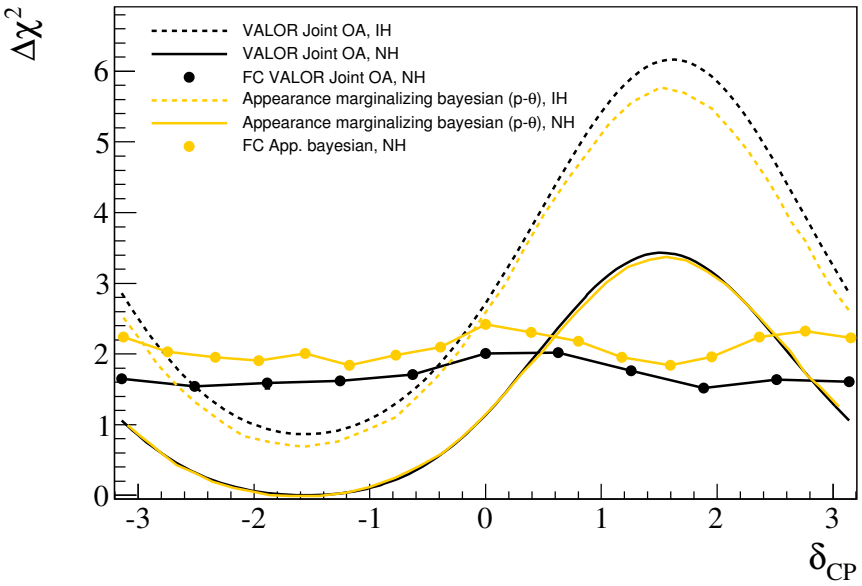


Figure 76: Marginalized $\Delta\chi^2$ vs δ_{CP} for the Run1+2+3+4 dataset joint fit with reactor constraint, compared to the result from the stand-alone appearance analysis in [19].

7.3. Comparisons of confidence regions

7.3.1. Comparisons of confidence regions with and without correlations for the SK detector efficiencies

In Fig. 7.3.1 confidence regions are compared for the Run1+2+3+4 dataset fit with Normal Hierarchy with and without correlations between the SK detector efficiencies for the single μ-like ring and single e-like ring samples.

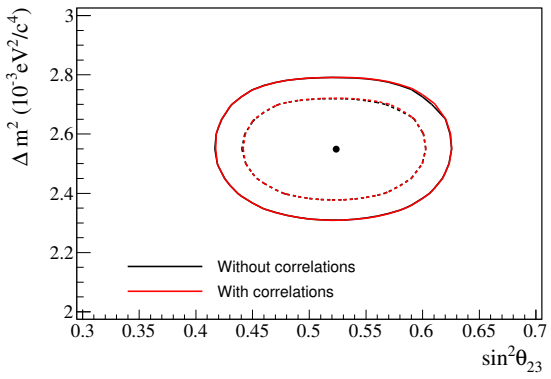


Figure 77: Comparison of confidence regions in the ($sin^2\theta_{23}$, $|\Delta m^2_{32}|$) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy with and without correlations between the SK detector efficiencies for the single μ-like ring and single e-like ring samples.

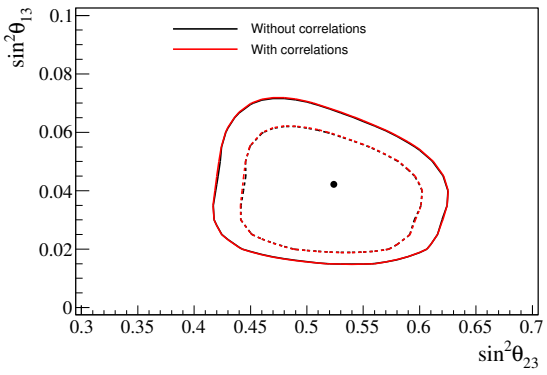


Figure 78: Comparison of confidence regions in the ($sin^2\theta_{23}$, $sin^2\theta_{13}$) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy with and without correlations between the SK detector efficiencies for the single μ-like ring and single e-like ring samples.

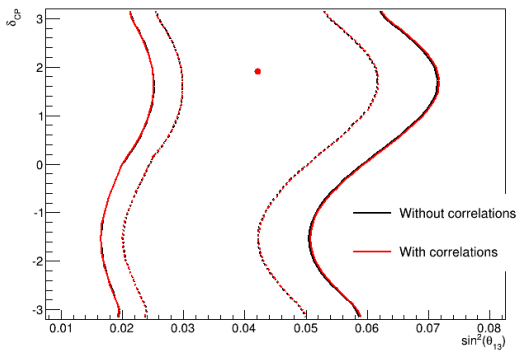


Figure 79: Comparison of confidence regions in the ($sin^2\theta_{13}$, δ_{CP}) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy with and without correlations between the SK detector efficiencies for the single μ-like ring and single e-like ring samples.

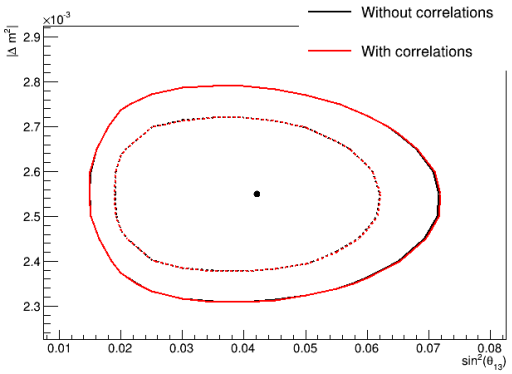


Figure 80: Comparison of confidence regions in the ($sin^2\theta_{13}$, $|\Delta m^2_{32}|$) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy with and without correlations between the SK detector efficiencies for the single μ-like ring and single e-like ring samples.

7.3.2. Comparisons of confidence regions with stand-alone analyses

In this section, results of the Run1+2+3+4 dataset joint fit will be compared with the fit to the same dataset in stand-alone ν_μ-disappearance and ν_e-appearance modes.

In Fig. 81 the confidence regions in the ($sin^2\theta_{23}$, $|\Delta m^2_{32}|$) parameters space for Normal Hierarchy and in the ($sin^2\theta_{23}$, $|\Delta m^2_{13}|$) in Fig. 82 for Inverted Hierarchy result of the Run1+2+3+4 dataset joint fit are compared with the results of the *VALOR* Run1+2+3+4 ν_μ-disappearance analysis. In both cases the same constant $\Delta\chi^2$ method was used. Clearly, confidence regions are comparable in size, indicating that, although $sin^2\theta_{13}$ and δ_{CP} are also included in the fit, the constraint from the single e-like ring sample prevents the confidence regions to become larger. Also, there is a small shift to larger values of $sin^2\theta_{23}$. The reason for this shift is that the best-fit value of $sin^2\theta_{13}$ for the joint fit is larger than the typical value used for this parameter (as it is as well for the stand-alone ν_e appearance analysis). Thus, the value for maximal disappearance becomes larger, and the result for the joint fit is also showing a preference for maximal disappearance for the best-fit point.

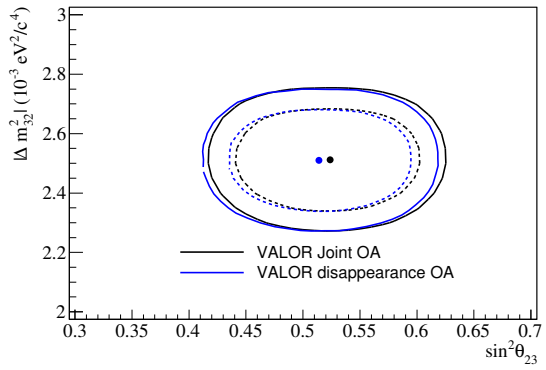


Figure 81: Comparison of confidence regions in the ($sin^2\theta_{23}$, $|\Delta m^2_{32}|$) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy for the joint oscillation analysis and the *VALOR* Run1+2+3+4 ν_μ-disappearance analysis. In both cases the same constant $\Delta\chi^2$ method was used.

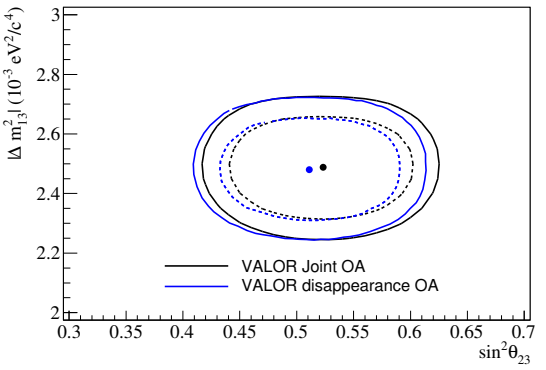


Figure 82: Comparison of confidence regions in the ($sin^2\theta_{23}$, $|\Delta m^2_{32}|$) parameter space for the Run1+2+3+4 dataset fit assuming Inverted Hierarchy for the joint oscillation analysis and the *VALOR* Run1+2+3+4 ν_μ-disappearance analysis. In both cases the same constant $\Delta\chi^2$ method was used.

In Fig. 83 the confidence regions in the ($sin^2 2\theta_{13}$, δ_{CP}) parameters space for the Run1+2+3+4 dataset fit with Normal Hierarchy for the joint oscillation analysis are compared with the results for the stand-alone appearance analysis using reconstructed energy, presented in [20] (plots were digitalized) using a fixed value for $|\Delta m^2_{32}|$ and $sin^2\theta_{23}$=0.5 for the stand-alone appearance analysis. Similarly for Inverted Hierarchy in Fig. 84. In Fig. 85 the same comparison is made with the results of the stand-alone ν_e appearance analysis in [20] marginalizing over $|\Delta m^2_{32}|$ and $sin^2\theta_{23}$. Similarly for Inverted Hierarchy in Fig. 86. In all these plots, the raster scan method was used and the joint fit was done in the variable $sin^2 2\theta_{13}$ to match the stand-alone appearance results. In the first Figs 83 and 84, the confidence regions of the joint oscillation analysis are approximately of the same size as the ones for the stand-alone appearance analysis. Although the total effect of systematics is smaller as shown in 21, in the joint fit it is included the uncertainty in $|\Delta m^2_{32}|$ and $sin^2\theta_{23}$, which are fitted, whereas fixed in the stand-alone analysis. On the other hand, when $|\Delta m^2_{32}|$ and $sin^2\theta_{23}$ are marginalized in the stand-alone appearance analysis, the confidence regions are larger than the ones for the joint fit, clearly visible in Figs. 85 and 86 because the later is including the single μ-like ring sample to constraint those parameters. In all cases, however, the best-fit values (and therefore confidence regions) are shifted for the joint analysis towards smaller values of $sin^2 2\theta_{13}$. This is due to the fact that the best-fit values of $sin^2\theta_{23}$ are larger than 0.5 along the region of the best-fit points in ($sin^2 2\theta_{13}$, δ_{CP}) as can be observed in Figs. 87 and 88, where the line of best-fit values in the ($sin^2 2\theta_{13}$, δ_{CP}) space is overlaid on the best-fit values of $sin^2\theta_{23}$ for Normal and Inverted Hierarchy respectively. Thus, as it was explained in the stand-alone appearance analysis like [20], because $sin^2\theta_{23}$¿0.5, the line of best-fit points moves to smaller values of $sin^2 2\theta_{13}$.

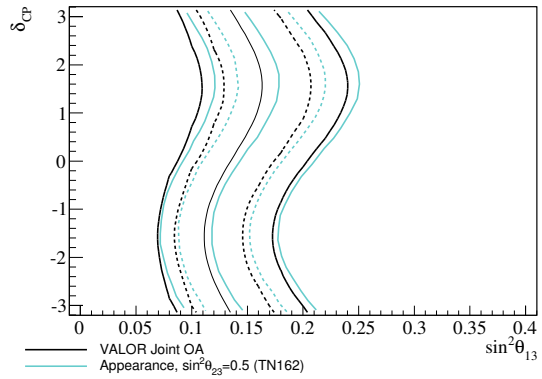


Figure 83: Comparison of confidence regions in the ($sin^2 2\theta_{13}$, δ_{CP}) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy for the joint oscillation analysis and the Run1+2+3+4 ν_e-appearance analysis reported in [20] using a fixed value for $|\Delta m^2_{32}|$ and $sin^2\theta_{23}$=0.5 for the stand-alone appearance analysis (plots digitalized). In both cases the raster scan method was used.

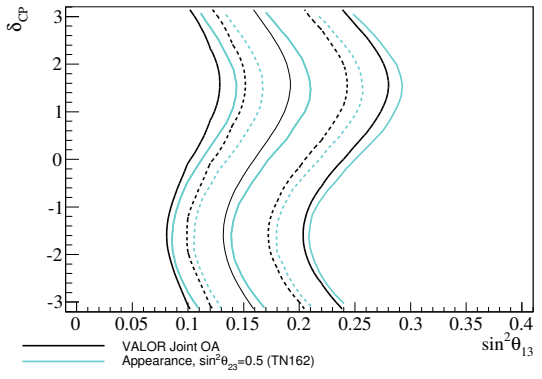


Figure 84: Comparison of confidence regions in the ($sin^2 2\theta_{13}$, δ_{CP}) parameter space for the Run1+2+3+4 dataset fit assuming Inverted Hierarchy for the joint oscillation analysis and the Run1+2+3+4 ν_e-appearance analysis reported in [20] using a fixed value for $|\Delta m^2_{32}|$ and $sin^2\theta_{23}$=0.5 for the stand-alone appearance analysis (plots digitalized). In both cases the raster scan method was used.

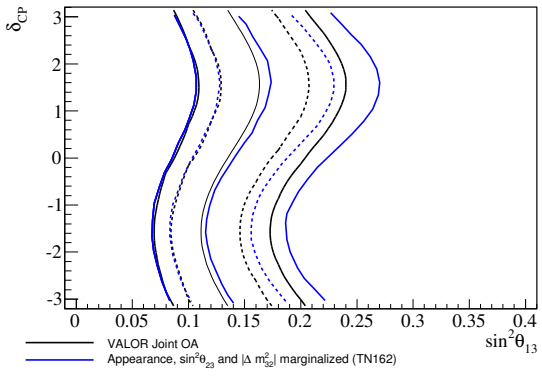


Figure 85: Comparison of confidence regions in the ($sin^2 2\theta_{13}$, δ_{CP}) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy for the joint oscillation analysis and the Run1+2+3+4 ν_e-appearance analysis reported in [20] marginalizing over $|\Delta m^2_{32}|$ and $sin^2\theta_{23}$ for the stand-alone appearance analysis (plots digitalized). In both cases the raster scan method was used.

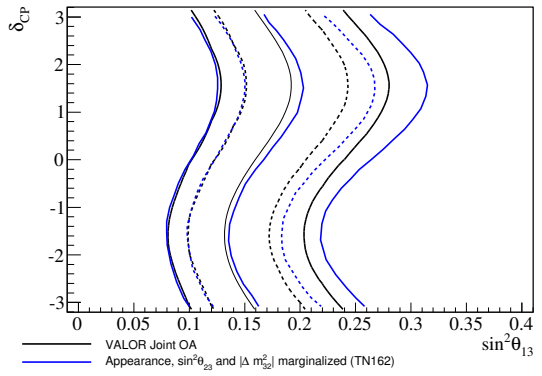


Figure 86: Comparison of confidence regions in the ($sin^2 2\theta_{13}$, δ_{CP}) parameter space for the Run1+2+3+4 dataset fit assuming Inverted Hierarchy for the joint oscillation analysis and the Run1+2+3+4 ν_e-appearance analysis reported in [20] marginalizing over $|\Delta m^2_{32}|$ and $sin^2\theta_{23}$ for the stand-alone appearance analysis (plots digitalized). In both cases the raster scan method was used.

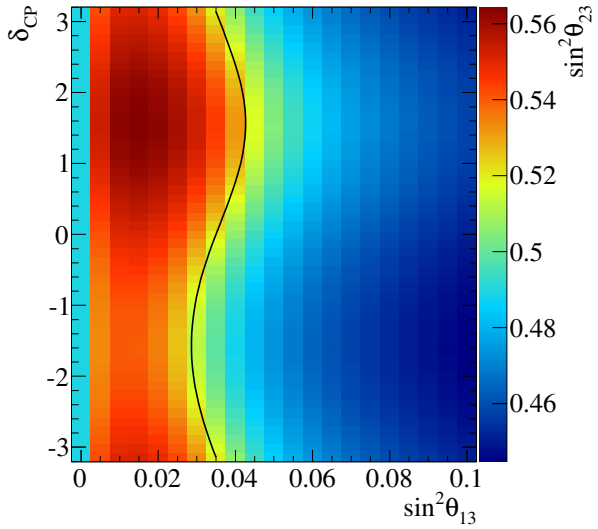


Figure 87: Line of best-fit values in the ($sin^2 2\theta_{13}$, δ_{CP}) space overlaid on the best-fit values of $sin^2\theta_{23}$ for Normal Hierarchy.

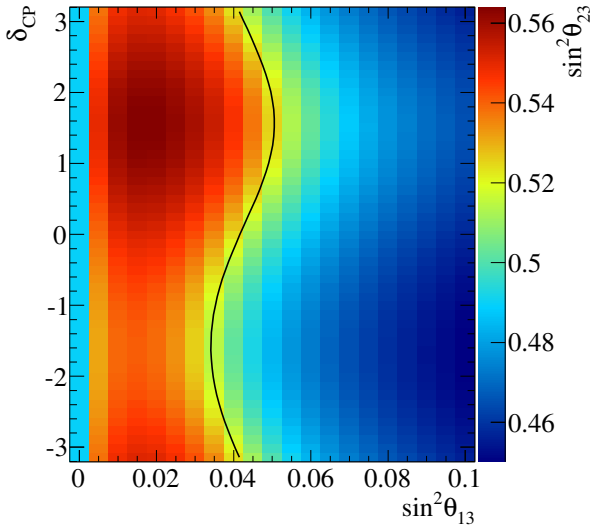


Figure 88: Line of best-fit values in the ($sin^2 2\theta_{13}$, δ_{CP}) space overlaid on the best-fit values of $sin^2\theta_{23}$ for Inverted Hierarchy.

7.3.3. Comparisons of confidence regions with results from other experiments

In Fig. 89 the confidence regions in the ($sin^2\theta_{23}$, Δm^2_{32}) parameters space for the Run1+2+3+4 dataset joint fit (T2K data only, without reactor constraint) for the two mass hierarchies are compared with the preliminary results of a joint oscillation analysis, also combining ν_μ disappearance and ν_e appearance, of the MINOS experiment, as presented in [34]. MINOS' result is fitting $sin^2\theta_{13}$ with reactor constraint (PDG2013) and fitting as well δ_{CP}.

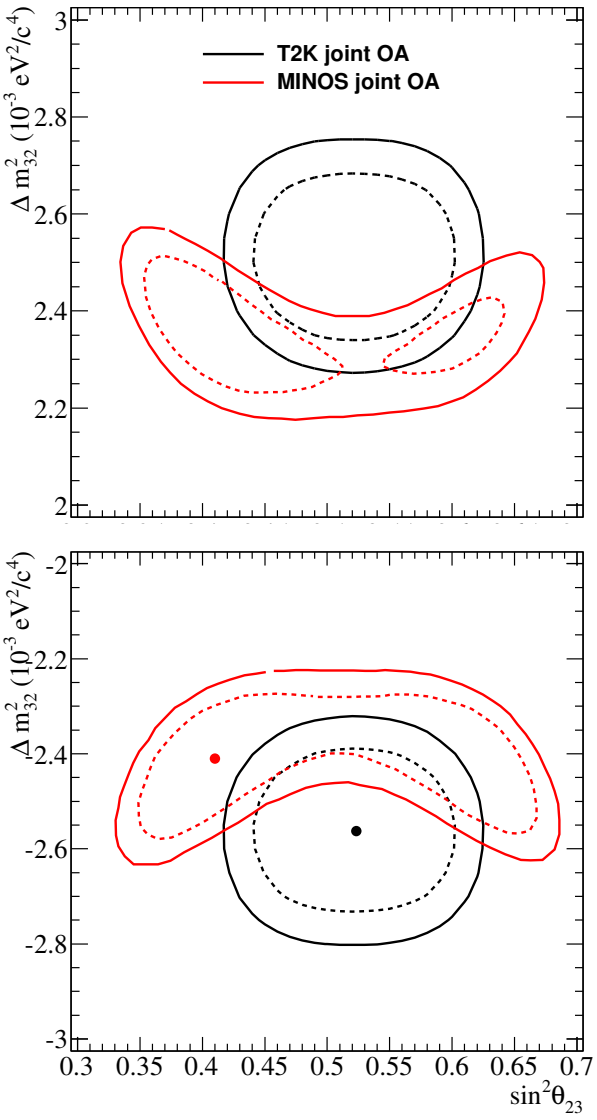


Figure 89: Confidence regions in the ($sin^2\theta_{23}$, Δm^2_{32}) parameters space for the Run1+2+3+4 dataset joint fit (T2K data only, wihtout reactor constraint) for the two mass hierarchies compared with the preliminary reults of a joint oscillation analysis, also combining ν_μ disappearance and ν_e appearance, of the MINOS experiment, as presented in [34]

In Fig. 90 the confidence regions in the ($sin^2\theta_{23}$, Δm^2_{32}) parameters space for the Run1+2+3+4 dataset joint fit (T2K data only, without reactor constraint) for the two mass hierarchies are compared with the preliminary results of a 3-flavour atmospheric neutrino fit from the Super-Kamiokande experiment [64]. This Super-K 3-flavour joint fit is a zenith angle oscillation fit combining as well ν_μ disappearance and ν_e appearance, and it was done using SK I-IV atmospheric data fixing $sin^2\theta_{13} = 0.098$ (PDG2012) and fitting δ_{CP}. Notice that the results from the Super-K 3-flavour fit are done using independent best-fit points for each mass hierarchy.

Finally Fig. 91 presents the comparison of the confidence regions in the ($sin^2\theta_{23}$, Δm^2_{32}) parameters space for the Run1+2+3+4 dataset joint fit (T2K data only, wihtout reactor constraint) for the two mass hierarchies with the results from both MINOS and Super-K 3-flavour oscillation fits. Fig. 92 shows the same plot with the result for Run1+2+3+4 dataset joint

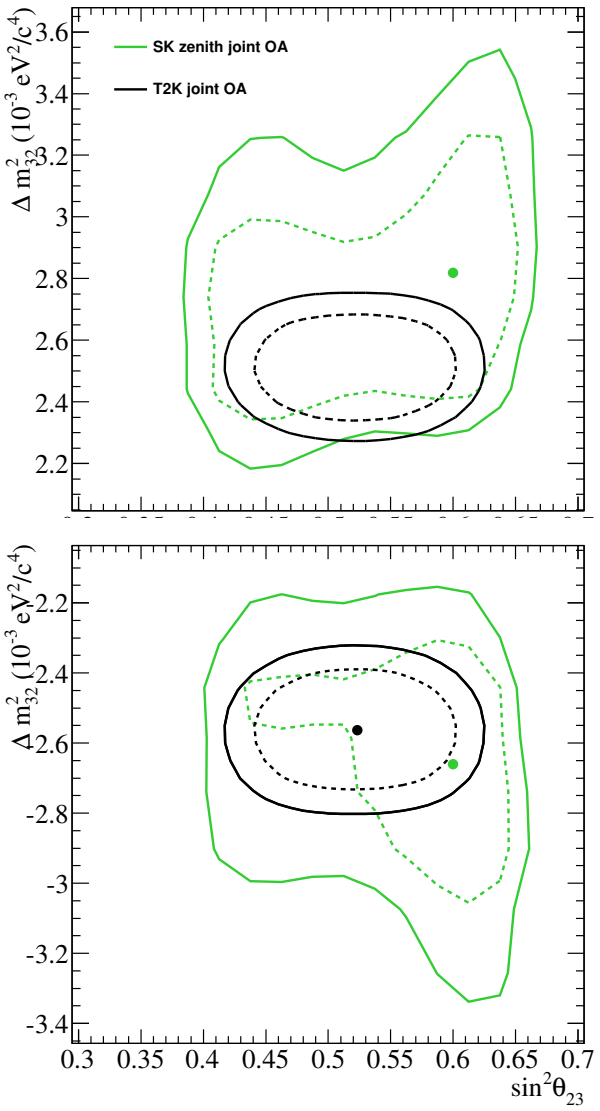


Figure 90: Confidence regions in the ($sin^2\theta_{23}$, Δm^2_{32}) parameters space for the Run1+2+3+4 dataset joint fit (T2K data only, wihtout reactor constraint) for the two mass hierarchies compared with the preliminary results of a 3-flavour atmospheric neutrino fit from the Super-Kamiokande experiment [64]. This Super-K 3-flavour joint fit is a zenith angle oscillation fit combining as well ν_μ disappearance and ν_e appearance, and it was done using SK I-IV atmospheric data fixing $sin^2\theta_{13} = 0.098$. Notice that the results from the Super-K 3-flavour fit are done using independent best-fit points for each mass hierarchy.

1634 fit including the reactor constraint (PDG2013).

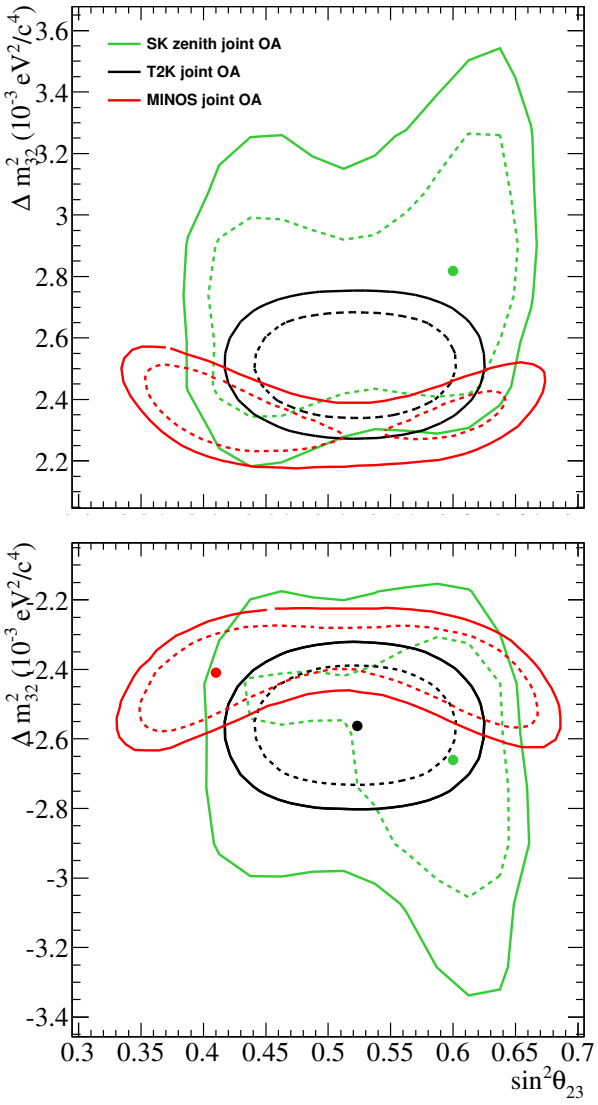


Figure 91: Confidence regions in the ($sin^2\theta_{23}$, Δm^2_{32}) parameters space for the Run1+2+3+4 dataset joint fit (T2K data only, without reactor constraint) for the two mass hierarchies compared with the preliminary results of a 3-flavour atmospheric neutrino fit from the Super-Kamiokande experiment [64] and preliminary results of a joint oscillation analysis, also combining ν_μ disappearance and ν_e appearance, of the MINOS experiment, as presented in [34]. Notice that the results from the Super-K 3-flavour fit are done using independent best-fit points for each mass hierarchy.

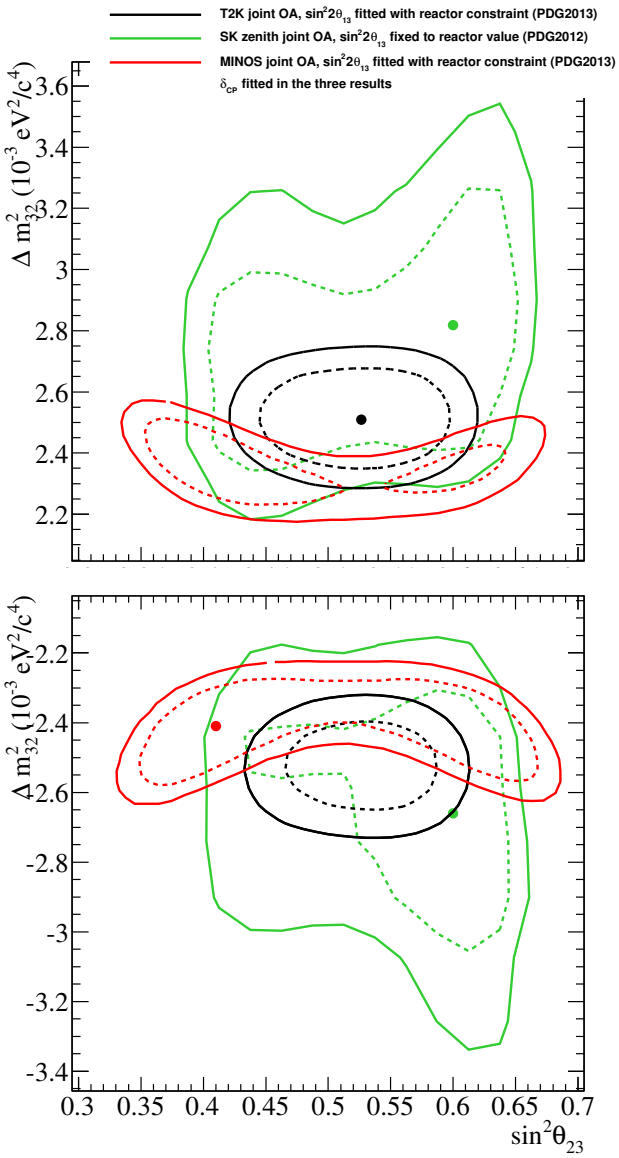


Figure 92: Confidence regions in the ($sin^2\theta_{23}$, Δm^2_{32}) parameters space for the Run1+2+3+4 dataset joint fit with reactor constraint (PDG2013) for the two mass hierarchies compared with the preliminary results of a 3-flavour atmospheric neutrino fit from the Super-Kamiokande experiment [64] and preliminary results of a joint oscillation analysis, also combining ν_μ disappearance and ν_e appearance, of the MINOS experiment, as presented in [34]. Notice that the results from the Super-K 3-flavour fit are done using independent best-fit points for each mass hierarchy.

8. Summary

This technical note reports the first joint 3-flavour oscillation analysis by the VaLOR group including correlations between the SK detector efficiencies for the single μ-like ring and single e-like ring events. Results presented in this technical note were made on the combined Run 1+2+3+4 dataset, which corresponds to an integrated neutrino beam exposure of 6.57×10^{20} POT. The joint analysis determines $|\Delta m^2|$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$ and δ_{CP} together by making a simultaneous fit of the reconstructed energy spectra of the single μ-like ring and single e-like ring SuperK event samples.

This fit is performed in a 3-flavour framework including constant-density matter effects. Official 2013 T2K analysis inputs are used, and all 64 systematic parameters considered in the analysis (see Sec. 5) are allowed to float in the fit. The Fogli-Lisi convention was used for the "atmospheric" mass-squared splitting $|\Delta m^2|$ (see Sec. 3) during this technical note, although it was requested to use $|\Delta m^2_{32}|$ (NH) and $|\Delta m^2_{13}|$ (IH) for the final results. The "solar" oscillation parameters $\sin^2\theta_{12}$ and Δm^2_{21} have been found to have a negligible effect (see appendix B), and are fixed at 0.306 and 7.5×10^{-5} eV^2/c^4 respectively.

This analysis has been extensively validated. The ν_μ-disappearance analysis is identical to the 2013 VALOR analysis reported in Ref. [6], and the ν_e-appearance analysis has been validated against the official 2013 analyses (see appendix A). Studies have been made of fitter residuals (appendix C), fits of 6.57×10^{20} POT fake datasets (appendix D). These studies have shown that the joint analysis fitter is performing as expected.

For the normal hierarchy (NH), the joint analysis fit of the reconstructed energy spectra of the single μ-like ring and single e-like ring SuperK event samples gives $|\Delta m^2_{32}| = 2.512\times10^{-3}\ eV^2/c^4$, $\sin^2\theta_{23} = 0.524$, $\sin^2\theta_{13} = 0.0422$ and $\delta_{CP} = 1.909$ (p-value = 0.921). The corresponding best-fit values for the inverted hierarchy (IH) are $|\Delta m^2_{13}| = 2.488\times10^{-3}\ eV^2/c^4$, $\sin^2\theta_{23} = 0.523$, $\sin^2\theta_{13} = 0.0491$ and $\delta_{CP} = 1.005$ (p-value = 0.952).

Confidence regions are constructed using the constant-$\Delta\chi^2$ method. For the NH, the 90% (68%) C.L. allowed region for the four fitted oscillation parameters can be summarized as $2.320\ (2.394)\times10^{-3}\ eV^2/c^4 < |\Delta m^2_{32}| < 2.697\ (2.623)\times10^{-3}\ eV^2/c^4$, $0.435\ (0.465) < sin^2\theta_{23} < 0.608\ (0.581)$, $0.018\ (0.021) < sin^2\theta_{13} < 0.063\ (0.055)$ and $-\pi\ (-\pi) < \delta_{CP} < \pi\ (\pi)$ For the IH, the corresponding 90% (68%) C.L. allowed regions are $2.292\ (2.370)\times10^{-3}\ eV^2/c^4 < |\Delta m^2_{13}| < 2.672\ (2.605)\times10^{-3}\ eV^2/c^4$, $0.434\ (0.458) < sin^2\theta_{23} < 0.578\ (0.596)$, $0.022\ (0.028) < sin^2\theta_{13} < 0.076\ (0.064)$ and $-\pi\ (-\pi) < \delta_{CP} < \pi\ (\pi)$ The joint analysis has the advantage compared with the stand-alone analyses that it does not need to assume the values of any of the oscillation parameters $|\Delta m^2|$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$ and δ_{CP}. In particular, the value of $\sin^2\theta_{23}$ is not assumed as in the stand-alone ν_e-appearance analysis, but is constrained by the fit of the reconstructed energy spectrum of the single μ-like ring SuperK event sample. It was shown that the confidence regions of the joint fit were smaller than the ones found with the stand-alone appearance analysis marginalizing $\sin^2\theta_{23}$.

In this technical note, results were also presented using the reactor constraint: one dimensional plots and the six combinations of two dimensional confidence regions. Finally, the marginalized $\Delta\chi^2$ distribution vs δ_{CP} was presented together with a Feldman-Cousins study, demonstrating a better sensitivity to δ_{CP} for the results of the joint analysis with respect to the results of the stand-alone appearance analysis using the same marginalization method.

References

[1] C.Andreopoulos, A.Cervera, T.Dealtry, L.Escudero, N.Grant, M. Sorel and P. Stamoulis, T2K 3.010×10^{20}-POT Joint 3-Flavour Oscillation Analysis T2K-TN-154(v4), http://www.t2k.org/docs/technotes/154

[2] C.Andreopoulos, T.Dealtry, J.Dobson, N.Grant, M.Haigh, J.Ilic, G.Pearce and A.Weber, T2K 3.23×10^{19}-POT Muon-Neutrino Disappearance Analysis, T2K-TN-036(v4), http://www.t2k.org/docs/technotes/036

[3] C.Andreopoulos, T.Dealtry, J.Dobson, N.Grant, M.Haigh, J.Ilic, G.Pearce and A.Weber, T2K 1.431×10^{20}-POT Muon-Neutrino Disappearance Analysis, T2K-TN-064(v3), http://www.t2k.org/docs/technotes/064

[4] C.Andreopoulos, T.Dealtry, L.Escudero and N.Grant, T2K 1.431×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis, T2K-TN-087(v1), http://www.t2k.org/docs/technotes/087

[5] C.Andreopoulos, T.Dealtry, L.Escudero and N.Grant, T2K 3.010×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-141(v5), http://www.t2k.org/docs/technotes/141

[6] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A. Weber, T2K 6.3933×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-155, http://www.t2k.org/docs/technotes/155

[7] M. Wilking, Status of the 2013 ND280 ν_μ analysis, http://www.t2k.org/asg/meeting/20130411/nd280numustatus/view.

[8] M. Hartz, A. Kaboth, and K. Mahn, Constraining the Flux and Cross Section Models with Data from the ND280 Detector for the 2013 Oscillation Analysis, T2K-TN-166(v3.0) http://www.t2k.org/docs/technotes/166.

[9] J. Kameda, Updated study of the systematic error in ν_μ disappearance analysis from Super-Kamiokande, T2K-TN-159(v2), http://www.t2k.org/docs/technotes/159.

[10] Y. Nishimura et al., SK nue systematic error for 2013a oscillation analysis, T2K-TN-157, http://www.t2k.org/docs/technotes/157.

[11] S. Mine et al., SK π^0 systematic errors for the ν_e analysis with T2K 6.4×10^{20} POT (RUN1-4) data, T2K-TN-156, http://www.t2k.org/docs/technotes/156.

[12] J. Hignight et al., Super-Kamiokande events and data quality studies for T2K Run4 T2K-TN-148(v2r3), http://www.t2k.org/docs/technotes/148.

[13] Beam WG, Flux tuning (11b v3.2), http://www.t2k.org/beam/NuFlux/FluxRelease/11brelease/11btuned-v3.2

[14] N Abgrall et al. Measurements of Cross Sections and Charged Pion Spectra in Proton-Carbon Interactions at 31 GeV/c. Phys.Rev., C84:034604, 2011; N. Abgrall et al. Measurement of production properties of positively charged kaons in proton-carbon interactions at 31 GeV/c. arXiv:hep-ex/1112.0150, 2011.

[15] Beam group, http://www.t2k.org/beam/NuFlux/FluxRelease/11arelease/11atuned-v2.3.

[16] P. de Perio et al., Cross section parameters for the 2012a oscillation analysis T2K-TN-108(v1.3), http://www.t2k.org/docs/technotes/108

[17] P. de Perio et al., Implementation of the NIWG cross-section parametrization, T2K-TN-113(v1.2)

[18] Y.Hayato, K.McFarland, and M.Wascko (Private communication)

[19] C. Bronner et al., Run1-4 p-theta electron neutrino appearance analysis, T2K-TN-161(v2.6), http://www.t2k.org/docs/technotes/161

[20] Y. Nishimura, J. Hignight, and S. Nakayama ν_e appearance analysis of RUN1-4 data with neutrino energy spectrum, T2K-TN-162(v2.2), http://www.t2k.org/docs/technotes/162

[21] T2K Collaboration (K.Abe et al.), First Muon-Neutrino Disappearance Study with an Off-Axis Beam, Phys.Rev.D85:031103, 2012; arXiv:1201.1386

[22] T2K Collaboration (K.Abe et al.), Measurement of Neutrino Oscillation Parameters from Muon Neutrino Disappearance with an Off-axis Beam, arXiv:1308.0465 [hep-ex], 2013.

[23]

[24] M. Ikeda, private communication.

[25] T2K-TN-186 http://www.t2k.org/docs/technotes/186

[26] P. de Perio, plenary talk at T2K Collaboration Meeting http://www.t2k.org/meet/collab/201401/talks/fripm/140124_deperio_skerror.pdf

[27] J. Imber, http://www.t2k.org/asg/meeting/2014-01-09/ASG20140109_SKJointError_Status_Imber.pdf

[28] Particle Data Group, http://pdg.lbl.gov/2013/tables/rpp2013-sum-leptons.pdf

[29] S. Berkman, P. de Perio, A. Konaka, Al Missert, H. Tanaka, S. Tobayama, M.J. Wiling and E.D. Zimmerman, fiTQun: A New Reconstruction Algorith for Super-K

[30] J. Beringer et al. (Particle Data Group), PR D86, 010001 (2012) (URL: http://pdg.lbl.gov)

[31] G.L.Fogli, E.Lisi et al., Evidence of theta(13)>0 from global neutrino data analysis, arXiv:1106.6028 [hep-ph]

[32] G.Fogli's talk (pages 4,5,6) in NNN05 conference, http://nnn05.in2p3.fr/schedule.html

[33] M. C. Gonzalez-Garcia, Michele Maltoni, Jordi Salvado, Thomas Schwetz, Global fit to three neutrino mixing: critical look at present precision, arXiv:1209.3023 [hep-ph]

[34] MINOS Collaboration, Talk at NNN 2013 http://indico.ipmu.jp/indico/getFile.py/access?contribId=52&sessionId=2&resId=0&materia

[35] The KamLAND collaboration, Precision Measurement of Neutrino Oscillation Parameters with KamLAND, arXiv:0801.4589 [hep-ex]

[36] C.Andreopoulos, A.Cervera, T.Dealtry, L.Escudero, N.Grant, M. Sorel and P. Stamoulis, T2K 6.570×10^{20}-POT Joint 3-Flavour Oscillation Analysis - Additional material http://www.t2k.org/docs/technotes/175/T2KTN175additional/view

[37] L. Escudero http://mural.uv.es/loesan/toys/

[38] M. Otani and J. Kameda, Super K events and updated systematic errors with 1.43×10^{20} POT, T2K-TN-065(v8), http://www.t2k.org/docs/technotes/065

[39] J. Kameda and R. Wendell, Study on Super-K events and systematic errors relevant for the ν_μ disappearance analysis with T2K 3.23×10^{19} POT data T2K-TN-034, http://www.t2k.org/docs/technotes/034

[40] ND280 number of selected events and systematics (10dv3 flux), http://www.t2k.org/asg/oagroup/nd280-offaxis See also T2K-TN-015(v5) at http://www.t2k.org/docs/technotes/015

[41] Beam working group, NSK/NND normalisation uncertainty v3.1, http://www.t2k.org/beam/NuFlux/FluxRelease/10d/nsknnderrv3

[42] Beam working group, 10dv3 shape error for SK ν_μ flux, http://www.t2k.org/beam/NuFlux/FluxRelease/10d/shapeerrorv31 See also T2K-TN-066(v3) at http://www.t2k.org/docs/technotes/066

[43] P. de Perio et al., NEUT nuclear effects (FSI) T2K-TN-033(v2.0), http://www.t2k.org/docs/technotes/033

[44] NIWG group, NIWG errors for 10a analysis, http://www.t2k.org/asg/oagroup/oa_niwg

[45] P. de Perio et al., Cross section parameters for the 2012a oscillation analysis, T2K-TN-108(v1.3a)

[46] P. de Perio et al., Implementation of the NIWG cross-section parametrization, T2K-TN-113(v1.2)

[47] J. Albert and K. Okumura, First ν_e oscillation analysis with T2K 2010a beam data, T2K-TN-042(v1.2.1), http://www.t2k.org/docs/technotes/042

[48] J. Albert and K. Okumura, Updated ν_e oscillation analysis with Run 1+2 data, T2K-TN-052(v1.1), http://www.t2k.org/docs/technotes/052

[49] S. Nakayama, ν_e appearance analysis with maximum likelihood fit to neutrino energy spectrum, T2K-TN-109(v7.1), http://www.t2k.org/docs/technotes/109

[50] K. Sakashita et al., Electron neutrino appearance analysis with electron momentum and angle distribution T2K-TN-114(v8), http://www.t2k.org/docs/technotes/114

[51] K.Abe, et al., Indication of Electron Neutrino Appearance from an Accelerator-produced Off-axis Muon Neutrino Beam, Phys. Rev. Lett. 107, 041801 (2011).

[52] Official release of electron neutrino appearance code and results (internal) http://www.t2k.org/asg/oagroup/anacode

[53] Study on SK ν_e candidates and systematic errors with T2K 3.23 $\times 10^{19}$ POT data http://www.t2k.org/docs/technotes/028, T2K-TN-028

[54] Y.Hayato, A neutrino interaction simulation program library NEUT, Acta Phys.Polon.B40:2477-2489,2009.

[55] Numerical recipes: the art of scientific computing, Cambridge University Press, 3rd edition, ISBN 978-0-521-88068-0, pp100-102

[56] G.J. Feldman and R.D. Cousins, A Unified approach to the classical statistical analysis of small signals, Phys.Rev.D57:3873-3889,1998; physics/9711021

[57] R.D. Cousins and V.L.Highland, Nucl.Instr. and Meth. A320:331035, 1992.

[58] G. Cowan (1998) Statistical data analysis, Oxford University Press, ISBN 0-19-850155-2.

[59] F.James, M.Roos, Minuit: A System for Function Minimization and Analysis of the Parameter Errors and Correlations, Comput.Phys.Commun. 10:343 (1975)

[60] http://www.t2k.org/asg/oagroup/meeting/20120416/lowstat/view

[61] http://www.t2k.org/asg/oagroup/meeting/2013/20131121/joint21nov

[62] Prob3++ 3-flavour oscillation probability software, http://www.phy.duke.edu/ raw22/public/Prob3++/

[63] J. Beringer et al. (Particle Data Group), Phys. Rev. D86, 010001 (2012)

[64] A. Himmel for the SK collaboration, Recent Atmospheric Neutrino Results from Super-Kamiokande http://arxiv.org/abs/1310.6677

A. Validation of the ν_e-appearance analysis implementation in the *VALOR* analysis framework

This section contains the validation of the *VALOR* ν_e-appearance implementation, which has been made reproducing results from the official 2013 ν_e-appearance analyses in [19] and [20] (mainly in [20], the ν_e-appearance analysis using reconstructed energy spectrum). Notice that although the framework used is the one for the joint oscillation analysis, the results in this section were made with stand-alone appearance analysis mode only. For the following comparisons, the values of the oscillation parameters used are indicated in 17. Digitalized plots used to compare with official results were taken mainly from [20] using fiTQun cut and "number of events + E_{reco} shape" information. However, additional checks were done comparing with results in [19] as well.

Parameter	Value
$\sin^2 2\theta_{12}$	0.8495
$\sin^2 2\theta_{23}$	1.0
Δm^2_{21}	7.6×10^{-5} eV^2/c^4
Δm^2_{32}	2.4×10^{-3} eV^2/c^4
δ_{CP}	0

Table 17: Fixed values of the oscillation parameters used for the comparison with the official stand-alone appearance analysis.

A.1. Differences with respect to the official analyses

For the following comparisons it is necessary to keep in mind that some differences between the analysis presented here and the official 2013 ν_e-appearance exist. Those differences are:

- The response functions (splines) for the systematic errors that need them are different since they were created by each group independently.
- *VALOR* analysis framework uses its own package to compute the oscillation probabilities, while the probability for the official analyses was found with the Prob3++ package[6].
- Unless otherwise stated, the spectral function was tweaked in a limited range between 0 (off) and 1 (on) in the *VALOR* analysis, whereas a normal distribution was used in the official analyses.
- The addition of extra templates, oscillated $\nu_e \rightarrow \nu_\mu$ and oscillated $\bar{\nu_\mu} \rightarrow \bar{\nu_e}$ as explained in 2, was considered in order to see the effect in the results.

A.2. Validation of the predicted ν_e candidate events

Table18 shows the nominal predicted number of ν_e candidate events corresponding to Run1+2+3+4 POT for the *VALOR* ν_e-appearance analysis and from the official analyses in references [19] and [20], and table 19 shows the predicted number of ν_e candidate events after the BANFF tuning. There is a very good agreement in the predicted number of events between the different analyses.

A.3. Validation of the effect of systematics

The validation of the effect of systematics was performed with the Run1+2+3+4a POT, comparing results with the corresponding ones for the official analyses, that can be found in early versions of the referenced technical notes, and was not re-done with full Run4 since there were almost no differences. Table 20 shows the effect of the different categories of systematic parameters in the predicted number of events for the 2013 *VALOR* appearance analysis (black) compared with the official results (blue) taken from Table 14 in [19]. The errors in the predicted number of events are given in percentage and they were calculated throwing 20k toy experiments varying only the systematics in the selected category (fixed the rest to nominal), and finding the RMS/mean of the distribution of number of events of the 20k toy experiments. The effects of the different systematics on the predicted number of events are similar, only small differences exist due to the different treatment of the spectral function systematic as shown in Tab.21. The effect due to the spectral function is still different between both analyses even when using the same treatment of the spectral function (a gaussian distribution without limits) as can be observed in Tab.21, because the response functions for this parameter were created in different ways in both analyses, as it was reported in [61] and presented in Tab. 22: the peculiarity about this systematic is that only weights at values 0 and $+1\sigma$ are available by definition,

[6] A detailed comparison of the two oscillation probability codes can be found in Appendix B of [5]

Events	$sin^2(2\theta_{13})$ = 0 (official)	$sin^2(2\theta_{13})$ = 0 (*VALOR*)	$sin^2(2\theta_{13})$ = 0.1 (official)	$sin^2(2\theta_{13})$ = 0.1 (*VALOR*)
Total	–	**5.081**	–	**22.476**
Total (without extra templates)	**5.07**	**5.074**	**22.26**	**22.247**
ν_μ	1.08	1.078	1.09	1.079
$\nu_e \rightarrow \nu_\mu$	–	3.E-6	–	0.0002
$\bar{\nu_\mu}$	0.06	0.062	0.06	0.062
ν_e	3.37	3.368	3.12	3.118
$\nu_\mu \rightarrow \nu_e$	0.41	0.406	17.85	17.838
$\bar{\nu_e}$	0.16	0.161	0.15	0.151
$\bar{\nu_\mu} \rightarrow \bar{\nu_e}$	–	0.007	–	0.228

Table 18: Predicted number of ν_e candidate events for the Run1+2+3+4 POT before BANFF tuning (that is, after external flux re-weighting using default 13a SK MC with the 11b tuned version 3.2), for *VALOR* appearance analysis (black) compared to official appearance analysis (blue) as in Tables 11 and 12 of [19] and Table 8 of [20], using $sin^2(2\theta_{12}) = 0.8495$, $\Delta m^2_{21} = 7.6 \times 10^{-5}$, $sin^2(2\theta_{23}) = 1.0$, $|\Delta m^2_{32}| = 2.4 \times 10^{-3}$, $\delta_{CP} = 0$ and normal hierarchy.

Events	$sin^2(2\theta_{13})$ = 0 (official)	$sin^2(2\theta_{13})$ = 0 (*VALOR*)	$sin^2(2\theta_{13})$ = 0.1 (official)	$sin^2(2\theta_{13})$ = 0.1 (*VALOR*)
Total	–	**4.937**	–	**21.818**
Total (without extra templates)	**4.92**	**4.930**	**21.56**	**21.586**
ν_μ	0.94	0.944	0.94	0.944
$\nu_e \rightarrow \nu_\mu$	–	3.E-6	–	0.0002
$\bar{\nu_\mu}$	0.05	0.055	0.05	0.055
ν_e	3.37	3.371	3.12	3.120
$\nu_\mu \rightarrow \nu_e$	0.40	0.399	17.30	17.315
$\bar{\nu_e}$	0.16	0.161	0.15	0.152
$\bar{\nu_\mu} \rightarrow \bar{\nu_e}$	–	0.007	–	0.232

Table 19: Predicted number of ν_e candidate events for the Run1+2+3+4 POT applying BANFF tuning, for *VALOR* appearance analysis (black) compared to official appearance analysis (blue) as in Tables 11 and 12 of [19] and Table 8 of [20], using $sin^2(2\theta_{12}) = 0.8495$, $\Delta m^2_{21} = 7.6 \times 10^{-5}$, $sin^2(2\theta_{23}) = 1.0$, $|\Delta m^2_{32}| = 2.4 \times 10^{-3}$, $\delta_{CP} = 0$ and normal hierarchy.

so the *VALOR* response functions are simply a linear function calculated with these two weight values and extrapolated to any other tweaking value; however, in the official analyses, although only weights at values 0 and +1σ were used, the extrapolation was performed using a cubic spline, i.e. a three degree polynomial function.

Figure 93 shows the total error envelope of the single e-like ring event reconstructed energy spectrum with $sin^2(2\theta_{13})$ = 0.1 and δ_{CP} = 0. This figure is compared with the digitalization of Fig. 9 in [20].

A.4. Validation of the 2013 official results

In this subsection the results of the validation of the 2013 *VALOR* appearance analysis with Run1+2+3+4 dataset are presented compared with official results from [20] when possible. A uniform distribution was assumed to tweak the spectral function systematic and the extra templates were also included in order to match the final treatment and inputs to be used for

Category	$sin^2(2\theta_{13}) = 0$ (official)	$sin^2(2\theta_{13}) = 0$ (*VALOR*)	$sin^2(2\theta_{13}) = 0.1$ (official)	$sin^2(2\theta_{13}) = 0.1$ (*VALOR*)
Total	**11.4%**	**10.291 %**	**8.9%**	**6.553 %**
BANFF	4.8%	4.799 %	2.9%	2.926 %
Uncorrelated cross sections	7.1%	4.886 %	7.6%	4.575 %
SK+FSI+PN	7.3%	7.331 %	3.5%	3.454 %

Table 20: Effect of the different categories of systematic parameters in the predicted number of events for the 2013 *VALOR* appearance analysis (black) compared with the official results (blue) taken from Table 14 in [19].The errors in the predicted number of events are given in percentage and they were calculated throwing 20k toy experiments, varying only the systematics in the selected category (fixed the rest to nominal), and finding the RMS/mean of the distribution of number of events of the 20k toy experiments.

Category		$sin^2(2\theta_{13}) = 0.1$ (official)	$sin^2(2\theta_{13}) = 0.1$ (*VALOR* - SF whole gaussian)	$sin^2(2\theta_{13}) = 0.1$ (*VALOR* - SF uniform [0,1])
Total		**8.9%**	**7.355 %**	**6.553 %**
BANFF		2.9%	2.926 %	
SK+FSI+PN		3.5%	3.454 %	
Uncorrelated cross sections		7.6%	5.832%	4.575 %
	Spectral function	6.0%	3.833%	1.132%
	CC other shape	0.1%	0.124 %	
	Fermi momentum	0.1%	0.094%	
	W shape	0.2%	0.229%	
	π-less Δ decay	3.7%	3.235%	
	CC coh norm	0.2%	0.274%	
	NC coh norm	0.2%	0.242%	
	NC other norm	0.5%	0.499%	
	σ_{ν_e/ν_μ}	2.9%	2.847%	

Table 21: Effect of the different categories of systematic parameters in the predicted number of events for the 2013 *VALOR* appearance analysis (black) compared with the official results (blue) taken from Table 14 in [19].The errors in the predicted number of events are given in percentage and they were calculated throwing 20k toy experiments, varying only the systematics in the selected category (fixed the rest to nominal), and finding the RMS/mean of the distribution of number of events of the 20k toy experiments.

the joint analysis[7]. There is an excellent agreement between best-fit values and spectra and confidence intervals are almost identical.

Table 23 shows the best-fit points and confidence intervals for $\sin^2 2\theta_{13}$ for the Run1+2+3+4 dataset assuming δ_{CP}=0, $|\Delta m^2|$=2.36$\times 10^{-3}$ eV^2/c^4 and $\sin^2 2\theta_{23}$=1.0 for the *VALOR* analysis with and without the extra templates, compared with the official results from Table 17 in [20].

Figure 94 (95) shows the systematic parameter pulls for the 51 systematics included in the stand-alone analysis for the Run 1+2+3+4 dataset fit with Normal (Inverted) Hierarchy.

Figure 96 (97) shows the Run 1+2+3+4 best-fit single e-like ring event reconstructed energy spectrum for the fit with Normal (Inverted) Hierarchy for the *VALOR* stand-alone ν_e appearance analysis (black) compared to the official stand-alone ν_e

[7] A later discussion within the oscillation analysis group and people from the NIWG group finally concluded in the change of the treatment of the spectral function to a gaussian distribution limited between 0 and 1 for the final joint oscillation analysis result

	Official	Official splines		VALOR splines	
		whole gaussian	bounded gaussian	whole gaussian	bounded gaussian
Total	8.9 %	8.77 %	6.57 %	7.56 %	6.55 %
SF	6.0%	5.82 %	1.10%	3.83 %	1.11 %

Table 22: Effect of the systematic parameter related to the spectral function in the different treatments followed by the *VALOR* group, matching the recommendation for disappearance analysis, and the official appearance analysis, comparing as well the effect of the different extrapolation methods used to create each set of response functions (splines) in the different groups.

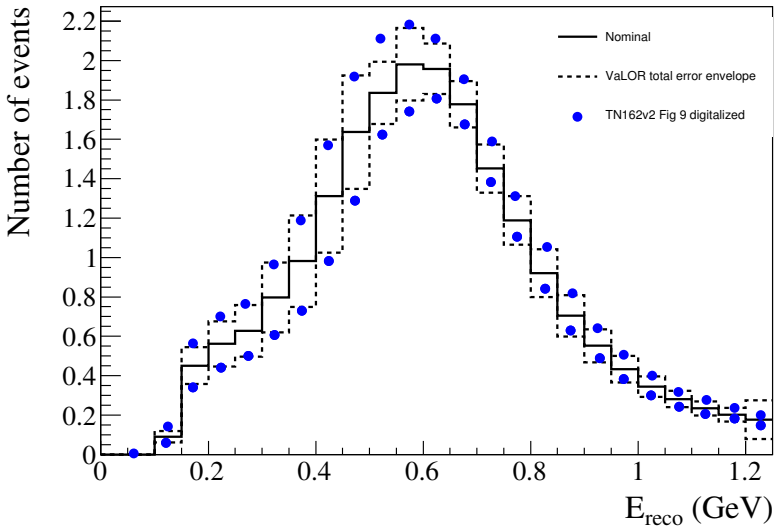


Figure 93: Total error envelope of the single e-like ring event reconstructed energy spectrum for oscillations with $|\Delta m^2|$=2.36×10^{-3} eV^2/c^4. $\sin^2 2\theta_{23}$=1.0, $\sin^2 2\theta_{13}$=0.1, δ_{CP}=0, calculated for an exposure of 6.3933×10^{20} POT with BANFF-tuned MC templates.

	Official (TN162)	*VALOR*	*VALOR extra templates*
Best-fit Normal	**0.144**	**0.143**	**0.141**
68 % C.L. (normal)	0.110 - 0.181	0.111-0.181	0.109-0.179
90 % C.L. (normal)	0.092 - 0.209	0.092-0.209	0.091-0.206
Best-fit Inverted	**0.172**	**0.174**	**0.172**
68 % C.L. (inverted)	0.134 - 0.217	0.136-0.219	0.134-0.215
90 % C.L. (inverted)	0.112 - 0.249	0.114-0.251	0.112-0.247

Table 23: Best-fit points and confidence intervals for $\sin^2 2\theta_{13}$ for the Run1+2+3+4 dataset assuming δ_{CP}=0, $|\Delta m^2|$=2.36×10^{-3} eV^2/c^4 and $\sin^2 2\theta_{23}$=1.0.

appearance analysis digitalized from Fig. 19 in [20].

Figure 98 (99) shows the $\Delta\chi^2$ distribution as a result of the Run 1+2+3+4 data fit with Normal (Inverted) Hierarchy vs different values of $\sin^2 2\theta_{13}$ fixing δ_{CP}=0, $|\Delta m^2|$=2.36×10^{-3} eV^2/c^4 and $\sin^2 2\theta_{23}$=1.0 for the *VALOR* stand-alone ν_e appearance analysis (black) with (solid lines) and without extra templates (dashed lines), compared to the official stand-alone ν_e appearance analysis digitalized from Fig. 18 in [20] (blue).

Figure 100 (101) shows the 68 % (dashed lines) and 90% (solid lines) C.L. in the ($\sin^2 2\theta_{13}$,δ_{CP}) parameter space as a result of the Run 1+2+3+4 data fit with Normal (Inverted) Hierarchy fixing $|\Delta m^2|$=2.36×10^{-3} eV^2/c^4 and $\sin^2 2\theta_{23}$=1.0 for the *VALOR* stand-alone ν_e appearance analysis (black) compared to the official stand-alone ν_e appearance analysis digitalized

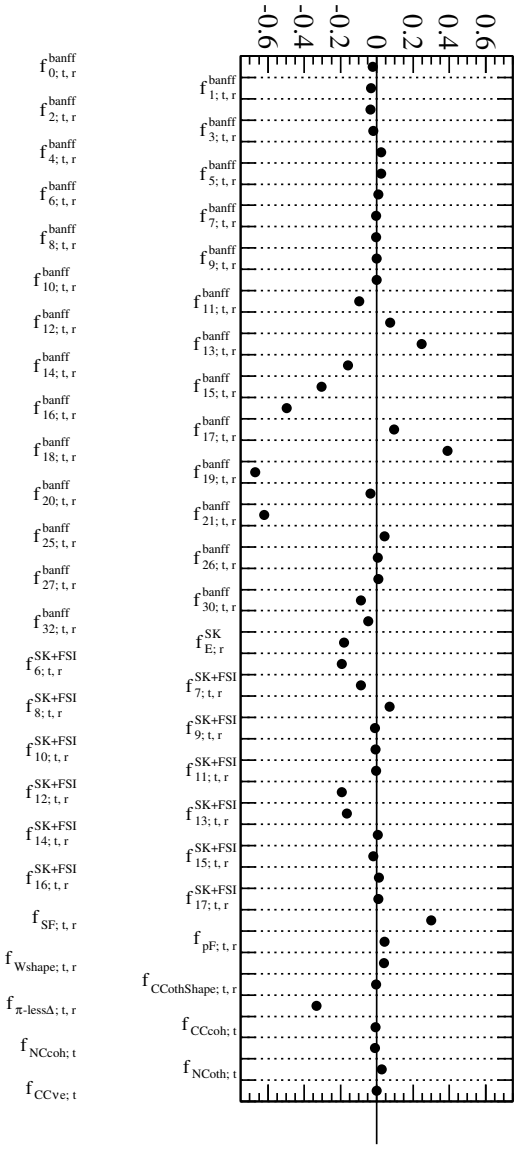


Figure 94: Systematic parameter pulls for the 51 systematics included in the stand-alone analysis for the Run 1+2+3+4 dataset fit with Normal Hierarchy.

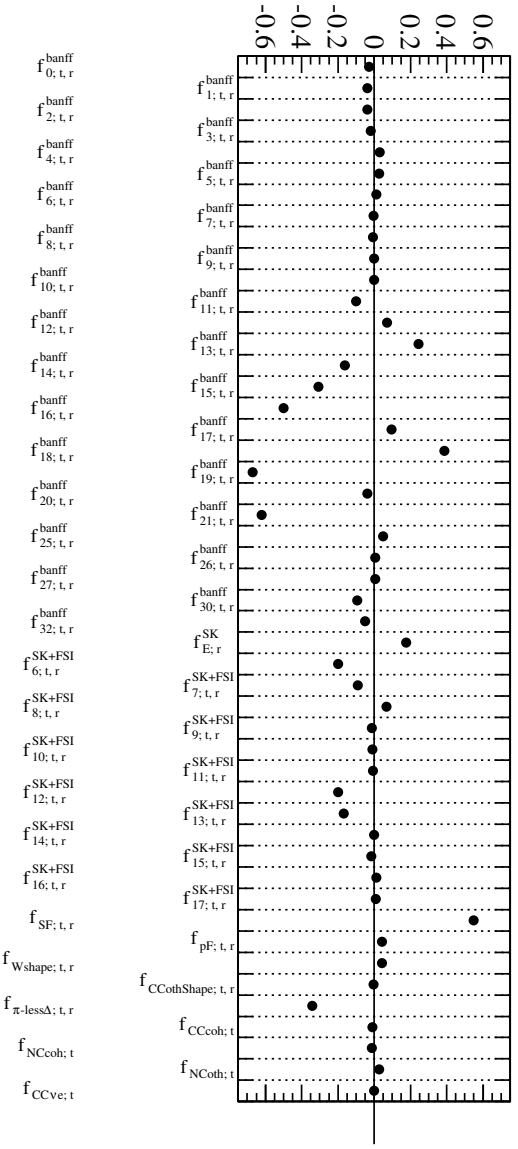


Figure 95: Systematic parameter pulls for the 51 systematics included in the stand-alone analysis for the Run 1+2+3+4 dataset fit with Inverted Hierarchy.

from Fig. 20 in [20] using the constant $\Delta\chi^2$ method and raster scan in both analyses.

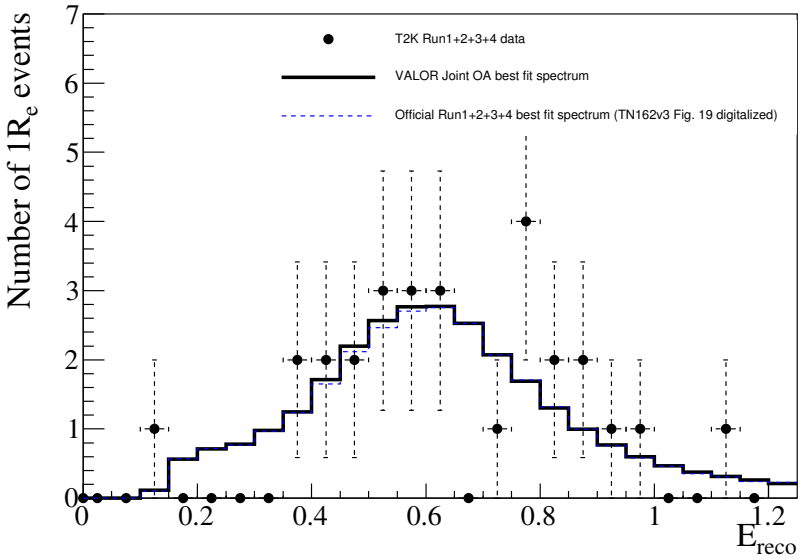


Figure 96: Run 1+2+3+4 best-fit single e-like ring event reconstructed energy spectrum for the fit with Normal Hierarchy for the *VALOR* stand-alone ν_e appearance analysis (black) compared to the official stand-alone ν_e appearance analysis digitalized from Fig. 19 in [20] (blue).

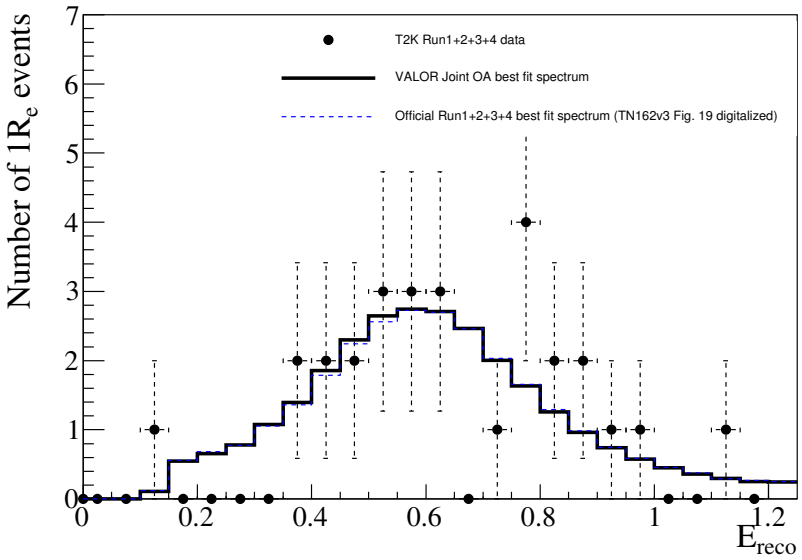


Figure 97: Run 1+2+3+4 best-fit single e-like ring event reconstructed energy spectrum for the fit with Inverted Hierarchy for the *VALOR* stand-alone ν_e appearance analysis (black) compared to the official stand-alone ν_e appearance analysis digitalized from Fig. 19 in [20] (blue).

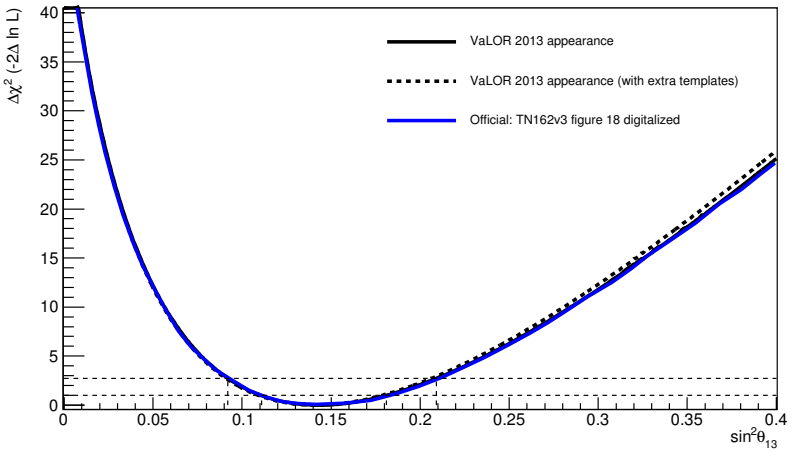


Figure 98: $\Delta\chi^2$ distribution as a result of the Run 1+2+3+4 data fit with Normal Hierarchy vs different values of $\sin^2 2\theta_{13}$ fixing δ_{CP}=0, $|\Delta m^2|$=2.36×10^{-3} eV^2/c^4 and $\sin^2 2\theta_{23}$=1.0 for the *VALOR* stand-alone ν_e appearance analysis (black) with (solid lines) and without (dashed lines), compared to the official stand-alone ν_e appearance analysis digitalized from Fig. 18 in [20] (blue).

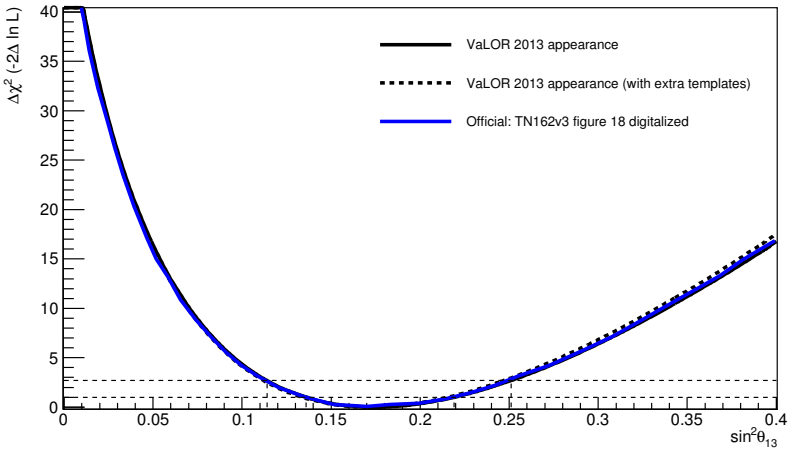


Figure 99: $\Delta\chi^2$ distribution as a result of the Run 1+2+3+4 data fit with Inverted Hierarchy vs different values of $\sin^2 2\theta_{13}$ fixing δ_{CP}=0, $|\Delta m^2|$=2.36×10^{-3} eV^2/c^4 and $\sin^2 2\theta_{23}$=1.0 for the *VALOR* stand-alone ν_e appearance analysis (black) with (solid lines) and without (dashed lines), compared to the official stand-alone ν_e appearance analysis digitalized from Fig. 18 in [20] (blue).

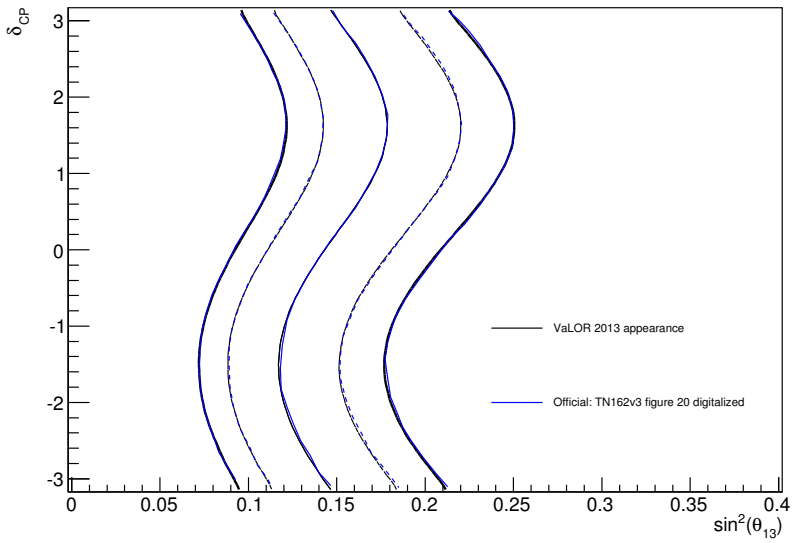


Figure 100: 68 % (dashed lines) and 90% (solid lines) C.L. in the ($\sin^2 2\theta_{13}$,δ_{CP}) parameter space as a result of the Run 1+2+3+4 data fit with Normal Hierarchy fixing $|\Delta m^2|$=2.36$\times 10^{-3}$ eV^2/c^4 and $\sin^2 2\theta_{23}$=1. for the *VALOR* stand-alone ν_e appearance analysis (black) compared to the official stand-alone ν_e appearance analysis digitalized from Fig. 20 in [20] (blue) using the constant $\Delta\chi^2$ method and raster scan in both analyses.

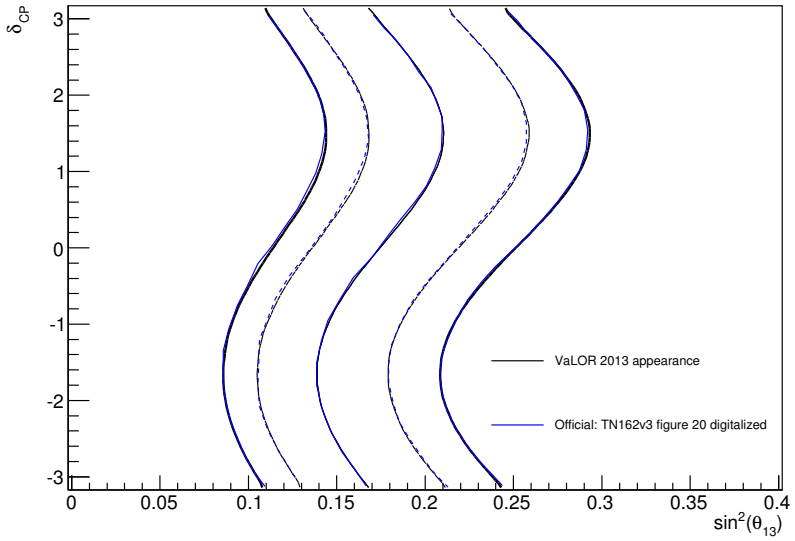


Figure 101: 68 % (dashed lines) and 90% (solid lines) C.L. in the ($\sin^2 2\theta_{13}$,δ_{CP}) parameter space as a result of the Run 1+2+3+4 data fit with Inverted Hierarchy fixing $|\Delta m^2|$=2.36$\times 10^{-3}$ eV^2/c^4 and $\sin^2 2\theta_{23}$=1.0 for the *VALOR* stand-alone ν_e appearance analysis (black) compared to the official stand-alone ν_e appearance analysis digitalized from Fig. 20 in [20] (blue) using the constant $\Delta\chi^2$ method and raster scan in both analyses.

B. Study of the effect of the 12-sector (solar) oscillation parameters in the Run 1+2+3+4 joint oscillation analysis

Several checks were done in order to test the effect of fitting the solar oscillation parameters $sin^2\theta_{12}$ and δm^2. Already the previous technical note with the Run1+2+3 joint oscillation analysis contained an appendix showing the effect of such parameters (See App. F in [1]). For this analysis it was demonstrated that the effect in the confidence regions is negligible when allowing the solar parameters to float in the fit with the constraint known for them and found in different literature sources. Figs. 102- 105 shows a comparison of the confidence regions with and without fitting for solar parameters. (Caveat: these results were made before the final SK matrix was available, so they contain no correlations between SK detector efficiencies). The values used for the results in this section are: $sin^2\theta_{12}$ = 0.306, $\sigma(sin^2\theta_{12})$ = 0.018, $\delta m^2 = 7.5 \times 10^5 (eV^2/c^4)$ and $\sigma(\delta m^2) = 0.2 \times 10^5 (eV^2/c^4)$ Finally, since there is no visible effect it was decided by the oscillation analysis group to keep the solar parameters fixed.

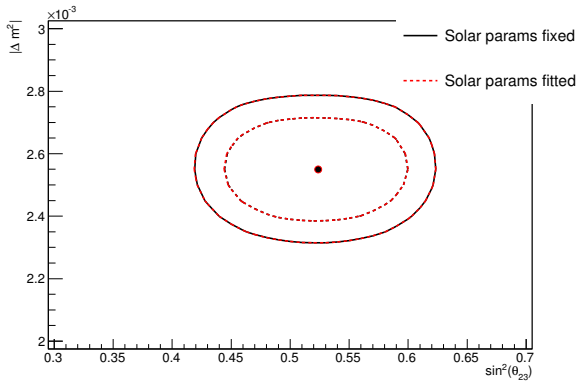


Figure 102: Comparison of confidence regions in the ($sin^2\theta_{23}$, $|\Delta m^2_{32}|$) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy with and without including the solar parameters $sin^2\theta_{12}$ and δm^2 in the fit.

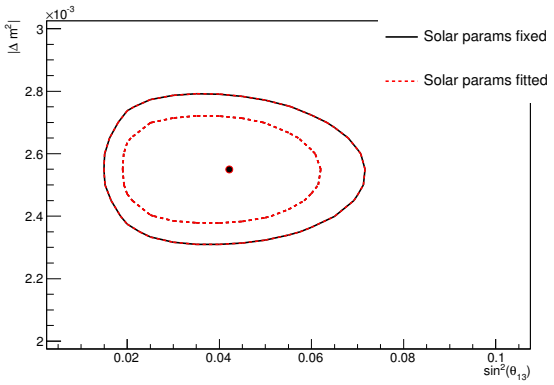


Figure 103: Comparison of confidence regions in the ($sin^2\theta_{13}$, $|\Delta m^2_{32}|$) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy with and without including the solar parameters $sin^2\theta_{12}$ and δm^2 in the fit.

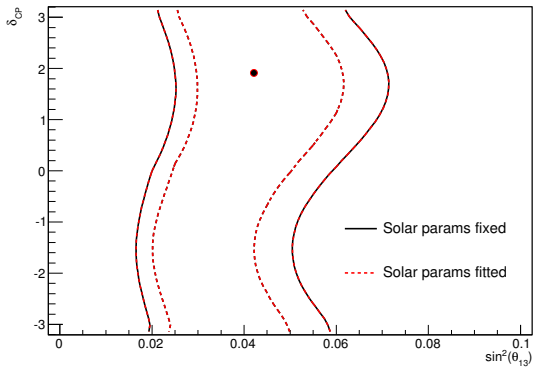


Figure 104: Comparison of confidence regions in the ($sin^2\theta_{13}$, δ_{CP}) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy with and without including the solar parameters $sin^2\theta_{12}$ and δm^2 in the fit.

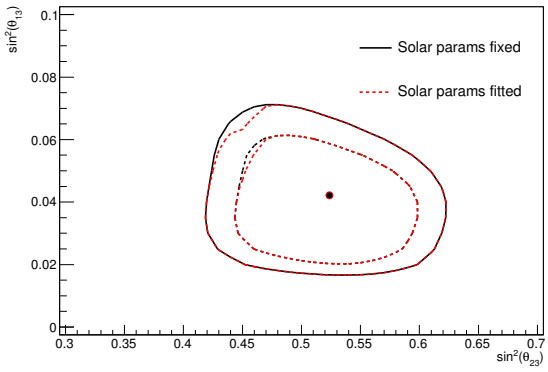


Figure 105: Comparison of confidence regions in the ($sin^2\theta_{23}$, $sin^2\theta_{13}$) parameter space for the Run1+2+3+4 dataset fit assuming Normal Hierarchy with and without including the solar parameters $sin^2\theta_{12}$ and δm^2 in the fit.

C. Validation of the *VALOR* Run1+2+3+4 joint 3-flavour oscillation analysis: study of fitter performance at 6.57 $\times 10^{20}$ POT

Different studies were made in order to validate the *VALOR* Run1+2+3+4 joint 3-flavour oscillation analysis. In this section, a summary of the studies concerning the fitter performance are presented. The implementation of each systematic and its effect in the proper modes and templates has been carefully studied and reported in an additional document that can be found together with this technical note [36].

C.1. Residuals and pulls for the oscillation parameters

The residuals and pulls presented in this section were obtained using 5k toy MC experiments generated with statistical fluctuations and variations of all 64 systematic parameters, fitting them using MINUIT and computing the MINUIT HESSE errors. In order to avoid effects of physical boundaries, the following study was made at the oscillation point $sin^2\theta_{23} = 0.5263$ and $sin^2\theta_{13} = 0.05$. The value for $sin^2\theta_{13}$ was selected to be large enough to avoid the effect of the physical boundary at $sin^2\theta_{13} = 0$ that could be observed as an asymmetry in the distribution of residuals if a small value of $sin^2\theta_{13}$ is used. For the other angle $sin^2\theta_{23}$ the value was selected to be in the point of maximal disappearance given that $sin^2\theta_{13} = 0.05$, which is calculated as $\approx (2(1 - sin^2\theta_{13}))^{-1}$ according to the explanation in Appendix E of [1]. The other oscillation parameters were set to $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$, and normal hierarchy was assumed.

Figures 106 and 107 show the distribution of residuals and pulls respectively of the four oscillation parameters at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ and normal hierarchy for an ensemble of 5k toy MC experiments. Residuals and pulls of only the events which are at the central peak in $sin^2\theta_{23}$ are presented in Figs. 108 and 109. For the other two peaks away from the true value of $sin^2\theta_{23}$ residuals and pulls are shown in Figs. 110 and 111. Since one of the two peaks is reduced in the joint fit, these figures show the distributions from the two independent fist performed with an initial value in each octant, before selecting the one with minimal χ^2.

Fig 112 shows the distribution of residuals and pulls respectively of the four oscillation parameters at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ and normal hierarchy for an ensemble of 4k toy MC experiments fixing δ_{CP} to its true value, and Fig. 113 selecting only the events which are at the central peak in $sin^2\theta_{23}$.

From the study of residuals and pulls the following features can be observed:

- **Three peak structure in $sin^2\theta_{23}$**

  The three peak structure in the residuals for $sin^2\theta_{23}$ observed in Fig. 106 and the specific location of the two extra peaks away from the true value is a known feature of this parameter which has been explained in detail in the validation of the *VALOR* Run1+2+3+4 ν_μ disappearance analysis in [6]. One of the two peaks away from the true value (the one corresponding to the solution in the second octant) is suppressed when the contribution to the total χ^2 coming from the single e-like ring sample is added (it was checked that when using only the χ^2 contribution from the single μ-like ring sample both peaks appeared). The explanation for this suppression will be given in a later bullet.

  Comparing Fig. 108 with Fig. 110 and Fig. 109 with Fig. 111 it is clear that the main source for the biases in the distribution of residuals for Δm^2 and $sin^2\theta_{13}$ comes from the extra peaks away from the true value.

- **Intrinsic asymmetry in $sin^2\theta_{13}$-δ_{CP}**

  It was observed that an intrinsic asymmetry exist in the $sin^2\theta_{13}$-δ_{CP} space. This asymmetry is clear when looking at the true $\Delta\chi^2$ distribution for the single e-like ring sample in the ($sin^2\theta_{13}$,δ_{CP}) space for a toy created at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ assuming normal hierarchy without statistical fluctuations nor systematic variations in Fig. 114. In this figure, the region of minimal $\Delta\chi^2$ is clearly not centered at $\delta_{CP} = 0$, $sin^2\theta_{13} = 0.05$ (the true value) but it is asymmetric, centered at a value $\delta_{CP} > 0$ and $sin^2\theta_{13} > 0.05$. This explains the fact that when creating toy experiments with statistical and systematic variations, the center of the distribution of residuals for δ_{CP} is not center at 0 but at $\delta_{CP} \approx 0.6$ even when removing other sources that could create a bias (fixing $sin^2\theta_{23}$ to its true value for instance) as observed for example in Fig. 106. This asymmetry and consequent bias in δ_{CP} causes also a bias to values $sin^2\theta_{13} > 0.05$.

  This is a direct consequence of the energy spectrum of T2K event candidates convoluted with the oscillation probability $P(\nu_\mu \rightarrow \nu_e)$. Figure 115 shows on the left the true energy spectrum for the single e-like ring event candidates for the oscillated ν_e category in the T2K SKMC, zooming around the peak of the spectrum. The oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ is selected, and the oscillation probability at that point is calculated and used as reference for each one of the energies at the peak on the left plot $P(\nu_\mu \rightarrow \nu_e)_{true}(E_i)$. Now for each one of the energies at the peak of the spectrum on the left plot, from 0.45 GeV to 0.85 GeV, a line is drawn on the right plot for the points in the ($sin^2\theta_{13}$,δ_{CP}) space for which $\Delta(P(\nu_\mu \rightarrow \nu_e))(E_i)$ = $P(\nu_\mu \rightarrow \nu_e)_{point}(E_i)$ -$P(\nu_\mu \rightarrow \nu_e)_{true}(E_i) < 0.005$. Red regions indicate points where lines of $\Delta(P(\nu_\mu \rightarrow \nu_e))(E_i)$ for all the energies at

the peak crossed. That translates into points where the $\Delta\chi^2$ value will be approximately the same, minimal with respect to the true oscillation point selected. The shape and position of the red region in the left side of Fig. 115 describes exactly the asymmetry observed in Fig. 114.

The true values of other oscillation parameters except δ_{CP} do not affect the position of this region as can be observed in Fig. 116, where the oscillation point is $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.03$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ on the left plot and $sin^2\theta_{23} = 0.3$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ on the right plot.

If the true value of δ_{CP} changes then the distribution is different. For instance, when doing this test with $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = -1$ the position of minimal $\Delta\chi^2$ values along the ($sin^2\theta_{13}$,δ_{CP}) space change as can be observed in Fig. 117.

This inherent bias in $sin^2\theta_{13}$-δ_{CP} is the cause of the suppression of one of the peaks in the distribution of residuals for $sin^2\theta_{23}$: due to the correlation between $sin^2\theta_{23}$ and $sin^2\theta_{13}$, a positive shift in $sin^2\theta_{23}$ is compensated by a negative shift in $sin^2\theta_{13}$ as can be observed in Fig. 110. However, the asymmetry in the minimal $\Delta\chi^2$ values in the ($sin^2\theta_{13}$,δ_{CP}) space produces a tension such that a negative shift of $sin^2\theta_{13}$ is penalized (it gets out of the distribution of minimal $\Delta\chi^2$), and therefore the solution in the second octant of $sin^2\theta_{23}$ is suppressed due to this fact. This can be also observed in Fig. 110, where the two distributions of residuals for $sin^2\theta_{13}$ are not symmetric as it would be expected if there would be not such intrinsic bias.

The asymmetry observed in the $sin^2\theta_{13}$ distributions even when $sin^2\theta_{23}$ is fixed to its true value, with a larger tail to positive values of $\Delta(sin^2\theta_{13})$ is another consequence of this intrinsic bias in ($sin^2\theta_{13}$,δ_{CP}) space, disappearing when δ_{CP} is fixed to its true value as can be observed comparing the bottom left plot in Fig. 106 with Fig. 112.

- **Larger bias in $|\Delta m^2|$ with respect to stand-alone disappearance analysis**

The mean of the residuals for the parameter $|\Delta m^2|$ from Fig. 106 is $\approx 3.5 \times 10^{-5} eV^2/c^4$, which seems too large compared with the mean of the residuals for a similar oscillation parameter point for the ***VALOR*** stand-alone disappearance analysis like the one quoted in Fig. 15 of [6], which is $\approx 1.4 \times 10^{-5} eV^2/c^4$ at current POT.

This difference disappears when the parameter δ_{CP} is fixed to its true value, being approximately the same in that case, $\approx 1.4 \times 10^{-5} eV^2/c^4$ as can be observed in the top right plot in Fig. 112.

The remaining bias of $\approx 1 \times 10^{-5} eV^2/c^4$ is directly related to the two peaks far from the true value appearing for $sin^2\theta_{23}$ residuals as can be observed when comparing top left plots of Fig. 108 and 110.

In conclusion, there is not a real increment in the bias for the joint fit with respect to the disappearance analysis when δ_{CP} is fixed according to the distribution of residuals.

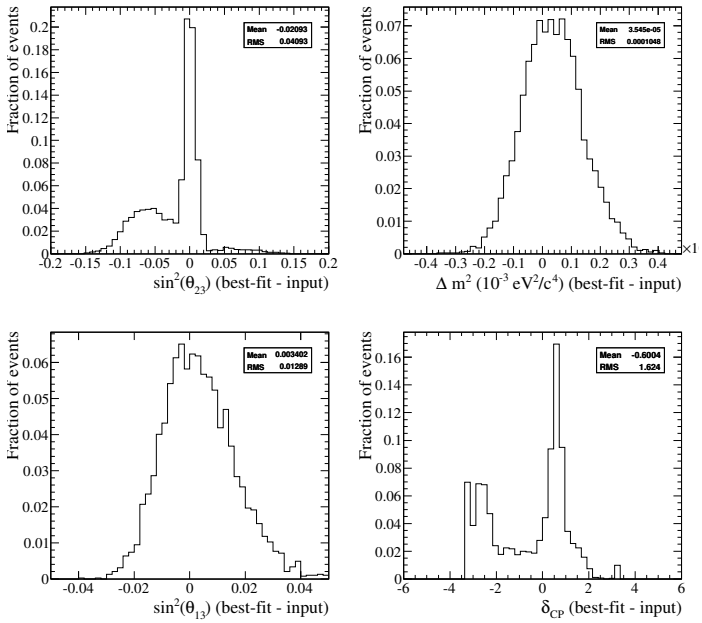


Figure 106: Distribution of residuals of the four oscillation parameters at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ and normal hierarchy for an ensemble of 5k toy MC experiments.

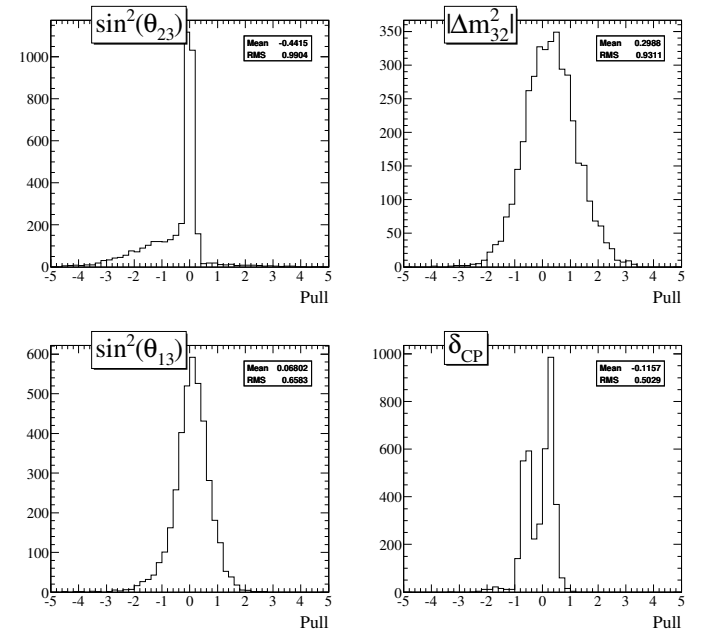


Figure 107: Pulls of the four oscillation parameters at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ and normal hierarchy for an ensemble of 5k toy MC experiments.

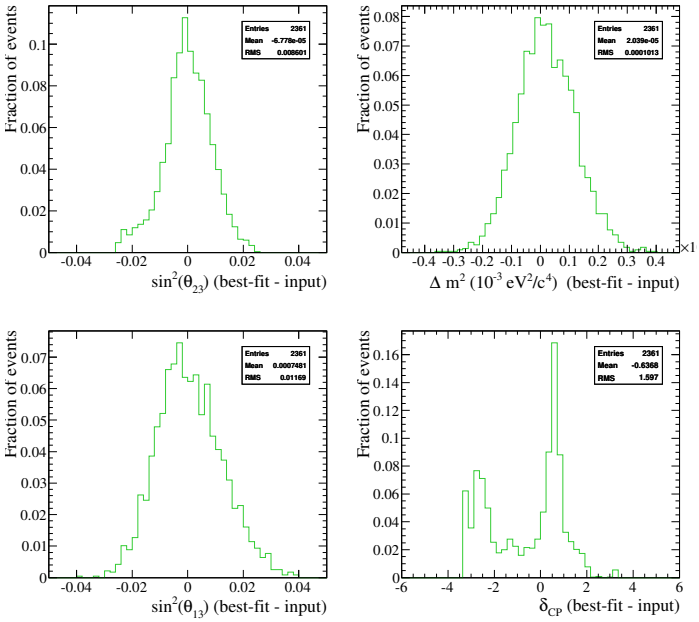


Figure 108: Distribution of residuals of the four oscillation parameters at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ and normal hierarchy for an ensemble of 5k toy MC experiments selecting only those experiments with a best-fit value of $sin^2\theta_{23}$ in the central peak.

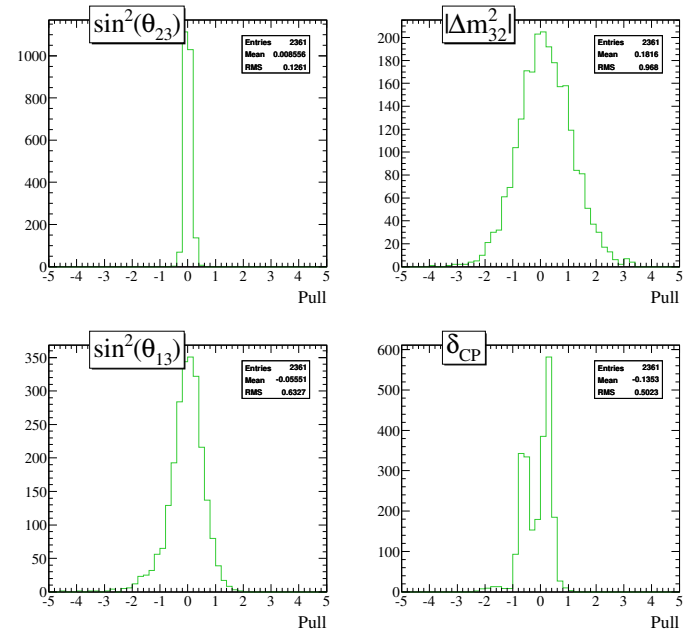


Figure 109: Pulls of the four oscillation parameters at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ and normal hierarchy for an ensemble of 5k toy MC experiments selecting only those experiments with a best-fit value of $sin^2\theta_{23}$ in the central peak.

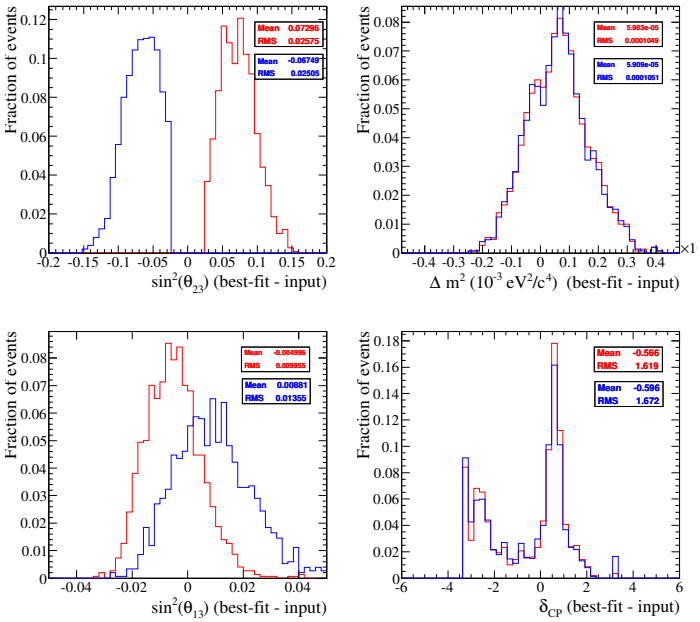


Figure 110: Distribution of residuals of the four oscillation parameters at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ and normal hierarchy for an ensemble of 5k toy MC experiments selecting only those experiments with a best-fit value of $sin^2\theta_{23}$ in the peaks away from the true value. Since one of the two peaks is reduced in the joint fit, these figures show the distributions from the two independent fist performed with an initial value in each octant, before selecting the one with minimal χ^2.

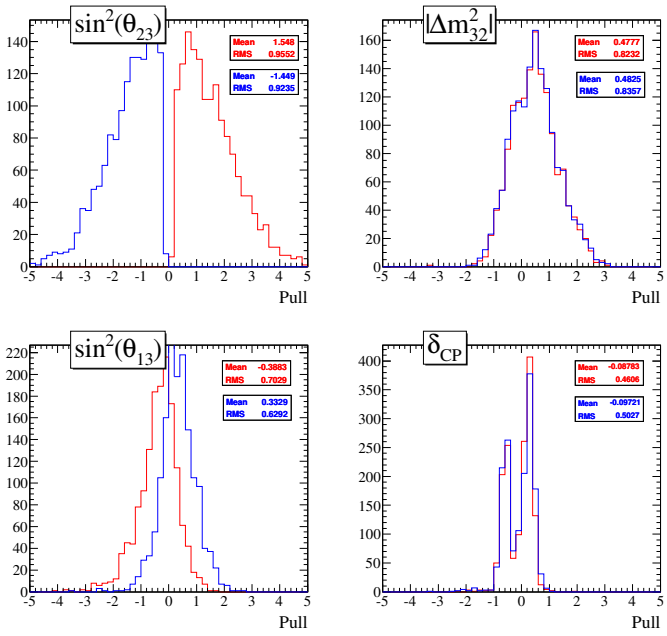


Figure 111: Pulls of the four oscillation parameters at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ and normal hierarchy for an ensemble of 5k toy MC experiments selecting only those experiments with a best-fit value of $sin^2\theta_{23}$ in the peaks away from the true value. Since one of the two peaks is reduced in the joint fit, these figures show the distributions from the two independent fist performed with an initial value in each octant, before selecting the one with minimal χ^2.

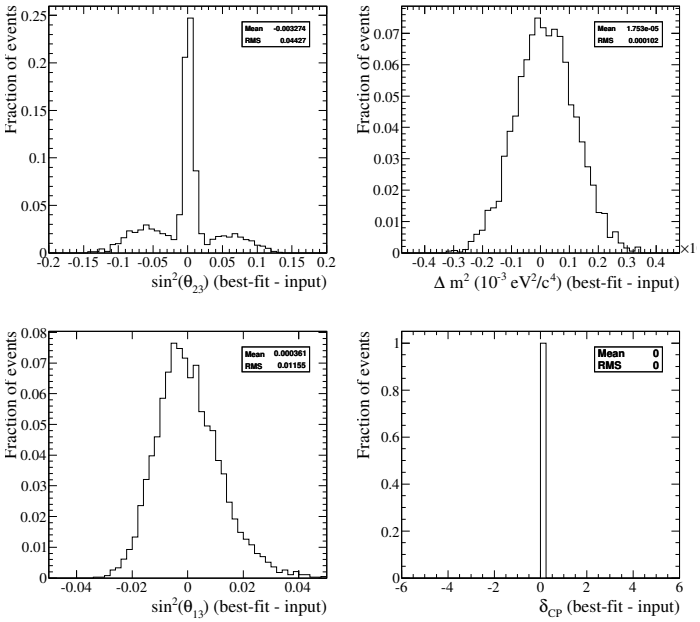


Figure 112: Distribution of residuals of the four oscillation parameters at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ and normal hierarchy for an ensemble of 4k toy MC experiments and fixing δ_{CP} to its true value.

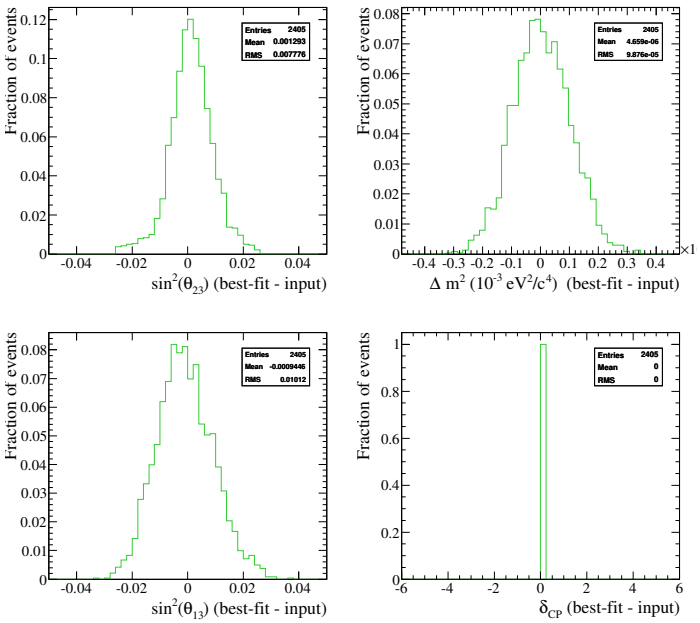


Figure 113: Distribution of residuals of the four oscillation parameters at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ and normal hierarchy for an ensemble of 4k toy MC experiments and fixing δ_{CP} to its true value, selecting only those experiments with a best-fit value of $sin^2\theta_{23}$ in the central peak.

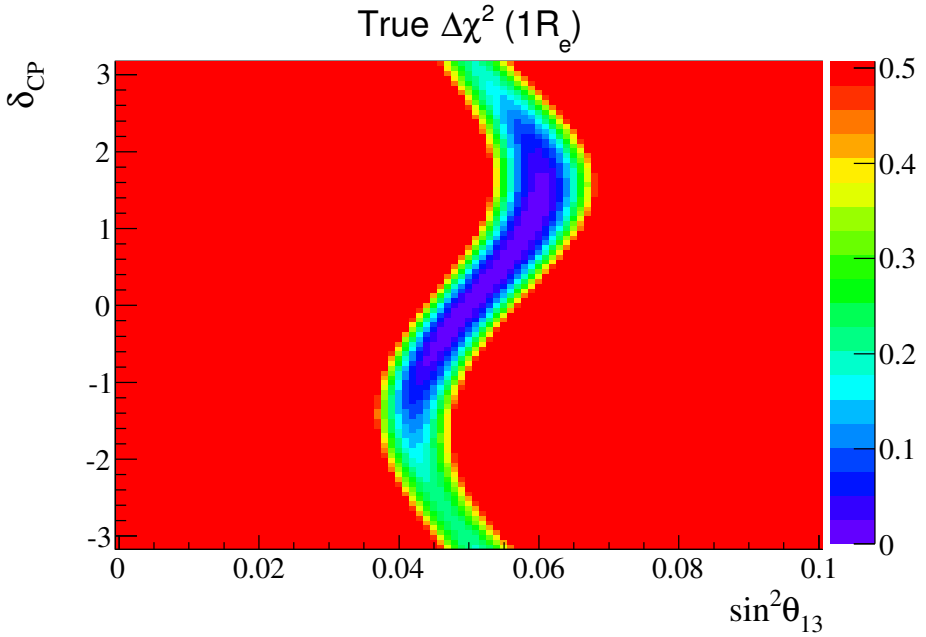


Figure 114: True $\Delta\chi^2$ distribution for the single e-like ring sample in the ($sin^2\theta_{13}$,δ_{CP}) space for a toy created at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ assuming normal hierarchy without statistical fluctuations nor systematic variations.

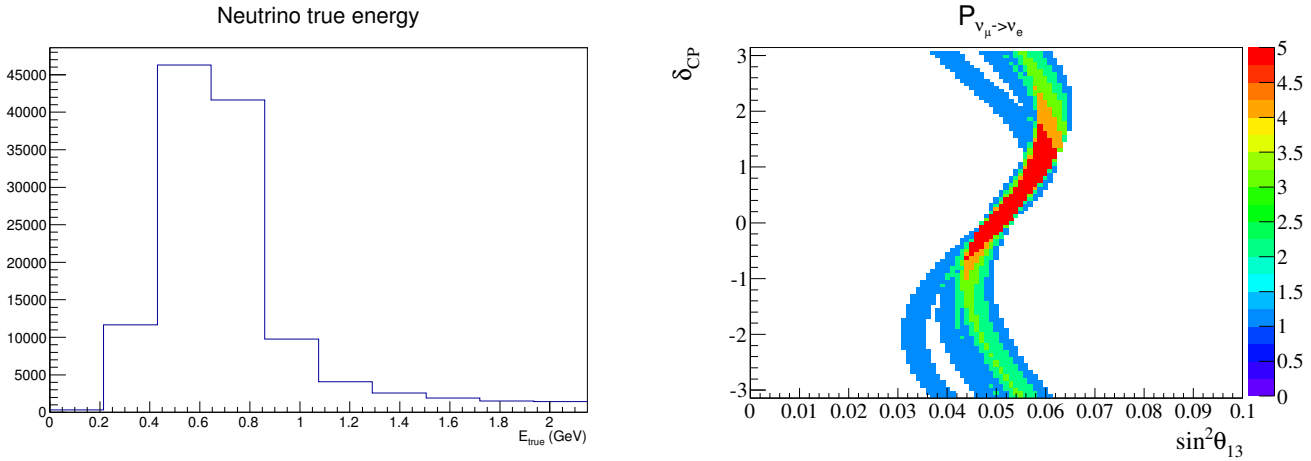


Figure 115: On the left: true energy spectrum for the single e-like ring event candidates for the oscillated ν_e category in the T2K SKMC, zooming around the peak of the spectrum. On the right: The oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ is selected, and the oscillation probability at that point is calculated and used as reference for each one of the energies at the peak on the left plot $P(\nu_\mu \rightarrow \nu_e)_{true}(E_i)$. Now for each one of the energies at the peak of the spectrum on the left plot, from 0.45 GeV to 0.85 GeV, a line is drawn on the right plot for the points in the ($sin^2\theta_{13}$,δ_{CP}) space for which $\Delta(P(\nu_\mu \rightarrow \nu_e))(E_i)$ = $P(\nu_\mu \rightarrow \nu_e)_{point}(E_i)$ -$P(\nu_\mu \rightarrow \nu_e)_{true}(E_i) < 0.005$.

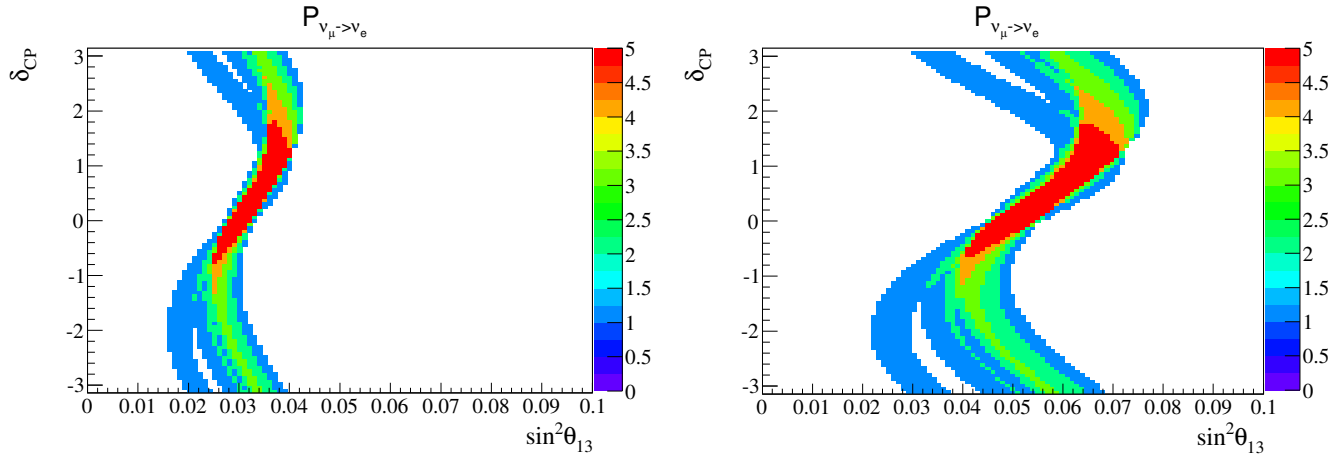


Figure 116: Lines of equal $P(\nu_\mu \rightarrow \nu_e)$ are drawn for the different values of the true energy at the peak of the spectrum on the left side of Fig. 115 (from 0.45 GeV to 0.85 GeV) corresponding to the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.03$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ (left) and $sin^2\theta_{23} = 0.3$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ (right).

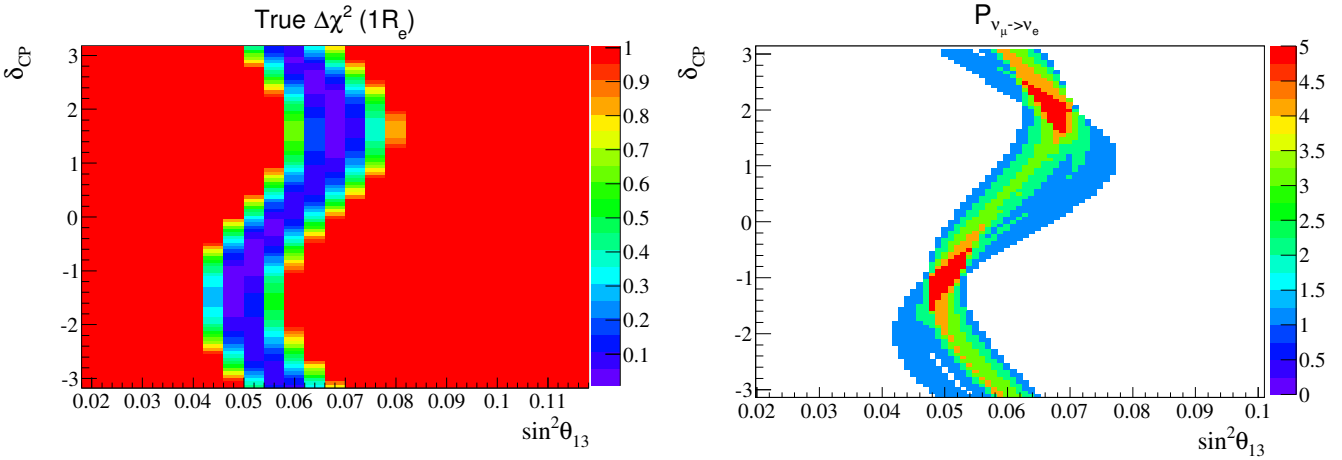


Figure 117: On the left: true $\Delta\chi^2$ distribution for the single e-like ring sample in the ($sin^2\theta_{13}$,δ_{CP}) space for a toy created at the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.05$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = -1$ assuming normal hierarchy without statistical fluctuations nor systematic variations. On the right: lines of equal $P(\nu_\mu \rightarrow \nu_e)$ are drawn for the different values of the true energy at the peak of the spectrum on the left side of Fig. 115 (from 0.45 GeV to 0.85 GeV) corresponding to the oscillation point $sin^2\theta_{23} = 0.5263$, $sin^2\theta_{13} = 0.03$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = -1$.

C.2. Residuals and pulls for the systematic parameters

Distributions of pulls of the 64 systematic parameters are presented in Figs. 118-124 for an oscillation point with $sin^2\theta_{23} = 0.513$, $sin^2\theta_{13} = 0.0251$, $\Delta m^2 = 2.4 \times 10^{-3} eV^2/c^4$ and $\delta_{CP} = 0$ including correlations between the SK efficiencies for the two samples. A summary of the pull mean and RMS is given in Fig. 125 comparing results with and without correlations. From this figure it is clear that only small changes appear when introducing the correlations between the SK efficiencies for the two samples.

No problems were identified in the tests done to the fitter, which performed as expected. In general all systematic parameters present a good behaviour with a mean in ≈ 0 and RMS ≈ 1. There are some small biases in the pull distributions for some systematic parameters, which are understood:

- **Spectral function**

  A special treatment is followed for this parameter, which is varied following a Gaussian distribution with mean at 0 and $\sigma = 1$ bounded between [0,1] when creating toy experiments. When fitting this parameter, it is limited to the same range [0,1] and a Gaussian constraint in the penalty term is applied. Therefore, a Gaussian distribution of residuals centered at 0 with RMS 1 is not expected for this parameter.

- **π-less Δ decay**

  The distribution of pulls for this parameter appears to be biased due to the special procedure followed to calculate the weights for the response functions (splines) for this parameter. In NEUT, 20% of the Δs decay within this process, being therefore the uncertainty equal to 20%. That means that this parameter cannot be tweaked to values smaller than -1σ, what would be unphysical, and instead the same weight found for a tweak of -1σ is applied to any tweak smaller that this value, creating an assymmetry and a constant plateau at negative values. Thus, a small negative bias appears for this parameter.

- **SuperK reconstructed energy scale**

  This systematic is different from the others since its effect is not translated into a weight of the number of events in a bin, but a migration of events from one bin to the adjacent ones. The single μ-like ring spectrum is not symmetric and therefore a positive tweak of the reconstructed energy scale parameter has a different effect in the spectrum than a negative one as already explained in the *VALOR* Run1+2+3+4 ν_μ disappearance analysis in [6]. There is also an asymmetric effect in the single e-like ring spectrum, more evident in the first and last bins (as can be seen in Fig. 46 in the additional material accompanying this technical note [36]). The RMS of this parameter is larger than 1, issue that can be solved if MINOS errors are computed instead of the ones calculated with the HESSE algorithm, since MINOS errors are different for positive and negative values of the distribution of residuals and they therefore account for the asymmetric effect of this parameter.

- **$1R_\mu$ NC SK efficiency + FSI + SI**

  There distribution of pulls for this parameter is biased to a negative value due to the low statistics at current POT, which makes that a downward Poisson fluctuation of the small number of events in the bins around the oscillation dip in the single μ-like ring spectrum is more likely than an upward fluctuation. Since NC events are mainly distributed around the oscillation dip, and downward fluctuations are more likely to happen in that zone of the spectrum, then a movement to a smaller value of this systematic parameter is more probable and creates a bias to negative values, which disappears when increasing the POT and the number of events at the oscillation dip, reducing the predominance of downward fluctuations.

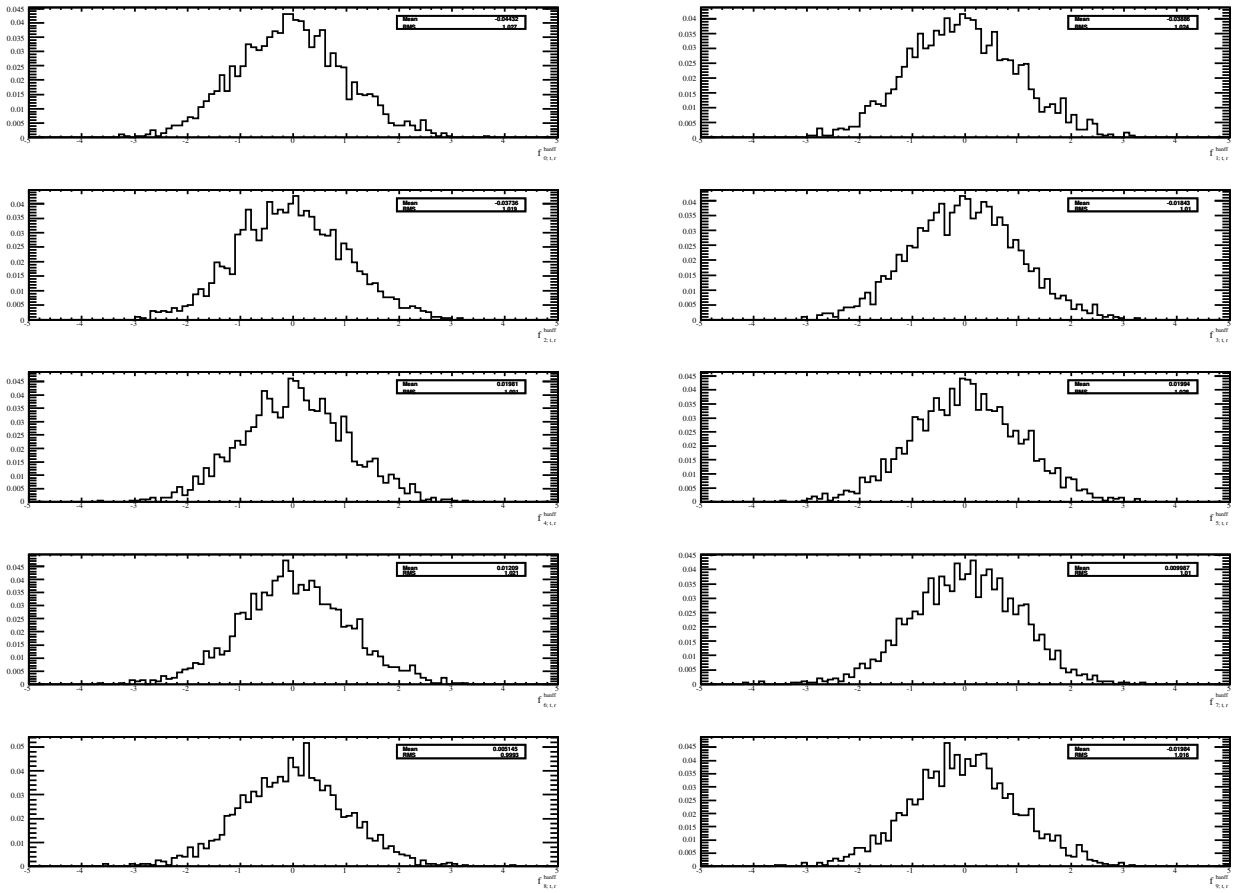


Figure 118: Distribution of pulls for the systematic parameters BANFF 0 - 9.

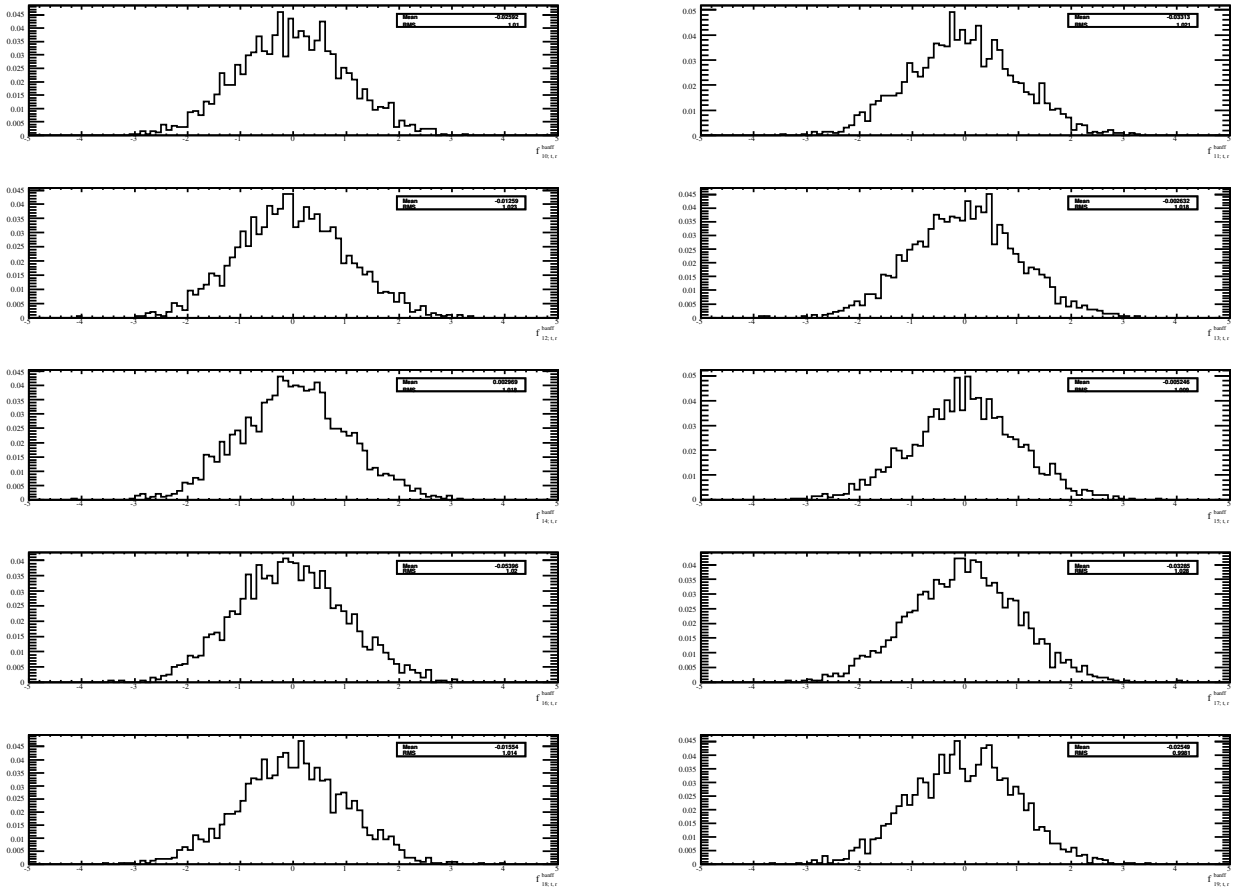


Figure 119: Distribution of pulls for the systematic parameters BANFF 10 - 19.

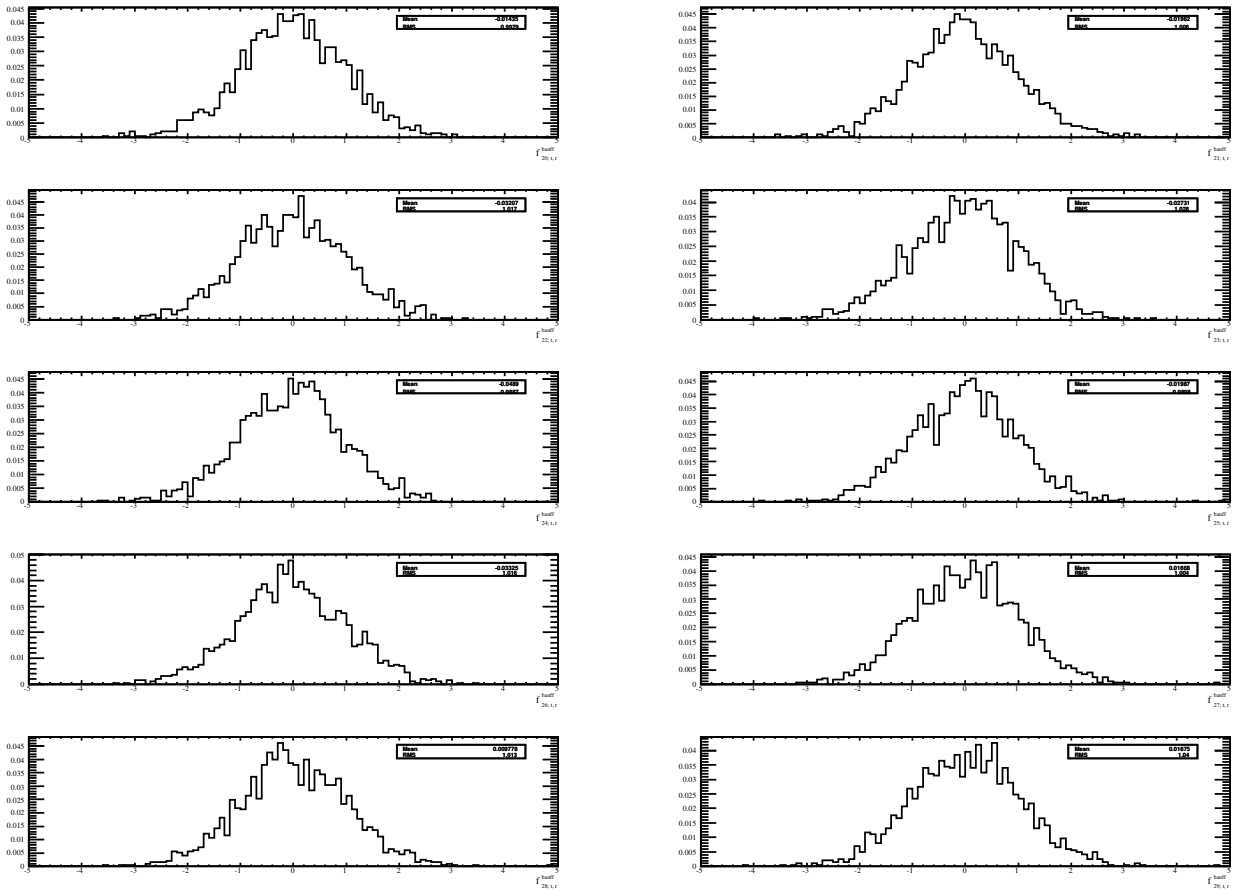


Figure 120: Distribution of pulls for the systematic parameters BANFF 20 - 29.

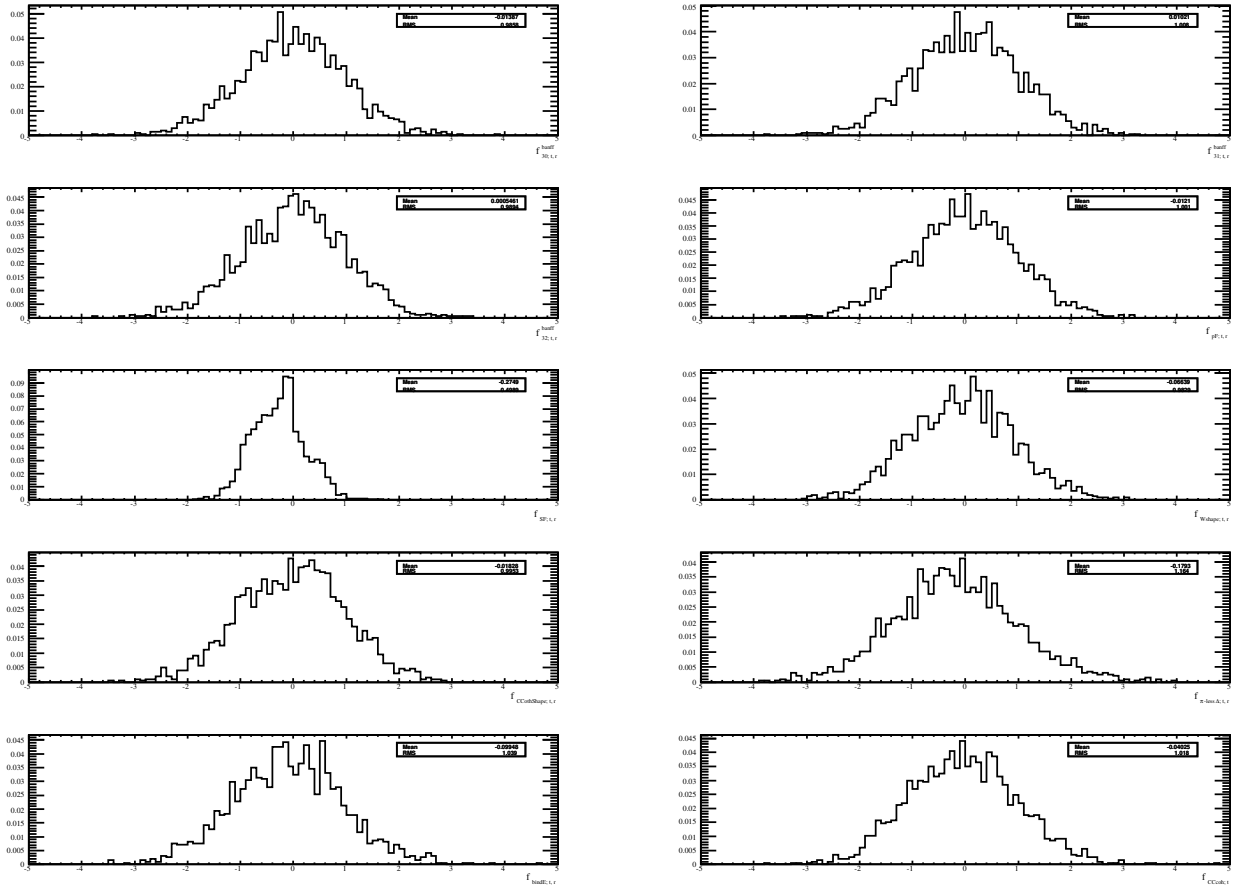


Figure 121: Distribution of pulls for the systematic parameters BANFF 30 - 32 and some of the independent cross section systematics.

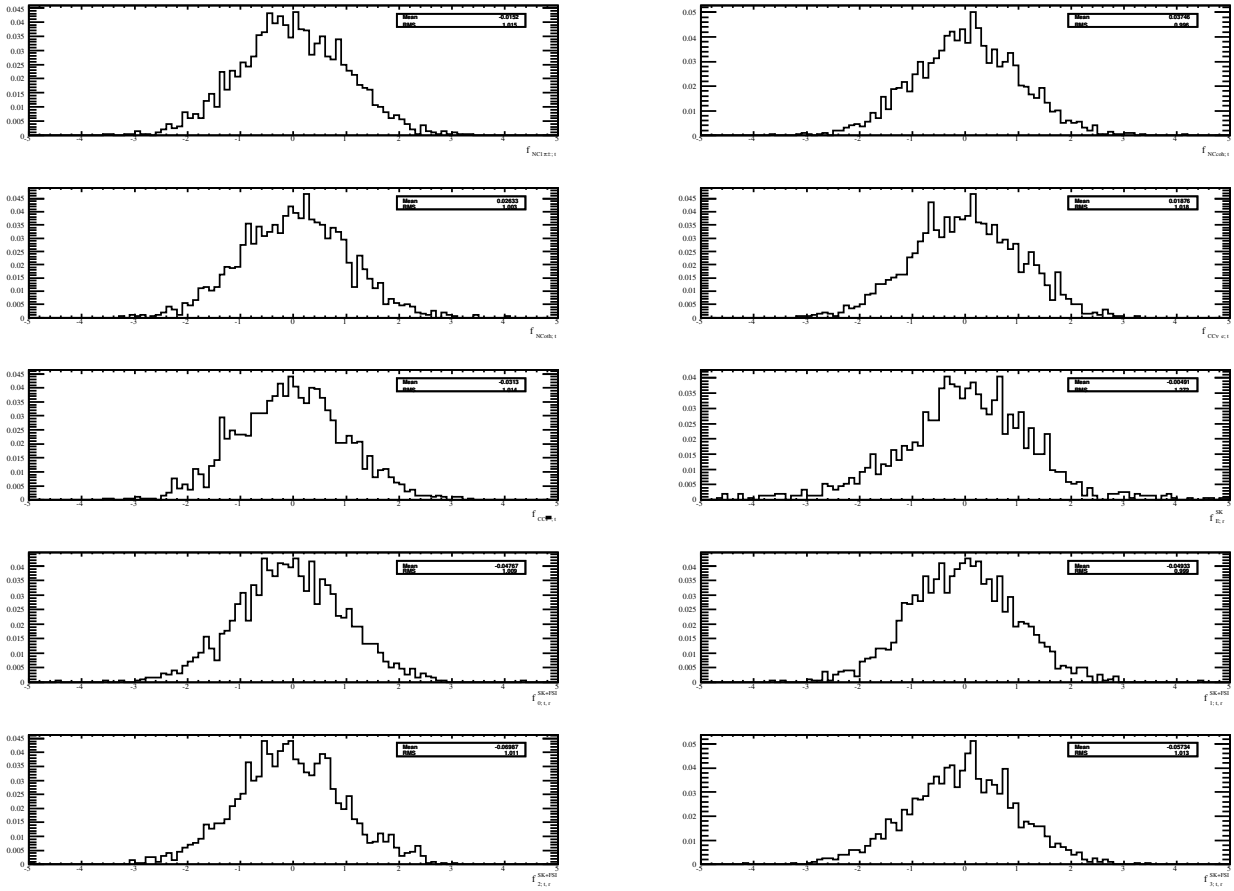


Figure 122: Distribution of pulls for some of the independent cross section systematics, reconstructed Energy scale and SK+FSI parameters 0-3.

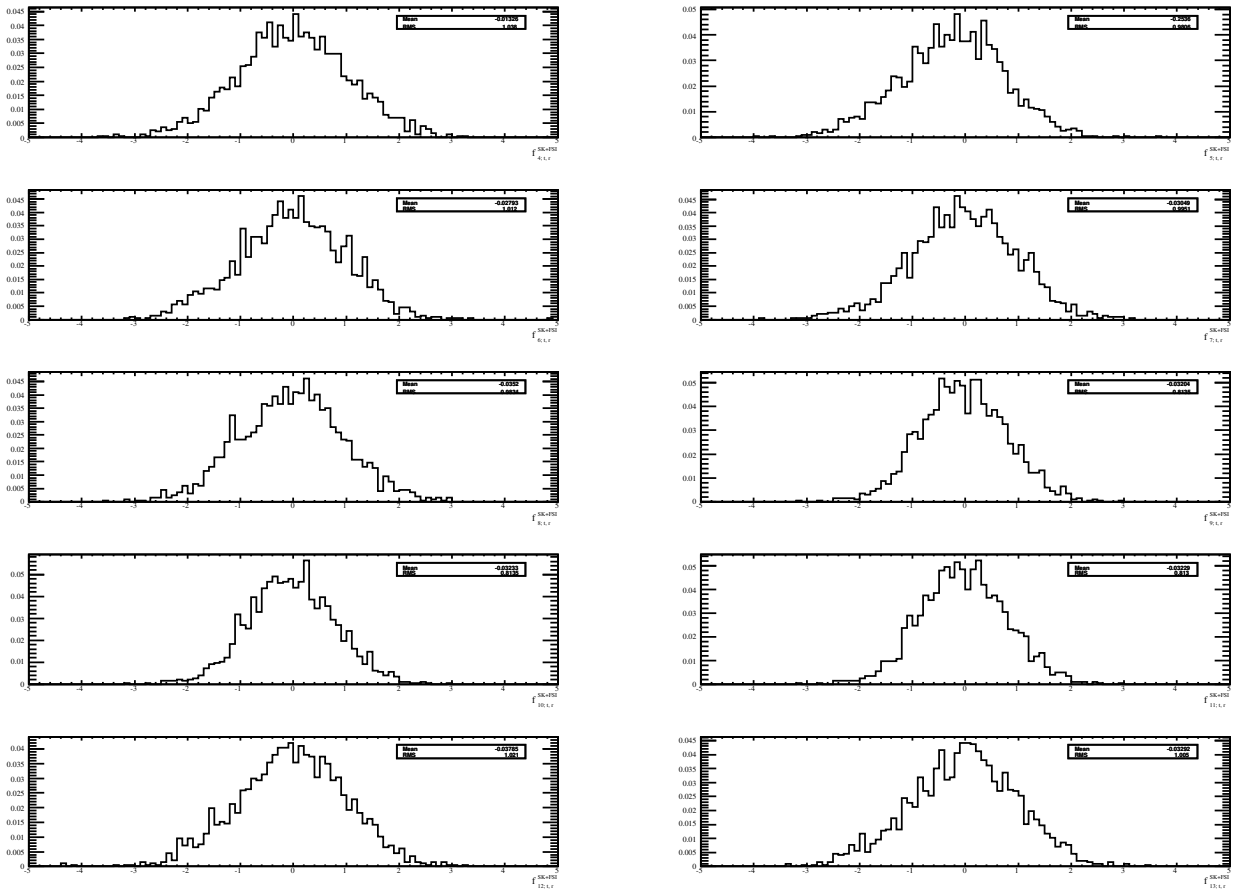


Figure 123: Distribution of pulls for the systematic parameters BANFF 4 - 13.

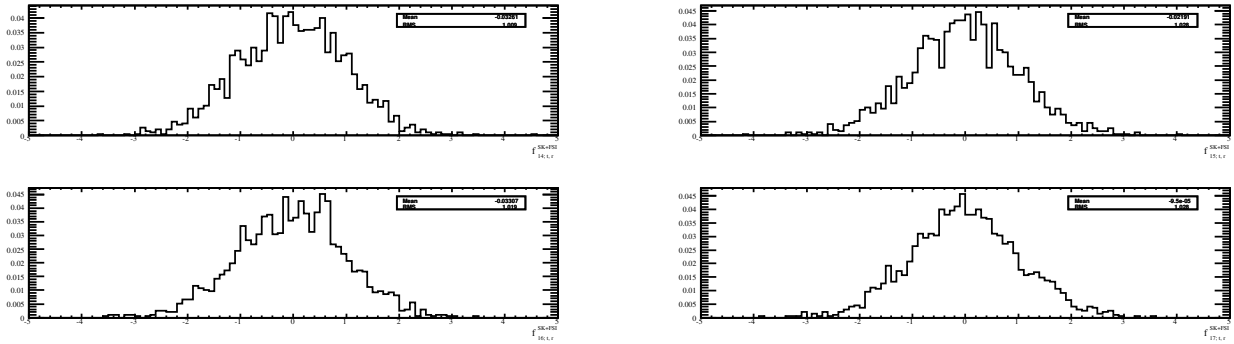


Figure 124: Distribution of pulls for the systematic parameters BANFF 14 - 17.

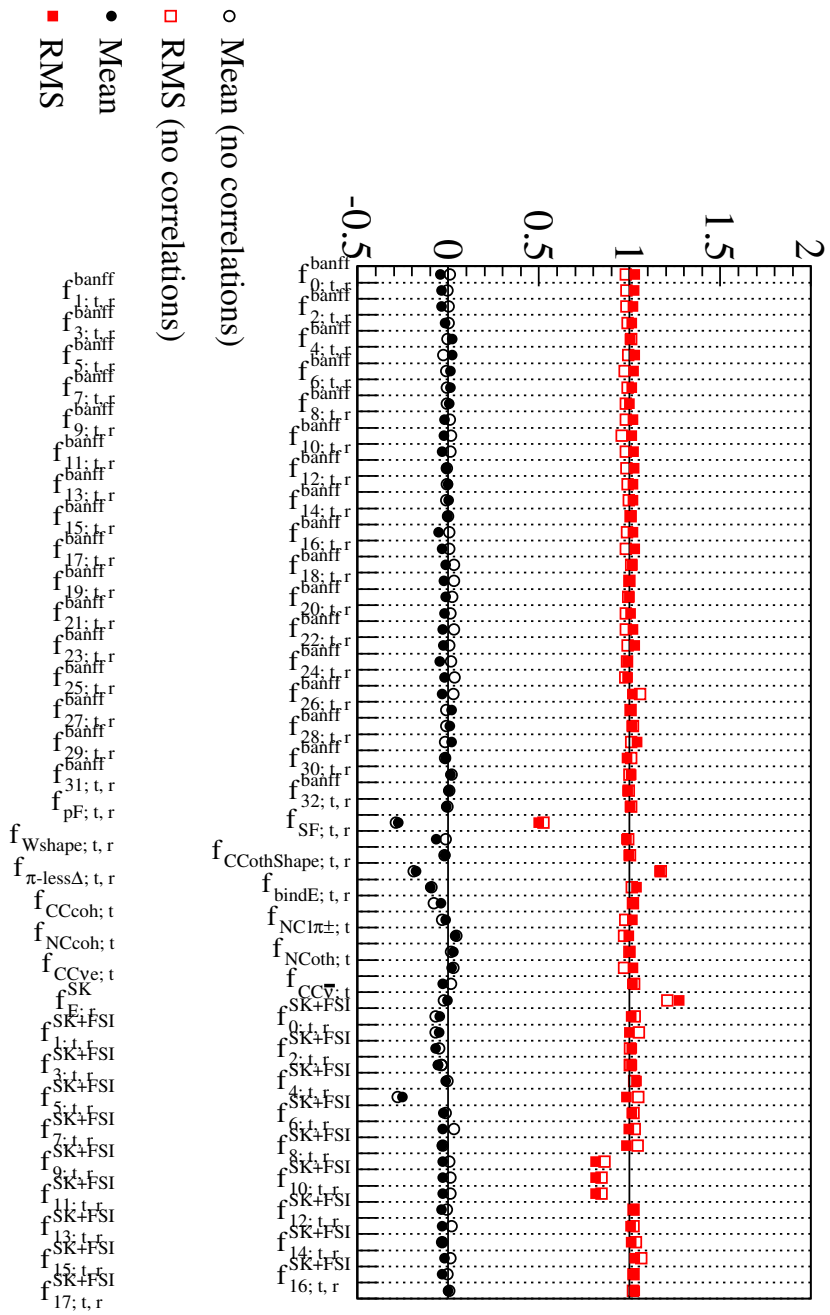


Figure 125: Summary of mean and RMS for the distribution of pulls of the 64 systematic parameters with and without correlations.

D. Validation of the *VALOR* Run1+2+3+4 joint 3-flavour oscillation analysis: 6.57 $\times 10^{20}$ POT toy dataset fits

Different fake single μ-like ring and e-like ring datasets were created for an exposure of 6.57 $\times 10^{20}$ POT in order to validate the *VALOR* Run1+2+3+4 joint 3-flavour oscillation analysis. The values of the oscillation parameters used to create those toy experiments are summarized in Tab. 24. All the fake datasets were created including statistical fluctuations and some of them included also random variations of all systematic parameters, as indicated in Tab. 24. The fake datasets can be found in [37].

The best-fit point obtained for each fake dataset is also presented in Tab. 24. Figs. 126-133 show the best-fit spectra obtained for each toy dataset fit. The pulls of the systematic parameters at the best-fit point are presented in Figs. 134-141.

The profile $\Delta\chi^2$ distributions vs each one of the oscillation parameters, $|\Delta m^2|$, $sin^2\theta_{23}$, $sin^2\theta_{13}$ and δ_{CP}, are presented for each toy dataset fit in Figures like 142, showing with a vertical line the best-fit value reported in Tab. 24. In addition, the 68% and 90% C.L. regions are given in the following four 2-dimensional oscillation parameter spaces: ($sin^2\theta_{23}$,$|\Delta m^2|$), ($sin^2\theta_{13}$,$|\Delta m^2|$), ($sin^2\theta_{13}$,δ_{CP}) and ($sin^2\theta_{23}$,$sin^2\theta_{13}$). They are presented in Figs.144, etc.

In all profile $\Delta\chi^2$ and confidence region plots presented for the joint 3-flavour analysis, the value of χ^2 was minimized with respect to all 64 systematic parameters and the oscillation parameters from ($|\Delta m^2|$,$\sin^2\theta_{23}$,$\sin^2\theta_{13}$,δ_{CP}) not shown on the plot (as explained in App. B, $\sin^2\theta_{12}$ and δm^2 have a negligible effect and they were kept fixed to the value indicated in Tab. 1). The best-fit values of the oscillation parameters not shown on the confidence region plots are included in a set of extra plots in App. E. The confidence regions for ($|\Delta m^2|$,$\sin^2\theta_{23}$) and ($\sin^2\theta_{13}$,δ_{CP}) are presented for one toy experiment, #0, comparing the fit with and without systematics in Fig. 162

Confidence regions were found with respect to the χ^2 global minimum reported in Tab. 24 (except for the result in ($\sin^2\theta_{13}$, δ_{CP}) space where the raster scan was used) using the constant-$\Delta\chi^2$ method (see Section 6).

The last toy dataset, #5, was fitted assuming inverted hierarchy (the true one) and also normal hierarchy in order to compare results assuming each mass hierarchy. Best-fit spectra and pulls for the fit with normal hierarchy are presented in Fig. 132 and 140 with the best-fit values also included in Tab. 24. The comparison between both fits is presented in Fig. 160.

Furthermore, one of the toy datasets, #1, was also fitted using $sin^2 2\theta_{13}$ instead of $sin^2\theta_{13}$, and the best-fit spectra and pulls of this new fit are presented in Figs. 133 and 141. The comparison between both fits, with the correct re-scaling of the $sin^2\theta_{13}$ axis to $sin^2 2\theta_{13}$, is presented in Fig. 161 and clearly shows that there is no difference between both fitting methods and results can be converted from one to another with the corresponding re-scaling.

The results in this subsection confirm that the fitter performs as expected. No unexpected issues appeared.

Set	Fit	MH	$\|\Delta m^2\|$ ($\times 10^{-3}$ eV^2/c^4)	$\sin^2\theta_{23}$	$\sin^2\theta_{13}$	δ_{CP}	$N_{SK}^{1R\mu}$	N_{SK}^{1Re}	χ^2	Sytematics	Figures
0	true	NH	**2.4375**	**0.513**	**0.0251**	**0.0**	**123**	**31**	-	Nominal	-
	$\sin^2\theta_{13}$	NH	2.413	0.513	0.0364	-0.0825	127.264	30.104	87.877	-	126,134, 142 - E
	$\sin^2 2\theta_{13}$	NH	2.413	0.513	0.1403	-0.0825	127.264	30.104	87.877	-	-
	wo systematics	NH	2.340	0.514	0.0364	-0.105	126.234	29.920	88.942	-	-
1	true	NH	**2.4375**	**0.37**	**0.0251**	**0.0**	**138**	**13**	-	Random	-
	$\sin^2\theta_{13}$	NH	2.327	0.619	0.0152	1.585	139.277	13.817	104.765	-	127,135, 145 - E
	$\sin^2 2\theta_{13}$	NH	2.327	0.619	0.0599	1.592	139.274	13.816	104.764	-	133,141, 161
2	true	NH	**2.750**	**0.513**	**0.0251**	**0.0**	**120**	**17**	-	Random	-
	$\sin^2\theta_{13}$	NH	2.578	0.508	0.0185	-0.0179	120.354	17.298	90.792	-	128,136, 148 - E
	$\sin^2 2\theta_{13}$	NH	2.578	0.508	0.0728	-0.0179	120.354	17.298	90.792	-	-
3	true	NH	**2.4375**	**0.513**	**0.04**	**0.0**	**112**	**39**	-	Random	-
	$\sin^2\theta_{13}$	NH	2.583	0.568	0.0572	1.087	116.335	38.972	73.952	-	129,137, 151 - E
	$\sin^2 2\theta_{13}$	NH	2.583	0.568	0.2157	1.087	116.335	38.972	73.952	-	-
4	true	NH	**2.4375**	**0.513**	**0.0251**	**-$\pi/2$**	**117**	**38**	-	Nominal	-
	$\sin^2\theta_{13}$	NH	2.466	0.526	0.0464	-2.564	120.035	38.414	76.768	-	130,138, 154 - E
	$\sin^2 2\theta_{13}$	NH	2.466	0.526	0.1768	-2.564	120.035	38.414	76.768	-	-
5	true	IH	**2.4375**	**0.513**	**0.0251**	**0.0**	**123**	**17**	-	Nominal	-
	$\sin^2\theta_{13}$	IH	2.530	0.511	0.0246	2.367	121.802	16.279	111.201	-	131,139, 157 - E
	$\sin^2 2\theta_{13}$	IH	2.530	0.511	0.0962	2.367	121.783	16.278	111.203	-	-
	$\sin^2\theta_{13}$	NH	2.557	0.511	0.0205	0.722	121.816	16.226	111.375	-	132,140, 160
	$\sin^2 2\theta_{13}$	NH	2.557	0.511	0.0802	0.723	121.814	16.227	111.375	-	-

Table 24: True input values (shown in bold) and VALOR best-fit values for a test ensemble of 6 fake datasets.

In order to check also the fitting including reactor constraint for $\sin^2 2\theta_{13}$, the toy datasets were also fitted with reactor constraint, using the true input value for the constraint and the error that will be finally used for applying the real reactor constraint to the data, coming from PDG2013. Table 25 summarizes these fits, comparing results of best-fit values with and

without constraint and Figs. 163 - 168 show the contours in the ($\sin^2 2\theta_{13}$,δ_{CP}) space for the six toy datasets. As expected, when the original best-fit value (without contraint) of $\sin^2 2\theta_{13}$ is larger (smaller) than the true value, when the constraint using the true value is applied, the value of $\sin^2 2\theta_{13}$ is forced to get smaller (larger), forcing in turn the value of $\sin^2\theta_{23}$ to be larger (smaller) and the value of δ_{CP} to be negative (positive), and obviously, best-fit values of χ^2 are always larger when the constraint in $\sin^2 2\theta_{13}$ is impossed.

Set	Fit	Constraint	MH	$\lvert\Delta m^2\rvert$ ($\times 10^{-3}$ eV^2/c^4)	$\sin^2\theta_{23}$	$\sin^2\theta_{13}$	δ_{CP}	χ^2	Sytematics	Figure
0	true	-	NH	**2.4375**	**0.513**	**0.098**	**0.0**	-	Nominal	-
	$\sin^2 2\theta_{13}$	NO	NH	2.413	0.513	0.1403	-0.0825	87.877	-	-
	$\sin^2 2\theta_{13}$	0.098 ± 0.01	NH	2.435	0.550	0.0996	-1.263	88.249	-	163
1	true	-	NH	**2.4375**	**0.37**	**0.098**	**0.0**	-	Random	-
	$\sin^2 2\theta_{13}$	NO	NH	2.327	0.619	0.0599	1.592	104.764	-	-
	$\sin^2 2\theta_{13}$	0.098 ± 0.01	NH	2.327	0.402	0.0981	1.599	104.77	-	164
2	true	-	NH	**2.750**	**0.513**	**0.098**	**0.0**	-	Random	-
	$\sin^2\theta_{13}$	NO	NH	2.578	0.508	0.0728	-0.0179	90.792	-	165
	$\sin^2\theta_{13}$	0.098 ± 0.01	NH	2.580	0.498	0.0973	0.802	90.977	-	165
3	true	-	NH	**2.4375**	**0.513**	**0.1536**	**0.0**	-	Random	-
	$\sin^2 2\theta_{13}$	NO	NH	2.583	0.568	0.2157	1.087	73.952	-	-
	$\sin^2 2\theta_{13}$	0.1536 ± 0.01	NH	2.590	0.569	0.1547	-1.013	74.533	-	166
4	true	-	NH	**2.4375**	**0.513**	**0.098**	**-**$\pi/2$	-	Nominal	-
	$\sin^2 2\theta_{13}$	NO	NH	2.466	0.526	0.1768	-2.564	76.768	-	-
	$\sin^2 2\theta_{13}$	0.098 ± 0.01	NH	2.464	0.542	0.105	-1.615	81.003	-	167
5	true	-	IH	**2.4375**	**0.513**	**0.098**	**0.0**	-	Nominal	-
	$\sin^2 2\theta_{13}$	NO	IH	2.530	0.511	0.0962	2.367	111.203	-	-
	$\sin^2 2\theta_{13}$	0.098 ± 0.01	IH	2.530	0.511	0.0979	2.323	111.214	-	168

Table 25: True input values (shown in bold) and VALOR best-fit values for a test ensemble of 6 fake datasets, comparing results with and without including constraint in $\sin^2 2\theta_{13}$.

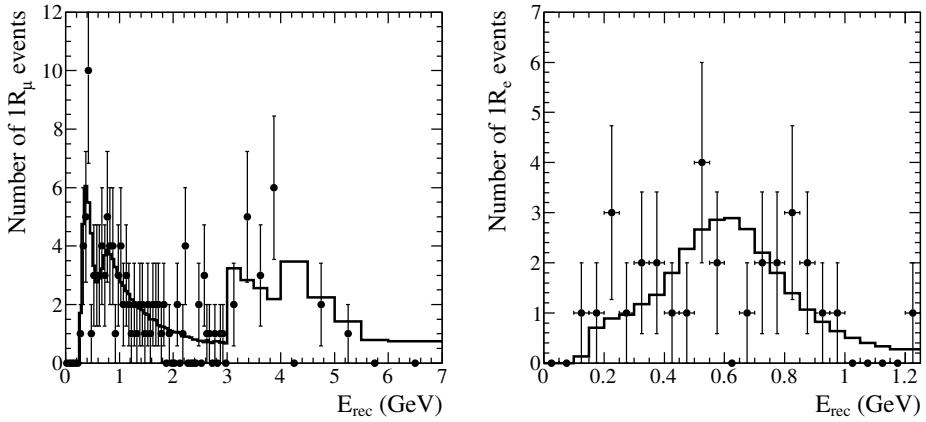


Figure 126: Single μ-like ring event (left) and single e-like ring event (right) best-fit reconstructed energy spectra to fake dataset #0.

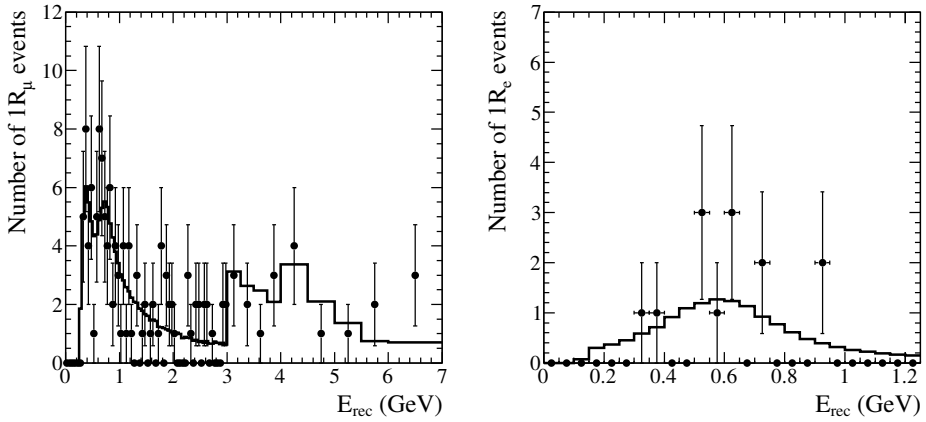


Figure 127: Single μ-like ring event (left) and single e-like ring event (right) best-fit reconstructed energy spectra to fake dataset #1.

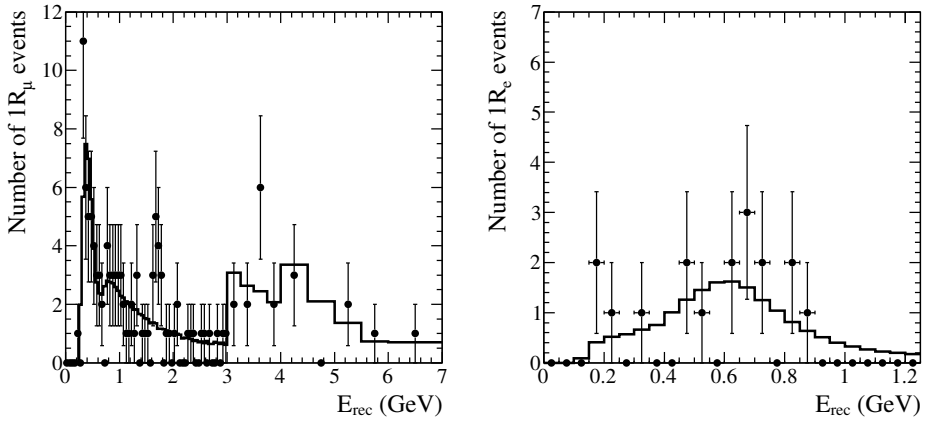


Figure 128: Single μ-like ring event (left) and single e-like ring event (right) best-fit reconstructed energy spectra to fake dataset #2.

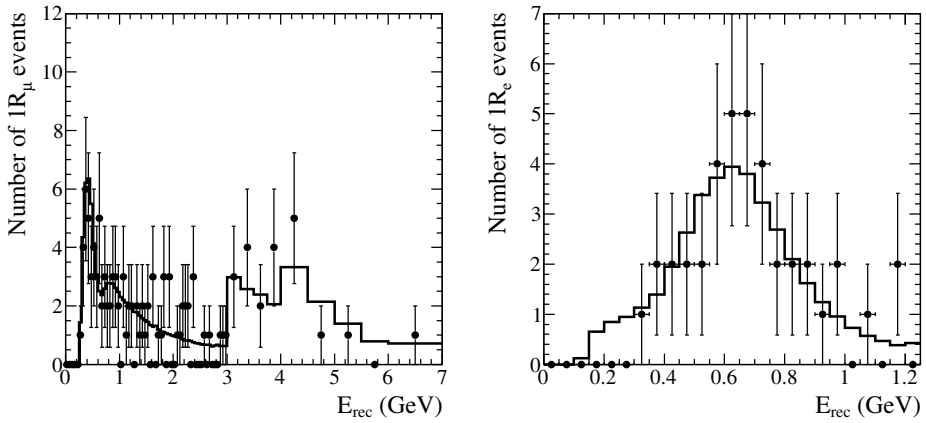


Figure 129: Single μ-like ring event (left) and single e-like ring event (right) best-fit reconstructed energy spectra to fake dataset #3.

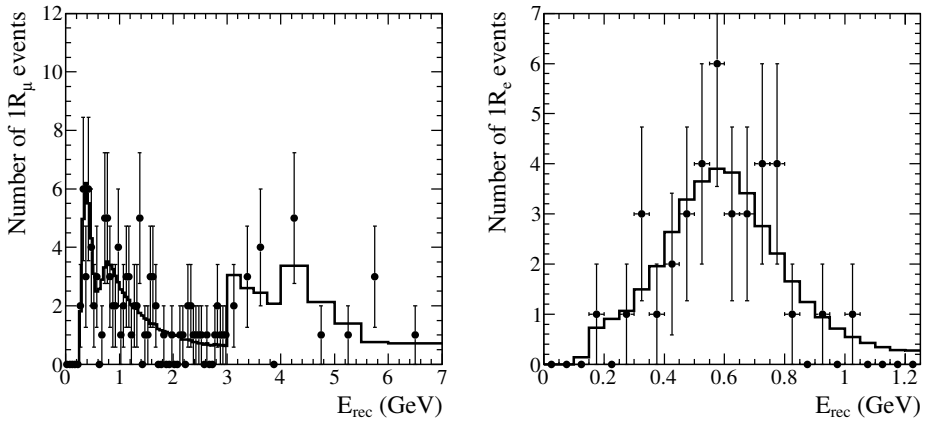


Figure 130: Single μ-like ring event (left) and single e-like ring event (right) best-fit reconstructed energy spectra to fake dataset #4.

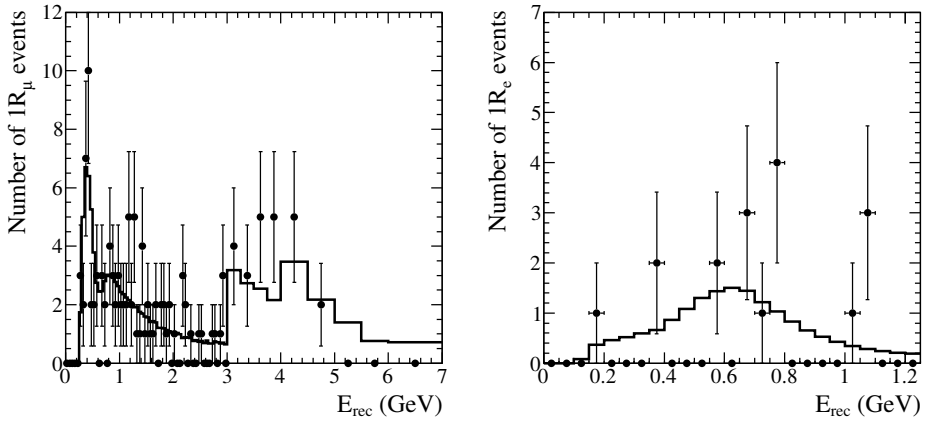


Figure 131: Single μ-like ring event (left) and single e-like ring event (right) best-fit reconstructed energy spectra to fake dataset #5 assuming inverted hierarchy (true one).

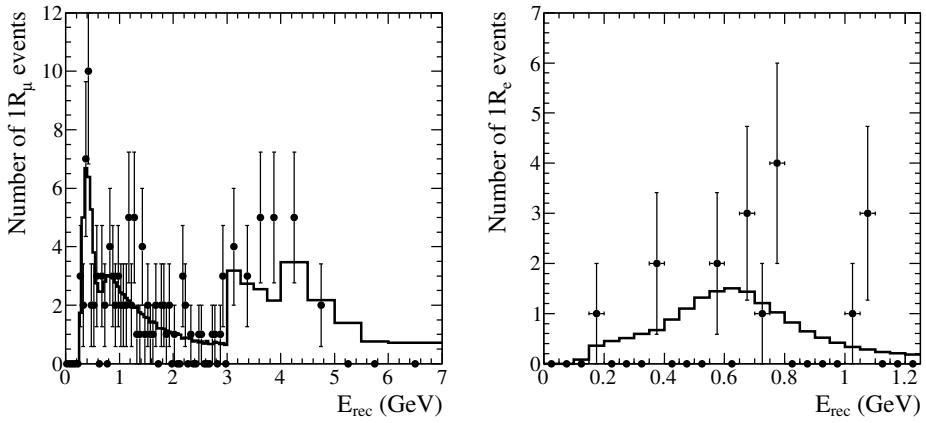


Figure 132: Single μ-like ring event (left) and single e-like ring event (right) best-fit reconstructed energy spectra to fake dataset #5 assuming normal hierarchy.

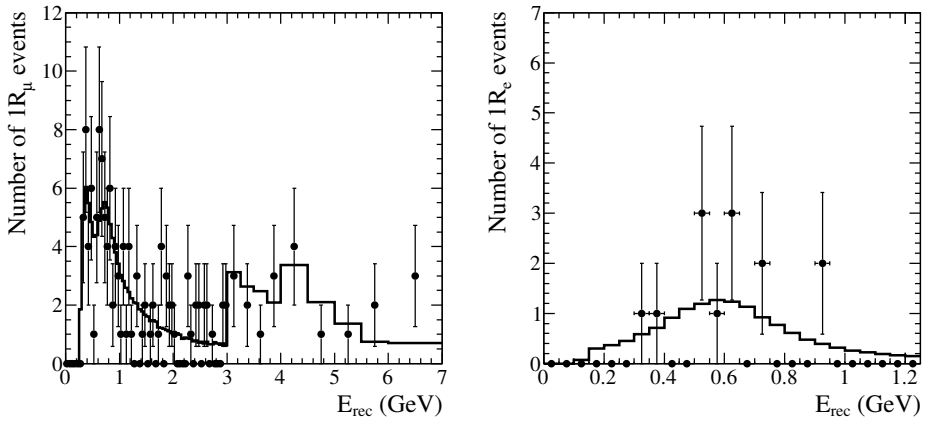


Figure 133: Single μ-like ring event (left) and single e-like ring event (right) best-fit reconstructed energy spectra to fake dataset #1 using $sin^2 2\theta_{13}$ instead of $sin^2\theta_{13}$.

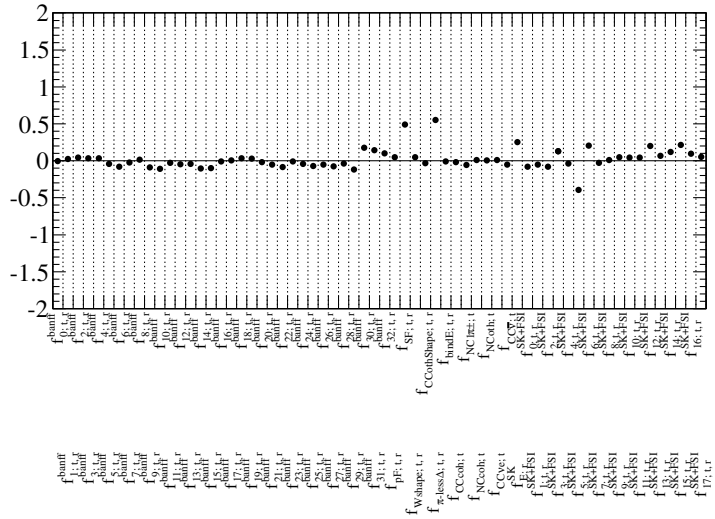


Figure 134: Systematic parameter pulls for the oscillation fit to fake dataset #0.

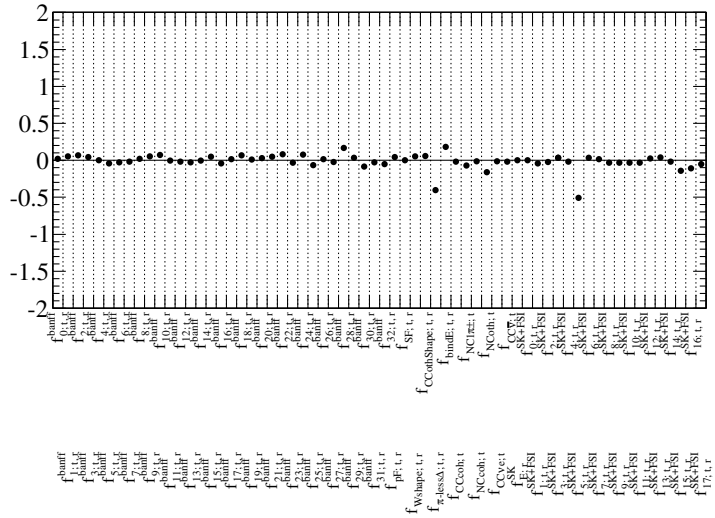


Figure 135: Systematic parameter pulls for the oscillation fit to fake dataset #1.

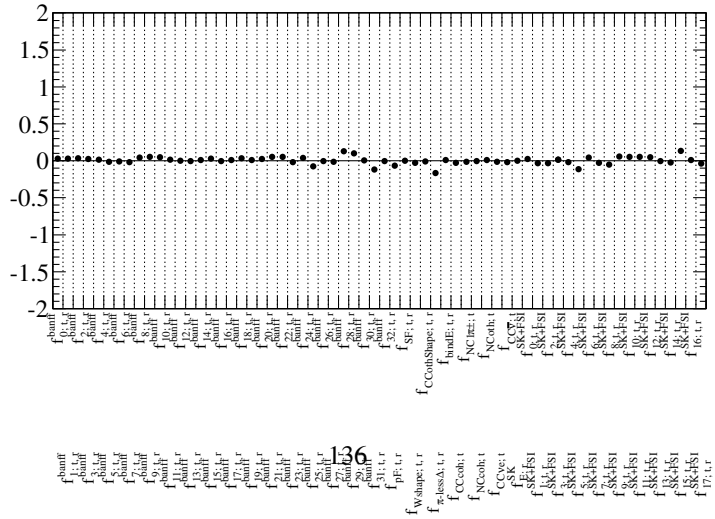


Figure 136: Systematic parameter pulls for the oscillation fit to fake dataset #2.

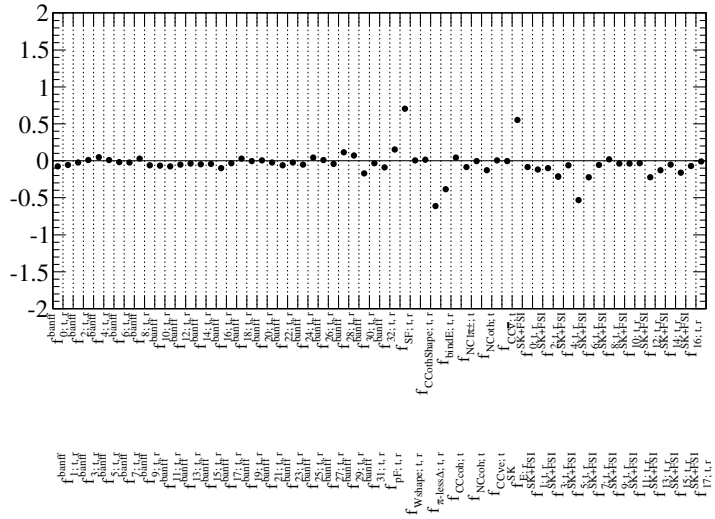


Figure 137: Systematic parameter pulls for the oscillation fit to fake dataset #3.

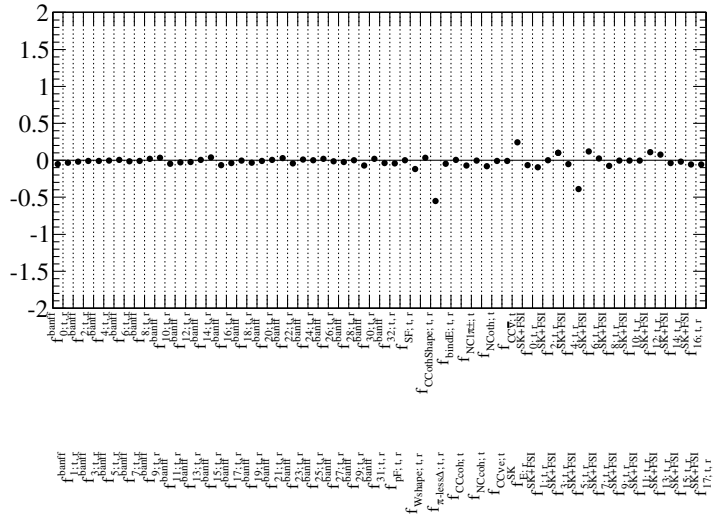


Figure 138: Systematic parameter pulls for the oscillation fit to fake dataset #4.

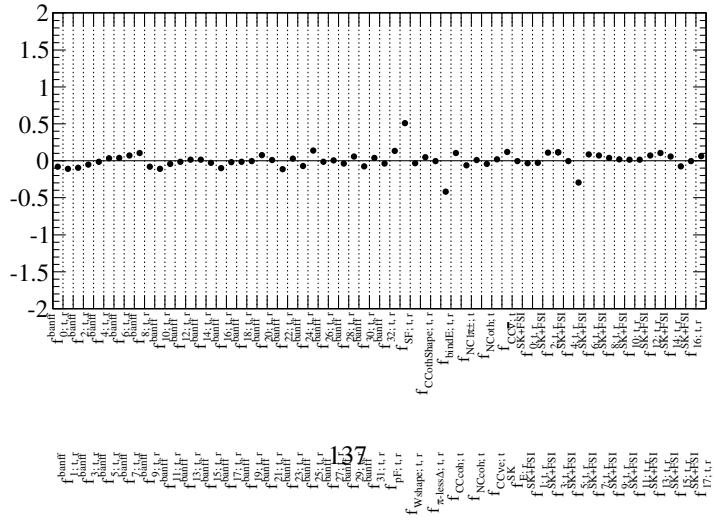


Figure 139: Systematic parameter pulls for the oscillation fit to fake dataset #5 assuming inverted hierarchy (true one).

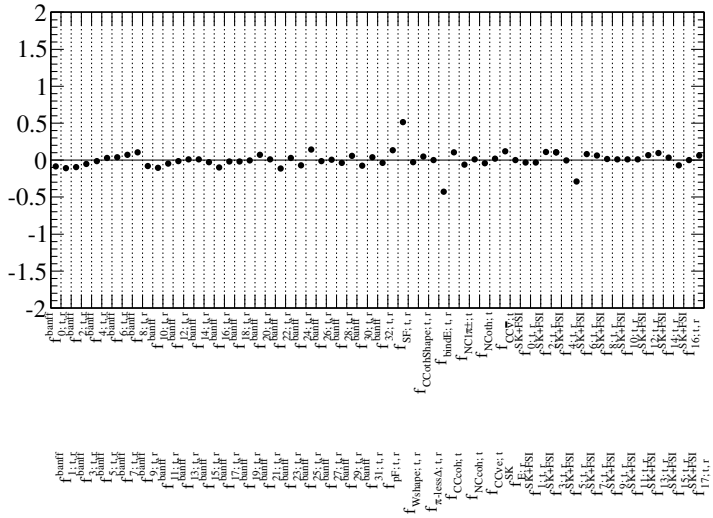


Figure 140: Systematic parameter pulls for the oscillation fit to fake dataset #5 assuming normal hierarchy.

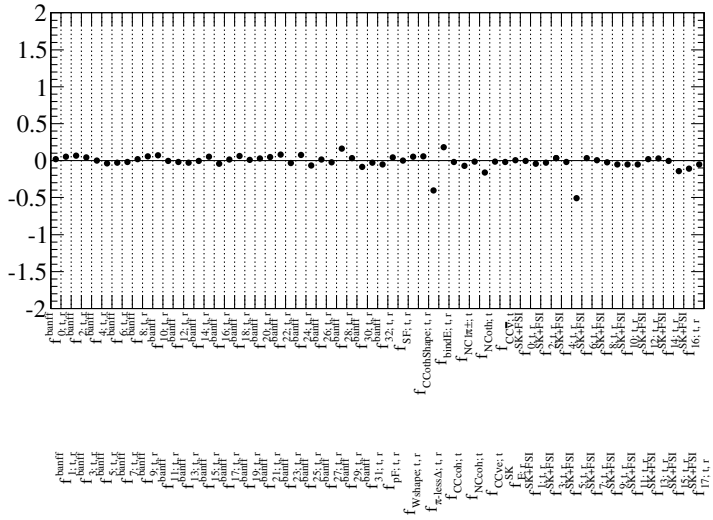


Figure 141: Systematic parameter pulls for the oscillation fit to fake dataset #1 using $sin^2 2\theta_{13}$ instead of $sin^2 \theta_{13}$.

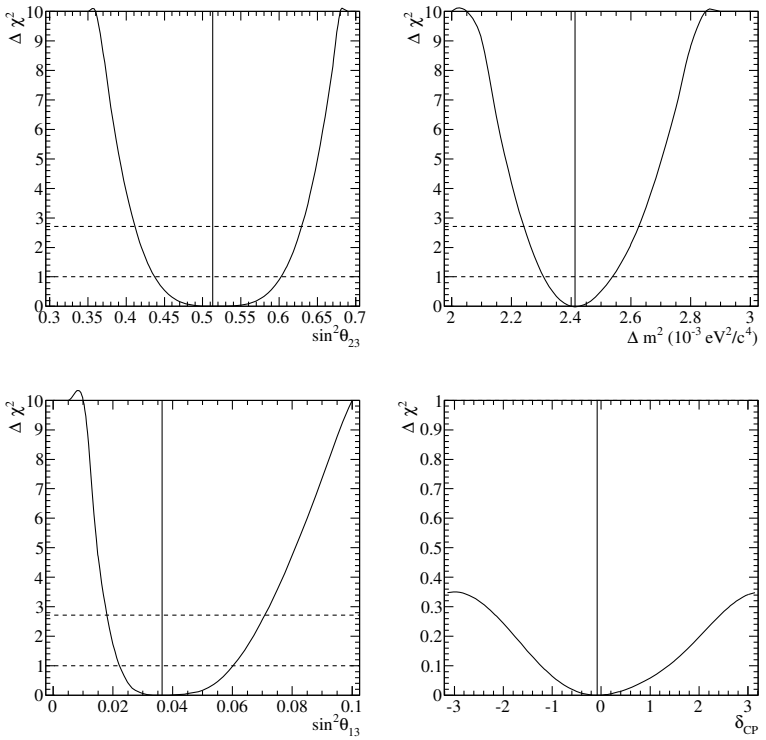


Figure 142: $\Delta\chi^2$ as function of $\sin^2\theta_{23}$,$|\Delta m^2|$,$\sin^2\theta_{13}$ and δ_{CP} for fake dataset #0. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

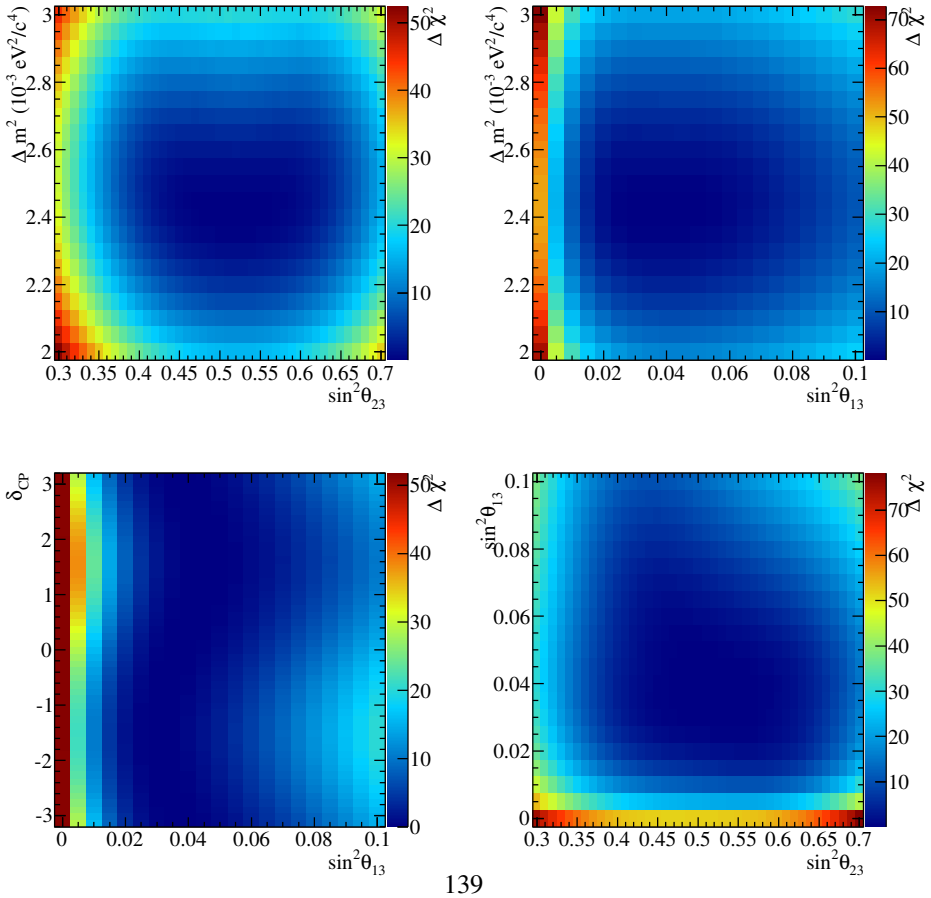


Figure 143: $\Delta\chi^2$ surface in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #0. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

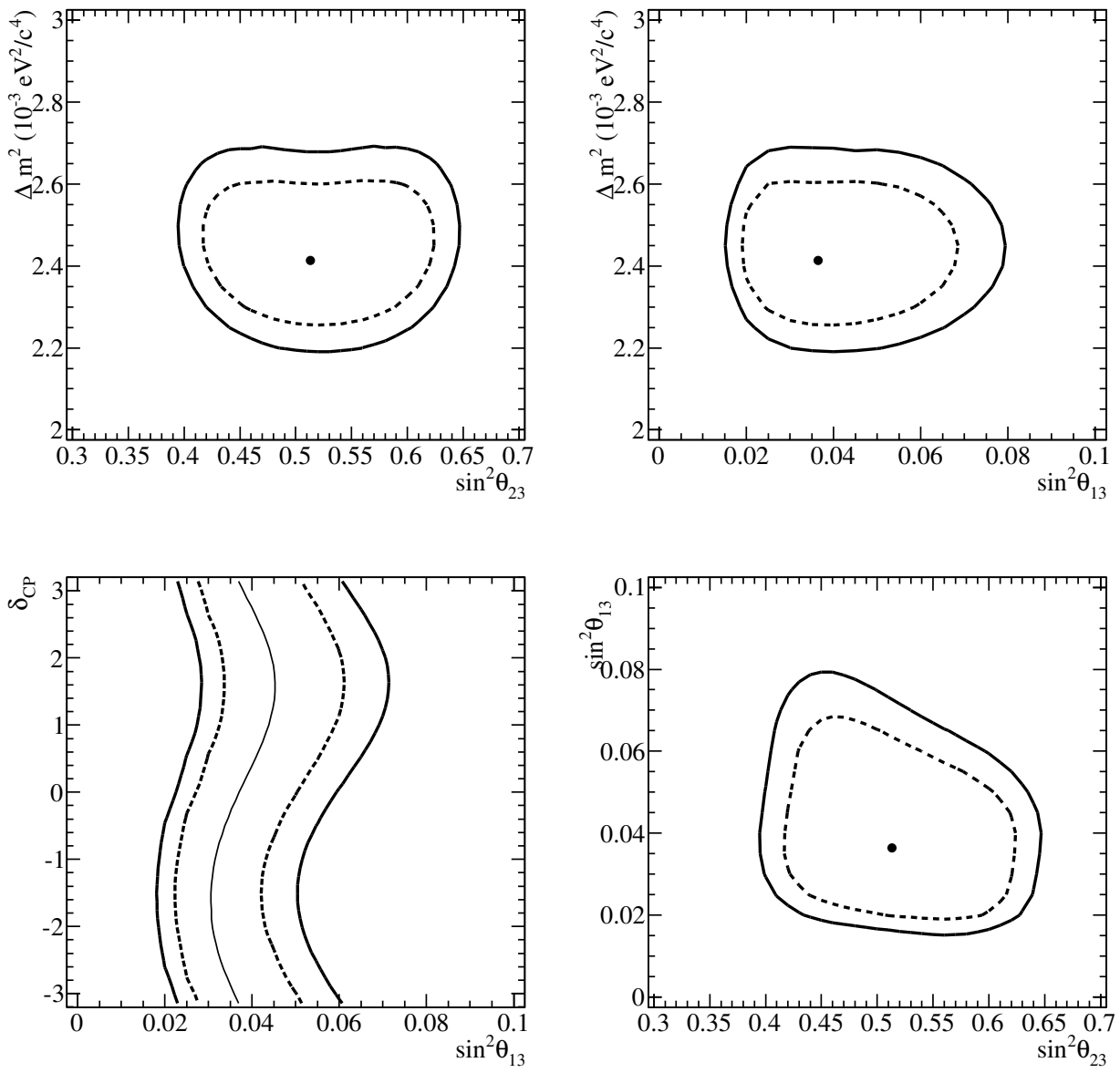


Figure 144: 68% and 90% CL regions in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #0. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

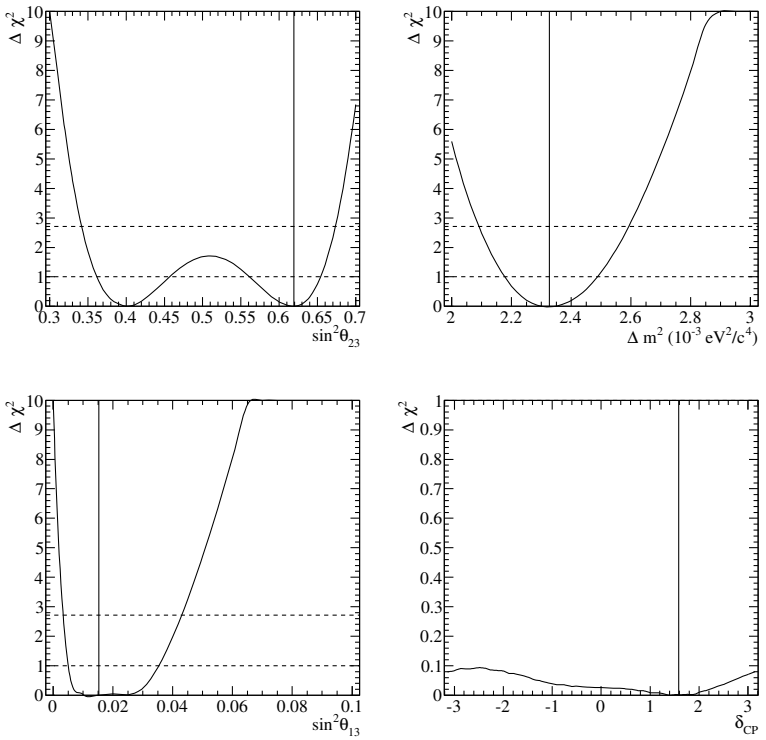


Figure 145: $\Delta\chi^2$ as function of $\sin^2\theta_{23}$,$|\Delta m^2|$,$\sin^2\theta_{13}$ and δ_{CP} for fake dataset #1. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

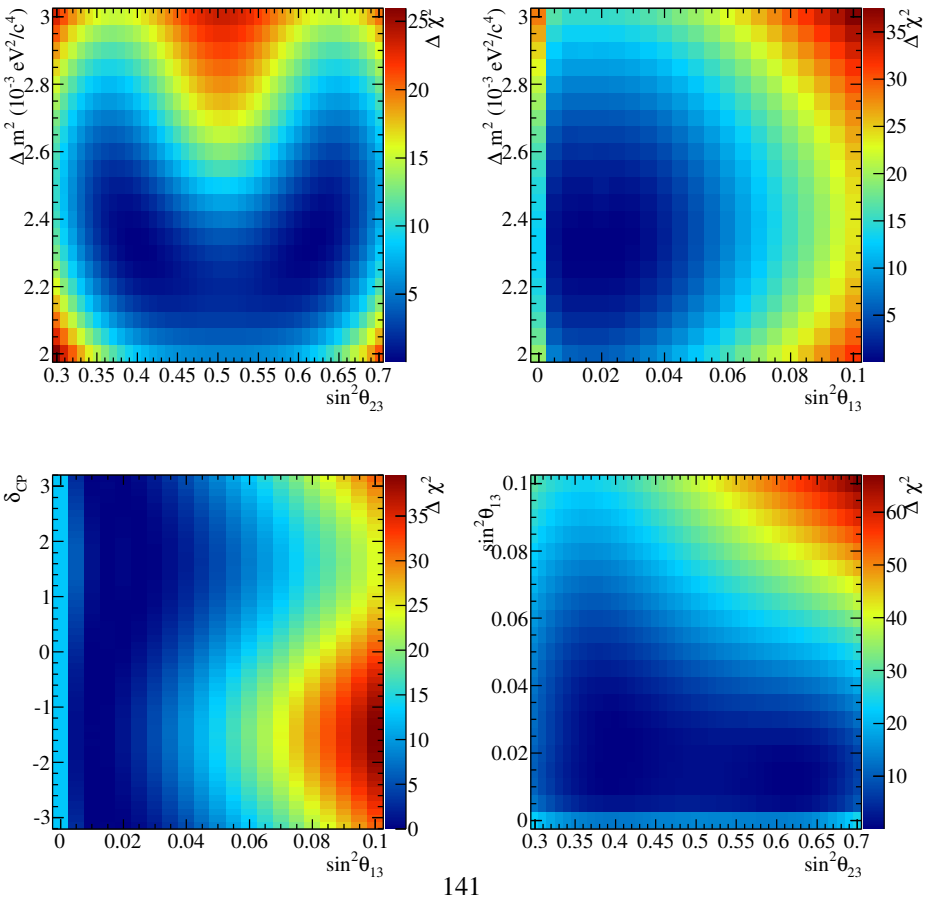


Figure 146: $\Delta\chi^2$ surface in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #1. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

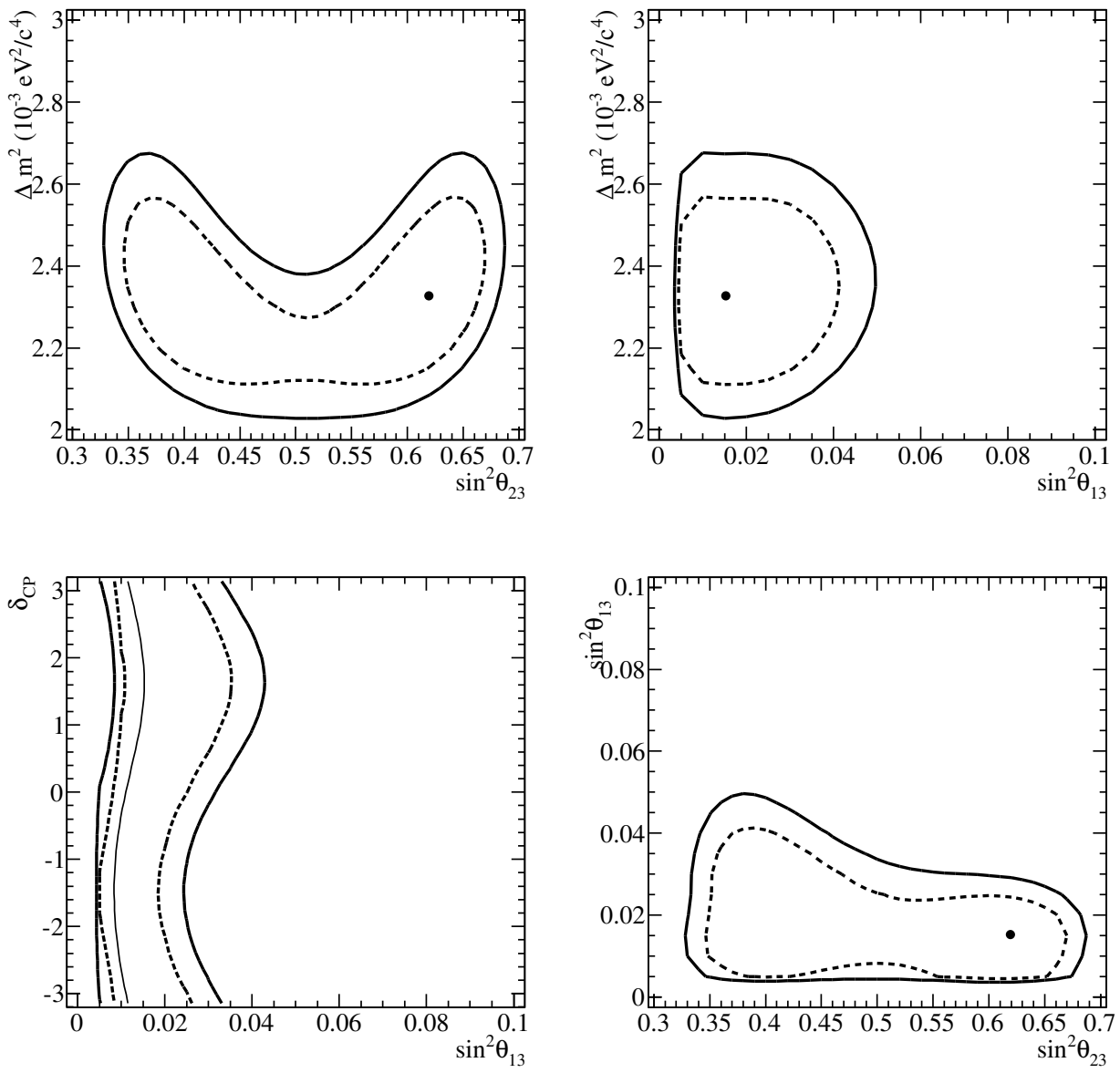


Figure 147: 68% and 90% CL regions in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #1. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

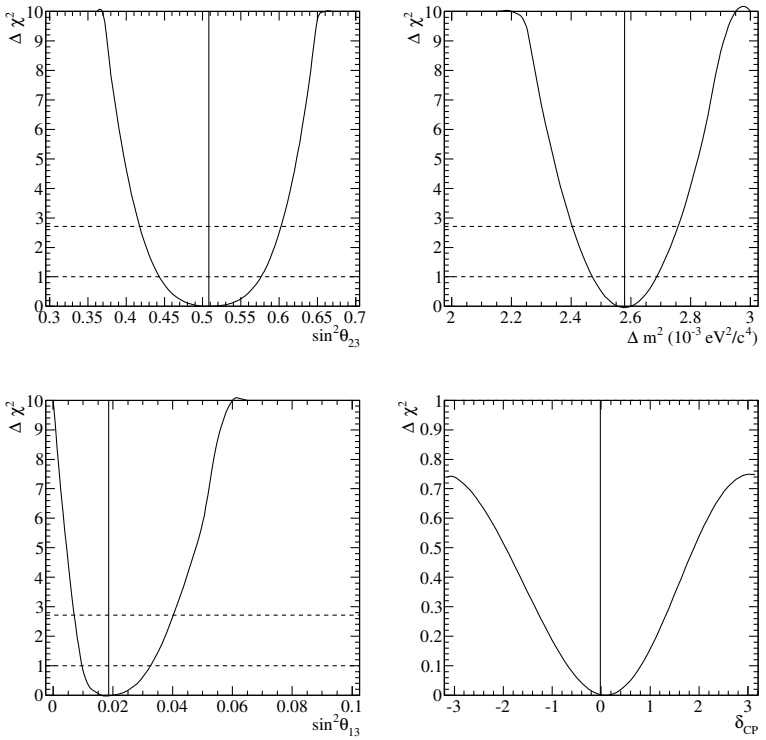


Figure 148: $\Delta\chi^2$ as function of $\sin^2\theta_{23}$,$|\Delta m^2|$,$\sin^2\theta_{13}$ and δ_{CP} for fake dataset #2. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

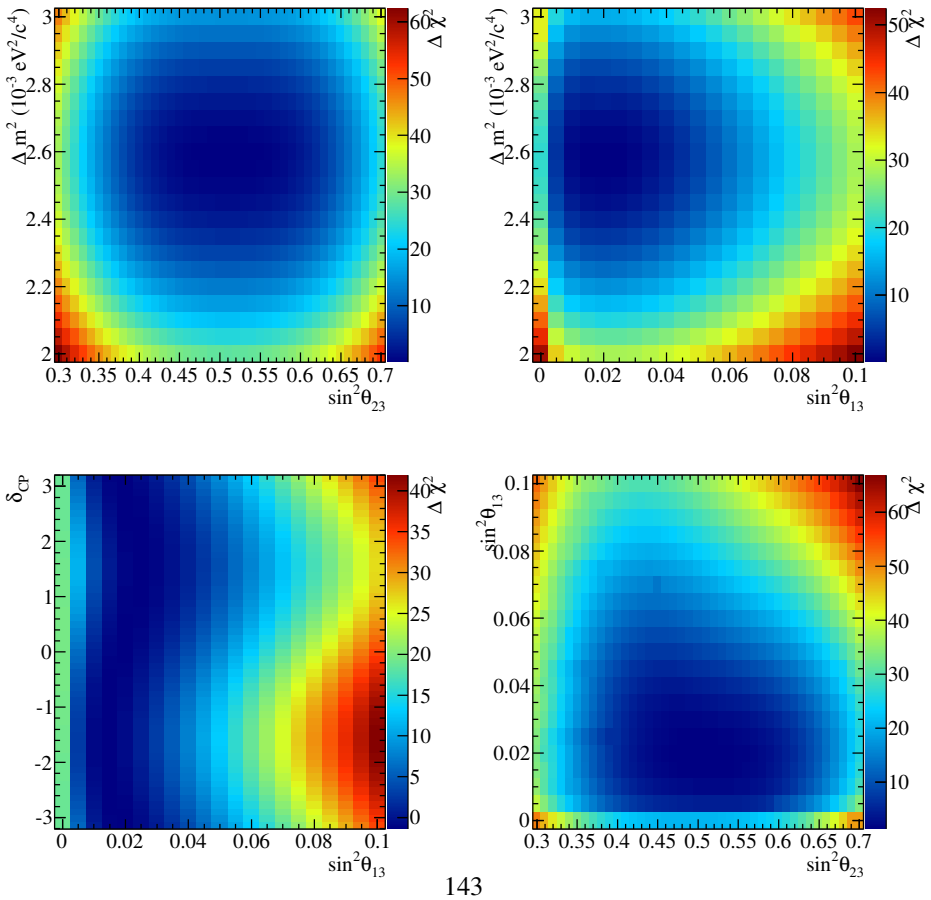


Figure 149: $\Delta\chi^2$ surface in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #2. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

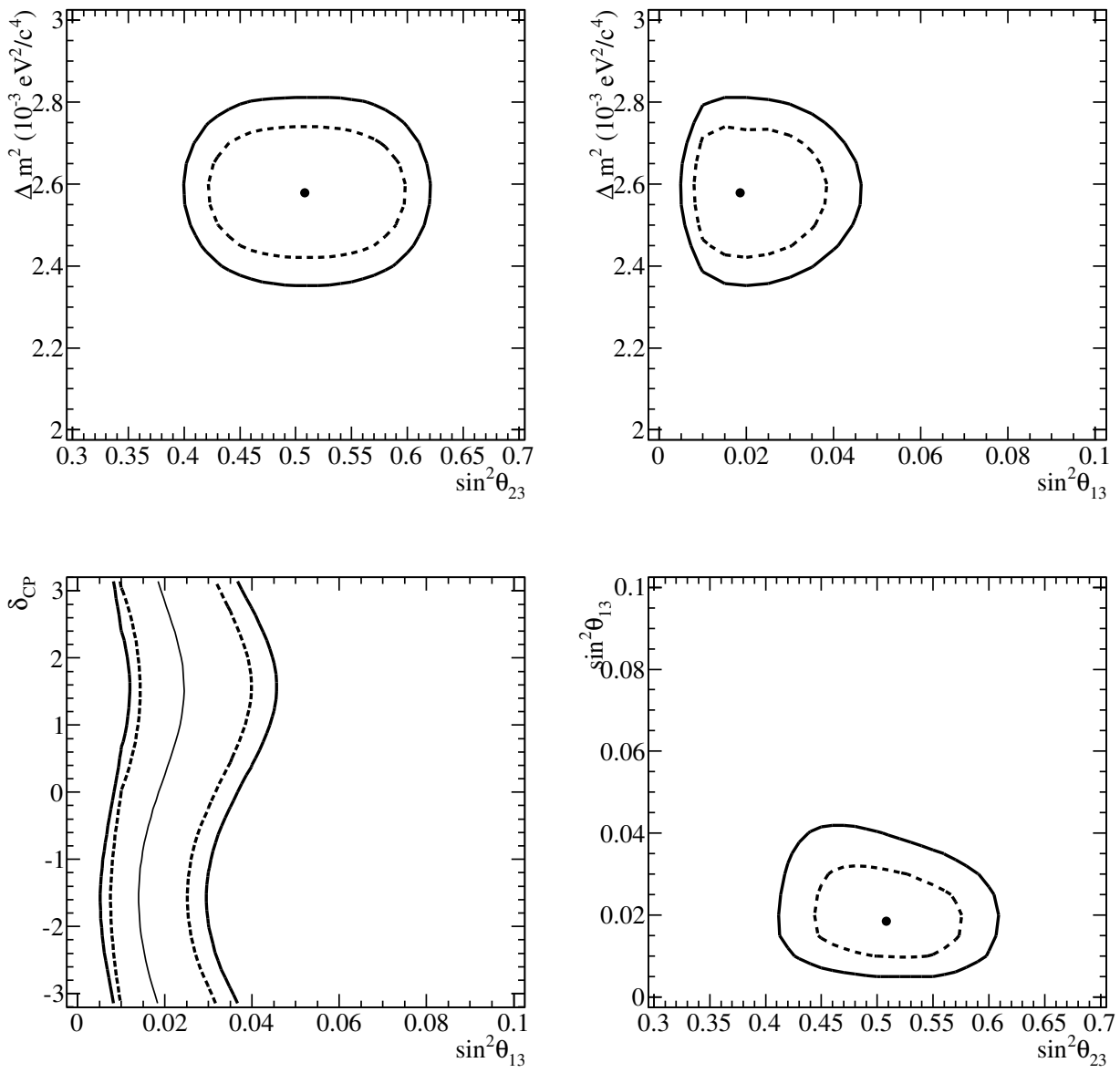


Figure 150: 68% and 90% CL regions in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #2. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

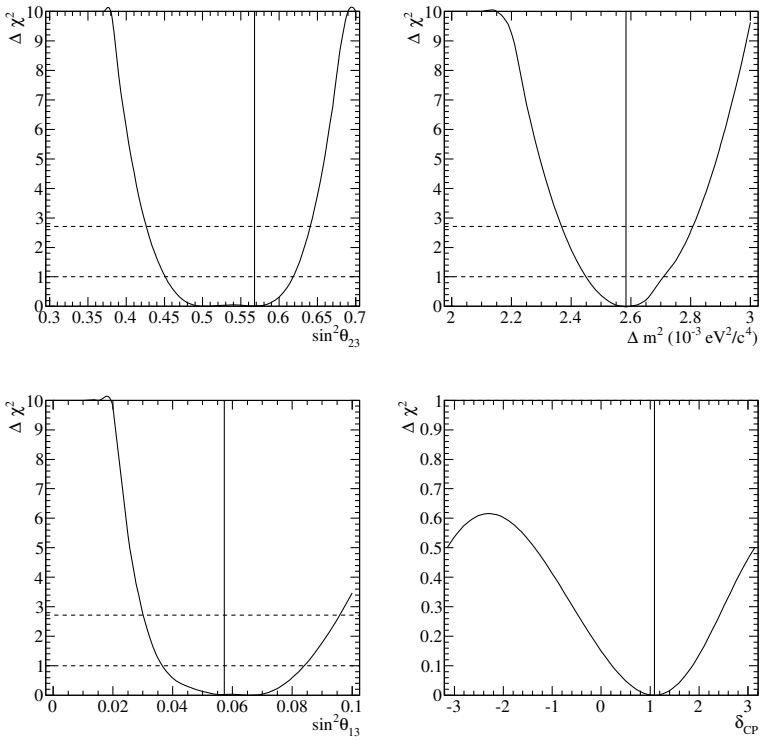


Figure 151: $\Delta\chi^2$ as function of $\sin^2\theta_{23}$,$|\Delta m^2|$,$\sin^2\theta_{13}$ and δ_{CP} for fake dataset #3. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

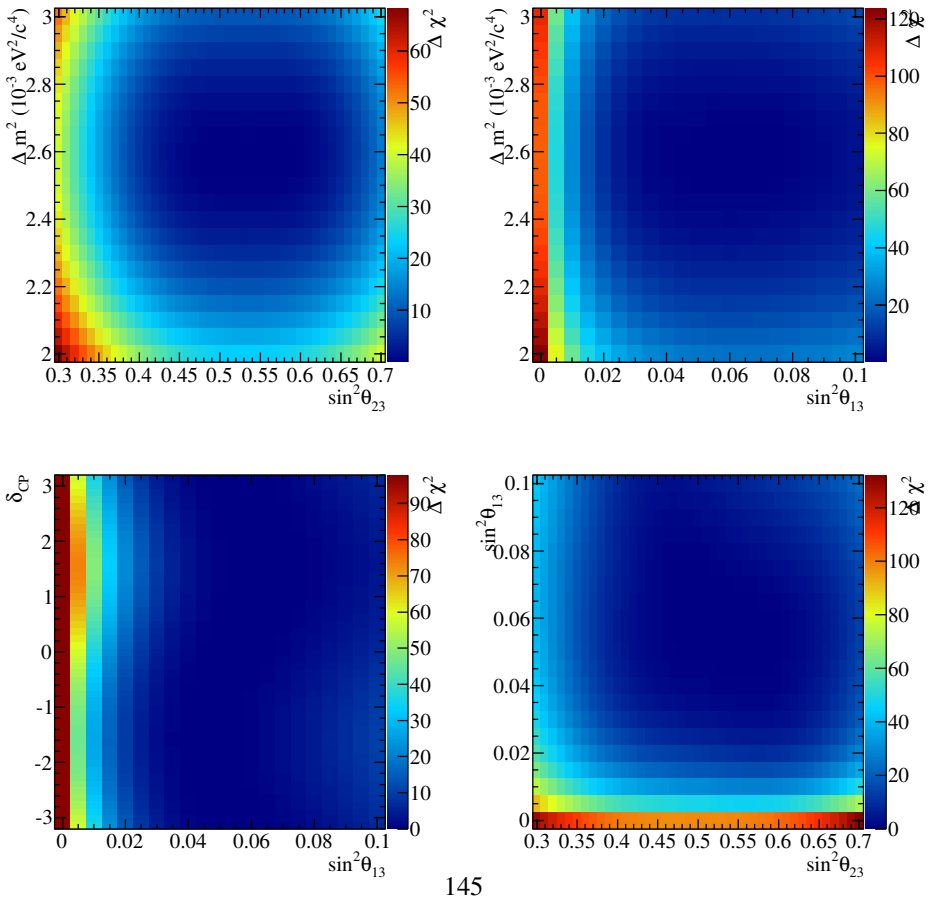


Figure 152: $\Delta\chi^2$ surface in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #3. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

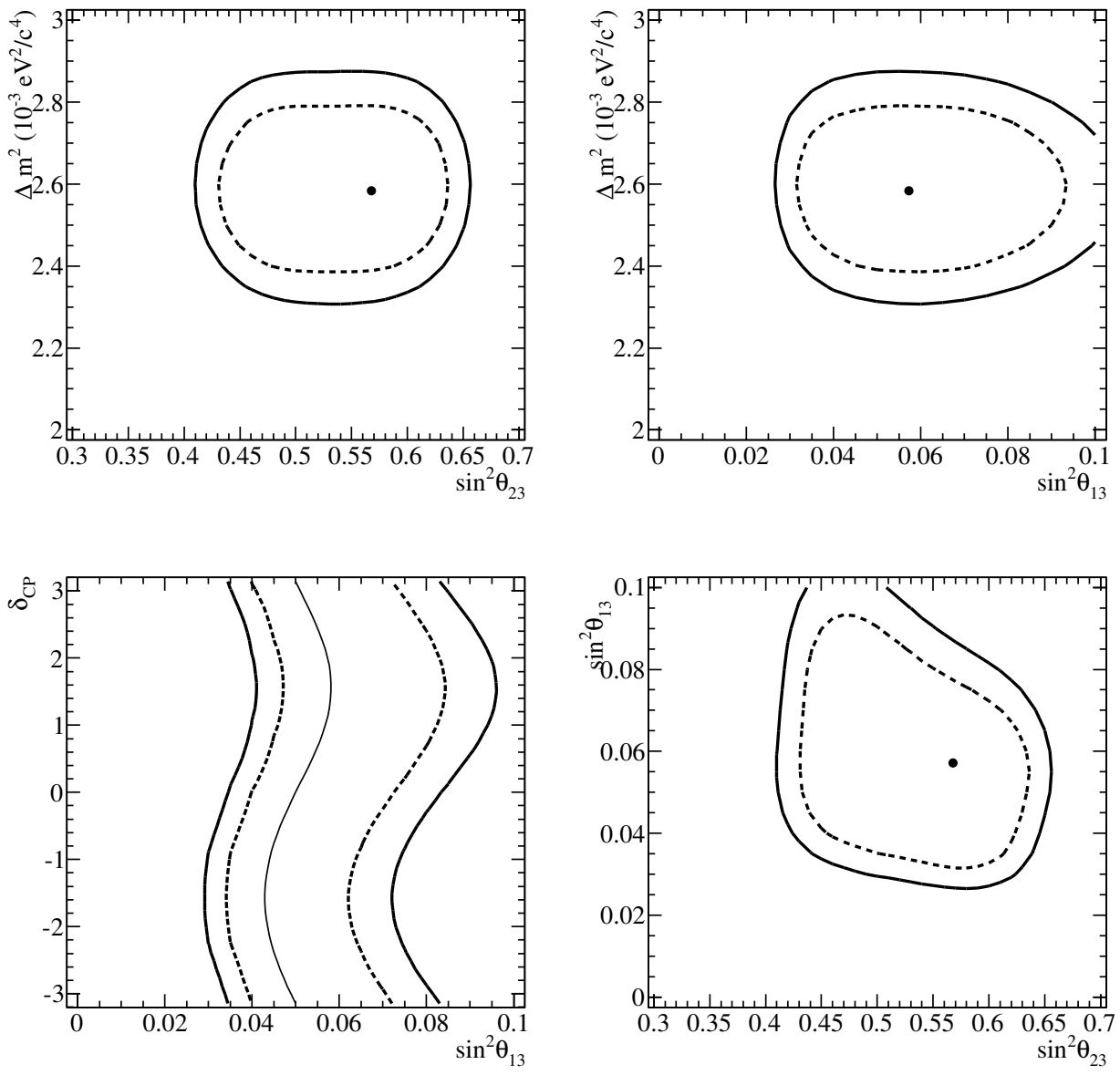


Figure 153: 68% and 90% CL regions in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #3. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

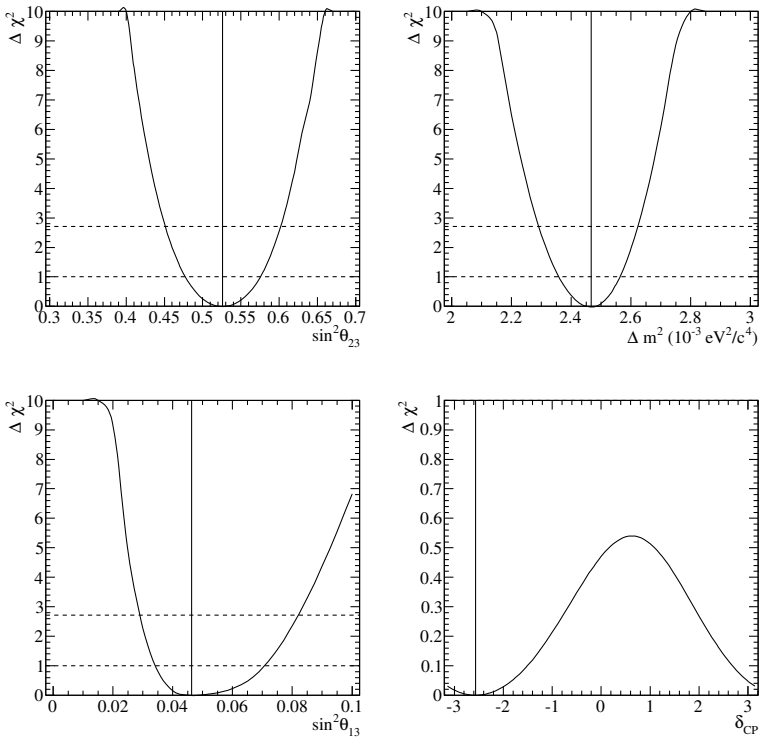


Figure 154: $\Delta\chi^2$ as function of $\sin^2\theta_{23}$,$|\Delta m^2|$,$\sin^2\theta_{13}$ and δ_{CP} for fake dataset #4. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

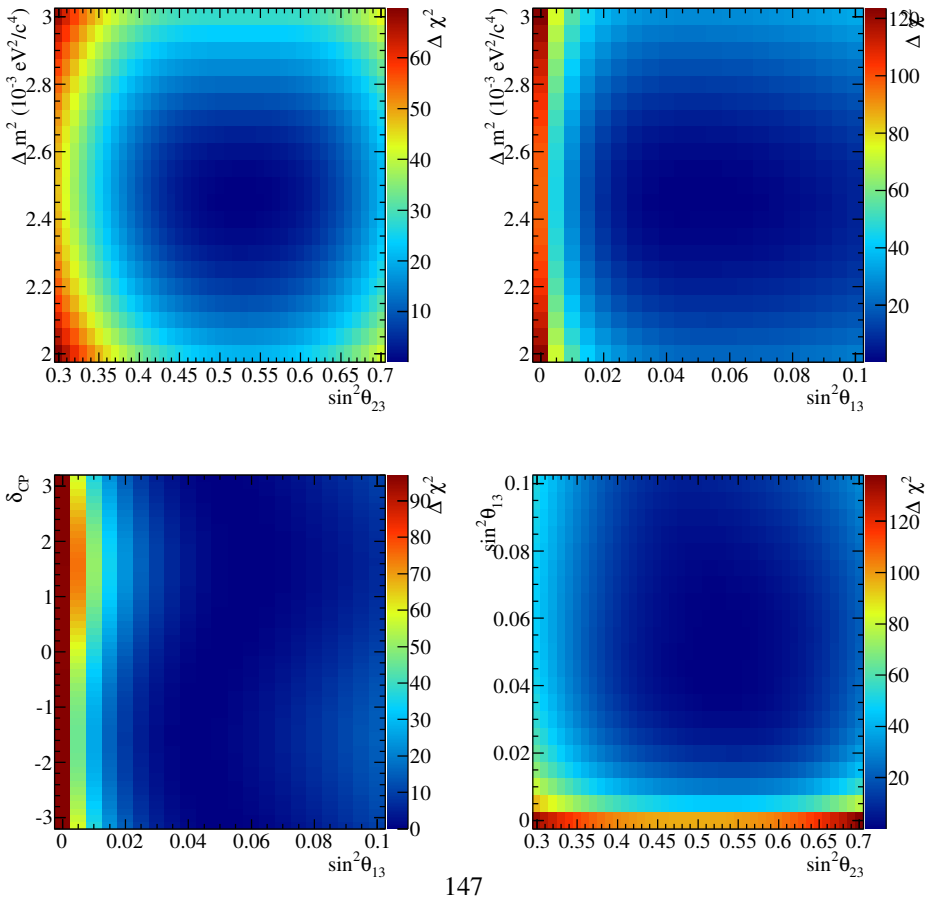


Figure 155: $\Delta\chi^2$ surface in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #4. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

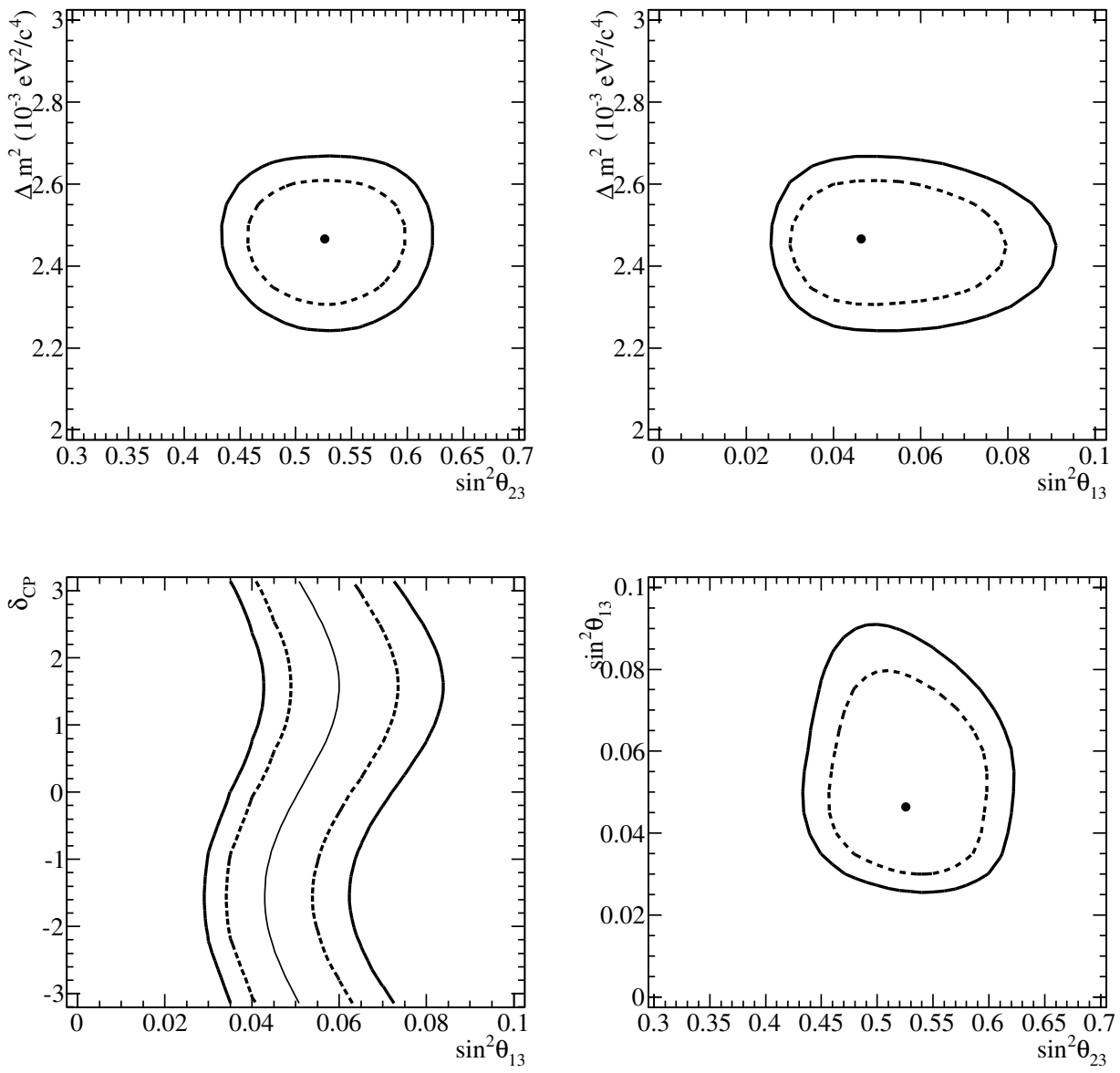


Figure 156: 68% and 90% CL regions in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #4. At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

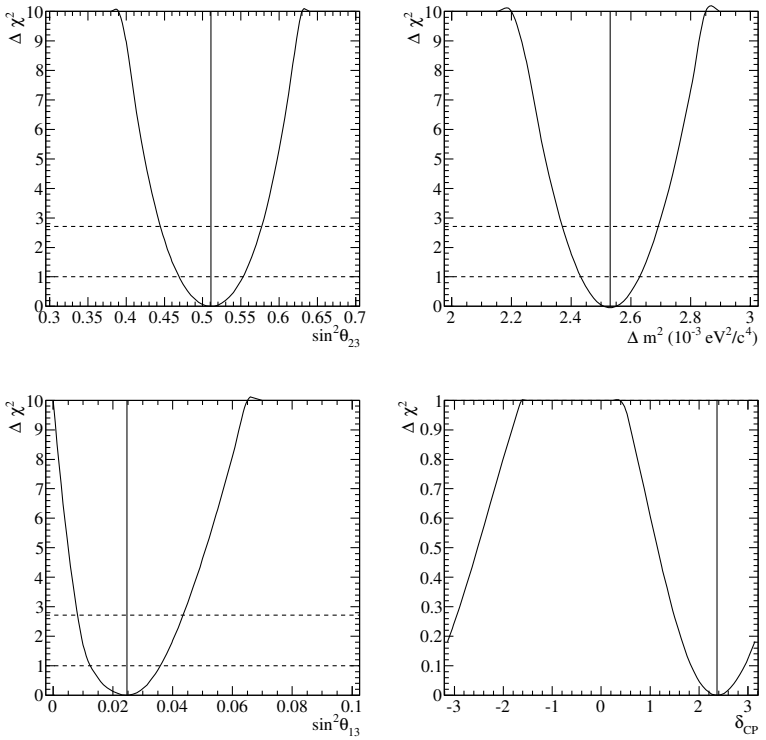


Figure 157: $\Delta\chi^2$ as function of $\sin^2\theta_{23}$,$|\Delta m^2|$,$\sin^2\theta_{13}$ and δ_{CP} for fake dataset #5 with inverted hierarchy (true one). At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

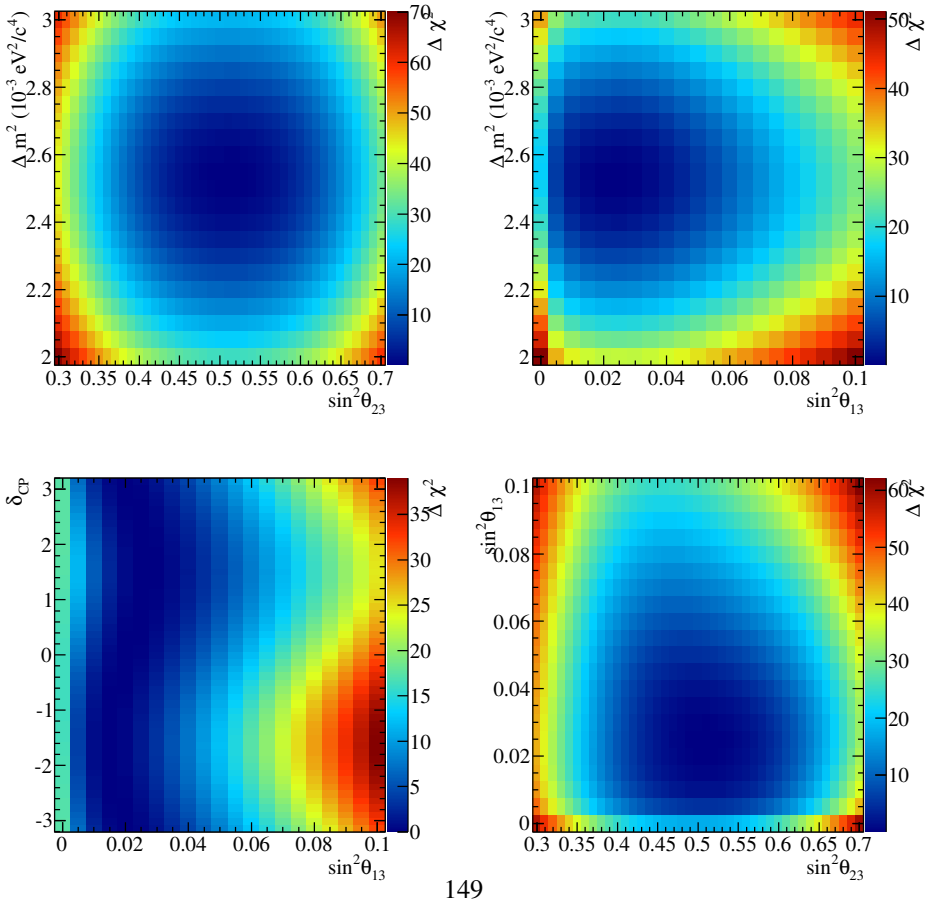


Figure 158: $\Delta\chi^2$ surface in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #5 with inverted hierarchy (true one). At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

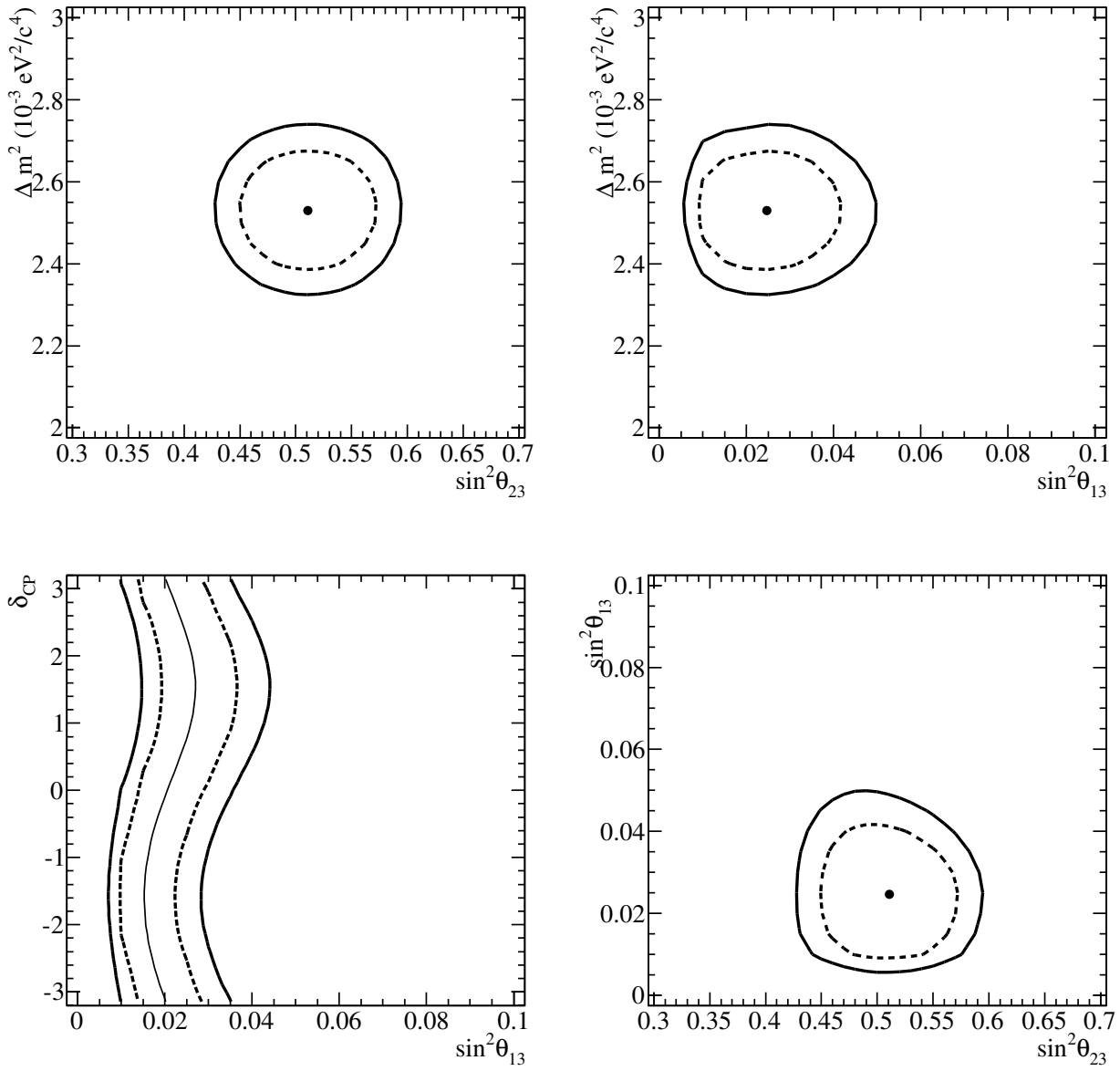


Figure 159: 68% and 90% CL regions in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #5 with inverted hierarchy (true one). At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

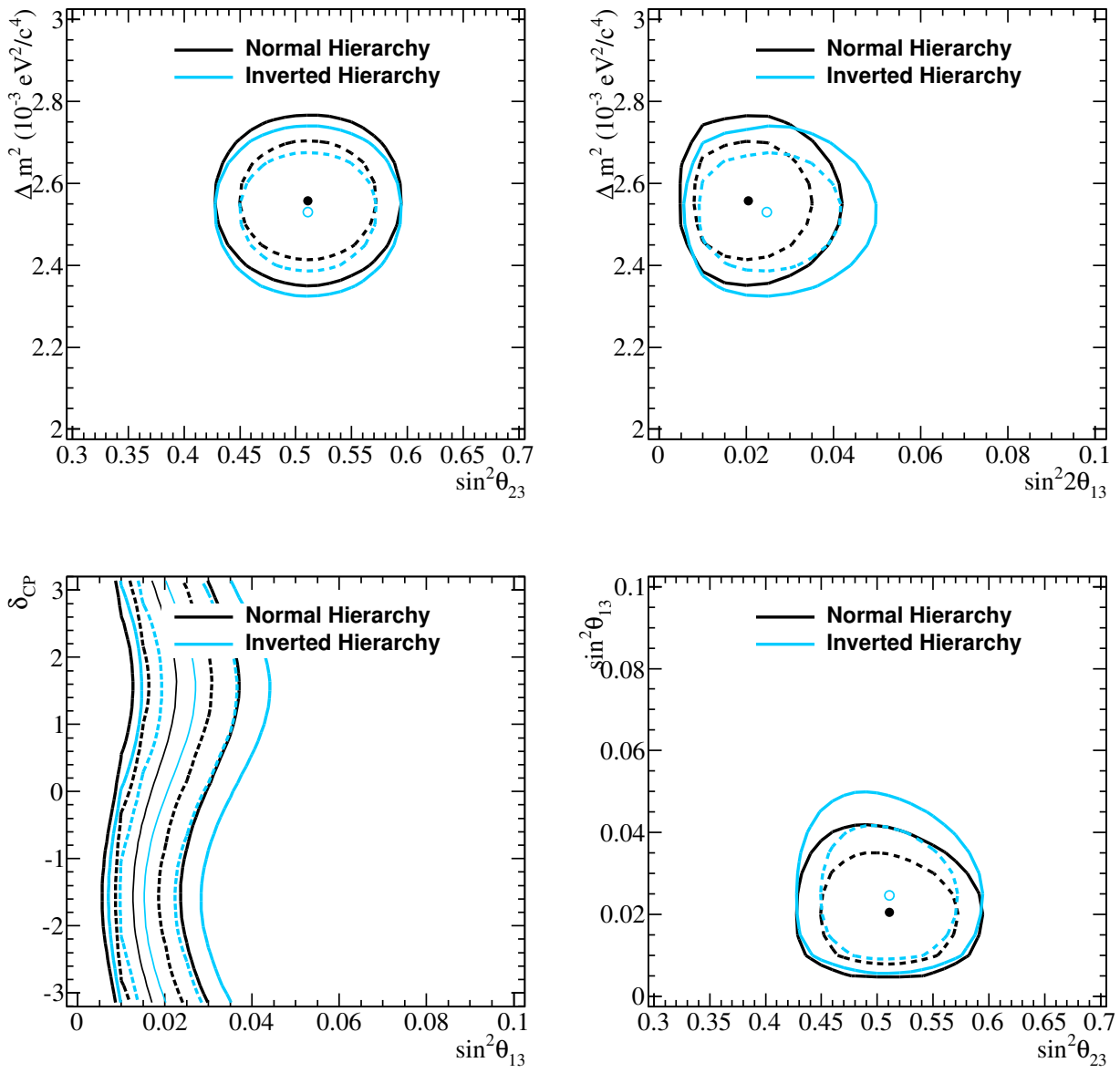


Figure 160: 68% and 90% CL regions in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #5 comparing results in inverted hierarchy (azure) and normal hierarchy (black). At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

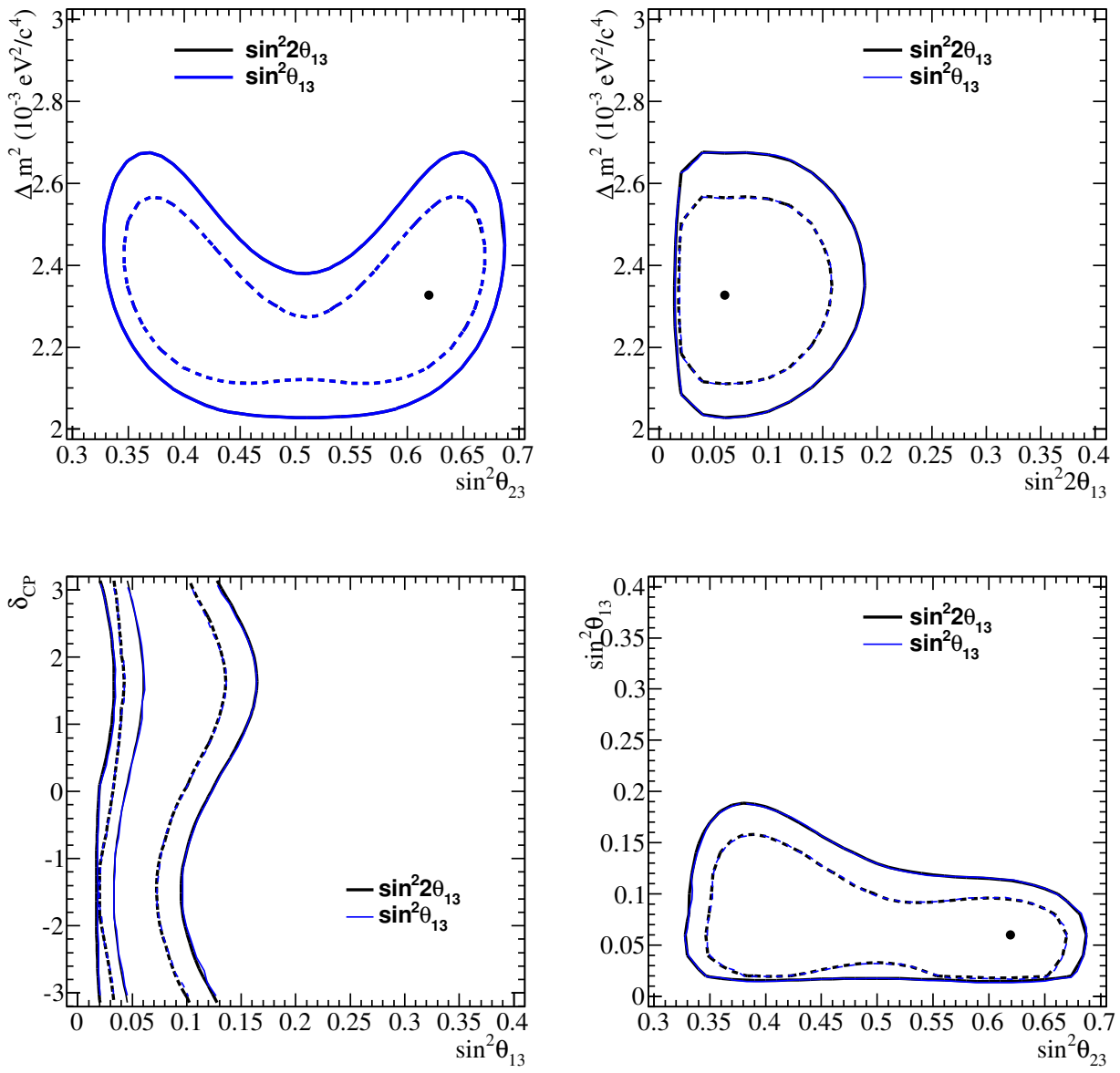


Figure 161: 68% and 90% CL regions in ($|\Delta m^2|$,$sin^2\theta_{23}$), ($|\Delta m^2|$,$sin^2\theta_{13}$), (δ_{CP},$sin^2\theta_{13}$) and ($sin^2\theta_{13}$,$sin^2\theta_{23}$) spaces for fake dataset #1 comparing results using $sin^2\theta_{13}$ (blue) and $sin^2 2\theta_{13}$ (black). At each point, χ^2 was minimized with respect to the oscillation parameters not shown in each figure and all 64 systematic parameters.

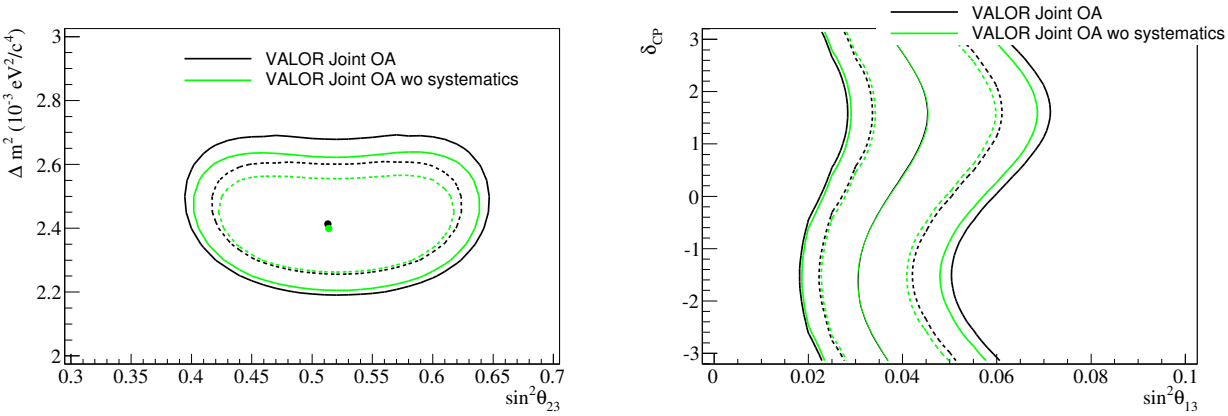


Figure 162: 68% and 90% CL regions in ($|\Delta m^2|$,$sin^2\theta_{23}$) and (δ_{CP},$sin^2\theta_{13}$) spaces for fake dataset #0 comparing results with (black) and without (green) including systematic parameters in the fit.

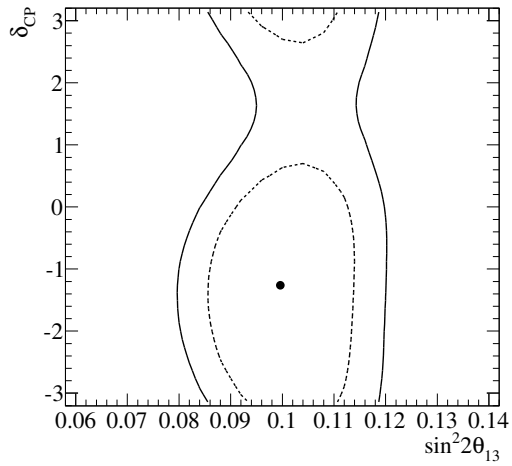


Figure 163: Confidence regions in the ($sin^2 2\theta_{13}$,δ_{CP}) parameter space for fake dataset #0 including constraint in $sin^2 2\theta_{13}$, assuming normal hierarchy.

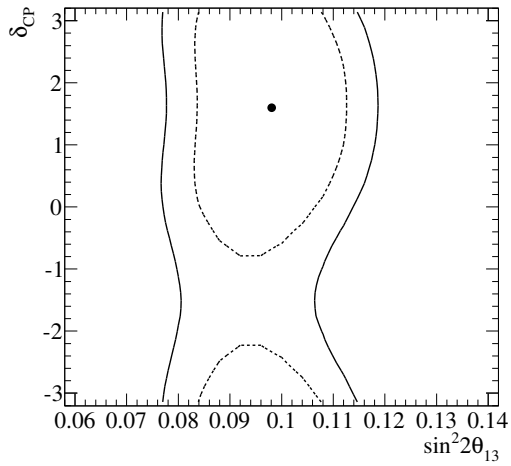


Figure 164: Confidence regions in the ($sin^2 2\theta_{13}$,δ_{CP}) parameter space for fake dataset #1 including constraint in $sin^2 2\theta_{13}$, assuming normal hierarchy.

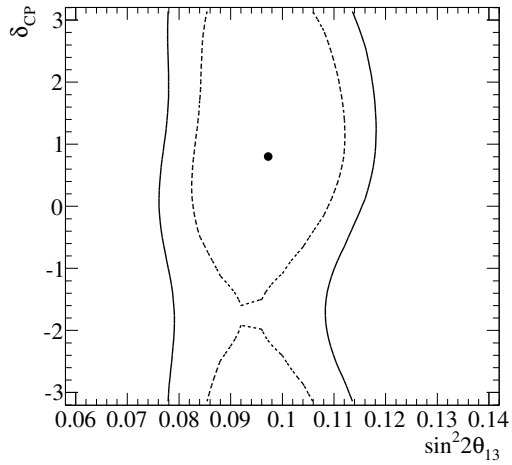


Figure 165: Confidence regions in the ($sin^2 2\theta_{13}$,δ_{CP}) parameter space for fake dataset #2 including constraint in $sin^2 2\theta_{13}$, assuming normal hierarchy.

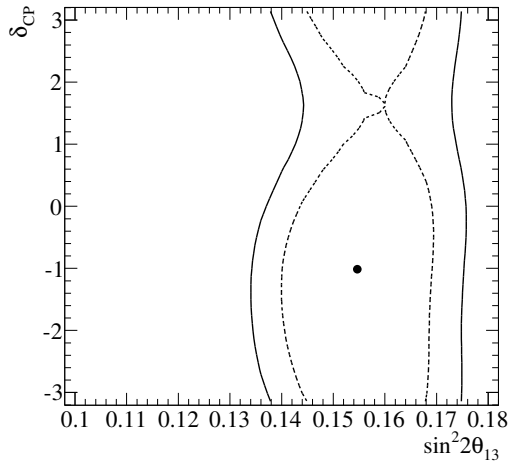


Figure 166: Confidence regions in the ($sin^2 2\theta_{13}$,δ_{CP}) parameter space for fake dataset #3 including constraint in $sin^2 2\theta_{13}$, assuming normal hierarchy.

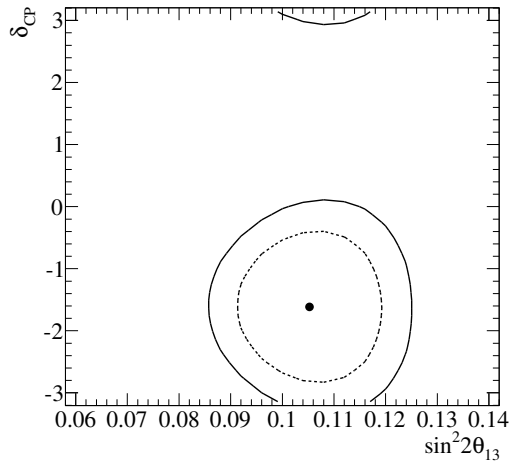


Figure 167: Confidence regions in the ($sin^2 2\theta_{13}$,δ_{CP}) parameter space for fake dataset #4 including constraint in $sin^2 2\theta_{13}$, assuming normal hierarchy.

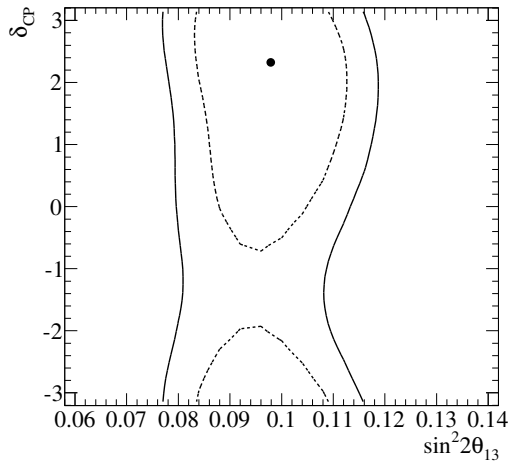


Figure 168: Confidence regions in the ($sin^2 2\theta_{13}$,δ_{CP}) parameter space for fake dataset #5 including constraint in $sin^2 2\theta_{13}$, assuming inverted hierarchy.

E. Extra plots for the toy dataset fits presented in D

This appendix includes plots with the best-fit values of the oscillation parameters not shown on each one of the two dimensional confidence region plots from the toy dataset fits presented in D.

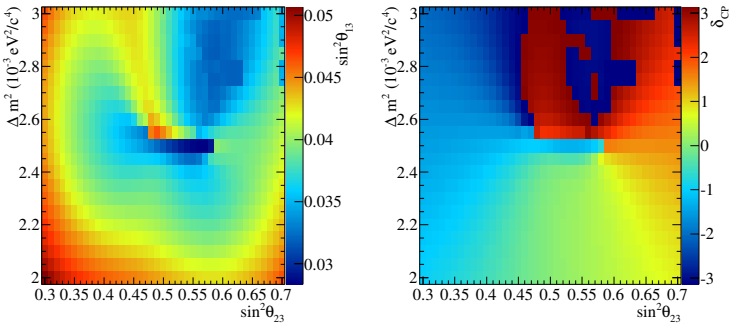


Figure 169: Best fit value of $sin^2\theta_{13}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{23}$) space for fake dataset #0.

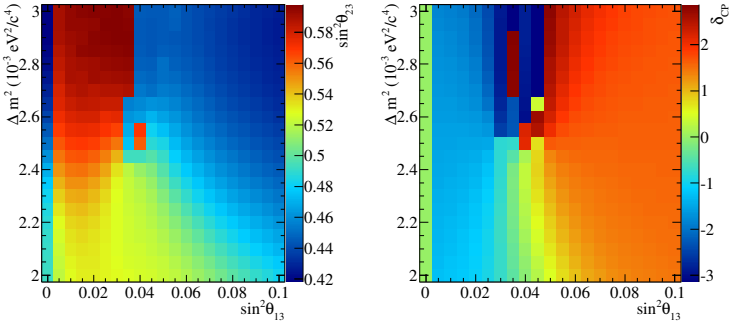


Figure 170: Best fit value of $sin^2\theta_{23}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{13}$) space for fake dataset #0.

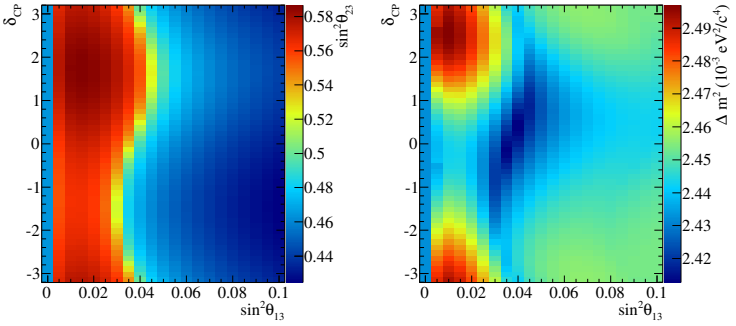
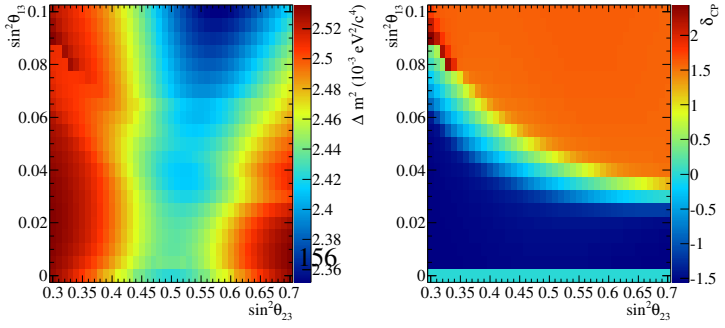


Figure 171: Best fit value of $sin^2\theta_{23}$ and $|\Delta m^2|$ for the fit in the (δ_{CP},$sin^2\theta_{23}$) space for fake dataset #0.



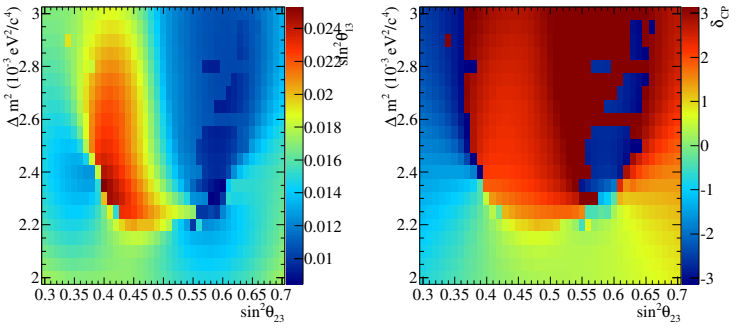


Figure 173: Best fit value of $sin^2\theta_{13}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{23}$) space for fake dataset #1.

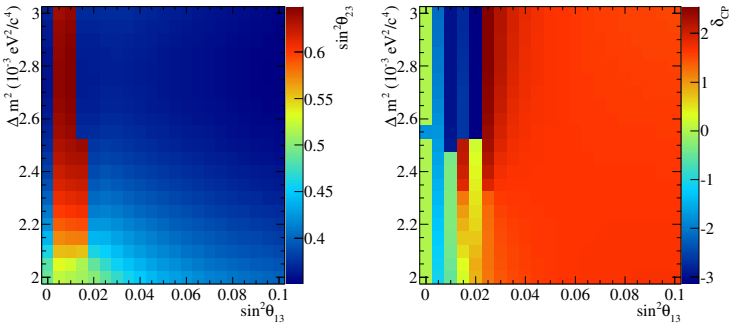


Figure 174: Best fit value of $sin^2\theta_{23}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{13}$) space for fake dataset #1.

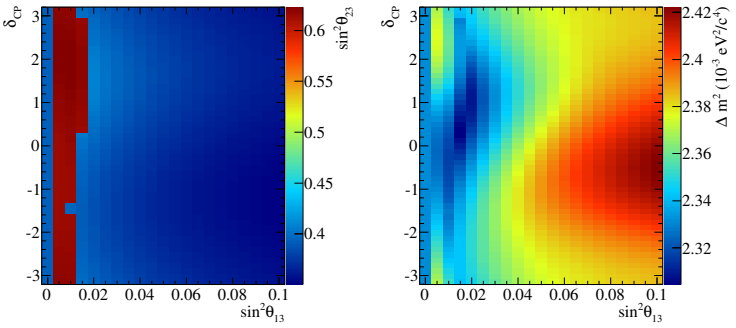
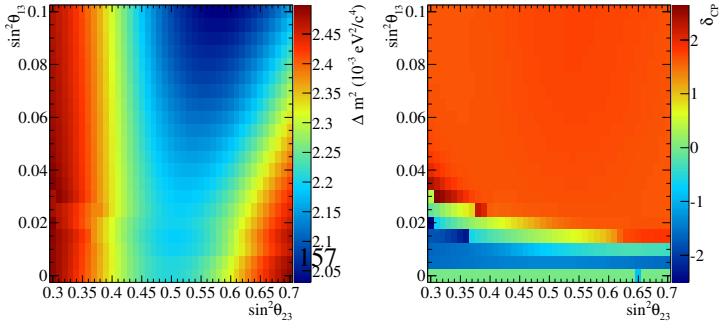


Figure 175: Best fit value of $sin^2\theta_{23}$ and $|\Delta m^2|$ for the fit in the (δ_{CP},$sin^2\theta_{23}$) space for fake dataset #1.



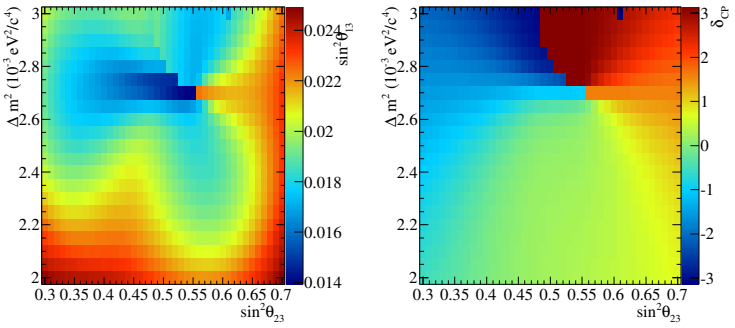


Figure 177: Best fit value of $sin^2\theta_{13}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{23}$) space for fake dataset #2.

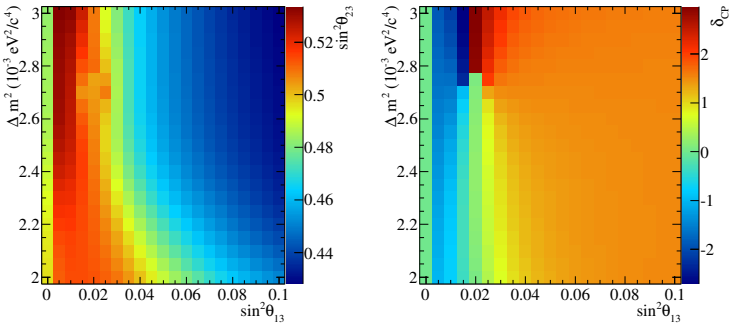


Figure 178: Best fit value of $sin^2\theta_{23}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{13}$) space for fake dataset #2.

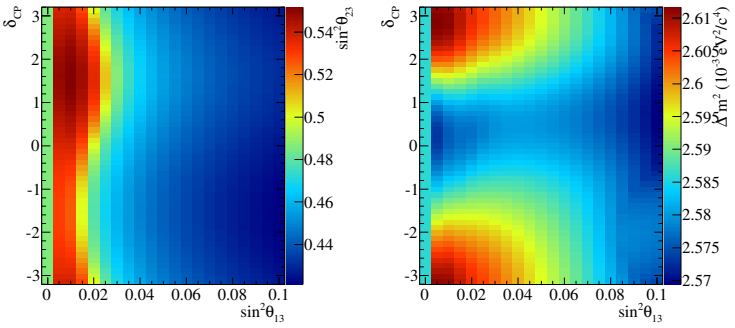
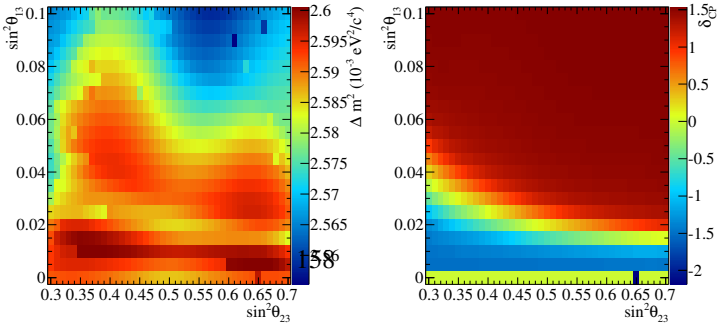


Figure 179: Best fit value of $sin^2\theta_{23}$ and $|\Delta m^2|$ for the fit in the (δ_{CP},$sin^2\theta_{23}$) space for fake dataset #2.



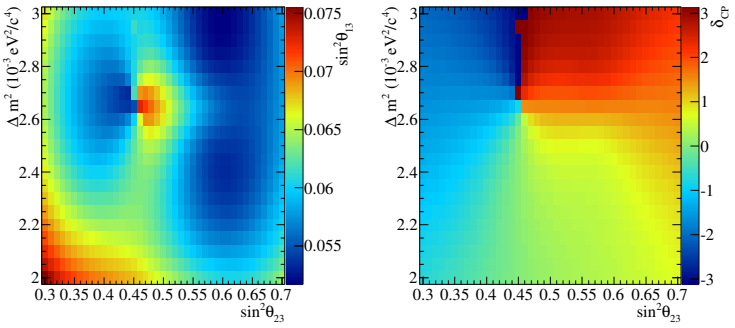


Figure 181: Best fit value of $sin^2\theta_{13}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{23}$) space for fake dataset #3.

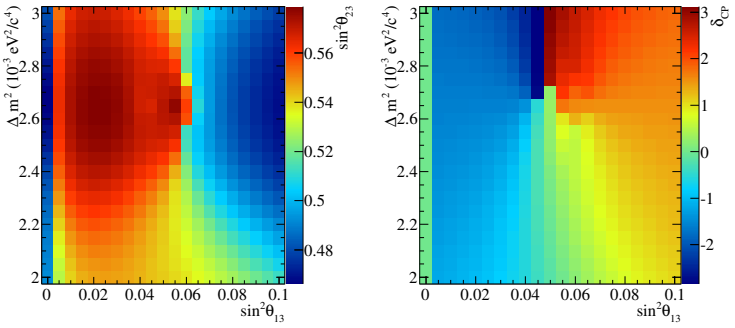


Figure 182: Best fit value of $sin^2\theta_{23}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{13}$) space for fake dataset #3.

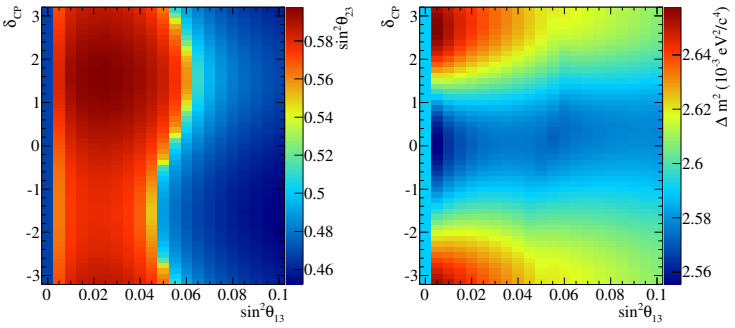
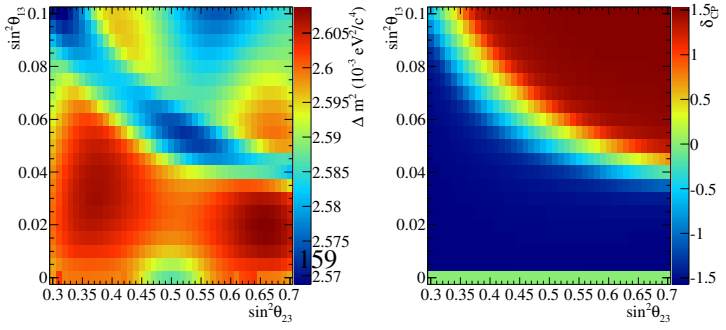


Figure 183: Best fit value of $sin^2\theta_{23}$ and $|\Delta m^2|$ for the fit in the (δ_{CP},$sin^2\theta_{23}$) space for fake dataset #3.



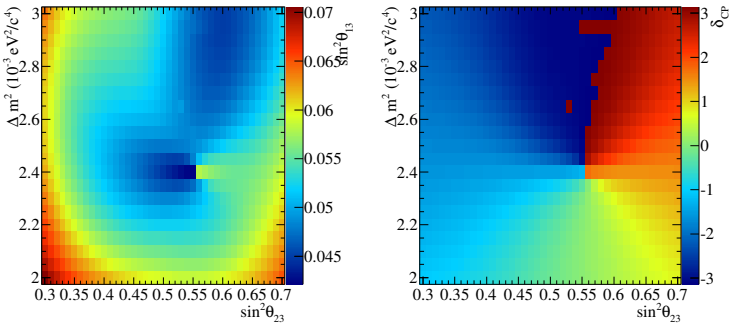


Figure 185: Best fit value of $sin^2\theta_{13}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{23}$) space for fake dataset #4.

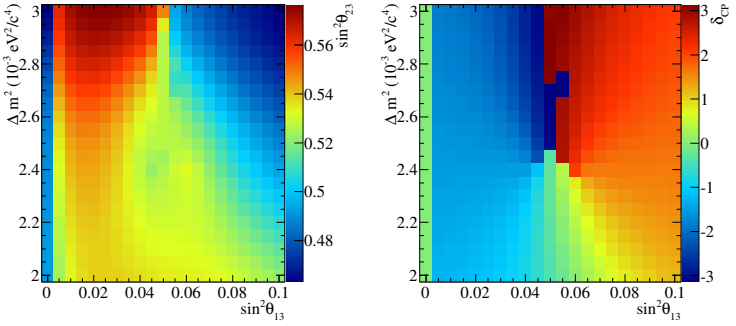


Figure 186: Best fit value of $sin^2\theta_{23}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{13}$) space for fake dataset #4.

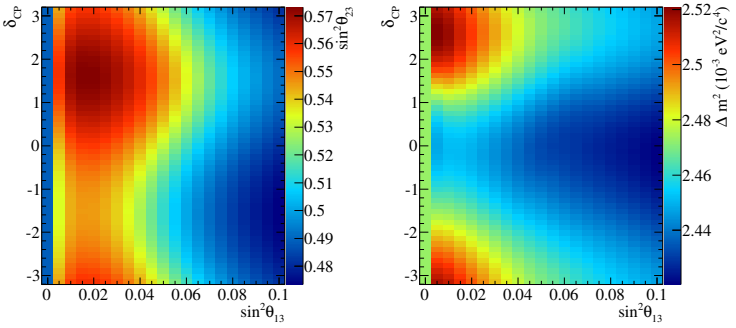
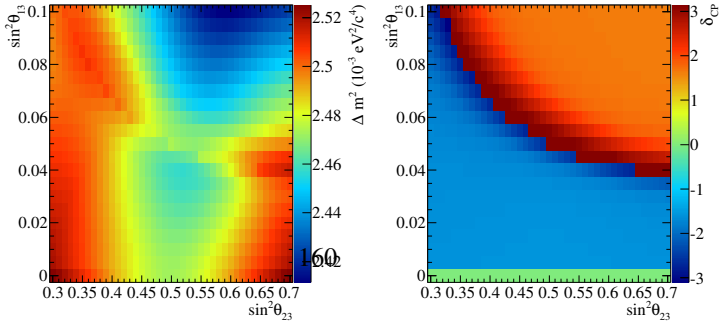


Figure 187: Best fit value of $sin^2\theta_{23}$ and $|\Delta m^2|$ for the fit in the (δ_{CP},$sin^2\theta_{23}$) space for fake dataset #4.



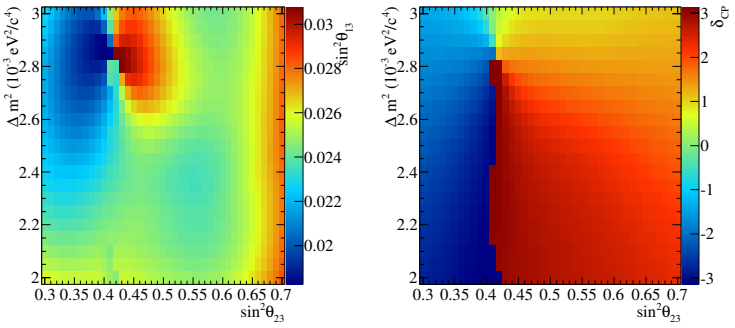


Figure 189: Best fit value of $sin^2\theta_{13}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{23}$) space for fake dataset #5 with inverted hierarchy (true one).

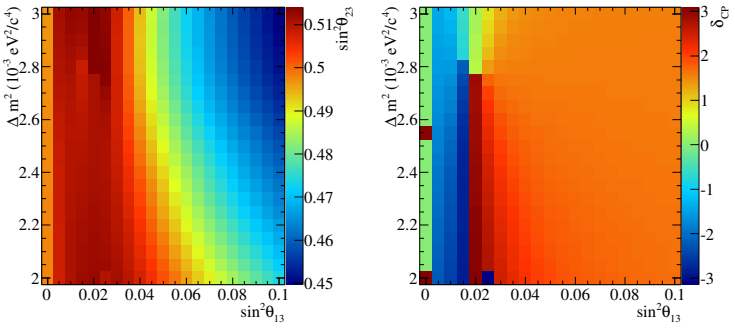


Figure 190: Best fit value of $sin^2\theta_{23}$ and δ_{CP} for the fit in the ($|\Delta m^2|$,$sin^2\theta_{13}$) space for fake dataset #5 with inverted hierarchy (true one).

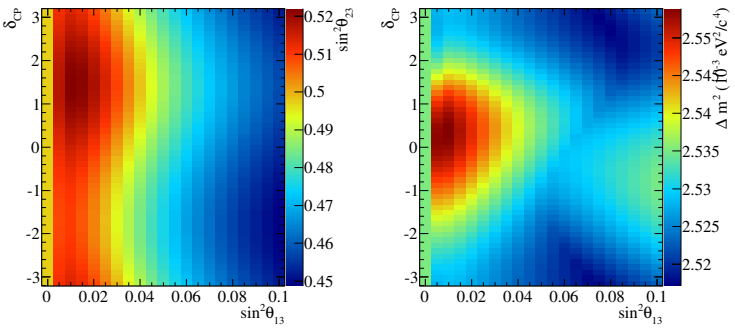


Figure 191: Best fit value of $sin^2\theta_{23}$ and $|\Delta m^2|$ for the fit in the (δ_{CP},$sin^2\theta_{23}$) space for fake dataset #5 with inverted hierarchy (true one).

